AEROSPACE OEM ENGINEERING COMPENDIUM — VOLUME 1

Covers 25 Certified Machine Models (Models #1 to #25)
350 Full-Page Fault SOPs | 125 Pages PLC Registers | 75 Pages BOM & PM Matrices

INCLUDED MACHINE MODELS IN THIS VOLUME:

• Model #1: Siemens-DMG-X150 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #2: Siemens-DMG-X200 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #3: Siemens-DMG-X250 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #4: Siemens-DMG-X300 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #5: Siemens-DMG-X350 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #6: Siemens-DMG-X400 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #7: Siemens-DMG-X450 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #8: Siemens-DMG-X500 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #9: Siemens-DMG-X550 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #10: Siemens-DMG-X600 (5-Axis Titanium Gantry Mill) — Controller: SINUMERIK ONE v6.15 — Prefix: CNC-MILL-XXXX
• Model #11: Makino-Aero-T150 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #12: Makino-Aero-T200 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #13: Makino-Aero-T250 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #14: Makino-Aero-T300 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #15: Makino-Aero-T350 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #16: Makino-Aero-T400 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #17: Makino-Aero-T450 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #18: Makino-Aero-T500 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #19: Makino-Aero-T550 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #20: Makino-Aero-T600 (Horizontal Aerospace Spar Mill) — Controller: Makino Professional 6 — Prefix: TI-PROFILER-XXXX
• Model #21: Mazak-Integrex-i150 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #22: Mazak-Integrex-i200 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #23: Mazak-Integrex-i250 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #24: Mazak-Integrex-i300 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-XXXX
• Model #25: Mazak-Integrex-i350 (Multi-Tasking Jet Shaft Center) — Controller: MAZATROL SmoothAi — Prefix: MT-TURN-

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OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 1/9)

MACHINE MODEL: Siemens-DMG-X150 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X150"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Siemens-DMG-X200 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X200"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: Siemens-DMG-X250 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X250"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 2/9)

MACHINE MODEL: Siemens-DMG-X300 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X300"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

MACHINE MODEL: Siemens-DMG-X350 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X350"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

MACHINE MODEL: Siemens-DMG-X400 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X400"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 3/9)

MACHINE MODEL: Siemens-DMG-X450 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X450"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: Siemens-DMG-X500 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X500"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

MACHINE MODEL: Siemens-DMG-X550 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X550"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 4/9)

MACHINE MODEL: Siemens-DMG-X600 (Asset Prefix: CNC-MILL)

• RAG Metadata Filter Value: "Siemens-DMG-X600"
• Equipment Classification: 5-Axis Titanium Gantry Mill | Manufacturer: Siemens-DMG
• CNC / PLC Control System: SINUMERIK ONE v6.15
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Makino-Aero-T150 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T150"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Makino-Aero-T200 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T200"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 5/9)

MACHINE MODEL: Makino-Aero-T250 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T250"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

MACHINE MODEL: Makino-Aero-T300 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T300"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

MACHINE MODEL: Makino-Aero-T350 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T350"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 6/9)

MACHINE MODEL: Makino-Aero-T400 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T400"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 21000 RPM
• Coolant / Process Supply Pressure: 118 Bar High-Pressure Loop
• Thermal Operating Limit: 54°C Max Continuous | Electrical Load: 81 kW
• Volumetric Positioning Accuracy: ±0.0021 mm

MACHINE MODEL: Makino-Aero-T450 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T450"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 22500 RPM
• Coolant / Process Supply Pressure: 126 Bar High-Pressure Loop
• Thermal Operating Limit: 55°C Max Continuous | Electrical Load: 87 kW
• Volumetric Positioning Accuracy: ±0.0022 mm

MACHINE MODEL: Makino-Aero-T500 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T500"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 24000 RPM
• Coolant / Process Supply Pressure: 134 Bar High-Pressure Loop
• Thermal Operating Limit: 56°C Max Continuous | Electrical Load: 93 kW
• Volumetric Positioning Accuracy: ±0.0023 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 7/9)

MACHINE MODEL: Makino-Aero-T550 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T550"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 25500 RPM
• Coolant / Process Supply Pressure: 142 Bar High-Pressure Loop
• Thermal Operating Limit: 57°C Max Continuous | Electrical Load: 99 kW
• Volumetric Positioning Accuracy: ±0.0015 mm

MACHINE MODEL: Makino-Aero-T600 (Asset Prefix: TI-PROFILER)

• RAG Metadata Filter Value: "Makino-Aero-T600"
• Equipment Classification: Horizontal Aerospace Spar Mill | Manufacturer: Makino-Aero
• CNC / PLC Control System: Makino Professional 6
• Max Spindle / Primary Drive Rating: 27000 RPM
• Coolant / Process Supply Pressure: 150 Bar High-Pressure Loop
• Thermal Operating Limit: 48°C Max Continuous | Electrical Load: 105 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

MACHINE MODEL: Mazak-Integrex-i150 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i150"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 13500 RPM
• Coolant / Process Supply Pressure: 78 Bar High-Pressure Loop
• Thermal Operating Limit: 49°C Max Continuous | Electrical Load: 51 kW
• Volumetric Positioning Accuracy: ±0.0016 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 8/9)

MACHINE MODEL: Mazak-Integrex-i200 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i200"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 15000 RPM
• Coolant / Process Supply Pressure: 86 Bar High-Pressure Loop
• Thermal Operating Limit: 50°C Max Continuous | Electrical Load: 57 kW
• Volumetric Positioning Accuracy: ±0.0017 mm

MACHINE MODEL: Mazak-Integrex-i250 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i250"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 16500 RPM
• Coolant / Process Supply Pressure: 94 Bar High-Pressure Loop
• Thermal Operating Limit: 51°C Max Continuous | Electrical Load: 63 kW
• Volumetric Positioning Accuracy: ±0.0018 mm

MACHINE MODEL: Mazak-Integrex-i300 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i300"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 18000 RPM
• Coolant / Process Supply Pressure: 102 Bar High-Pressure Loop
• Thermal Operating Limit: 52°C Max Continuous | Electrical Load: 69 kW
• Volumetric Positioning Accuracy: ±0.0019 mm

OPERATING ENVELOPES & METADATA SPECIFICATIONS (PART 9/9)

MACHINE MODEL: Mazak-Integrex-i350 (Asset Prefix: MT-TURN)

• RAG Metadata Filter Value: "Mazak-Integrex-i350"
• Equipment Classification: Multi-Tasking Jet Shaft Center | Manufacturer: Mazak-Integrex
• CNC / PLC Control System: MAZATROL SmoothAi
• Max Spindle / Primary Drive Rating: 19500 RPM
• Coolant / Process Supply Pressure: 110 Bar High-Pressure Loop
• Thermal Operating Limit: 53°C Max Continuous | Electrical Load: 75 kW
• Volumetric Positioning Accuracy: ±0.0020 mm

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-SPN-1

Target Machine Model: Siemens-DMG-X150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE150-1
- PLC Hex Register Address: DB20.DBW_0x2328
- Nominal Operating Range: 15 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-BRG-CER-101
- Overhaul Seal & Consumable Kit: KIT-SIE150-SPN-1
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-SPN-1 at PLC register 0x2328.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-BRG-CER-101) with seal kit (KIT-SIE150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-SPN-1 on Siemens-DMG-X150 by installing PN-SIE150-BRG-CER-101 & kit KIT-SIE150-SPN-1. Fasteners torqued to 169 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-ATC-2

Target Machine Model: Siemens-DMG-X150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE150-2
- PLC Hex Register Address: DB20.DBW_0x2330
- Nominal Operating Range: 18 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-CAM-GRIP-102
- Overhaul Seal & Consumable Kit: KIT-SIE150-ATC-2
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-ATC-2 at PLC register 0x2330.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-CAM-GRIP-102) with seal kit (KIT-SIE150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-ATC-2 on Siemens-DMG-X150 by installing PN-SIE150-CAM-GRIP-102 & kit KIT-SIE150-ATC-2. Fasteners torqued to 178 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-HYD-3

Target Machine Model: Siemens-DMG-X150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE150-3
- PLC Hex Register Address: DB20.DBW_0x2338
- Nominal Operating Range: 21 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-VLV-PROP-103
- Overhaul Seal & Consumable Kit: KIT-SIE150-HYD-3
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-HYD-3 at PLC register 0x2338.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-VLV-PROP-103) with seal kit (KIT-SIE150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-HYD-3 on Siemens-DMG-X150 by installing PN-SIE150-VLV-PROP-103 & kit KIT-SIE150-HYD-3. Fasteners torqued to 187 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-CLN-4

Target Machine Model: Siemens-DMG-X150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE150-4
- PLC Hex Register Address: DB20.DBW_0x2340
- Nominal Operating Range: 24 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-PMP-PLG-104
- Overhaul Seal & Consumable Kit: KIT-SIE150-CLN-4
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-CLN-4 at PLC register 0x2340.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-PMP-PLG-104) with seal kit (KIT-SIE150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-CLN-4 on Siemens-DMG-X150 by installing PN-SIE150-PMP-PLG-104 & kit KIT-SIE150-CLN-4. Fasteners torqued to 196 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-XAX-5

Target Machine Model: Siemens-DMG-X150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE150-5
- PLC Hex Register Address: DB20.DBW_0x2348
- Nominal Operating Range: 27 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-LIN-COIL-105
- Overhaul Seal & Consumable Kit: KIT-SIE150-XAX-5
- Critical Fastener Torque Value: 205 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-XAX-5 at PLC register 0x2348.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-LIN-COIL-105) with seal kit (KIT-SIE150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 205 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-XAX-5 on Siemens-DMG-X150 by installing PN-SIE150-LIN-COIL-105 & kit KIT-SIE150-XAX-5. Fasteners torqued to 205 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-YAX-6

Target Machine Model: Siemens-DMG-X150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE150-6
- PLC Hex Register Address: DB20.DBW_0x2350
- Nominal Operating Range: 30 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-BSW-NUT-106
- Overhaul Seal & Consumable Kit: KIT-SIE150-YAX-6
- Critical Fastener Torque Value: 24 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-YAX-6 at PLC register 0x2350.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-BSW-NUT-106) with seal kit (KIT-SIE150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 24 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-YAX-6 on Siemens-DMG-X150 by installing PN-SIE150-BSW-NUT-106 & kit KIT-SIE150-YAX-6. Fasteners torqued to 24 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-ZAX-7

Target Machine Model: Siemens-DMG-X150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE150-7
- PLC Hex Register Address: DB20.DBW_0x2358
- Nominal Operating Range: 33 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-CYL-SEAL-107
- Overhaul Seal & Consumable Kit: KIT-SIE150-ZAX-7
- Critical Fastener Torque Value: 33 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-ZAX-7 at PLC register 0x2358.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-CYL-SEAL-107) with seal kit (KIT-SIE150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 33 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-ZAX-7 on Siemens-DMG-X150 by installing PN-SIE150-CYL-SEAL-107 & kit KIT-SIE150-ZAX-7. Fasteners torqued to 33 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-BAX-8

Target Machine Model: Siemens-DMG-X150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE150-8
- PLC Hex Register Address: DB20.DBW_0x2360
- Nominal Operating Range: 36 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-TRQ-STAT-108
- Overhaul Seal & Consumable Kit: KIT-SIE150-BAX-8
- Critical Fastener Torque Value: 42 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-BAX-8 at PLC register 0x2360.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-TRQ-STAT-108) with seal kit (KIT-SIE150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 42 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-BAX-8 on Siemens-DMG-X150 by installing PN-SIE150-TRQ-STAT-108 & kit KIT-SIE150-BAX-8. Fasteners torqued to 42 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-CAX-9

Target Machine Model: Siemens-DMG-X150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE150-9
- PLC Hex Register Address: DB20.DBW_0x2368
- Nominal Operating Range: 39 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-BRK-DISC-109
- Overhaul Seal & Consumable Kit: KIT-SIE150-CAX-9
- Critical Fastener Torque Value: 51 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-CAX-9 at PLC register 0x2368.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-BRK-DISC-109) with seal kit (KIT-SIE150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 51 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-CAX-9 on Siemens-DMG-X150 by installing PN-SIE150-BRK-DISC-109 & kit KIT-SIE150-CAX-9. Fasteners torqued to 51 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-OPT-10

Target Machine Model: Siemens-DMG-X150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE150-10
- PLC Hex Register Address: DB20.DBW_0x2370
- Nominal Operating Range: 42 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-GLASS-RD-110
- Overhaul Seal & Consumable Kit: KIT-SIE150-OPT-10
- Critical Fastener Torque Value: 60 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-OPT-10 at PLC register 0x2370.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-GLASS-RD-110) with seal kit (KIT-SIE150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 60 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-OPT-10 on Siemens-DMG-X150 by installing PN-SIE150-GLASS-RD-110 & kit KIT-SIE150-OPT-10. Fasteners torqued to 60 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-PLC-11

Target Machine Model: Siemens-DMG-X150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE150-11
- PLC Hex Register Address: DB20.DBW_0x2378
- Nominal Operating Range: 45 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-IGBT-MOD-111
- Overhaul Seal & Consumable Kit: KIT-SIE150-PLC-11
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-PLC-11 at PLC register 0x2378.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-IGBT-MOD-111) with seal kit (KIT-SIE150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-PLC-11 on Siemens-DMG-X150 by installing PN-SIE150-IGBT-MOD-111 & kit KIT-SIE150-PLC-11. Fasteners torqued to 69 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-LUB-12

Target Machine Model: Siemens-DMG-X150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE150-12
- PLC Hex Register Address: DB20.DBW_0x2380
- Nominal Operating Range: 48 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-INJ-MET-112
- Overhaul Seal & Consumable Kit: KIT-SIE150-LUB-12
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-LUB-12 at PLC register 0x2380.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-INJ-MET-112) with seal kit (KIT-SIE150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-LUB-12 on Siemens-DMG-X150 by installing PN-SIE150-INJ-MET-112 & kit KIT-SIE150-LUB-12. Fasteners torqued to 78 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-VAC-13

Target Machine Model: Siemens-DMG-X150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE150-13
- PLC Hex Register Address: DB20.DBW_0x2388
- Nominal Operating Range: 51 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-GASK-VIT-113
- Overhaul Seal & Consumable Kit: KIT-SIE150-VAC-13
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-VAC-13 at PLC register 0x2388.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-GASK-VIT-113) with seal kit (KIT-SIE150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-VAC-13 on Siemens-DMG-X150 by installing PN-SIE150-GASK-VIT-113 & kit KIT-SIE150-VAC-13. Fasteners torqued to 87 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE150-THM-14

Target Machine Model: Siemens-DMG-X150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X150 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE150-14
- PLC Hex Register Address: DB20.DBW_0x2390
- Nominal Operating Range: 54 - 60 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 95 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE150-HEX-CORE-114
- Overhaul Seal & Consumable Kit: KIT-SIE150-THM-14
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X150 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE150-THM-14 at PLC register 0x2390.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE150-HEX-CORE-114) with seal kit (KIT-SIE150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X150, technician must log: "Resolved ERR-SIE150-THM-14 on Siemens-DMG-X150 by installing PN-SIE150-HEX-CORE-114 & kit KIT-SIE150-THM-14. Fasteners torqued to 96 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-SPN-1

Target Machine Model: Siemens-DMG-X200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE200-1
- PLC Hex Register Address: DB21.DBW_0x2648
- Nominal Operating Range: 15 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-BRG-CER-201
- Overhaul Seal & Consumable Kit: KIT-SIE200-SPN-1
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-SPN-1 at PLC register 0x2648.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-BRG-CER-201) with seal kit (KIT-SIE200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-SPN-1 on Siemens-DMG-X200 by installing PN-SIE200-BRG-CER-201 & kit KIT-SIE200-SPN-1. Fasteners torqued to 119 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-ATC-2

Target Machine Model: Siemens-DMG-X200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE200-2
- PLC Hex Register Address: DB21.DBW_0x2650
- Nominal Operating Range: 18 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-CAM-GRIP-202
- Overhaul Seal & Consumable Kit: KIT-SIE200-ATC-2
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-ATC-2 at PLC register 0x2650.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-CAM-GRIP-202) with seal kit (KIT-SIE200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-ATC-2 on Siemens-DMG-X200 by installing PN-SIE200-CAM-GRIP-202 & kit KIT-SIE200-ATC-2. Fasteners torqued to 128 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-HYD-3

Target Machine Model: Siemens-DMG-X200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE200-3
- PLC Hex Register Address: DB21.DBW_0x2658
- Nominal Operating Range: 21 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-VLV-PROP-203
- Overhaul Seal & Consumable Kit: KIT-SIE200-HYD-3
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-HYD-3 at PLC register 0x2658.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-VLV-PROP-203) with seal kit (KIT-SIE200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-HYD-3 on Siemens-DMG-X200 by installing PN-SIE200-VLV-PROP-203 & kit KIT-SIE200-HYD-3. Fasteners torqued to 137 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-CLN-4

Target Machine Model: Siemens-DMG-X200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE200-4
- PLC Hex Register Address: DB21.DBW_0x2660
- Nominal Operating Range: 24 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-PMP-PLG-204
- Overhaul Seal & Consumable Kit: KIT-SIE200-CLN-4
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-CLN-4 at PLC register 0x2660.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-PMP-PLG-204) with seal kit (KIT-SIE200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-CLN-4 on Siemens-DMG-X200 by installing PN-SIE200-PMP-PLG-204 & kit KIT-SIE200-CLN-4. Fasteners torqued to 146 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-XAX-5

Target Machine Model: Siemens-DMG-X200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE200-5
- PLC Hex Register Address: DB21.DBW_0x2668
- Nominal Operating Range: 27 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-LIN-COIL-205
- Overhaul Seal & Consumable Kit: KIT-SIE200-XAX-5
- Critical Fastener Torque Value: 155 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-XAX-5 at PLC register 0x2668.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-LIN-COIL-205) with seal kit (KIT-SIE200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 155 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-XAX-5 on Siemens-DMG-X200 by installing PN-SIE200-LIN-COIL-205 & kit KIT-SIE200-XAX-5. Fasteners torqued to 155 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-YAX-6

Target Machine Model: Siemens-DMG-X200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE200-6
- PLC Hex Register Address: DB21.DBW_0x2670
- Nominal Operating Range: 30 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-BSW-NUT-206
- Overhaul Seal & Consumable Kit: KIT-SIE200-YAX-6
- Critical Fastener Torque Value: 164 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-YAX-6 at PLC register 0x2670.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-BSW-NUT-206) with seal kit (KIT-SIE200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 164 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-YAX-6 on Siemens-DMG-X200 by installing PN-SIE200-BSW-NUT-206 & kit KIT-SIE200-YAX-6. Fasteners torqued to 164 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-ZAX-7

Target Machine Model: Siemens-DMG-X200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE200-7
- PLC Hex Register Address: DB21.DBW_0x2678
- Nominal Operating Range: 33 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-CYL-SEAL-207
- Overhaul Seal & Consumable Kit: KIT-SIE200-ZAX-7
- Critical Fastener Torque Value: 173 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-ZAX-7 at PLC register 0x2678.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-CYL-SEAL-207) with seal kit (KIT-SIE200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 173 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-ZAX-7 on Siemens-DMG-X200 by installing PN-SIE200-CYL-SEAL-207 & kit KIT-SIE200-ZAX-7. Fasteners torqued to 173 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-BAX-8

Target Machine Model: Siemens-DMG-X200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE200-8
- PLC Hex Register Address: DB21.DBW_0x2680
- Nominal Operating Range: 36 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-TRQ-STAT-208
- Overhaul Seal & Consumable Kit: KIT-SIE200-BAX-8
- Critical Fastener Torque Value: 182 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-BAX-8 at PLC register 0x2680.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-TRQ-STAT-208) with seal kit (KIT-SIE200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 182 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-BAX-8 on Siemens-DMG-X200 by installing PN-SIE200-TRQ-STAT-208 & kit KIT-SIE200-BAX-8. Fasteners torqued to 182 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-CAX-9

Target Machine Model: Siemens-DMG-X200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE200-9
- PLC Hex Register Address: DB21.DBW_0x2688
- Nominal Operating Range: 39 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-BRK-DISC-209
- Overhaul Seal & Consumable Kit: KIT-SIE200-CAX-9
- Critical Fastener Torque Value: 191 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-CAX-9 at PLC register 0x2688.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-BRK-DISC-209) with seal kit (KIT-SIE200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 191 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-CAX-9 on Siemens-DMG-X200 by installing PN-SIE200-BRK-DISC-209 & kit KIT-SIE200-CAX-9. Fasteners torqued to 191 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-OPT-10

Target Machine Model: Siemens-DMG-X200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE200-10
- PLC Hex Register Address: DB21.DBW_0x2690
- Nominal Operating Range: 42 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-GLASS-RD-210
- Overhaul Seal & Consumable Kit: KIT-SIE200-OPT-10
- Critical Fastener Torque Value: 200 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-OPT-10 at PLC register 0x2690.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-GLASS-RD-210) with seal kit (KIT-SIE200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 200 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-OPT-10 on Siemens-DMG-X200 by installing PN-SIE200-GLASS-RD-210 & kit KIT-SIE200-OPT-10. Fasteners torqued to 200 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-PLC-11

Target Machine Model: Siemens-DMG-X200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE200-11
- PLC Hex Register Address: DB21.DBW_0x2698
- Nominal Operating Range: 45 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-IGBT-MOD-211
- Overhaul Seal & Consumable Kit: KIT-SIE200-PLC-11
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-PLC-11 at PLC register 0x2698.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-IGBT-MOD-211) with seal kit (KIT-SIE200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-PLC-11 on Siemens-DMG-X200 by installing PN-SIE200-IGBT-MOD-211 & kit KIT-SIE200-PLC-11. Fasteners torqued to 209 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-LUB-12

Target Machine Model: Siemens-DMG-X200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE200-12
- PLC Hex Register Address: DB21.DBW_0x26A0
- Nominal Operating Range: 48 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-INJ-MET-212
- Overhaul Seal & Consumable Kit: KIT-SIE200-LUB-12
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-LUB-12 at PLC register 0x26A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-INJ-MET-212) with seal kit (KIT-SIE200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-LUB-12 on Siemens-DMG-X200 by installing PN-SIE200-INJ-MET-212 & kit KIT-SIE200-LUB-12. Fasteners torqued to 28 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-VAC-13

Target Machine Model: Siemens-DMG-X200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE200-13
- PLC Hex Register Address: DB21.DBW_0x26A8
- Nominal Operating Range: 51 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-GASK-VIT-213
- Overhaul Seal & Consumable Kit: KIT-SIE200-VAC-13
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-VAC-13 at PLC register 0x26A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-GASK-VIT-213) with seal kit (KIT-SIE200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-VAC-13 on Siemens-DMG-X200 by installing PN-SIE200-GASK-VIT-213 & kit KIT-SIE200-VAC-13. Fasteners torqued to 37 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE200-THM-14

Target Machine Model: Siemens-DMG-X200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X200 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE200-14
- PLC Hex Register Address: DB21.DBW_0x26B0
- Nominal Operating Range: 54 - 61 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 96 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE200-HEX-CORE-214
- Overhaul Seal & Consumable Kit: KIT-SIE200-THM-14
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X200 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE200-THM-14 at PLC register 0x26B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE200-HEX-CORE-214) with seal kit (KIT-SIE200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X200, technician must log: "Resolved ERR-SIE200-THM-14 on Siemens-DMG-X200 by installing PN-SIE200-HEX-CORE-214 & kit KIT-SIE200-THM-14. Fasteners torqued to 46 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-SPN-1

Target Machine Model: Siemens-DMG-X250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE250-1
- PLC Hex Register Address: DB22.DBW_0x2968
- Nominal Operating Range: 15 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-BRG-CER-301
- Overhaul Seal & Consumable Kit: KIT-SIE250-SPN-1
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-SPN-1 at PLC register 0x2968.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-BRG-CER-301) with seal kit (KIT-SIE250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-SPN-1 on Siemens-DMG-X250 by installing PN-SIE250-BRG-CER-301 & kit KIT-SIE250-SPN-1. Fasteners torqued to 69 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-ATC-2

Target Machine Model: Siemens-DMG-X250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE250-2
- PLC Hex Register Address: DB22.DBW_0x2970
- Nominal Operating Range: 18 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-CAM-GRIP-302
- Overhaul Seal & Consumable Kit: KIT-SIE250-ATC-2
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-ATC-2 at PLC register 0x2970.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-CAM-GRIP-302) with seal kit (KIT-SIE250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-ATC-2 on Siemens-DMG-X250 by installing PN-SIE250-CAM-GRIP-302 & kit KIT-SIE250-ATC-2. Fasteners torqued to 78 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-HYD-3

Target Machine Model: Siemens-DMG-X250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE250-3
- PLC Hex Register Address: DB22.DBW_0x2978
- Nominal Operating Range: 21 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-VLV-PROP-303
- Overhaul Seal & Consumable Kit: KIT-SIE250-HYD-3
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-HYD-3 at PLC register 0x2978.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-VLV-PROP-303) with seal kit (KIT-SIE250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-HYD-3 on Siemens-DMG-X250 by installing PN-SIE250-VLV-PROP-303 & kit KIT-SIE250-HYD-3. Fasteners torqued to 87 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-CLN-4

Target Machine Model: Siemens-DMG-X250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE250-4
- PLC Hex Register Address: DB22.DBW_0x2980
- Nominal Operating Range: 24 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-PMP-PLG-304
- Overhaul Seal & Consumable Kit: KIT-SIE250-CLN-4
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-CLN-4 at PLC register 0x2980.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-PMP-PLG-304) with seal kit (KIT-SIE250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-CLN-4 on Siemens-DMG-X250 by installing PN-SIE250-PMP-PLG-304 & kit KIT-SIE250-CLN-4. Fasteners torqued to 96 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-XAX-5

Target Machine Model: Siemens-DMG-X250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE250-5
- PLC Hex Register Address: DB22.DBW_0x2988
- Nominal Operating Range: 27 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-LIN-COIL-305
- Overhaul Seal & Consumable Kit: KIT-SIE250-XAX-5
- Critical Fastener Torque Value: 105 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-XAX-5 at PLC register 0x2988.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-LIN-COIL-305) with seal kit (KIT-SIE250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 105 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-XAX-5 on Siemens-DMG-X250 by installing PN-SIE250-LIN-COIL-305 & kit KIT-SIE250-XAX-5. Fasteners torqued to 105 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-YAX-6

Target Machine Model: Siemens-DMG-X250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE250-6
- PLC Hex Register Address: DB22.DBW_0x2990
- Nominal Operating Range: 30 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-BSW-NUT-306
- Overhaul Seal & Consumable Kit: KIT-SIE250-YAX-6
- Critical Fastener Torque Value: 114 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-YAX-6 at PLC register 0x2990.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-BSW-NUT-306) with seal kit (KIT-SIE250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 114 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-YAX-6 on Siemens-DMG-X250 by installing PN-SIE250-BSW-NUT-306 & kit KIT-SIE250-YAX-6. Fasteners torqued to 114 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-ZAX-7

Target Machine Model: Siemens-DMG-X250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE250-7
- PLC Hex Register Address: DB22.DBW_0x2998
- Nominal Operating Range: 33 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-CYL-SEAL-307
- Overhaul Seal & Consumable Kit: KIT-SIE250-ZAX-7
- Critical Fastener Torque Value: 123 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-ZAX-7 at PLC register 0x2998.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-CYL-SEAL-307) with seal kit (KIT-SIE250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 123 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-ZAX-7 on Siemens-DMG-X250 by installing PN-SIE250-CYL-SEAL-307 & kit KIT-SIE250-ZAX-7. Fasteners torqued to 123 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-BAX-8

Target Machine Model: Siemens-DMG-X250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE250-8
- PLC Hex Register Address: DB22.DBW_0x29A0
- Nominal Operating Range: 36 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-TRQ-STAT-308
- Overhaul Seal & Consumable Kit: KIT-SIE250-BAX-8
- Critical Fastener Torque Value: 132 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-BAX-8 at PLC register 0x29A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-TRQ-STAT-308) with seal kit (KIT-SIE250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 132 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-BAX-8 on Siemens-DMG-X250 by installing PN-SIE250-TRQ-STAT-308 & kit KIT-SIE250-BAX-8. Fasteners torqued to 132 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-CAX-9

Target Machine Model: Siemens-DMG-X250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE250-9
- PLC Hex Register Address: DB22.DBW_0x29A8
- Nominal Operating Range: 39 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-BRK-DISC-309
- Overhaul Seal & Consumable Kit: KIT-SIE250-CAX-9
- Critical Fastener Torque Value: 141 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-CAX-9 at PLC register 0x29A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-BRK-DISC-309) with seal kit (KIT-SIE250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 141 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-CAX-9 on Siemens-DMG-X250 by installing PN-SIE250-BRK-DISC-309 & kit KIT-SIE250-CAX-9. Fasteners torqued to 141 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-OPT-10

Target Machine Model: Siemens-DMG-X250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE250-10
- PLC Hex Register Address: DB22.DBW_0x29B0
- Nominal Operating Range: 42 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-GLASS-RD-310
- Overhaul Seal & Consumable Kit: KIT-SIE250-OPT-10
- Critical Fastener Torque Value: 150 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-OPT-10 at PLC register 0x29B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-GLASS-RD-310) with seal kit (KIT-SIE250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 150 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-OPT-10 on Siemens-DMG-X250 by installing PN-SIE250-GLASS-RD-310 & kit KIT-SIE250-OPT-10. Fasteners torqued to 150 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-PLC-11

Target Machine Model: Siemens-DMG-X250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE250-11
- PLC Hex Register Address: DB22.DBW_0x29B8
- Nominal Operating Range: 45 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-IGBT-MOD-311
- Overhaul Seal & Consumable Kit: KIT-SIE250-PLC-11
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-PLC-11 at PLC register 0x29B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-IGBT-MOD-311) with seal kit (KIT-SIE250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-PLC-11 on Siemens-DMG-X250 by installing PN-SIE250-IGBT-MOD-311 & kit KIT-SIE250-PLC-11. Fasteners torqued to 159 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-LUB-12

Target Machine Model: Siemens-DMG-X250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE250-12
- PLC Hex Register Address: DB22.DBW_0x29C0
- Nominal Operating Range: 48 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-INJ-MET-312
- Overhaul Seal & Consumable Kit: KIT-SIE250-LUB-12
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-LUB-12 at PLC register 0x29C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-INJ-MET-312) with seal kit (KIT-SIE250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-LUB-12 on Siemens-DMG-X250 by installing PN-SIE250-INJ-MET-312 & kit KIT-SIE250-LUB-12. Fasteners torqued to 168 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-VAC-13

Target Machine Model: Siemens-DMG-X250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE250-13
- PLC Hex Register Address: DB22.DBW_0x29C8
- Nominal Operating Range: 51 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-GASK-VIT-313
- Overhaul Seal & Consumable Kit: KIT-SIE250-VAC-13
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-VAC-13 at PLC register 0x29C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-GASK-VIT-313) with seal kit (KIT-SIE250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-VAC-13 on Siemens-DMG-X250 by installing PN-SIE250-GASK-VIT-313 & kit KIT-SIE250-VAC-13. Fasteners torqued to 177 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE250-THM-14

Target Machine Model: Siemens-DMG-X250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X250 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE250-14
- PLC Hex Register Address: DB22.DBW_0x29D0
- Nominal Operating Range: 54 - 62 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 97 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE250-HEX-CORE-314
- Overhaul Seal & Consumable Kit: KIT-SIE250-THM-14
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X250 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE250-THM-14 at PLC register 0x29D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE250-HEX-CORE-314) with seal kit (KIT-SIE250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X250, technician must log: "Resolved ERR-SIE250-THM-14 on Siemens-DMG-X250 by installing PN-SIE250-HEX-CORE-314 & kit KIT-SIE250-THM-14. Fasteners torqued to 186 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-SPN-1

Target Machine Model: Siemens-DMG-X300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE300-1
- PLC Hex Register Address: DB23.DBW_0x2C88
- Nominal Operating Range: 15 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-BRG-CER-401
- Overhaul Seal & Consumable Kit: KIT-SIE300-SPN-1
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-SPN-1 at PLC register 0x2C88.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-BRG-CER-401) with seal kit (KIT-SIE300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-SPN-1 on Siemens-DMG-X300 by installing PN-SIE300-BRG-CER-401 & kit KIT-SIE300-SPN-1. Fasteners torqued to 209 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-ATC-2

Target Machine Model: Siemens-DMG-X300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE300-2
- PLC Hex Register Address: DB23.DBW_0x2C90
- Nominal Operating Range: 18 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-CAM-GRIP-402
- Overhaul Seal & Consumable Kit: KIT-SIE300-ATC-2
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-ATC-2 at PLC register 0x2C90.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-CAM-GRIP-402) with seal kit (KIT-SIE300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-ATC-2 on Siemens-DMG-X300 by installing PN-SIE300-CAM-GRIP-402 & kit KIT-SIE300-ATC-2. Fasteners torqued to 28 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-HYD-3

Target Machine Model: Siemens-DMG-X300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE300-3
- PLC Hex Register Address: DB23.DBW_0x2C98
- Nominal Operating Range: 21 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-VLV-PROP-403
- Overhaul Seal & Consumable Kit: KIT-SIE300-HYD-3
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-HYD-3 at PLC register 0x2C98.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-VLV-PROP-403) with seal kit (KIT-SIE300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-HYD-3 on Siemens-DMG-X300 by installing PN-SIE300-VLV-PROP-403 & kit KIT-SIE300-HYD-3. Fasteners torqued to 37 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-CLN-4

Target Machine Model: Siemens-DMG-X300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE300-4
- PLC Hex Register Address: DB23.DBW_0x2CA0
- Nominal Operating Range: 24 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-PMP-PLG-404
- Overhaul Seal & Consumable Kit: KIT-SIE300-CLN-4
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-CLN-4 at PLC register 0x2CA0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-PMP-PLG-404) with seal kit (KIT-SIE300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-CLN-4 on Siemens-DMG-X300 by installing PN-SIE300-PMP-PLG-404 & kit KIT-SIE300-CLN-4. Fasteners torqued to 46 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-XAX-5

Target Machine Model: Siemens-DMG-X300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE300-5
- PLC Hex Register Address: DB23.DBW_0x2CA8
- Nominal Operating Range: 27 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-LIN-COIL-405
- Overhaul Seal & Consumable Kit: KIT-SIE300-XAX-5
- Critical Fastener Torque Value: 55 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-XAX-5 at PLC register 0x2CA8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-LIN-COIL-405) with seal kit (KIT-SIE300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 55 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-XAX-5 on Siemens-DMG-X300 by installing PN-SIE300-LIN-COIL-405 & kit KIT-SIE300-XAX-5. Fasteners torqued to 55 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-YAX-6

Target Machine Model: Siemens-DMG-X300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE300-6
- PLC Hex Register Address: DB23.DBW_0x2CB0
- Nominal Operating Range: 30 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-BSW-NUT-406
- Overhaul Seal & Consumable Kit: KIT-SIE300-YAX-6
- Critical Fastener Torque Value: 64 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-YAX-6 at PLC register 0x2CB0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-BSW-NUT-406) with seal kit (KIT-SIE300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 64 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-YAX-6 on Siemens-DMG-X300 by installing PN-SIE300-BSW-NUT-406 & kit KIT-SIE300-YAX-6. Fasteners torqued to 64 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-ZAX-7

Target Machine Model: Siemens-DMG-X300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE300-7
- PLC Hex Register Address: DB23.DBW_0x2CB8
- Nominal Operating Range: 33 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-CYL-SEAL-407
- Overhaul Seal & Consumable Kit: KIT-SIE300-ZAX-7
- Critical Fastener Torque Value: 73 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-ZAX-7 at PLC register 0x2CB8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-CYL-SEAL-407) with seal kit (KIT-SIE300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 73 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-ZAX-7 on Siemens-DMG-X300 by installing PN-SIE300-CYL-SEAL-407 & kit KIT-SIE300-ZAX-7. Fasteners torqued to 73 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-BAX-8

Target Machine Model: Siemens-DMG-X300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE300-8
- PLC Hex Register Address: DB23.DBW_0x2CC0
- Nominal Operating Range: 36 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-TRQ-STAT-408
- Overhaul Seal & Consumable Kit: KIT-SIE300-BAX-8
- Critical Fastener Torque Value: 82 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-BAX-8 at PLC register 0x2CC0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-TRQ-STAT-408) with seal kit (KIT-SIE300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 82 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-BAX-8 on Siemens-DMG-X300 by installing PN-SIE300-TRQ-STAT-408 & kit KIT-SIE300-BAX-8. Fasteners torqued to 82 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-CAX-9

Target Machine Model: Siemens-DMG-X300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE300-9
- PLC Hex Register Address: DB23.DBW_0x2CC8
- Nominal Operating Range: 39 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-BRK-DISC-409
- Overhaul Seal & Consumable Kit: KIT-SIE300-CAX-9
- Critical Fastener Torque Value: 91 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-CAX-9 at PLC register 0x2CC8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-BRK-DISC-409) with seal kit (KIT-SIE300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 91 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-CAX-9 on Siemens-DMG-X300 by installing PN-SIE300-BRK-DISC-409 & kit KIT-SIE300-CAX-9. Fasteners torqued to 91 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-OPT-10

Target Machine Model: Siemens-DMG-X300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE300-10
- PLC Hex Register Address: DB23.DBW_0x2CD0
- Nominal Operating Range: 42 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-GLASS-RD-410
- Overhaul Seal & Consumable Kit: KIT-SIE300-OPT-10
- Critical Fastener Torque Value: 100 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-OPT-10 at PLC register 0x2CD0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-GLASS-RD-410) with seal kit (KIT-SIE300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 100 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-OPT-10 on Siemens-DMG-X300 by installing PN-SIE300-GLASS-RD-410 & kit KIT-SIE300-OPT-10. Fasteners torqued to 100 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-PLC-11

Target Machine Model: Siemens-DMG-X300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE300-11
- PLC Hex Register Address: DB23.DBW_0x2CD8
- Nominal Operating Range: 45 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-IGBT-MOD-411
- Overhaul Seal & Consumable Kit: KIT-SIE300-PLC-11
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-PLC-11 at PLC register 0x2CD8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-IGBT-MOD-411) with seal kit (KIT-SIE300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-PLC-11 on Siemens-DMG-X300 by installing PN-SIE300-IGBT-MOD-411 & kit KIT-SIE300-PLC-11. Fasteners torqued to 109 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-LUB-12

Target Machine Model: Siemens-DMG-X300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE300-12
- PLC Hex Register Address: DB23.DBW_0x2CE0
- Nominal Operating Range: 48 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-INJ-MET-412
- Overhaul Seal & Consumable Kit: KIT-SIE300-LUB-12
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-LUB-12 at PLC register 0x2CE0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-INJ-MET-412) with seal kit (KIT-SIE300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-LUB-12 on Siemens-DMG-X300 by installing PN-SIE300-INJ-MET-412 & kit KIT-SIE300-LUB-12. Fasteners torqued to 118 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-VAC-13

Target Machine Model: Siemens-DMG-X300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE300-13
- PLC Hex Register Address: DB23.DBW_0x2CE8
- Nominal Operating Range: 51 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-GASK-VIT-413
- Overhaul Seal & Consumable Kit: KIT-SIE300-VAC-13
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-VAC-13 at PLC register 0x2CE8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-GASK-VIT-413) with seal kit (KIT-SIE300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-VAC-13 on Siemens-DMG-X300 by installing PN-SIE300-GASK-VIT-413 & kit KIT-SIE300-VAC-13. Fasteners torqued to 127 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE300-THM-14

Target Machine Model: Siemens-DMG-X300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X300 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE300-14
- PLC Hex Register Address: DB23.DBW_0x2CF0
- Nominal Operating Range: 54 - 63 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 98 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE300-HEX-CORE-414
- Overhaul Seal & Consumable Kit: KIT-SIE300-THM-14
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X300 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE300-THM-14 at PLC register 0x2CF0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE300-HEX-CORE-414) with seal kit (KIT-SIE300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X300, technician must log: "Resolved ERR-SIE300-THM-14 on Siemens-DMG-X300 by installing PN-SIE300-HEX-CORE-414 & kit KIT-SIE300-THM-14. Fasteners torqued to 136 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-SPN-1

Target Machine Model: Siemens-DMG-X350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE350-1
- PLC Hex Register Address: DB24.DBW_0x2FA8
- Nominal Operating Range: 15 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-BRG-CER-501
- Overhaul Seal & Consumable Kit: KIT-SIE350-SPN-1
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-SPN-1 at PLC register 0x2FA8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-BRG-CER-501) with seal kit (KIT-SIE350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-SPN-1 on Siemens-DMG-X350 by installing PN-SIE350-BRG-CER-501 & kit KIT-SIE350-SPN-1. Fasteners torqued to 159 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-ATC-2

Target Machine Model: Siemens-DMG-X350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE350-2
- PLC Hex Register Address: DB24.DBW_0x2FB0
- Nominal Operating Range: 18 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-CAM-GRIP-502
- Overhaul Seal & Consumable Kit: KIT-SIE350-ATC-2
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-ATC-2 at PLC register 0x2FB0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-CAM-GRIP-502) with seal kit (KIT-SIE350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-ATC-2 on Siemens-DMG-X350 by installing PN-SIE350-CAM-GRIP-502 & kit KIT-SIE350-ATC-2. Fasteners torqued to 168 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-HYD-3

Target Machine Model: Siemens-DMG-X350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE350-3
- PLC Hex Register Address: DB24.DBW_0x2FB8
- Nominal Operating Range: 21 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-VLV-PROP-503
- Overhaul Seal & Consumable Kit: KIT-SIE350-HYD-3
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-HYD-3 at PLC register 0x2FB8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-VLV-PROP-503) with seal kit (KIT-SIE350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-HYD-3 on Siemens-DMG-X350 by installing PN-SIE350-VLV-PROP-503 & kit KIT-SIE350-HYD-3. Fasteners torqued to 177 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-CLN-4

Target Machine Model: Siemens-DMG-X350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE350-4
- PLC Hex Register Address: DB24.DBW_0x2FC0
- Nominal Operating Range: 24 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-PMP-PLG-504
- Overhaul Seal & Consumable Kit: KIT-SIE350-CLN-4
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-CLN-4 at PLC register 0x2FC0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-PMP-PLG-504) with seal kit (KIT-SIE350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-CLN-4 on Siemens-DMG-X350 by installing PN-SIE350-PMP-PLG-504 & kit KIT-SIE350-CLN-4. Fasteners torqued to 186 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-XAX-5

Target Machine Model: Siemens-DMG-X350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE350-5
- PLC Hex Register Address: DB24.DBW_0x2FC8
- Nominal Operating Range: 27 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-LIN-COIL-505
- Overhaul Seal & Consumable Kit: KIT-SIE350-XAX-5
- Critical Fastener Torque Value: 195 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-XAX-5 at PLC register 0x2FC8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-LIN-COIL-505) with seal kit (KIT-SIE350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 195 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-XAX-5 on Siemens-DMG-X350 by installing PN-SIE350-LIN-COIL-505 & kit KIT-SIE350-XAX-5. Fasteners torqued to 195 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-YAX-6

Target Machine Model: Siemens-DMG-X350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE350-6
- PLC Hex Register Address: DB24.DBW_0x2FD0
- Nominal Operating Range: 30 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-BSW-NUT-506
- Overhaul Seal & Consumable Kit: KIT-SIE350-YAX-6
- Critical Fastener Torque Value: 204 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-YAX-6 at PLC register 0x2FD0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-BSW-NUT-506) with seal kit (KIT-SIE350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 204 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-YAX-6 on Siemens-DMG-X350 by installing PN-SIE350-BSW-NUT-506 & kit KIT-SIE350-YAX-6. Fasteners torqued to 204 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-ZAX-7

Target Machine Model: Siemens-DMG-X350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE350-7
- PLC Hex Register Address: DB24.DBW_0x2FD8
- Nominal Operating Range: 33 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-CYL-SEAL-507
- Overhaul Seal & Consumable Kit: KIT-SIE350-ZAX-7
- Critical Fastener Torque Value: 23 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-ZAX-7 at PLC register 0x2FD8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-CYL-SEAL-507) with seal kit (KIT-SIE350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 23 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-ZAX-7 on Siemens-DMG-X350 by installing PN-SIE350-CYL-SEAL-507 & kit KIT-SIE350-ZAX-7. Fasteners torqued to 23 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-BAX-8

Target Machine Model: Siemens-DMG-X350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE350-8
- PLC Hex Register Address: DB24.DBW_0x2FE0
- Nominal Operating Range: 36 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-TRQ-STAT-508
- Overhaul Seal & Consumable Kit: KIT-SIE350-BAX-8
- Critical Fastener Torque Value: 32 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-BAX-8 at PLC register 0x2FE0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-TRQ-STAT-508) with seal kit (KIT-SIE350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 32 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-BAX-8 on Siemens-DMG-X350 by installing PN-SIE350-TRQ-STAT-508 & kit KIT-SIE350-BAX-8. Fasteners torqued to 32 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-CAX-9

Target Machine Model: Siemens-DMG-X350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE350-9
- PLC Hex Register Address: DB24.DBW_0x2FE8
- Nominal Operating Range: 39 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-BRK-DISC-509
- Overhaul Seal & Consumable Kit: KIT-SIE350-CAX-9
- Critical Fastener Torque Value: 41 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-CAX-9 at PLC register 0x2FE8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-BRK-DISC-509) with seal kit (KIT-SIE350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 41 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-CAX-9 on Siemens-DMG-X350 by installing PN-SIE350-BRK-DISC-509 & kit KIT-SIE350-CAX-9. Fasteners torqued to 41 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-OPT-10

Target Machine Model: Siemens-DMG-X350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE350-10
- PLC Hex Register Address: DB24.DBW_0x2FF0
- Nominal Operating Range: 42 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-GLASS-RD-510
- Overhaul Seal & Consumable Kit: KIT-SIE350-OPT-10
- Critical Fastener Torque Value: 50 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-OPT-10 at PLC register 0x2FF0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-GLASS-RD-510) with seal kit (KIT-SIE350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 50 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-OPT-10 on Siemens-DMG-X350 by installing PN-SIE350-GLASS-RD-510 & kit KIT-SIE350-OPT-10. Fasteners torqued to 50 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-PLC-11

Target Machine Model: Siemens-DMG-X350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE350-11
- PLC Hex Register Address: DB24.DBW_0x2FF8
- Nominal Operating Range: 45 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-IGBT-MOD-511
- Overhaul Seal & Consumable Kit: KIT-SIE350-PLC-11
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-PLC-11 at PLC register 0x2FF8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-IGBT-MOD-511) with seal kit (KIT-SIE350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-PLC-11 on Siemens-DMG-X350 by installing PN-SIE350-IGBT-MOD-511 & kit KIT-SIE350-PLC-11. Fasteners torqued to 59 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-LUB-12

Target Machine Model: Siemens-DMG-X350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE350-12
- PLC Hex Register Address: DB24.DBW_0x3000
- Nominal Operating Range: 48 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-INJ-MET-512
- Overhaul Seal & Consumable Kit: KIT-SIE350-LUB-12
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-LUB-12 at PLC register 0x3000.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-INJ-MET-512) with seal kit (KIT-SIE350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-LUB-12 on Siemens-DMG-X350 by installing PN-SIE350-INJ-MET-512 & kit KIT-SIE350-LUB-12. Fasteners torqued to 68 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-VAC-13

Target Machine Model: Siemens-DMG-X350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE350-13
- PLC Hex Register Address: DB24.DBW_0x3008
- Nominal Operating Range: 51 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-GASK-VIT-513
- Overhaul Seal & Consumable Kit: KIT-SIE350-VAC-13
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-VAC-13 at PLC register 0x3008.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-GASK-VIT-513) with seal kit (KIT-SIE350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-VAC-13 on Siemens-DMG-X350 by installing PN-SIE350-GASK-VIT-513 & kit KIT-SIE350-VAC-13. Fasteners torqued to 77 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE350-THM-14

Target Machine Model: Siemens-DMG-X350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X350 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE350-14
- PLC Hex Register Address: DB24.DBW_0x3010
- Nominal Operating Range: 54 - 64 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 99 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE350-HEX-CORE-514
- Overhaul Seal & Consumable Kit: KIT-SIE350-THM-14
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X350 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE350-THM-14 at PLC register 0x3010.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE350-HEX-CORE-514) with seal kit (KIT-SIE350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X350, technician must log: "Resolved ERR-SIE350-THM-14 on Siemens-DMG-X350 by installing PN-SIE350-HEX-CORE-514 & kit KIT-SIE350-THM-14. Fasteners torqued to 86 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-SPN-1

Target Machine Model: Siemens-DMG-X400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE400-1
- PLC Hex Register Address: DB25.DBW_0x32C8
- Nominal Operating Range: 15 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-BRG-CER-601
- Overhaul Seal & Consumable Kit: KIT-SIE400-SPN-1
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-SPN-1 at PLC register 0x32C8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-BRG-CER-601) with seal kit (KIT-SIE400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-SPN-1 on Siemens-DMG-X400 by installing PN-SIE400-BRG-CER-601 & kit KIT-SIE400-SPN-1. Fasteners torqued to 109 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-ATC-2

Target Machine Model: Siemens-DMG-X400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE400-2
- PLC Hex Register Address: DB25.DBW_0x32D0
- Nominal Operating Range: 18 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-CAM-GRIP-602
- Overhaul Seal & Consumable Kit: KIT-SIE400-ATC-2
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-ATC-2 at PLC register 0x32D0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-CAM-GRIP-602) with seal kit (KIT-SIE400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-ATC-2 on Siemens-DMG-X400 by installing PN-SIE400-CAM-GRIP-602 & kit KIT-SIE400-ATC-2. Fasteners torqued to 118 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-HYD-3

Target Machine Model: Siemens-DMG-X400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE400-3
- PLC Hex Register Address: DB25.DBW_0x32D8
- Nominal Operating Range: 21 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-VLV-PROP-603
- Overhaul Seal & Consumable Kit: KIT-SIE400-HYD-3
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-HYD-3 at PLC register 0x32D8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-VLV-PROP-603) with seal kit (KIT-SIE400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-HYD-3 on Siemens-DMG-X400 by installing PN-SIE400-VLV-PROP-603 & kit KIT-SIE400-HYD-3. Fasteners torqued to 127 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-CLN-4

Target Machine Model: Siemens-DMG-X400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE400-4
- PLC Hex Register Address: DB25.DBW_0x32E0
- Nominal Operating Range: 24 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-PMP-PLG-604
- Overhaul Seal & Consumable Kit: KIT-SIE400-CLN-4
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-CLN-4 at PLC register 0x32E0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-PMP-PLG-604) with seal kit (KIT-SIE400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-CLN-4 on Siemens-DMG-X400 by installing PN-SIE400-PMP-PLG-604 & kit KIT-SIE400-CLN-4. Fasteners torqued to 136 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-XAX-5

Target Machine Model: Siemens-DMG-X400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE400-5
- PLC Hex Register Address: DB25.DBW_0x32E8
- Nominal Operating Range: 27 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-LIN-COIL-605
- Overhaul Seal & Consumable Kit: KIT-SIE400-XAX-5
- Critical Fastener Torque Value: 145 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-XAX-5 at PLC register 0x32E8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-LIN-COIL-605) with seal kit (KIT-SIE400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 145 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-XAX-5 on Siemens-DMG-X400 by installing PN-SIE400-LIN-COIL-605 & kit KIT-SIE400-XAX-5. Fasteners torqued to 145 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-YAX-6

Target Machine Model: Siemens-DMG-X400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE400-6
- PLC Hex Register Address: DB25.DBW_0x32F0
- Nominal Operating Range: 30 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-BSW-NUT-606
- Overhaul Seal & Consumable Kit: KIT-SIE400-YAX-6
- Critical Fastener Torque Value: 154 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-YAX-6 at PLC register 0x32F0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-BSW-NUT-606) with seal kit (KIT-SIE400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 154 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-YAX-6 on Siemens-DMG-X400 by installing PN-SIE400-BSW-NUT-606 & kit KIT-SIE400-YAX-6. Fasteners torqued to 154 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-ZAX-7

Target Machine Model: Siemens-DMG-X400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE400-7
- PLC Hex Register Address: DB25.DBW_0x32F8
- Nominal Operating Range: 33 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-CYL-SEAL-607
- Overhaul Seal & Consumable Kit: KIT-SIE400-ZAX-7
- Critical Fastener Torque Value: 163 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-ZAX-7 at PLC register 0x32F8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-CYL-SEAL-607) with seal kit (KIT-SIE400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 163 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-ZAX-7 on Siemens-DMG-X400 by installing PN-SIE400-CYL-SEAL-607 & kit KIT-SIE400-ZAX-7. Fasteners torqued to 163 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-BAX-8

Target Machine Model: Siemens-DMG-X400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE400-8
- PLC Hex Register Address: DB25.DBW_0x3300
- Nominal Operating Range: 36 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-TRQ-STAT-608
- Overhaul Seal & Consumable Kit: KIT-SIE400-BAX-8
- Critical Fastener Torque Value: 172 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-BAX-8 at PLC register 0x3300.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-TRQ-STAT-608) with seal kit (KIT-SIE400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 172 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-BAX-8 on Siemens-DMG-X400 by installing PN-SIE400-TRQ-STAT-608 & kit KIT-SIE400-BAX-8. Fasteners torqued to 172 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-CAX-9

Target Machine Model: Siemens-DMG-X400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE400-9
- PLC Hex Register Address: DB25.DBW_0x3308
- Nominal Operating Range: 39 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-BRK-DISC-609
- Overhaul Seal & Consumable Kit: KIT-SIE400-CAX-9
- Critical Fastener Torque Value: 181 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-CAX-9 at PLC register 0x3308.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-BRK-DISC-609) with seal kit (KIT-SIE400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 181 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-CAX-9 on Siemens-DMG-X400 by installing PN-SIE400-BRK-DISC-609 & kit KIT-SIE400-CAX-9. Fasteners torqued to 181 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-OPT-10

Target Machine Model: Siemens-DMG-X400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE400-10
- PLC Hex Register Address: DB25.DBW_0x3310
- Nominal Operating Range: 42 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-GLASS-RD-610
- Overhaul Seal & Consumable Kit: KIT-SIE400-OPT-10
- Critical Fastener Torque Value: 190 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-OPT-10 at PLC register 0x3310.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-GLASS-RD-610) with seal kit (KIT-SIE400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 190 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-OPT-10 on Siemens-DMG-X400 by installing PN-SIE400-GLASS-RD-610 & kit KIT-SIE400-OPT-10. Fasteners torqued to 190 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-PLC-11

Target Machine Model: Siemens-DMG-X400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE400-11
- PLC Hex Register Address: DB25.DBW_0x3318
- Nominal Operating Range: 45 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-IGBT-MOD-611
- Overhaul Seal & Consumable Kit: KIT-SIE400-PLC-11
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-PLC-11 at PLC register 0x3318.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-IGBT-MOD-611) with seal kit (KIT-SIE400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-PLC-11 on Siemens-DMG-X400 by installing PN-SIE400-IGBT-MOD-611 & kit KIT-SIE400-PLC-11. Fasteners torqued to 199 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-LUB-12

Target Machine Model: Siemens-DMG-X400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE400-12
- PLC Hex Register Address: DB25.DBW_0x3320
- Nominal Operating Range: 48 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-INJ-MET-612
- Overhaul Seal & Consumable Kit: KIT-SIE400-LUB-12
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-LUB-12 at PLC register 0x3320.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-INJ-MET-612) with seal kit (KIT-SIE400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-LUB-12 on Siemens-DMG-X400 by installing PN-SIE400-INJ-MET-612 & kit KIT-SIE400-LUB-12. Fasteners torqued to 208 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-VAC-13

Target Machine Model: Siemens-DMG-X400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE400-13
- PLC Hex Register Address: DB25.DBW_0x3328
- Nominal Operating Range: 51 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-GASK-VIT-613
- Overhaul Seal & Consumable Kit: KIT-SIE400-VAC-13
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-VAC-13 at PLC register 0x3328.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-GASK-VIT-613) with seal kit (KIT-SIE400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-VAC-13 on Siemens-DMG-X400 by installing PN-SIE400-GASK-VIT-613 & kit KIT-SIE400-VAC-13. Fasteners torqued to 27 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE400-THM-14

Target Machine Model: Siemens-DMG-X400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X400 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE400-14
- PLC Hex Register Address: DB25.DBW_0x3330
- Nominal Operating Range: 54 - 65 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 100 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE400-HEX-CORE-614
- Overhaul Seal & Consumable Kit: KIT-SIE400-THM-14
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X400 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE400-THM-14 at PLC register 0x3330.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE400-HEX-CORE-614) with seal kit (KIT-SIE400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X400, technician must log: "Resolved ERR-SIE400-THM-14 on Siemens-DMG-X400 by installing PN-SIE400-HEX-CORE-614 & kit KIT-SIE400-THM-14. Fasteners torqued to 36 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-SPN-1

Target Machine Model: Siemens-DMG-X450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE450-1
- PLC Hex Register Address: DB26.DBW_0x35E8
- Nominal Operating Range: 15 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-BRG-CER-701
- Overhaul Seal & Consumable Kit: KIT-SIE450-SPN-1
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-SPN-1 at PLC register 0x35E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-BRG-CER-701) with seal kit (KIT-SIE450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-SPN-1 on Siemens-DMG-X450 by installing PN-SIE450-BRG-CER-701 & kit KIT-SIE450-SPN-1. Fasteners torqued to 59 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-ATC-2

Target Machine Model: Siemens-DMG-X450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE450-2
- PLC Hex Register Address: DB26.DBW_0x35F0
- Nominal Operating Range: 18 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-CAM-GRIP-702
- Overhaul Seal & Consumable Kit: KIT-SIE450-ATC-2
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-ATC-2 at PLC register 0x35F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-CAM-GRIP-702) with seal kit (KIT-SIE450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-ATC-2 on Siemens-DMG-X450 by installing PN-SIE450-CAM-GRIP-702 & kit KIT-SIE450-ATC-2. Fasteners torqued to 68 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-HYD-3

Target Machine Model: Siemens-DMG-X450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE450-3
- PLC Hex Register Address: DB26.DBW_0x35F8
- Nominal Operating Range: 21 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-VLV-PROP-703
- Overhaul Seal & Consumable Kit: KIT-SIE450-HYD-3
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-HYD-3 at PLC register 0x35F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-VLV-PROP-703) with seal kit (KIT-SIE450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-HYD-3 on Siemens-DMG-X450 by installing PN-SIE450-VLV-PROP-703 & kit KIT-SIE450-HYD-3. Fasteners torqued to 77 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-CLN-4

Target Machine Model: Siemens-DMG-X450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE450-4
- PLC Hex Register Address: DB26.DBW_0x3600
- Nominal Operating Range: 24 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-PMP-PLG-704
- Overhaul Seal & Consumable Kit: KIT-SIE450-CLN-4
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-CLN-4 at PLC register 0x3600.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-PMP-PLG-704) with seal kit (KIT-SIE450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-CLN-4 on Siemens-DMG-X450 by installing PN-SIE450-PMP-PLG-704 & kit KIT-SIE450-CLN-4. Fasteners torqued to 86 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-XAX-5

Target Machine Model: Siemens-DMG-X450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE450-5
- PLC Hex Register Address: DB26.DBW_0x3608
- Nominal Operating Range: 27 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-LIN-COIL-705
- Overhaul Seal & Consumable Kit: KIT-SIE450-XAX-5
- Critical Fastener Torque Value: 95 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-XAX-5 at PLC register 0x3608.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-LIN-COIL-705) with seal kit (KIT-SIE450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 95 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-XAX-5 on Siemens-DMG-X450 by installing PN-SIE450-LIN-COIL-705 & kit KIT-SIE450-XAX-5. Fasteners torqued to 95 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-YAX-6

Target Machine Model: Siemens-DMG-X450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE450-6
- PLC Hex Register Address: DB26.DBW_0x3610
- Nominal Operating Range: 30 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-BSW-NUT-706
- Overhaul Seal & Consumable Kit: KIT-SIE450-YAX-6
- Critical Fastener Torque Value: 104 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-YAX-6 at PLC register 0x3610.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-BSW-NUT-706) with seal kit (KIT-SIE450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 104 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-YAX-6 on Siemens-DMG-X450 by installing PN-SIE450-BSW-NUT-706 & kit KIT-SIE450-YAX-6. Fasteners torqued to 104 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-ZAX-7

Target Machine Model: Siemens-DMG-X450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE450-7
- PLC Hex Register Address: DB26.DBW_0x3618
- Nominal Operating Range: 33 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-CYL-SEAL-707
- Overhaul Seal & Consumable Kit: KIT-SIE450-ZAX-7
- Critical Fastener Torque Value: 113 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-ZAX-7 at PLC register 0x3618.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-CYL-SEAL-707) with seal kit (KIT-SIE450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 113 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-ZAX-7 on Siemens-DMG-X450 by installing PN-SIE450-CYL-SEAL-707 & kit KIT-SIE450-ZAX-7. Fasteners torqued to 113 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-BAX-8

Target Machine Model: Siemens-DMG-X450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE450-8
- PLC Hex Register Address: DB26.DBW_0x3620
- Nominal Operating Range: 36 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-TRQ-STAT-708
- Overhaul Seal & Consumable Kit: KIT-SIE450-BAX-8
- Critical Fastener Torque Value: 122 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-BAX-8 at PLC register 0x3620.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-TRQ-STAT-708) with seal kit (KIT-SIE450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 122 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-BAX-8 on Siemens-DMG-X450 by installing PN-SIE450-TRQ-STAT-708 & kit KIT-SIE450-BAX-8. Fasteners torqued to 122 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-CAX-9

Target Machine Model: Siemens-DMG-X450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE450-9
- PLC Hex Register Address: DB26.DBW_0x3628
- Nominal Operating Range: 39 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-BRK-DISC-709
- Overhaul Seal & Consumable Kit: KIT-SIE450-CAX-9
- Critical Fastener Torque Value: 131 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-CAX-9 at PLC register 0x3628.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-BRK-DISC-709) with seal kit (KIT-SIE450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 131 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-CAX-9 on Siemens-DMG-X450 by installing PN-SIE450-BRK-DISC-709 & kit KIT-SIE450-CAX-9. Fasteners torqued to 131 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-OPT-10

Target Machine Model: Siemens-DMG-X450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE450-10
- PLC Hex Register Address: DB26.DBW_0x3630
- Nominal Operating Range: 42 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-GLASS-RD-710
- Overhaul Seal & Consumable Kit: KIT-SIE450-OPT-10
- Critical Fastener Torque Value: 140 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-OPT-10 at PLC register 0x3630.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-GLASS-RD-710) with seal kit (KIT-SIE450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 140 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-OPT-10 on Siemens-DMG-X450 by installing PN-SIE450-GLASS-RD-710 & kit KIT-SIE450-OPT-10. Fasteners torqued to 140 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-PLC-11

Target Machine Model: Siemens-DMG-X450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE450-11
- PLC Hex Register Address: DB26.DBW_0x3638
- Nominal Operating Range: 45 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-IGBT-MOD-711
- Overhaul Seal & Consumable Kit: KIT-SIE450-PLC-11
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-PLC-11 at PLC register 0x3638.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-IGBT-MOD-711) with seal kit (KIT-SIE450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-PLC-11 on Siemens-DMG-X450 by installing PN-SIE450-IGBT-MOD-711 & kit KIT-SIE450-PLC-11. Fasteners torqued to 149 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-LUB-12

Target Machine Model: Siemens-DMG-X450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE450-12
- PLC Hex Register Address: DB26.DBW_0x3640
- Nominal Operating Range: 48 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-INJ-MET-712
- Overhaul Seal & Consumable Kit: KIT-SIE450-LUB-12
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-LUB-12 at PLC register 0x3640.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-INJ-MET-712) with seal kit (KIT-SIE450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-LUB-12 on Siemens-DMG-X450 by installing PN-SIE450-INJ-MET-712 & kit KIT-SIE450-LUB-12. Fasteners torqued to 158 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-VAC-13

Target Machine Model: Siemens-DMG-X450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE450-13
- PLC Hex Register Address: DB26.DBW_0x3648
- Nominal Operating Range: 51 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-GASK-VIT-713
- Overhaul Seal & Consumable Kit: KIT-SIE450-VAC-13
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-VAC-13 at PLC register 0x3648.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-GASK-VIT-713) with seal kit (KIT-SIE450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-VAC-13 on Siemens-DMG-X450 by installing PN-SIE450-GASK-VIT-713 & kit KIT-SIE450-VAC-13. Fasteners torqued to 167 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE450-THM-14

Target Machine Model: Siemens-DMG-X450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X450 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE450-14
- PLC Hex Register Address: DB26.DBW_0x3650
- Nominal Operating Range: 54 - 66 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 101 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE450-HEX-CORE-714
- Overhaul Seal & Consumable Kit: KIT-SIE450-THM-14
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X450 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE450-THM-14 at PLC register 0x3650.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE450-HEX-CORE-714) with seal kit (KIT-SIE450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X450, technician must log: "Resolved ERR-SIE450-THM-14 on Siemens-DMG-X450 by installing PN-SIE450-HEX-CORE-714 & kit KIT-SIE450-THM-14. Fasteners torqued to 176 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-SPN-1

Target Machine Model: Siemens-DMG-X500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE500-1
- PLC Hex Register Address: DB27.DBW_0x3908
- Nominal Operating Range: 15 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-BRG-CER-801
- Overhaul Seal & Consumable Kit: KIT-SIE500-SPN-1
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-SPN-1 at PLC register 0x3908.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-BRG-CER-801) with seal kit (KIT-SIE500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-SPN-1 on Siemens-DMG-X500 by installing PN-SIE500-BRG-CER-801 & kit KIT-SIE500-SPN-1. Fasteners torqued to 199 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-ATC-2

Target Machine Model: Siemens-DMG-X500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE500-2
- PLC Hex Register Address: DB27.DBW_0x3910
- Nominal Operating Range: 18 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-CAM-GRIP-802
- Overhaul Seal & Consumable Kit: KIT-SIE500-ATC-2
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-ATC-2 at PLC register 0x3910.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-CAM-GRIP-802) with seal kit (KIT-SIE500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-ATC-2 on Siemens-DMG-X500 by installing PN-SIE500-CAM-GRIP-802 & kit KIT-SIE500-ATC-2. Fasteners torqued to 208 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-HYD-3

Target Machine Model: Siemens-DMG-X500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE500-3
- PLC Hex Register Address: DB27.DBW_0x3918
- Nominal Operating Range: 21 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-VLV-PROP-803
- Overhaul Seal & Consumable Kit: KIT-SIE500-HYD-3
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-HYD-3 at PLC register 0x3918.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-VLV-PROP-803) with seal kit (KIT-SIE500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-HYD-3 on Siemens-DMG-X500 by installing PN-SIE500-VLV-PROP-803 & kit KIT-SIE500-HYD-3. Fasteners torqued to 27 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-CLN-4

Target Machine Model: Siemens-DMG-X500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE500-4
- PLC Hex Register Address: DB27.DBW_0x3920
- Nominal Operating Range: 24 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-PMP-PLG-804
- Overhaul Seal & Consumable Kit: KIT-SIE500-CLN-4
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-CLN-4 at PLC register 0x3920.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-PMP-PLG-804) with seal kit (KIT-SIE500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-CLN-4 on Siemens-DMG-X500 by installing PN-SIE500-PMP-PLG-804 & kit KIT-SIE500-CLN-4. Fasteners torqued to 36 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-XAX-5

Target Machine Model: Siemens-DMG-X500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE500-5
- PLC Hex Register Address: DB27.DBW_0x3928
- Nominal Operating Range: 27 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-LIN-COIL-805
- Overhaul Seal & Consumable Kit: KIT-SIE500-XAX-5
- Critical Fastener Torque Value: 45 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-XAX-5 at PLC register 0x3928.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-LIN-COIL-805) with seal kit (KIT-SIE500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 45 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-XAX-5 on Siemens-DMG-X500 by installing PN-SIE500-LIN-COIL-805 & kit KIT-SIE500-XAX-5. Fasteners torqued to 45 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-YAX-6

Target Machine Model: Siemens-DMG-X500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE500-6
- PLC Hex Register Address: DB27.DBW_0x3930
- Nominal Operating Range: 30 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-BSW-NUT-806
- Overhaul Seal & Consumable Kit: KIT-SIE500-YAX-6
- Critical Fastener Torque Value: 54 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-YAX-6 at PLC register 0x3930.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-BSW-NUT-806) with seal kit (KIT-SIE500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 54 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-YAX-6 on Siemens-DMG-X500 by installing PN-SIE500-BSW-NUT-806 & kit KIT-SIE500-YAX-6. Fasteners torqued to 54 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-ZAX-7

Target Machine Model: Siemens-DMG-X500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE500-7
- PLC Hex Register Address: DB27.DBW_0x3938
- Nominal Operating Range: 33 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-CYL-SEAL-807
- Overhaul Seal & Consumable Kit: KIT-SIE500-ZAX-7
- Critical Fastener Torque Value: 63 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-ZAX-7 at PLC register 0x3938.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-CYL-SEAL-807) with seal kit (KIT-SIE500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 63 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-ZAX-7 on Siemens-DMG-X500 by installing PN-SIE500-CYL-SEAL-807 & kit KIT-SIE500-ZAX-7. Fasteners torqued to 63 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-BAX-8

Target Machine Model: Siemens-DMG-X500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE500-8
- PLC Hex Register Address: DB27.DBW_0x3940
- Nominal Operating Range: 36 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-TRQ-STAT-808
- Overhaul Seal & Consumable Kit: KIT-SIE500-BAX-8
- Critical Fastener Torque Value: 72 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-BAX-8 at PLC register 0x3940.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-TRQ-STAT-808) with seal kit (KIT-SIE500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 72 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-BAX-8 on Siemens-DMG-X500 by installing PN-SIE500-TRQ-STAT-808 & kit KIT-SIE500-BAX-8. Fasteners torqued to 72 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-CAX-9

Target Machine Model: Siemens-DMG-X500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE500-9
- PLC Hex Register Address: DB27.DBW_0x3948
- Nominal Operating Range: 39 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-BRK-DISC-809
- Overhaul Seal & Consumable Kit: KIT-SIE500-CAX-9
- Critical Fastener Torque Value: 81 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-CAX-9 at PLC register 0x3948.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-BRK-DISC-809) with seal kit (KIT-SIE500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 81 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-CAX-9 on Siemens-DMG-X500 by installing PN-SIE500-BRK-DISC-809 & kit KIT-SIE500-CAX-9. Fasteners torqued to 81 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-OPT-10

Target Machine Model: Siemens-DMG-X500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE500-10
- PLC Hex Register Address: DB27.DBW_0x3950
- Nominal Operating Range: 42 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-GLASS-RD-810
- Overhaul Seal & Consumable Kit: KIT-SIE500-OPT-10
- Critical Fastener Torque Value: 90 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-OPT-10 at PLC register 0x3950.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-GLASS-RD-810) with seal kit (KIT-SIE500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 90 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-OPT-10 on Siemens-DMG-X500 by installing PN-SIE500-GLASS-RD-810 & kit KIT-SIE500-OPT-10. Fasteners torqued to 90 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-PLC-11

Target Machine Model: Siemens-DMG-X500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE500-11
- PLC Hex Register Address: DB27.DBW_0x3958
- Nominal Operating Range: 45 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-IGBT-MOD-811
- Overhaul Seal & Consumable Kit: KIT-SIE500-PLC-11
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-PLC-11 at PLC register 0x3958.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-IGBT-MOD-811) with seal kit (KIT-SIE500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-PLC-11 on Siemens-DMG-X500 by installing PN-SIE500-IGBT-MOD-811 & kit KIT-SIE500-PLC-11. Fasteners torqued to 99 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-LUB-12

Target Machine Model: Siemens-DMG-X500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE500-12
- PLC Hex Register Address: DB27.DBW_0x3960
- Nominal Operating Range: 48 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-INJ-MET-812
- Overhaul Seal & Consumable Kit: KIT-SIE500-LUB-12
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-LUB-12 at PLC register 0x3960.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-INJ-MET-812) with seal kit (KIT-SIE500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-LUB-12 on Siemens-DMG-X500 by installing PN-SIE500-INJ-MET-812 & kit KIT-SIE500-LUB-12. Fasteners torqued to 108 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-VAC-13

Target Machine Model: Siemens-DMG-X500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE500-13
- PLC Hex Register Address: DB27.DBW_0x3968
- Nominal Operating Range: 51 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-GASK-VIT-813
- Overhaul Seal & Consumable Kit: KIT-SIE500-VAC-13
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-VAC-13 at PLC register 0x3968.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-GASK-VIT-813) with seal kit (KIT-SIE500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-VAC-13 on Siemens-DMG-X500 by installing PN-SIE500-GASK-VIT-813 & kit KIT-SIE500-VAC-13. Fasteners torqued to 117 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE500-THM-14

Target Machine Model: Siemens-DMG-X500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X500 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE500-14
- PLC Hex Register Address: DB27.DBW_0x3970
- Nominal Operating Range: 54 - 67 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 102 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE500-HEX-CORE-814
- Overhaul Seal & Consumable Kit: KIT-SIE500-THM-14
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X500 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE500-THM-14 at PLC register 0x3970.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE500-HEX-CORE-814) with seal kit (KIT-SIE500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X500, technician must log: "Resolved ERR-SIE500-THM-14 on Siemens-DMG-X500 by installing PN-SIE500-HEX-CORE-814 & kit KIT-SIE500-THM-14. Fasteners torqued to 126 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-SPN-1

Target Machine Model: Siemens-DMG-X550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE550-1
- PLC Hex Register Address: DB28.DBW_0x3C28
- Nominal Operating Range: 15 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-BRG-CER-901
- Overhaul Seal & Consumable Kit: KIT-SIE550-SPN-1
- Critical Fastener Torque Value: 149 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-SPN-1 at PLC register 0x3C28.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-BRG-CER-901) with seal kit (KIT-SIE550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 149 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-SPN-1 on Siemens-DMG-X550 by installing PN-SIE550-BRG-CER-901 & kit KIT-SIE550-SPN-1. Fasteners torqued to 149 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-ATC-2

Target Machine Model: Siemens-DMG-X550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE550-2
- PLC Hex Register Address: DB28.DBW_0x3C30
- Nominal Operating Range: 18 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-CAM-GRIP-902
- Overhaul Seal & Consumable Kit: KIT-SIE550-ATC-2
- Critical Fastener Torque Value: 158 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-ATC-2 at PLC register 0x3C30.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-CAM-GRIP-902) with seal kit (KIT-SIE550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 158 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-ATC-2 on Siemens-DMG-X550 by installing PN-SIE550-CAM-GRIP-902 & kit KIT-SIE550-ATC-2. Fasteners torqued to 158 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-HYD-3

Target Machine Model: Siemens-DMG-X550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE550-3
- PLC Hex Register Address: DB28.DBW_0x3C38
- Nominal Operating Range: 21 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-VLV-PROP-903
- Overhaul Seal & Consumable Kit: KIT-SIE550-HYD-3
- Critical Fastener Torque Value: 167 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-HYD-3 at PLC register 0x3C38.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-VLV-PROP-903) with seal kit (KIT-SIE550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 167 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-HYD-3 on Siemens-DMG-X550 by installing PN-SIE550-VLV-PROP-903 & kit KIT-SIE550-HYD-3. Fasteners torqued to 167 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-CLN-4

Target Machine Model: Siemens-DMG-X550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE550-4
- PLC Hex Register Address: DB28.DBW_0x3C40
- Nominal Operating Range: 24 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-PMP-PLG-904
- Overhaul Seal & Consumable Kit: KIT-SIE550-CLN-4
- Critical Fastener Torque Value: 176 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-CLN-4 at PLC register 0x3C40.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-PMP-PLG-904) with seal kit (KIT-SIE550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 176 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-CLN-4 on Siemens-DMG-X550 by installing PN-SIE550-PMP-PLG-904 & kit KIT-SIE550-CLN-4. Fasteners torqued to 176 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-XAX-5

Target Machine Model: Siemens-DMG-X550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE550-5
- PLC Hex Register Address: DB28.DBW_0x3C48
- Nominal Operating Range: 27 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-LIN-COIL-905
- Overhaul Seal & Consumable Kit: KIT-SIE550-XAX-5
- Critical Fastener Torque Value: 185 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-XAX-5 at PLC register 0x3C48.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-LIN-COIL-905) with seal kit (KIT-SIE550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 185 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-XAX-5 on Siemens-DMG-X550 by installing PN-SIE550-LIN-COIL-905 & kit KIT-SIE550-XAX-5. Fasteners torqued to 185 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-YAX-6

Target Machine Model: Siemens-DMG-X550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE550-6
- PLC Hex Register Address: DB28.DBW_0x3C50
- Nominal Operating Range: 30 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-BSW-NUT-906
- Overhaul Seal & Consumable Kit: KIT-SIE550-YAX-6
- Critical Fastener Torque Value: 194 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-YAX-6 at PLC register 0x3C50.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-BSW-NUT-906) with seal kit (KIT-SIE550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 194 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-YAX-6 on Siemens-DMG-X550 by installing PN-SIE550-BSW-NUT-906 & kit KIT-SIE550-YAX-6. Fasteners torqued to 194 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-ZAX-7

Target Machine Model: Siemens-DMG-X550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE550-7
- PLC Hex Register Address: DB28.DBW_0x3C58
- Nominal Operating Range: 33 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-CYL-SEAL-907
- Overhaul Seal & Consumable Kit: KIT-SIE550-ZAX-7
- Critical Fastener Torque Value: 203 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-ZAX-7 at PLC register 0x3C58.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-CYL-SEAL-907) with seal kit (KIT-SIE550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 203 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-ZAX-7 on Siemens-DMG-X550 by installing PN-SIE550-CYL-SEAL-907 & kit KIT-SIE550-ZAX-7. Fasteners torqued to 203 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-BAX-8

Target Machine Model: Siemens-DMG-X550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE550-8
- PLC Hex Register Address: DB28.DBW_0x3C60
- Nominal Operating Range: 36 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-TRQ-STAT-908
- Overhaul Seal & Consumable Kit: KIT-SIE550-BAX-8
- Critical Fastener Torque Value: 22 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-BAX-8 at PLC register 0x3C60.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-TRQ-STAT-908) with seal kit (KIT-SIE550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 22 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-BAX-8 on Siemens-DMG-X550 by installing PN-SIE550-TRQ-STAT-908 & kit KIT-SIE550-BAX-8. Fasteners torqued to 22 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-CAX-9

Target Machine Model: Siemens-DMG-X550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE550-9
- PLC Hex Register Address: DB28.DBW_0x3C68
- Nominal Operating Range: 39 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-BRK-DISC-909
- Overhaul Seal & Consumable Kit: KIT-SIE550-CAX-9
- Critical Fastener Torque Value: 31 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-CAX-9 at PLC register 0x3C68.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-BRK-DISC-909) with seal kit (KIT-SIE550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 31 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-CAX-9 on Siemens-DMG-X550 by installing PN-SIE550-BRK-DISC-909 & kit KIT-SIE550-CAX-9. Fasteners torqued to 31 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-OPT-10

Target Machine Model: Siemens-DMG-X550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE550-10
- PLC Hex Register Address: DB28.DBW_0x3C70
- Nominal Operating Range: 42 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-GLASS-RD-910
- Overhaul Seal & Consumable Kit: KIT-SIE550-OPT-10
- Critical Fastener Torque Value: 40 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-OPT-10 at PLC register 0x3C70.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-GLASS-RD-910) with seal kit (KIT-SIE550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 40 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-OPT-10 on Siemens-DMG-X550 by installing PN-SIE550-GLASS-RD-910 & kit KIT-SIE550-OPT-10. Fasteners torqued to 40 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-PLC-11

Target Machine Model: Siemens-DMG-X550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE550-11
- PLC Hex Register Address: DB28.DBW_0x3C78
- Nominal Operating Range: 45 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-IGBT-MOD-911
- Overhaul Seal & Consumable Kit: KIT-SIE550-PLC-11
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-PLC-11 at PLC register 0x3C78.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-IGBT-MOD-911) with seal kit (KIT-SIE550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-PLC-11 on Siemens-DMG-X550 by installing PN-SIE550-IGBT-MOD-911 & kit KIT-SIE550-PLC-11. Fasteners torqued to 49 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-LUB-12

Target Machine Model: Siemens-DMG-X550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE550-12
- PLC Hex Register Address: DB28.DBW_0x3C80
- Nominal Operating Range: 48 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-INJ-MET-912
- Overhaul Seal & Consumable Kit: KIT-SIE550-LUB-12
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-LUB-12 at PLC register 0x3C80.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-INJ-MET-912) with seal kit (KIT-SIE550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-LUB-12 on Siemens-DMG-X550 by installing PN-SIE550-INJ-MET-912 & kit KIT-SIE550-LUB-12. Fasteners torqued to 58 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-VAC-13

Target Machine Model: Siemens-DMG-X550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE550-13
- PLC Hex Register Address: DB28.DBW_0x3C88
- Nominal Operating Range: 51 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-GASK-VIT-913
- Overhaul Seal & Consumable Kit: KIT-SIE550-VAC-13
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-VAC-13 at PLC register 0x3C88.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-GASK-VIT-913) with seal kit (KIT-SIE550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-VAC-13 on Siemens-DMG-X550 by installing PN-SIE550-GASK-VIT-913 & kit KIT-SIE550-VAC-13. Fasteners torqued to 67 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE550-THM-14

Target Machine Model: Siemens-DMG-X550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X550 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE550-14
- PLC Hex Register Address: DB28.DBW_0x3C90
- Nominal Operating Range: 54 - 68 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 103 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE550-HEX-CORE-914
- Overhaul Seal & Consumable Kit: KIT-SIE550-THM-14
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X550 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE550-THM-14 at PLC register 0x3C90.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE550-HEX-CORE-914) with seal kit (KIT-SIE550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X550, technician must log: "Resolved ERR-SIE550-THM-14 on Siemens-DMG-X550 by installing PN-SIE550-HEX-CORE-914 & kit KIT-SIE550-THM-14. Fasteners torqued to 76 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-SPN-1

Target Machine Model: Siemens-DMG-X600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PT-SPN-SIE600-1
- PLC Hex Register Address: DB29.DBW_0x3F48
- Nominal Operating Range: 15 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-BRG-CER-1001
- Overhaul Seal & Consumable Kit: KIT-SIE600-SPN-1
- Critical Fastener Torque Value: 99 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-SPN-1 at PLC register 0x3F48.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-BRG-CER-1001) with seal kit (KIT-SIE600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 99 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-SPN-1 on Siemens-DMG-X600 by installing PN-SIE600-BRG-CER-1001 & kit KIT-SIE600-SPN-1. Fasteners torqued to 99 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-ATC-2

Target Machine Model: Siemens-DMG-X600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PX-ATC-SIE600-2
- PLC Hex Register Address: DB29.DBW_0x3F50
- Nominal Operating Range: 18 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-CAM-GRIP-1002
- Overhaul Seal & Consumable Kit: KIT-SIE600-ATC-2
- Critical Fastener Torque Value: 108 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-ATC-2 at PLC register 0x3F50.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-CAM-GRIP-1002) with seal kit (KIT-SIE600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 108 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-ATC-2 on Siemens-DMG-X600 by installing PN-SIE600-CAM-GRIP-1002 & kit KIT-SIE600-ATC-2. Fasteners torqued to 108 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-HYD-3

Target Machine Model: Siemens-DMG-X600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-HYD-SIE600-3
- PLC Hex Register Address: DB29.DBW_0x3F58
- Nominal Operating Range: 21 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-VLV-PROP-1003
- Overhaul Seal & Consumable Kit: KIT-SIE600-HYD-3
- Critical Fastener Torque Value: 117 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-HYD-3 at PLC register 0x3F58.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-VLV-PROP-1003) with seal kit (KIT-SIE600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 117 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-HYD-3 on Siemens-DMG-X600 by installing PN-SIE600-VLV-PROP-1003 & kit KIT-SIE600-HYD-3. Fasteners torqued to 117 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-CLN-4

Target Machine Model: Siemens-DMG-X600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-CLN-SIE600-4
- PLC Hex Register Address: DB29.DBW_0x3F60
- Nominal Operating Range: 24 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-PMP-PLG-1004
- Overhaul Seal & Consumable Kit: KIT-SIE600-CLN-4
- Critical Fastener Torque Value: 126 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-CLN-4 at PLC register 0x3F60.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-PMP-PLG-1004) with seal kit (KIT-SIE600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 126 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-CLN-4 on Siemens-DMG-X600 by installing PN-SIE600-PMP-PLG-1004 & kit KIT-SIE600-CLN-4. Fasteners torqued to 126 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-XAX-5

Target Machine Model: Siemens-DMG-X600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-XAX-SIE600-5
- PLC Hex Register Address: DB29.DBW_0x3F68
- Nominal Operating Range: 27 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-LIN-COIL-1005
- Overhaul Seal & Consumable Kit: KIT-SIE600-XAX-5
- Critical Fastener Torque Value: 135 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-XAX-5 at PLC register 0x3F68.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-LIN-COIL-1005) with seal kit (KIT-SIE600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 135 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-XAX-5 on Siemens-DMG-X600 by installing PN-SIE600-LIN-COIL-1005 & kit KIT-SIE600-XAX-5. Fasteners torqued to 135 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-YAX-6

Target Machine Model: Siemens-DMG-X600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-YAX-SIE600-6
- PLC Hex Register Address: DB29.DBW_0x3F70
- Nominal Operating Range: 30 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-BSW-NUT-1006
- Overhaul Seal & Consumable Kit: KIT-SIE600-YAX-6
- Critical Fastener Torque Value: 144 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-YAX-6 at PLC register 0x3F70.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-BSW-NUT-1006) with seal kit (KIT-SIE600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 144 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-YAX-6 on Siemens-DMG-X600 by installing PN-SIE600-BSW-NUT-1006 & kit KIT-SIE600-YAX-6. Fasteners torqued to 144 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-ZAX-7

Target Machine Model: Siemens-DMG-X600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-ZAX-SIE600-7
- PLC Hex Register Address: DB29.DBW_0x3F78
- Nominal Operating Range: 33 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-CYL-SEAL-1007
- Overhaul Seal & Consumable Kit: KIT-SIE600-ZAX-7
- Critical Fastener Torque Value: 153 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-ZAX-7 at PLC register 0x3F78.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-CYL-SEAL-1007) with seal kit (KIT-SIE600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 153 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-ZAX-7 on Siemens-DMG-X600 by installing PN-SIE600-CYL-SEAL-1007 & kit KIT-SIE600-ZAX-7. Fasteners torqued to 153 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-BAX-8

Target Machine Model: Siemens-DMG-X600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: EN-BAX-SIE600-8
- PLC Hex Register Address: DB29.DBW_0x3F80
- Nominal Operating Range: 36 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-TRQ-STAT-1008
- Overhaul Seal & Consumable Kit: KIT-SIE600-BAX-8
- Critical Fastener Torque Value: 162 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-BAX-8 at PLC register 0x3F80.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-TRQ-STAT-1008) with seal kit (KIT-SIE600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 162 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-BAX-8 on Siemens-DMG-X600 by installing PN-SIE600-TRQ-STAT-1008 & kit KIT-SIE600-BAX-8. Fasteners torqued to 162 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-CAX-9

Target Machine Model: Siemens-DMG-X600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: PR-CAX-SIE600-9
- PLC Hex Register Address: DB29.DBW_0x3F88
- Nominal Operating Range: 39 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-BRK-DISC-1009
- Overhaul Seal & Consumable Kit: KIT-SIE600-CAX-9
- Critical Fastener Torque Value: 171 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-CAX-9 at PLC register 0x3F88.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-BRK-DISC-1009) with seal kit (KIT-SIE600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 171 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-CAX-9 on Siemens-DMG-X600 by installing PN-SIE600-BRK-DISC-1009 & kit KIT-SIE600-CAX-9. Fasteners torqued to 171 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-OPT-10

Target Machine Model: Siemens-DMG-X600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: OP-SCL-SIE600-10
- PLC Hex Register Address: DB29.DBW_0x3F90
- Nominal Operating Range: 42 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-GLASS-RD-1010
- Overhaul Seal & Consumable Kit: KIT-SIE600-OPT-10
- Critical Fastener Torque Value: 180 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-OPT-10 at PLC register 0x3F90.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-GLASS-RD-1010) with seal kit (KIT-SIE600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 180 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-OPT-10 on Siemens-DMG-X600 by installing PN-SIE600-GLASS-RD-1010 & kit KIT-SIE600-OPT-10. Fasteners torqued to 180 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-PLC-11

Target Machine Model: Siemens-DMG-X600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VT-DRV-SIE600-11
- PLC Hex Register Address: DB29.DBW_0x3F98
- Nominal Operating Range: 45 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-IGBT-MOD-1011
- Overhaul Seal & Consumable Kit: KIT-SIE600-PLC-11
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-PLC-11 at PLC register 0x3F98.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-IGBT-MOD-1011) with seal kit (KIT-SIE600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-PLC-11 on Siemens-DMG-X600 by installing PN-SIE600-IGBT-MOD-1011 & kit KIT-SIE600-PLC-11. Fasteners torqued to 189 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-LUB-12

Target Machine Model: Siemens-DMG-X600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: FL-LUB-SIE600-12
- PLC Hex Register Address: DB29.DBW_0x3FA0
- Nominal Operating Range: 48 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-INJ-MET-1012
- Overhaul Seal & Consumable Kit: KIT-SIE600-LUB-12
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-LUB-12 at PLC register 0x3FA0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-INJ-MET-1012) with seal kit (KIT-SIE600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-LUB-12 on Siemens-DMG-X600 by installing PN-SIE600-INJ-MET-1012 & kit KIT-SIE600-LUB-12. Fasteners torqued to 198 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-VAC-13

Target Machine Model: Siemens-DMG-X600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: VC-MNF-SIE600-13
- PLC Hex Register Address: DB29.DBW_0x3FA8
- Nominal Operating Range: 51 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-GASK-VIT-1013
- Overhaul Seal & Consumable Kit: KIT-SIE600-VAC-13
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-VAC-13 at PLC register 0x3FA8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-GASK-VIT-1013) with seal kit (KIT-SIE600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-VAC-13 on Siemens-DMG-X600 by installing PN-SIE600-GASK-VIT-1013 & kit KIT-SIE600-VAC-13. Fasteners torqued to 207 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-SIE600-THM-14

Target Machine Model: Siemens-DMG-X600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Siemens-DMG-X600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Siemens-DMG-X600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Siemens-DMG-X600 (SINUMERIK ONE v6.15)
- Telemetry Sensor ID: TM-CHL-SIE600-14
- PLC Hex Register Address: DB29.DBW_0x3FB0
- Nominal Operating Range: 54 - 69 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 104 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-SIE600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-SIE600-HEX-CORE-1014
- Overhaul Seal & Consumable Kit: KIT-SIE600-THM-14
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Siemens-DMG-X600 (SINUMERIK ONE v6.15) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to SINUMERIK ONE v6.15 and verify fault latch ERR-SIE600-THM-14 at PLC register 0x3FB0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Siemens-DMG-X600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-SIE600-HEX-CORE-1014) with seal kit (KIT-SIE600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-SIE600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Siemens-DMG-X600, technician must log: "Resolved ERR-SIE600-THM-14 on Siemens-DMG-X600 by installing PN-SIE600-HEX-CORE-1014 & kit KIT-SIE600-THM-14. Fasteners torqued to 26 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-SPN-1

Target Machine Model: Makino-Aero-T150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK150-1
- PLC Hex Register Address: DB30.DBW_0x4268
- Nominal Operating Range: 15 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-BRG-CER-1101
- Overhaul Seal & Consumable Kit: KIT-MAK150-SPN-1
- Critical Fastener Torque Value: 49 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-SPN-1 at PLC register 0x4268.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-BRG-CER-1101) with seal kit (KIT-MAK150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 49 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-SPN-1 on Makino-Aero-T150 by installing PN-MAK150-BRG-CER-1101 & kit KIT-MAK150-SPN-1. Fasteners torqued to 49 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-ATC-2

Target Machine Model: Makino-Aero-T150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK150-2
- PLC Hex Register Address: DB30.DBW_0x4270
- Nominal Operating Range: 18 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-CAM-GRIP-1102
- Overhaul Seal & Consumable Kit: KIT-MAK150-ATC-2
- Critical Fastener Torque Value: 58 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-ATC-2 at PLC register 0x4270.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-CAM-GRIP-1102) with seal kit (KIT-MAK150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 58 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-ATC-2 on Makino-Aero-T150 by installing PN-MAK150-CAM-GRIP-1102 & kit KIT-MAK150-ATC-2. Fasteners torqued to 58 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-HYD-3

Target Machine Model: Makino-Aero-T150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK150-3
- PLC Hex Register Address: DB30.DBW_0x4278
- Nominal Operating Range: 21 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-VLV-PROP-1103
- Overhaul Seal & Consumable Kit: KIT-MAK150-HYD-3
- Critical Fastener Torque Value: 67 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-HYD-3 at PLC register 0x4278.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-VLV-PROP-1103) with seal kit (KIT-MAK150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 67 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-HYD-3 on Makino-Aero-T150 by installing PN-MAK150-VLV-PROP-1103 & kit KIT-MAK150-HYD-3. Fasteners torqued to 67 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-CLN-4

Target Machine Model: Makino-Aero-T150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK150-4
- PLC Hex Register Address: DB30.DBW_0x4280
- Nominal Operating Range: 24 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-PMP-PLG-1104
- Overhaul Seal & Consumable Kit: KIT-MAK150-CLN-4
- Critical Fastener Torque Value: 76 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-CLN-4 at PLC register 0x4280.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-PMP-PLG-1104) with seal kit (KIT-MAK150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 76 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-CLN-4 on Makino-Aero-T150 by installing PN-MAK150-PMP-PLG-1104 & kit KIT-MAK150-CLN-4. Fasteners torqued to 76 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-XAX-5

Target Machine Model: Makino-Aero-T150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK150-5
- PLC Hex Register Address: DB30.DBW_0x4288
- Nominal Operating Range: 27 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-LIN-COIL-1105
- Overhaul Seal & Consumable Kit: KIT-MAK150-XAX-5
- Critical Fastener Torque Value: 85 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-XAX-5 at PLC register 0x4288.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-LIN-COIL-1105) with seal kit (KIT-MAK150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 85 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-XAX-5 on Makino-Aero-T150 by installing PN-MAK150-LIN-COIL-1105 & kit KIT-MAK150-XAX-5. Fasteners torqued to 85 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-YAX-6

Target Machine Model: Makino-Aero-T150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK150-6
- PLC Hex Register Address: DB30.DBW_0x4290
- Nominal Operating Range: 30 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-BSW-NUT-1106
- Overhaul Seal & Consumable Kit: KIT-MAK150-YAX-6
- Critical Fastener Torque Value: 94 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-YAX-6 at PLC register 0x4290.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-BSW-NUT-1106) with seal kit (KIT-MAK150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 94 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-YAX-6 on Makino-Aero-T150 by installing PN-MAK150-BSW-NUT-1106 & kit KIT-MAK150-YAX-6. Fasteners torqued to 94 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-ZAX-7

Target Machine Model: Makino-Aero-T150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK150-7
- PLC Hex Register Address: DB30.DBW_0x4298
- Nominal Operating Range: 33 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-CYL-SEAL-1107
- Overhaul Seal & Consumable Kit: KIT-MAK150-ZAX-7
- Critical Fastener Torque Value: 103 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-ZAX-7 at PLC register 0x4298.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-CYL-SEAL-1107) with seal kit (KIT-MAK150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 103 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-ZAX-7 on Makino-Aero-T150 by installing PN-MAK150-CYL-SEAL-1107 & kit KIT-MAK150-ZAX-7. Fasteners torqued to 103 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-BAX-8

Target Machine Model: Makino-Aero-T150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK150-8
- PLC Hex Register Address: DB30.DBW_0x42A0
- Nominal Operating Range: 36 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-TRQ-STAT-1108
- Overhaul Seal & Consumable Kit: KIT-MAK150-BAX-8
- Critical Fastener Torque Value: 112 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-BAX-8 at PLC register 0x42A0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-TRQ-STAT-1108) with seal kit (KIT-MAK150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 112 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-BAX-8 on Makino-Aero-T150 by installing PN-MAK150-TRQ-STAT-1108 & kit KIT-MAK150-BAX-8. Fasteners torqued to 112 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-CAX-9

Target Machine Model: Makino-Aero-T150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK150-9
- PLC Hex Register Address: DB30.DBW_0x42A8
- Nominal Operating Range: 39 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-BRK-DISC-1109
- Overhaul Seal & Consumable Kit: KIT-MAK150-CAX-9
- Critical Fastener Torque Value: 121 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-CAX-9 at PLC register 0x42A8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-BRK-DISC-1109) with seal kit (KIT-MAK150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 121 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-CAX-9 on Makino-Aero-T150 by installing PN-MAK150-BRK-DISC-1109 & kit KIT-MAK150-CAX-9. Fasteners torqued to 121 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-OPT-10

Target Machine Model: Makino-Aero-T150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK150-10
- PLC Hex Register Address: DB30.DBW_0x42B0
- Nominal Operating Range: 42 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-GLASS-RD-1110
- Overhaul Seal & Consumable Kit: KIT-MAK150-OPT-10
- Critical Fastener Torque Value: 130 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-OPT-10 at PLC register 0x42B0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-GLASS-RD-1110) with seal kit (KIT-MAK150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 130 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-OPT-10 on Makino-Aero-T150 by installing PN-MAK150-GLASS-RD-1110 & kit KIT-MAK150-OPT-10. Fasteners torqued to 130 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-PLC-11

Target Machine Model: Makino-Aero-T150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK150-11
- PLC Hex Register Address: DB30.DBW_0x42B8
- Nominal Operating Range: 45 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-IGBT-MOD-1111
- Overhaul Seal & Consumable Kit: KIT-MAK150-PLC-11
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-PLC-11 at PLC register 0x42B8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-IGBT-MOD-1111) with seal kit (KIT-MAK150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-PLC-11 on Makino-Aero-T150 by installing PN-MAK150-IGBT-MOD-1111 & kit KIT-MAK150-PLC-11. Fasteners torqued to 139 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-LUB-12

Target Machine Model: Makino-Aero-T150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK150-12
- PLC Hex Register Address: DB30.DBW_0x42C0
- Nominal Operating Range: 48 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-INJ-MET-1112
- Overhaul Seal & Consumable Kit: KIT-MAK150-LUB-12
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-LUB-12 at PLC register 0x42C0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-INJ-MET-1112) with seal kit (KIT-MAK150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-LUB-12 on Makino-Aero-T150 by installing PN-MAK150-INJ-MET-1112 & kit KIT-MAK150-LUB-12. Fasteners torqued to 148 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-VAC-13

Target Machine Model: Makino-Aero-T150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK150-13
- PLC Hex Register Address: DB30.DBW_0x42C8
- Nominal Operating Range: 51 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-GASK-VIT-1113
- Overhaul Seal & Consumable Kit: KIT-MAK150-VAC-13
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-VAC-13 at PLC register 0x42C8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-GASK-VIT-1113) with seal kit (KIT-MAK150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-VAC-13 on Makino-Aero-T150 by installing PN-MAK150-GASK-VIT-1113 & kit KIT-MAK150-VAC-13. Fasteners torqued to 157 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK150-THM-14

Target Machine Model: Makino-Aero-T150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T150 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK150-14
- PLC Hex Register Address: DB30.DBW_0x42D0
- Nominal Operating Range: 54 - 70 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 105 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK150-HEX-CORE-1114
- Overhaul Seal & Consumable Kit: KIT-MAK150-THM-14
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T150 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK150-THM-14 at PLC register 0x42D0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK150-HEX-CORE-1114) with seal kit (KIT-MAK150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T150, technician must log: "Resolved ERR-MAK150-THM-14 on Makino-Aero-T150 by installing PN-MAK150-HEX-CORE-1114 & kit KIT-MAK150-THM-14. Fasteners torqued to 166 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-SPN-1

Target Machine Model: Makino-Aero-T200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK200-1
- PLC Hex Register Address: DB31.DBW_0x4588
- Nominal Operating Range: 15 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-BRG-CER-1201
- Overhaul Seal & Consumable Kit: KIT-MAK200-SPN-1
- Critical Fastener Torque Value: 189 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-SPN-1 at PLC register 0x4588.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-BRG-CER-1201) with seal kit (KIT-MAK200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 189 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-SPN-1 on Makino-Aero-T200 by installing PN-MAK200-BRG-CER-1201 & kit KIT-MAK200-SPN-1. Fasteners torqued to 189 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-ATC-2

Target Machine Model: Makino-Aero-T200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK200-2
- PLC Hex Register Address: DB31.DBW_0x4590
- Nominal Operating Range: 18 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-CAM-GRIP-1202
- Overhaul Seal & Consumable Kit: KIT-MAK200-ATC-2
- Critical Fastener Torque Value: 198 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-ATC-2 at PLC register 0x4590.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-CAM-GRIP-1202) with seal kit (KIT-MAK200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 198 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-ATC-2 on Makino-Aero-T200 by installing PN-MAK200-CAM-GRIP-1202 & kit KIT-MAK200-ATC-2. Fasteners torqued to 198 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-HYD-3

Target Machine Model: Makino-Aero-T200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK200-3
- PLC Hex Register Address: DB31.DBW_0x4598
- Nominal Operating Range: 21 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-VLV-PROP-1203
- Overhaul Seal & Consumable Kit: KIT-MAK200-HYD-3
- Critical Fastener Torque Value: 207 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-HYD-3 at PLC register 0x4598.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-VLV-PROP-1203) with seal kit (KIT-MAK200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 207 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-HYD-3 on Makino-Aero-T200 by installing PN-MAK200-VLV-PROP-1203 & kit KIT-MAK200-HYD-3. Fasteners torqued to 207 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-CLN-4

Target Machine Model: Makino-Aero-T200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK200-4
- PLC Hex Register Address: DB31.DBW_0x45A0
- Nominal Operating Range: 24 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-PMP-PLG-1204
- Overhaul Seal & Consumable Kit: KIT-MAK200-CLN-4
- Critical Fastener Torque Value: 26 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-CLN-4 at PLC register 0x45A0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-PMP-PLG-1204) with seal kit (KIT-MAK200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 26 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-CLN-4 on Makino-Aero-T200 by installing PN-MAK200-PMP-PLG-1204 & kit KIT-MAK200-CLN-4. Fasteners torqued to 26 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-XAX-5

Target Machine Model: Makino-Aero-T200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK200-5
- PLC Hex Register Address: DB31.DBW_0x45A8
- Nominal Operating Range: 27 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-LIN-COIL-1205
- Overhaul Seal & Consumable Kit: KIT-MAK200-XAX-5
- Critical Fastener Torque Value: 35 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-XAX-5 at PLC register 0x45A8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-LIN-COIL-1205) with seal kit (KIT-MAK200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 35 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-XAX-5 on Makino-Aero-T200 by installing PN-MAK200-LIN-COIL-1205 & kit KIT-MAK200-XAX-5. Fasteners torqued to 35 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-YAX-6

Target Machine Model: Makino-Aero-T200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK200-6
- PLC Hex Register Address: DB31.DBW_0x45B0
- Nominal Operating Range: 30 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-BSW-NUT-1206
- Overhaul Seal & Consumable Kit: KIT-MAK200-YAX-6
- Critical Fastener Torque Value: 44 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-YAX-6 at PLC register 0x45B0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-BSW-NUT-1206) with seal kit (KIT-MAK200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 44 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-YAX-6 on Makino-Aero-T200 by installing PN-MAK200-BSW-NUT-1206 & kit KIT-MAK200-YAX-6. Fasteners torqued to 44 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-ZAX-7

Target Machine Model: Makino-Aero-T200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK200-7
- PLC Hex Register Address: DB31.DBW_0x45B8
- Nominal Operating Range: 33 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-CYL-SEAL-1207
- Overhaul Seal & Consumable Kit: KIT-MAK200-ZAX-7
- Critical Fastener Torque Value: 53 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-ZAX-7 at PLC register 0x45B8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-CYL-SEAL-1207) with seal kit (KIT-MAK200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 53 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-ZAX-7 on Makino-Aero-T200 by installing PN-MAK200-CYL-SEAL-1207 & kit KIT-MAK200-ZAX-7. Fasteners torqued to 53 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-BAX-8

Target Machine Model: Makino-Aero-T200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK200-8
- PLC Hex Register Address: DB31.DBW_0x45C0
- Nominal Operating Range: 36 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-TRQ-STAT-1208
- Overhaul Seal & Consumable Kit: KIT-MAK200-BAX-8
- Critical Fastener Torque Value: 62 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-BAX-8 at PLC register 0x45C0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-TRQ-STAT-1208) with seal kit (KIT-MAK200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 62 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-BAX-8 on Makino-Aero-T200 by installing PN-MAK200-TRQ-STAT-1208 & kit KIT-MAK200-BAX-8. Fasteners torqued to 62 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-CAX-9

Target Machine Model: Makino-Aero-T200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK200-9
- PLC Hex Register Address: DB31.DBW_0x45C8
- Nominal Operating Range: 39 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-BRK-DISC-1209
- Overhaul Seal & Consumable Kit: KIT-MAK200-CAX-9
- Critical Fastener Torque Value: 71 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-CAX-9 at PLC register 0x45C8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-BRK-DISC-1209) with seal kit (KIT-MAK200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 71 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-CAX-9 on Makino-Aero-T200 by installing PN-MAK200-BRK-DISC-1209 & kit KIT-MAK200-CAX-9. Fasteners torqued to 71 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-OPT-10

Target Machine Model: Makino-Aero-T200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK200-10
- PLC Hex Register Address: DB31.DBW_0x45D0
- Nominal Operating Range: 42 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-GLASS-RD-1210
- Overhaul Seal & Consumable Kit: KIT-MAK200-OPT-10
- Critical Fastener Torque Value: 80 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-OPT-10 at PLC register 0x45D0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-GLASS-RD-1210) with seal kit (KIT-MAK200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 80 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-OPT-10 on Makino-Aero-T200 by installing PN-MAK200-GLASS-RD-1210 & kit KIT-MAK200-OPT-10. Fasteners torqued to 80 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-PLC-11

Target Machine Model: Makino-Aero-T200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK200-11
- PLC Hex Register Address: DB31.DBW_0x45D8
- Nominal Operating Range: 45 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-IGBT-MOD-1211
- Overhaul Seal & Consumable Kit: KIT-MAK200-PLC-11
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-PLC-11 at PLC register 0x45D8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-IGBT-MOD-1211) with seal kit (KIT-MAK200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-PLC-11 on Makino-Aero-T200 by installing PN-MAK200-IGBT-MOD-1211 & kit KIT-MAK200-PLC-11. Fasteners torqued to 89 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-LUB-12

Target Machine Model: Makino-Aero-T200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK200-12
- PLC Hex Register Address: DB31.DBW_0x45E0
- Nominal Operating Range: 48 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-INJ-MET-1212
- Overhaul Seal & Consumable Kit: KIT-MAK200-LUB-12
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-LUB-12 at PLC register 0x45E0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-INJ-MET-1212) with seal kit (KIT-MAK200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-LUB-12 on Makino-Aero-T200 by installing PN-MAK200-INJ-MET-1212 & kit KIT-MAK200-LUB-12. Fasteners torqued to 98 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-VAC-13

Target Machine Model: Makino-Aero-T200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK200-13
- PLC Hex Register Address: DB31.DBW_0x45E8
- Nominal Operating Range: 51 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-GASK-VIT-1213
- Overhaul Seal & Consumable Kit: KIT-MAK200-VAC-13
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-VAC-13 at PLC register 0x45E8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-GASK-VIT-1213) with seal kit (KIT-MAK200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-VAC-13 on Makino-Aero-T200 by installing PN-MAK200-GASK-VIT-1213 & kit KIT-MAK200-VAC-13. Fasteners torqued to 107 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK200-THM-14

Target Machine Model: Makino-Aero-T200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T200 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK200-14
- PLC Hex Register Address: DB31.DBW_0x45F0
- Nominal Operating Range: 54 - 71 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 106 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK200-HEX-CORE-1214
- Overhaul Seal & Consumable Kit: KIT-MAK200-THM-14
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T200 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK200-THM-14 at PLC register 0x45F0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK200-HEX-CORE-1214) with seal kit (KIT-MAK200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T200, technician must log: "Resolved ERR-MAK200-THM-14 on Makino-Aero-T200 by installing PN-MAK200-HEX-CORE-1214 & kit KIT-MAK200-THM-14. Fasteners torqued to 116 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-SPN-1

Target Machine Model: Makino-Aero-T250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK250-1
- PLC Hex Register Address: DB32.DBW_0x48A8
- Nominal Operating Range: 15 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-BRG-CER-1301
- Overhaul Seal & Consumable Kit: KIT-MAK250-SPN-1
- Critical Fastener Torque Value: 139 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-SPN-1 at PLC register 0x48A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-BRG-CER-1301) with seal kit (KIT-MAK250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 139 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-SPN-1 on Makino-Aero-T250 by installing PN-MAK250-BRG-CER-1301 & kit KIT-MAK250-SPN-1. Fasteners torqued to 139 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-ATC-2

Target Machine Model: Makino-Aero-T250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK250-2
- PLC Hex Register Address: DB32.DBW_0x48B0
- Nominal Operating Range: 18 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-CAM-GRIP-1302
- Overhaul Seal & Consumable Kit: KIT-MAK250-ATC-2
- Critical Fastener Torque Value: 148 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-ATC-2 at PLC register 0x48B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-CAM-GRIP-1302) with seal kit (KIT-MAK250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 148 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-ATC-2 on Makino-Aero-T250 by installing PN-MAK250-CAM-GRIP-1302 & kit KIT-MAK250-ATC-2. Fasteners torqued to 148 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-HYD-3

Target Machine Model: Makino-Aero-T250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK250-3
- PLC Hex Register Address: DB32.DBW_0x48B8
- Nominal Operating Range: 21 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-VLV-PROP-1303
- Overhaul Seal & Consumable Kit: KIT-MAK250-HYD-3
- Critical Fastener Torque Value: 157 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-HYD-3 at PLC register 0x48B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-VLV-PROP-1303) with seal kit (KIT-MAK250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 157 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-HYD-3 on Makino-Aero-T250 by installing PN-MAK250-VLV-PROP-1303 & kit KIT-MAK250-HYD-3. Fasteners torqued to 157 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-CLN-4

Target Machine Model: Makino-Aero-T250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK250-4
- PLC Hex Register Address: DB32.DBW_0x48C0
- Nominal Operating Range: 24 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-PMP-PLG-1304
- Overhaul Seal & Consumable Kit: KIT-MAK250-CLN-4
- Critical Fastener Torque Value: 166 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-CLN-4 at PLC register 0x48C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-PMP-PLG-1304) with seal kit (KIT-MAK250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 166 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-CLN-4 on Makino-Aero-T250 by installing PN-MAK250-PMP-PLG-1304 & kit KIT-MAK250-CLN-4. Fasteners torqued to 166 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-XAX-5

Target Machine Model: Makino-Aero-T250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK250-5
- PLC Hex Register Address: DB32.DBW_0x48C8
- Nominal Operating Range: 27 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-LIN-COIL-1305
- Overhaul Seal & Consumable Kit: KIT-MAK250-XAX-5
- Critical Fastener Torque Value: 175 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-XAX-5 at PLC register 0x48C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-LIN-COIL-1305) with seal kit (KIT-MAK250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 175 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-XAX-5 on Makino-Aero-T250 by installing PN-MAK250-LIN-COIL-1305 & kit KIT-MAK250-XAX-5. Fasteners torqued to 175 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-YAX-6

Target Machine Model: Makino-Aero-T250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK250-6
- PLC Hex Register Address: DB32.DBW_0x48D0
- Nominal Operating Range: 30 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-BSW-NUT-1306
- Overhaul Seal & Consumable Kit: KIT-MAK250-YAX-6
- Critical Fastener Torque Value: 184 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-YAX-6 at PLC register 0x48D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-BSW-NUT-1306) with seal kit (KIT-MAK250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 184 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-YAX-6 on Makino-Aero-T250 by installing PN-MAK250-BSW-NUT-1306 & kit KIT-MAK250-YAX-6. Fasteners torqued to 184 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-ZAX-7

Target Machine Model: Makino-Aero-T250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK250-7
- PLC Hex Register Address: DB32.DBW_0x48D8
- Nominal Operating Range: 33 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-CYL-SEAL-1307
- Overhaul Seal & Consumable Kit: KIT-MAK250-ZAX-7
- Critical Fastener Torque Value: 193 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-ZAX-7 at PLC register 0x48D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-CYL-SEAL-1307) with seal kit (KIT-MAK250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 193 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-ZAX-7 on Makino-Aero-T250 by installing PN-MAK250-CYL-SEAL-1307 & kit KIT-MAK250-ZAX-7. Fasteners torqued to 193 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-BAX-8

Target Machine Model: Makino-Aero-T250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK250-8
- PLC Hex Register Address: DB32.DBW_0x48E0
- Nominal Operating Range: 36 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-TRQ-STAT-1308
- Overhaul Seal & Consumable Kit: KIT-MAK250-BAX-8
- Critical Fastener Torque Value: 202 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-BAX-8 at PLC register 0x48E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-TRQ-STAT-1308) with seal kit (KIT-MAK250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 202 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-BAX-8 on Makino-Aero-T250 by installing PN-MAK250-TRQ-STAT-1308 & kit KIT-MAK250-BAX-8. Fasteners torqued to 202 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-CAX-9

Target Machine Model: Makino-Aero-T250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK250-9
- PLC Hex Register Address: DB32.DBW_0x48E8
- Nominal Operating Range: 39 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-BRK-DISC-1309
- Overhaul Seal & Consumable Kit: KIT-MAK250-CAX-9
- Critical Fastener Torque Value: 21 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-CAX-9 at PLC register 0x48E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-BRK-DISC-1309) with seal kit (KIT-MAK250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 21 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-CAX-9 on Makino-Aero-T250 by installing PN-MAK250-BRK-DISC-1309 & kit KIT-MAK250-CAX-9. Fasteners torqued to 21 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-OPT-10

Target Machine Model: Makino-Aero-T250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK250-10
- PLC Hex Register Address: DB32.DBW_0x48F0
- Nominal Operating Range: 42 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-GLASS-RD-1310
- Overhaul Seal & Consumable Kit: KIT-MAK250-OPT-10
- Critical Fastener Torque Value: 30 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-OPT-10 at PLC register 0x48F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-GLASS-RD-1310) with seal kit (KIT-MAK250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 30 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-OPT-10 on Makino-Aero-T250 by installing PN-MAK250-GLASS-RD-1310 & kit KIT-MAK250-OPT-10. Fasteners torqued to 30 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-PLC-11

Target Machine Model: Makino-Aero-T250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK250-11
- PLC Hex Register Address: DB32.DBW_0x48F8
- Nominal Operating Range: 45 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-IGBT-MOD-1311
- Overhaul Seal & Consumable Kit: KIT-MAK250-PLC-11
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-PLC-11 at PLC register 0x48F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-IGBT-MOD-1311) with seal kit (KIT-MAK250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-PLC-11 on Makino-Aero-T250 by installing PN-MAK250-IGBT-MOD-1311 & kit KIT-MAK250-PLC-11. Fasteners torqued to 39 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-LUB-12

Target Machine Model: Makino-Aero-T250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK250-12
- PLC Hex Register Address: DB32.DBW_0x4900
- Nominal Operating Range: 48 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-INJ-MET-1312
- Overhaul Seal & Consumable Kit: KIT-MAK250-LUB-12
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-LUB-12 at PLC register 0x4900.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-INJ-MET-1312) with seal kit (KIT-MAK250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-LUB-12 on Makino-Aero-T250 by installing PN-MAK250-INJ-MET-1312 & kit KIT-MAK250-LUB-12. Fasteners torqued to 48 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-VAC-13

Target Machine Model: Makino-Aero-T250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK250-13
- PLC Hex Register Address: DB32.DBW_0x4908
- Nominal Operating Range: 51 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-GASK-VIT-1313
- Overhaul Seal & Consumable Kit: KIT-MAK250-VAC-13
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-VAC-13 at PLC register 0x4908.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-GASK-VIT-1313) with seal kit (KIT-MAK250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-VAC-13 on Makino-Aero-T250 by installing PN-MAK250-GASK-VIT-1313 & kit KIT-MAK250-VAC-13. Fasteners torqued to 57 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK250-THM-14

Target Machine Model: Makino-Aero-T250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T250 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK250-14
- PLC Hex Register Address: DB32.DBW_0x4910
- Nominal Operating Range: 54 - 72 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 107 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK250-HEX-CORE-1314
- Overhaul Seal & Consumable Kit: KIT-MAK250-THM-14
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T250 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK250-THM-14 at PLC register 0x4910.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK250-HEX-CORE-1314) with seal kit (KIT-MAK250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T250, technician must log: "Resolved ERR-MAK250-THM-14 on Makino-Aero-T250 by installing PN-MAK250-HEX-CORE-1314 & kit KIT-MAK250-THM-14. Fasteners torqued to 66 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-SPN-1

Target Machine Model: Makino-Aero-T300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK300-1
- PLC Hex Register Address: DB33.DBW_0x4BC8
- Nominal Operating Range: 15 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-BRG-CER-1401
- Overhaul Seal & Consumable Kit: KIT-MAK300-SPN-1
- Critical Fastener Torque Value: 89 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-SPN-1 at PLC register 0x4BC8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-BRG-CER-1401) with seal kit (KIT-MAK300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 89 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-SPN-1 on Makino-Aero-T300 by installing PN-MAK300-BRG-CER-1401 & kit KIT-MAK300-SPN-1. Fasteners torqued to 89 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-ATC-2

Target Machine Model: Makino-Aero-T300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK300-2
- PLC Hex Register Address: DB33.DBW_0x4BD0
- Nominal Operating Range: 18 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-CAM-GRIP-1402
- Overhaul Seal & Consumable Kit: KIT-MAK300-ATC-2
- Critical Fastener Torque Value: 98 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-ATC-2 at PLC register 0x4BD0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-CAM-GRIP-1402) with seal kit (KIT-MAK300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 98 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-ATC-2 on Makino-Aero-T300 by installing PN-MAK300-CAM-GRIP-1402 & kit KIT-MAK300-ATC-2. Fasteners torqued to 98 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-HYD-3

Target Machine Model: Makino-Aero-T300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK300-3
- PLC Hex Register Address: DB33.DBW_0x4BD8
- Nominal Operating Range: 21 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-VLV-PROP-1403
- Overhaul Seal & Consumable Kit: KIT-MAK300-HYD-3
- Critical Fastener Torque Value: 107 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-HYD-3 at PLC register 0x4BD8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-VLV-PROP-1403) with seal kit (KIT-MAK300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 107 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-HYD-3 on Makino-Aero-T300 by installing PN-MAK300-VLV-PROP-1403 & kit KIT-MAK300-HYD-3. Fasteners torqued to 107 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-CLN-4

Target Machine Model: Makino-Aero-T300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK300-4
- PLC Hex Register Address: DB33.DBW_0x4BE0
- Nominal Operating Range: 24 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-PMP-PLG-1404
- Overhaul Seal & Consumable Kit: KIT-MAK300-CLN-4
- Critical Fastener Torque Value: 116 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-CLN-4 at PLC register 0x4BE0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-PMP-PLG-1404) with seal kit (KIT-MAK300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 116 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-CLN-4 on Makino-Aero-T300 by installing PN-MAK300-PMP-PLG-1404 & kit KIT-MAK300-CLN-4. Fasteners torqued to 116 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-XAX-5

Target Machine Model: Makino-Aero-T300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK300-5
- PLC Hex Register Address: DB33.DBW_0x4BE8
- Nominal Operating Range: 27 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-LIN-COIL-1405
- Overhaul Seal & Consumable Kit: KIT-MAK300-XAX-5
- Critical Fastener Torque Value: 125 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-XAX-5 at PLC register 0x4BE8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-LIN-COIL-1405) with seal kit (KIT-MAK300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 125 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-XAX-5 on Makino-Aero-T300 by installing PN-MAK300-LIN-COIL-1405 & kit KIT-MAK300-XAX-5. Fasteners torqued to 125 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-YAX-6

Target Machine Model: Makino-Aero-T300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK300-6
- PLC Hex Register Address: DB33.DBW_0x4BF0
- Nominal Operating Range: 30 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-BSW-NUT-1406
- Overhaul Seal & Consumable Kit: KIT-MAK300-YAX-6
- Critical Fastener Torque Value: 134 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-YAX-6 at PLC register 0x4BF0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-BSW-NUT-1406) with seal kit (KIT-MAK300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 134 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-YAX-6 on Makino-Aero-T300 by installing PN-MAK300-BSW-NUT-1406 & kit KIT-MAK300-YAX-6. Fasteners torqued to 134 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-ZAX-7

Target Machine Model: Makino-Aero-T300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK300-7
- PLC Hex Register Address: DB33.DBW_0x4BF8
- Nominal Operating Range: 33 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-CYL-SEAL-1407
- Overhaul Seal & Consumable Kit: KIT-MAK300-ZAX-7
- Critical Fastener Torque Value: 143 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-ZAX-7 at PLC register 0x4BF8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-CYL-SEAL-1407) with seal kit (KIT-MAK300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 143 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-ZAX-7 on Makino-Aero-T300 by installing PN-MAK300-CYL-SEAL-1407 & kit KIT-MAK300-ZAX-7. Fasteners torqued to 143 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-BAX-8

Target Machine Model: Makino-Aero-T300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK300-8
- PLC Hex Register Address: DB33.DBW_0x4C00
- Nominal Operating Range: 36 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-TRQ-STAT-1408
- Overhaul Seal & Consumable Kit: KIT-MAK300-BAX-8
- Critical Fastener Torque Value: 152 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-BAX-8 at PLC register 0x4C00.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-TRQ-STAT-1408) with seal kit (KIT-MAK300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 152 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-BAX-8 on Makino-Aero-T300 by installing PN-MAK300-TRQ-STAT-1408 & kit KIT-MAK300-BAX-8. Fasteners torqued to 152 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-CAX-9

Target Machine Model: Makino-Aero-T300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK300-9
- PLC Hex Register Address: DB33.DBW_0x4C08
- Nominal Operating Range: 39 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-BRK-DISC-1409
- Overhaul Seal & Consumable Kit: KIT-MAK300-CAX-9
- Critical Fastener Torque Value: 161 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-CAX-9 at PLC register 0x4C08.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-BRK-DISC-1409) with seal kit (KIT-MAK300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 161 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-CAX-9 on Makino-Aero-T300 by installing PN-MAK300-BRK-DISC-1409 & kit KIT-MAK300-CAX-9. Fasteners torqued to 161 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-OPT-10

Target Machine Model: Makino-Aero-T300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK300-10
- PLC Hex Register Address: DB33.DBW_0x4C10
- Nominal Operating Range: 42 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-GLASS-RD-1410
- Overhaul Seal & Consumable Kit: KIT-MAK300-OPT-10
- Critical Fastener Torque Value: 170 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-OPT-10 at PLC register 0x4C10.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-GLASS-RD-1410) with seal kit (KIT-MAK300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 170 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-OPT-10 on Makino-Aero-T300 by installing PN-MAK300-GLASS-RD-1410 & kit KIT-MAK300-OPT-10. Fasteners torqued to 170 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-PLC-11

Target Machine Model: Makino-Aero-T300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK300-11
- PLC Hex Register Address: DB33.DBW_0x4C18
- Nominal Operating Range: 45 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-IGBT-MOD-1411
- Overhaul Seal & Consumable Kit: KIT-MAK300-PLC-11
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-PLC-11 at PLC register 0x4C18.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-IGBT-MOD-1411) with seal kit (KIT-MAK300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-PLC-11 on Makino-Aero-T300 by installing PN-MAK300-IGBT-MOD-1411 & kit KIT-MAK300-PLC-11. Fasteners torqued to 179 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-LUB-12

Target Machine Model: Makino-Aero-T300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK300-12
- PLC Hex Register Address: DB33.DBW_0x4C20
- Nominal Operating Range: 48 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-INJ-MET-1412
- Overhaul Seal & Consumable Kit: KIT-MAK300-LUB-12
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-LUB-12 at PLC register 0x4C20.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-INJ-MET-1412) with seal kit (KIT-MAK300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-LUB-12 on Makino-Aero-T300 by installing PN-MAK300-INJ-MET-1412 & kit KIT-MAK300-LUB-12. Fasteners torqued to 188 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-VAC-13

Target Machine Model: Makino-Aero-T300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK300-13
- PLC Hex Register Address: DB33.DBW_0x4C28
- Nominal Operating Range: 51 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-GASK-VIT-1413
- Overhaul Seal & Consumable Kit: KIT-MAK300-VAC-13
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-VAC-13 at PLC register 0x4C28.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-GASK-VIT-1413) with seal kit (KIT-MAK300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-VAC-13 on Makino-Aero-T300 by installing PN-MAK300-GASK-VIT-1413 & kit KIT-MAK300-VAC-13. Fasteners torqued to 197 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK300-THM-14

Target Machine Model: Makino-Aero-T300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T300 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK300-14
- PLC Hex Register Address: DB33.DBW_0x4C30
- Nominal Operating Range: 54 - 73 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 108 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK300-HEX-CORE-1414
- Overhaul Seal & Consumable Kit: KIT-MAK300-THM-14
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T300 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK300-THM-14 at PLC register 0x4C30.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK300-HEX-CORE-1414) with seal kit (KIT-MAK300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T300, technician must log: "Resolved ERR-MAK300-THM-14 on Makino-Aero-T300 by installing PN-MAK300-HEX-CORE-1414 & kit KIT-MAK300-THM-14. Fasteners torqued to 206 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-SPN-1

Target Machine Model: Makino-Aero-T350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK350-1
- PLC Hex Register Address: DB34.DBW_0x4EE8
- Nominal Operating Range: 15 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-BRG-CER-1501
- Overhaul Seal & Consumable Kit: KIT-MAK350-SPN-1
- Critical Fastener Torque Value: 39 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-SPN-1 at PLC register 0x4EE8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-BRG-CER-1501) with seal kit (KIT-MAK350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 39 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-SPN-1 on Makino-Aero-T350 by installing PN-MAK350-BRG-CER-1501 & kit KIT-MAK350-SPN-1. Fasteners torqued to 39 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-ATC-2

Target Machine Model: Makino-Aero-T350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK350-2
- PLC Hex Register Address: DB34.DBW_0x4EF0
- Nominal Operating Range: 18 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-CAM-GRIP-1502
- Overhaul Seal & Consumable Kit: KIT-MAK350-ATC-2
- Critical Fastener Torque Value: 48 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-ATC-2 at PLC register 0x4EF0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-CAM-GRIP-1502) with seal kit (KIT-MAK350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 48 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-ATC-2 on Makino-Aero-T350 by installing PN-MAK350-CAM-GRIP-1502 & kit KIT-MAK350-ATC-2. Fasteners torqued to 48 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-HYD-3

Target Machine Model: Makino-Aero-T350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK350-3
- PLC Hex Register Address: DB34.DBW_0x4EF8
- Nominal Operating Range: 21 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-VLV-PROP-1503
- Overhaul Seal & Consumable Kit: KIT-MAK350-HYD-3
- Critical Fastener Torque Value: 57 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-HYD-3 at PLC register 0x4EF8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-VLV-PROP-1503) with seal kit (KIT-MAK350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 57 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-HYD-3 on Makino-Aero-T350 by installing PN-MAK350-VLV-PROP-1503 & kit KIT-MAK350-HYD-3. Fasteners torqued to 57 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-CLN-4

Target Machine Model: Makino-Aero-T350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK350-4
- PLC Hex Register Address: DB34.DBW_0x4F00
- Nominal Operating Range: 24 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-PMP-PLG-1504
- Overhaul Seal & Consumable Kit: KIT-MAK350-CLN-4
- Critical Fastener Torque Value: 66 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-CLN-4 at PLC register 0x4F00.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-PMP-PLG-1504) with seal kit (KIT-MAK350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 66 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-CLN-4 on Makino-Aero-T350 by installing PN-MAK350-PMP-PLG-1504 & kit KIT-MAK350-CLN-4. Fasteners torqued to 66 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-XAX-5

Target Machine Model: Makino-Aero-T350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK350-5
- PLC Hex Register Address: DB34.DBW_0x4F08
- Nominal Operating Range: 27 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-LIN-COIL-1505
- Overhaul Seal & Consumable Kit: KIT-MAK350-XAX-5
- Critical Fastener Torque Value: 75 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-XAX-5 at PLC register 0x4F08.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-LIN-COIL-1505) with seal kit (KIT-MAK350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 75 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-XAX-5 on Makino-Aero-T350 by installing PN-MAK350-LIN-COIL-1505 & kit KIT-MAK350-XAX-5. Fasteners torqued to 75 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-YAX-6

Target Machine Model: Makino-Aero-T350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK350-6
- PLC Hex Register Address: DB34.DBW_0x4F10
- Nominal Operating Range: 30 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-BSW-NUT-1506
- Overhaul Seal & Consumable Kit: KIT-MAK350-YAX-6
- Critical Fastener Torque Value: 84 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-YAX-6 at PLC register 0x4F10.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-BSW-NUT-1506) with seal kit (KIT-MAK350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 84 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-YAX-6 on Makino-Aero-T350 by installing PN-MAK350-BSW-NUT-1506 & kit KIT-MAK350-YAX-6. Fasteners torqued to 84 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-ZAX-7

Target Machine Model: Makino-Aero-T350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK350-7
- PLC Hex Register Address: DB34.DBW_0x4F18
- Nominal Operating Range: 33 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-CYL-SEAL-1507
- Overhaul Seal & Consumable Kit: KIT-MAK350-ZAX-7
- Critical Fastener Torque Value: 93 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-ZAX-7 at PLC register 0x4F18.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-CYL-SEAL-1507) with seal kit (KIT-MAK350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 93 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-ZAX-7 on Makino-Aero-T350 by installing PN-MAK350-CYL-SEAL-1507 & kit KIT-MAK350-ZAX-7. Fasteners torqued to 93 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-BAX-8

Target Machine Model: Makino-Aero-T350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK350-8
- PLC Hex Register Address: DB34.DBW_0x4F20
- Nominal Operating Range: 36 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-TRQ-STAT-1508
- Overhaul Seal & Consumable Kit: KIT-MAK350-BAX-8
- Critical Fastener Torque Value: 102 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-BAX-8 at PLC register 0x4F20.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-TRQ-STAT-1508) with seal kit (KIT-MAK350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 102 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-BAX-8 on Makino-Aero-T350 by installing PN-MAK350-TRQ-STAT-1508 & kit KIT-MAK350-BAX-8. Fasteners torqued to 102 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-CAX-9

Target Machine Model: Makino-Aero-T350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK350-9
- PLC Hex Register Address: DB34.DBW_0x4F28
- Nominal Operating Range: 39 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-BRK-DISC-1509
- Overhaul Seal & Consumable Kit: KIT-MAK350-CAX-9
- Critical Fastener Torque Value: 111 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-CAX-9 at PLC register 0x4F28.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-BRK-DISC-1509) with seal kit (KIT-MAK350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 111 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-CAX-9 on Makino-Aero-T350 by installing PN-MAK350-BRK-DISC-1509 & kit KIT-MAK350-CAX-9. Fasteners torqued to 111 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-OPT-10

Target Machine Model: Makino-Aero-T350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK350-10
- PLC Hex Register Address: DB34.DBW_0x4F30
- Nominal Operating Range: 42 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-GLASS-RD-1510
- Overhaul Seal & Consumable Kit: KIT-MAK350-OPT-10
- Critical Fastener Torque Value: 120 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-OPT-10 at PLC register 0x4F30.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-GLASS-RD-1510) with seal kit (KIT-MAK350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 120 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-OPT-10 on Makino-Aero-T350 by installing PN-MAK350-GLASS-RD-1510 & kit KIT-MAK350-OPT-10. Fasteners torqued to 120 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-PLC-11

Target Machine Model: Makino-Aero-T350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK350-11
- PLC Hex Register Address: DB34.DBW_0x4F38
- Nominal Operating Range: 45 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-IGBT-MOD-1511
- Overhaul Seal & Consumable Kit: KIT-MAK350-PLC-11
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-PLC-11 at PLC register 0x4F38.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-IGBT-MOD-1511) with seal kit (KIT-MAK350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-PLC-11 on Makino-Aero-T350 by installing PN-MAK350-IGBT-MOD-1511 & kit KIT-MAK350-PLC-11. Fasteners torqued to 129 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-LUB-12

Target Machine Model: Makino-Aero-T350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK350-12
- PLC Hex Register Address: DB34.DBW_0x4F40
- Nominal Operating Range: 48 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-INJ-MET-1512
- Overhaul Seal & Consumable Kit: KIT-MAK350-LUB-12
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-LUB-12 at PLC register 0x4F40.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-INJ-MET-1512) with seal kit (KIT-MAK350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-LUB-12 on Makino-Aero-T350 by installing PN-MAK350-INJ-MET-1512 & kit KIT-MAK350-LUB-12. Fasteners torqued to 138 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-VAC-13

Target Machine Model: Makino-Aero-T350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK350-13
- PLC Hex Register Address: DB34.DBW_0x4F48
- Nominal Operating Range: 51 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-GASK-VIT-1513
- Overhaul Seal & Consumable Kit: KIT-MAK350-VAC-13
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-VAC-13 at PLC register 0x4F48.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-GASK-VIT-1513) with seal kit (KIT-MAK350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-VAC-13 on Makino-Aero-T350 by installing PN-MAK350-GASK-VIT-1513 & kit KIT-MAK350-VAC-13. Fasteners torqued to 147 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK350-THM-14

Target Machine Model: Makino-Aero-T350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T350 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK350-14
- PLC Hex Register Address: DB34.DBW_0x4F50
- Nominal Operating Range: 54 - 74 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 109 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK350-HEX-CORE-1514
- Overhaul Seal & Consumable Kit: KIT-MAK350-THM-14
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T350 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK350-THM-14 at PLC register 0x4F50.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK350-HEX-CORE-1514) with seal kit (KIT-MAK350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T350, technician must log: "Resolved ERR-MAK350-THM-14 on Makino-Aero-T350 by installing PN-MAK350-HEX-CORE-1514 & kit KIT-MAK350-THM-14. Fasteners torqued to 156 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-SPN-1

Target Machine Model: Makino-Aero-T400 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK400-1
- PLC Hex Register Address: DB20.DBW_0x5208
- Nominal Operating Range: 15 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-BRG-CER-1601
- Overhaul Seal & Consumable Kit: KIT-MAK400-SPN-1
- Critical Fastener Torque Value: 179 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-SPN-1 at PLC register 0x5208.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-BRG-CER-1601) with seal kit (KIT-MAK400-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 179 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-SPN-1 on Makino-Aero-T400 by installing PN-MAK400-BRG-CER-1601 & kit KIT-MAK400-SPN-1. Fasteners torqued to 179 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-ATC-2

Target Machine Model: Makino-Aero-T400 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK400-2
- PLC Hex Register Address: DB20.DBW_0x5210
- Nominal Operating Range: 18 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-CAM-GRIP-1602
- Overhaul Seal & Consumable Kit: KIT-MAK400-ATC-2
- Critical Fastener Torque Value: 188 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-ATC-2 at PLC register 0x5210.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-CAM-GRIP-1602) with seal kit (KIT-MAK400-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 188 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-ATC-2 on Makino-Aero-T400 by installing PN-MAK400-CAM-GRIP-1602 & kit KIT-MAK400-ATC-2. Fasteners torqued to 188 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-HYD-3

Target Machine Model: Makino-Aero-T400 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK400-3
- PLC Hex Register Address: DB20.DBW_0x5218
- Nominal Operating Range: 21 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-VLV-PROP-1603
- Overhaul Seal & Consumable Kit: KIT-MAK400-HYD-3
- Critical Fastener Torque Value: 197 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-HYD-3 at PLC register 0x5218.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-VLV-PROP-1603) with seal kit (KIT-MAK400-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 197 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-HYD-3 on Makino-Aero-T400 by installing PN-MAK400-VLV-PROP-1603 & kit KIT-MAK400-HYD-3. Fasteners torqued to 197 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-CLN-4

Target Machine Model: Makino-Aero-T400 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK400-4
- PLC Hex Register Address: DB20.DBW_0x5220
- Nominal Operating Range: 24 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-PMP-PLG-1604
- Overhaul Seal & Consumable Kit: KIT-MAK400-CLN-4
- Critical Fastener Torque Value: 206 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-CLN-4 at PLC register 0x5220.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-PMP-PLG-1604) with seal kit (KIT-MAK400-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 206 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-CLN-4 on Makino-Aero-T400 by installing PN-MAK400-PMP-PLG-1604 & kit KIT-MAK400-CLN-4. Fasteners torqued to 206 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-XAX-5

Target Machine Model: Makino-Aero-T400 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK400-5
- PLC Hex Register Address: DB20.DBW_0x5228
- Nominal Operating Range: 27 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-LIN-COIL-1605
- Overhaul Seal & Consumable Kit: KIT-MAK400-XAX-5
- Critical Fastener Torque Value: 25 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-XAX-5 at PLC register 0x5228.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-LIN-COIL-1605) with seal kit (KIT-MAK400-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 25 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-XAX-5 on Makino-Aero-T400 by installing PN-MAK400-LIN-COIL-1605 & kit KIT-MAK400-XAX-5. Fasteners torqued to 25 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-YAX-6

Target Machine Model: Makino-Aero-T400 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK400-6
- PLC Hex Register Address: DB20.DBW_0x5230
- Nominal Operating Range: 30 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-BSW-NUT-1606
- Overhaul Seal & Consumable Kit: KIT-MAK400-YAX-6
- Critical Fastener Torque Value: 34 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-YAX-6 at PLC register 0x5230.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-BSW-NUT-1606) with seal kit (KIT-MAK400-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 34 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-YAX-6 on Makino-Aero-T400 by installing PN-MAK400-BSW-NUT-1606 & kit KIT-MAK400-YAX-6. Fasteners torqued to 34 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-ZAX-7

Target Machine Model: Makino-Aero-T400 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK400-7
- PLC Hex Register Address: DB20.DBW_0x5238
- Nominal Operating Range: 33 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-CYL-SEAL-1607
- Overhaul Seal & Consumable Kit: KIT-MAK400-ZAX-7
- Critical Fastener Torque Value: 43 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-ZAX-7 at PLC register 0x5238.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-CYL-SEAL-1607) with seal kit (KIT-MAK400-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 43 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-ZAX-7 on Makino-Aero-T400 by installing PN-MAK400-CYL-SEAL-1607 & kit KIT-MAK400-ZAX-7. Fasteners torqued to 43 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-BAX-8

Target Machine Model: Makino-Aero-T400 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK400-8
- PLC Hex Register Address: DB20.DBW_0x5240
- Nominal Operating Range: 36 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-TRQ-STAT-1608
- Overhaul Seal & Consumable Kit: KIT-MAK400-BAX-8
- Critical Fastener Torque Value: 52 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-BAX-8 at PLC register 0x5240.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-TRQ-STAT-1608) with seal kit (KIT-MAK400-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 52 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-BAX-8 on Makino-Aero-T400 by installing PN-MAK400-TRQ-STAT-1608 & kit KIT-MAK400-BAX-8. Fasteners torqued to 52 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-CAX-9

Target Machine Model: Makino-Aero-T400 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK400-9
- PLC Hex Register Address: DB20.DBW_0x5248
- Nominal Operating Range: 39 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-BRK-DISC-1609
- Overhaul Seal & Consumable Kit: KIT-MAK400-CAX-9
- Critical Fastener Torque Value: 61 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-CAX-9 at PLC register 0x5248.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-BRK-DISC-1609) with seal kit (KIT-MAK400-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 61 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-CAX-9 on Makino-Aero-T400 by installing PN-MAK400-BRK-DISC-1609 & kit KIT-MAK400-CAX-9. Fasteners torqued to 61 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-OPT-10

Target Machine Model: Makino-Aero-T400 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK400-10
- PLC Hex Register Address: DB20.DBW_0x5250
- Nominal Operating Range: 42 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-GLASS-RD-1610
- Overhaul Seal & Consumable Kit: KIT-MAK400-OPT-10
- Critical Fastener Torque Value: 70 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-OPT-10 at PLC register 0x5250.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-GLASS-RD-1610) with seal kit (KIT-MAK400-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 70 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-OPT-10 on Makino-Aero-T400 by installing PN-MAK400-GLASS-RD-1610 & kit KIT-MAK400-OPT-10. Fasteners torqued to 70 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-PLC-11

Target Machine Model: Makino-Aero-T400 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK400-11
- PLC Hex Register Address: DB20.DBW_0x5258
- Nominal Operating Range: 45 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-IGBT-MOD-1611
- Overhaul Seal & Consumable Kit: KIT-MAK400-PLC-11
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-PLC-11 at PLC register 0x5258.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-IGBT-MOD-1611) with seal kit (KIT-MAK400-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-PLC-11 on Makino-Aero-T400 by installing PN-MAK400-IGBT-MOD-1611 & kit KIT-MAK400-PLC-11. Fasteners torqued to 79 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-LUB-12

Target Machine Model: Makino-Aero-T400 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK400-12
- PLC Hex Register Address: DB20.DBW_0x5260
- Nominal Operating Range: 48 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-INJ-MET-1612
- Overhaul Seal & Consumable Kit: KIT-MAK400-LUB-12
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-LUB-12 at PLC register 0x5260.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-INJ-MET-1612) with seal kit (KIT-MAK400-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-LUB-12 on Makino-Aero-T400 by installing PN-MAK400-INJ-MET-1612 & kit KIT-MAK400-LUB-12. Fasteners torqued to 88 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-VAC-13

Target Machine Model: Makino-Aero-T400 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK400-13
- PLC Hex Register Address: DB20.DBW_0x5268
- Nominal Operating Range: 51 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-GASK-VIT-1613
- Overhaul Seal & Consumable Kit: KIT-MAK400-VAC-13
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-VAC-13 at PLC register 0x5268.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-GASK-VIT-1613) with seal kit (KIT-MAK400-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-VAC-13 on Makino-Aero-T400 by installing PN-MAK400-GASK-VIT-1613 & kit KIT-MAK400-VAC-13. Fasteners torqued to 97 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK400-THM-14

Target Machine Model: Makino-Aero-T400 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T400 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T400 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T400 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK400-14
- PLC Hex Register Address: DB20.DBW_0x5270
- Nominal Operating Range: 54 - 75 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 110 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK400-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK400-HEX-CORE-1614
- Overhaul Seal & Consumable Kit: KIT-MAK400-THM-14
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T400 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK400-THM-14 at PLC register 0x5270.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T400 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK400-HEX-CORE-1614) with seal kit (KIT-MAK400-THM-14).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK400-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T400, technician must log: "Resolved ERR-MAK400-THM-14 on Makino-Aero-T400 by installing PN-MAK400-HEX-CORE-1614 & kit KIT-MAK400-THM-14. Fasteners torqued to 106 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-SPN-1

Target Machine Model: Makino-Aero-T450 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK450-1
- PLC Hex Register Address: DB21.DBW_0x5528
- Nominal Operating Range: 15 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-BRG-CER-1701
- Overhaul Seal & Consumable Kit: KIT-MAK450-SPN-1
- Critical Fastener Torque Value: 129 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-SPN-1 at PLC register 0x5528.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-BRG-CER-1701) with seal kit (KIT-MAK450-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 129 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-SPN-1 on Makino-Aero-T450 by installing PN-MAK450-BRG-CER-1701 & kit KIT-MAK450-SPN-1. Fasteners torqued to 129 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-ATC-2

Target Machine Model: Makino-Aero-T450 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK450-2
- PLC Hex Register Address: DB21.DBW_0x5530
- Nominal Operating Range: 18 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-CAM-GRIP-1702
- Overhaul Seal & Consumable Kit: KIT-MAK450-ATC-2
- Critical Fastener Torque Value: 138 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-ATC-2 at PLC register 0x5530.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-CAM-GRIP-1702) with seal kit (KIT-MAK450-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 138 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-ATC-2 on Makino-Aero-T450 by installing PN-MAK450-CAM-GRIP-1702 & kit KIT-MAK450-ATC-2. Fasteners torqued to 138 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-HYD-3

Target Machine Model: Makino-Aero-T450 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK450-3
- PLC Hex Register Address: DB21.DBW_0x5538
- Nominal Operating Range: 21 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-VLV-PROP-1703
- Overhaul Seal & Consumable Kit: KIT-MAK450-HYD-3
- Critical Fastener Torque Value: 147 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-HYD-3 at PLC register 0x5538.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-VLV-PROP-1703) with seal kit (KIT-MAK450-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 147 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-HYD-3 on Makino-Aero-T450 by installing PN-MAK450-VLV-PROP-1703 & kit KIT-MAK450-HYD-3. Fasteners torqued to 147 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-CLN-4

Target Machine Model: Makino-Aero-T450 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK450-4
- PLC Hex Register Address: DB21.DBW_0x5540
- Nominal Operating Range: 24 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-PMP-PLG-1704
- Overhaul Seal & Consumable Kit: KIT-MAK450-CLN-4
- Critical Fastener Torque Value: 156 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-CLN-4 at PLC register 0x5540.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-PMP-PLG-1704) with seal kit (KIT-MAK450-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 156 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-CLN-4 on Makino-Aero-T450 by installing PN-MAK450-PMP-PLG-1704 & kit KIT-MAK450-CLN-4. Fasteners torqued to 156 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-XAX-5

Target Machine Model: Makino-Aero-T450 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK450-5
- PLC Hex Register Address: DB21.DBW_0x5548
- Nominal Operating Range: 27 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-LIN-COIL-1705
- Overhaul Seal & Consumable Kit: KIT-MAK450-XAX-5
- Critical Fastener Torque Value: 165 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-XAX-5 at PLC register 0x5548.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-LIN-COIL-1705) with seal kit (KIT-MAK450-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 165 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-XAX-5 on Makino-Aero-T450 by installing PN-MAK450-LIN-COIL-1705 & kit KIT-MAK450-XAX-5. Fasteners torqued to 165 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-YAX-6

Target Machine Model: Makino-Aero-T450 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK450-6
- PLC Hex Register Address: DB21.DBW_0x5550
- Nominal Operating Range: 30 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-BSW-NUT-1706
- Overhaul Seal & Consumable Kit: KIT-MAK450-YAX-6
- Critical Fastener Torque Value: 174 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-YAX-6 at PLC register 0x5550.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-BSW-NUT-1706) with seal kit (KIT-MAK450-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 174 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-YAX-6 on Makino-Aero-T450 by installing PN-MAK450-BSW-NUT-1706 & kit KIT-MAK450-YAX-6. Fasteners torqued to 174 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-ZAX-7

Target Machine Model: Makino-Aero-T450 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK450-7
- PLC Hex Register Address: DB21.DBW_0x5558
- Nominal Operating Range: 33 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-CYL-SEAL-1707
- Overhaul Seal & Consumable Kit: KIT-MAK450-ZAX-7
- Critical Fastener Torque Value: 183 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-ZAX-7 at PLC register 0x5558.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-CYL-SEAL-1707) with seal kit (KIT-MAK450-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 183 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-ZAX-7 on Makino-Aero-T450 by installing PN-MAK450-CYL-SEAL-1707 & kit KIT-MAK450-ZAX-7. Fasteners torqued to 183 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-BAX-8

Target Machine Model: Makino-Aero-T450 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK450-8
- PLC Hex Register Address: DB21.DBW_0x5560
- Nominal Operating Range: 36 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-TRQ-STAT-1708
- Overhaul Seal & Consumable Kit: KIT-MAK450-BAX-8
- Critical Fastener Torque Value: 192 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-BAX-8 at PLC register 0x5560.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-TRQ-STAT-1708) with seal kit (KIT-MAK450-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 192 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-BAX-8 on Makino-Aero-T450 by installing PN-MAK450-TRQ-STAT-1708 & kit KIT-MAK450-BAX-8. Fasteners torqued to 192 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-CAX-9

Target Machine Model: Makino-Aero-T450 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK450-9
- PLC Hex Register Address: DB21.DBW_0x5568
- Nominal Operating Range: 39 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-BRK-DISC-1709
- Overhaul Seal & Consumable Kit: KIT-MAK450-CAX-9
- Critical Fastener Torque Value: 201 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-CAX-9 at PLC register 0x5568.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-BRK-DISC-1709) with seal kit (KIT-MAK450-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 201 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-CAX-9 on Makino-Aero-T450 by installing PN-MAK450-BRK-DISC-1709 & kit KIT-MAK450-CAX-9. Fasteners torqued to 201 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-OPT-10

Target Machine Model: Makino-Aero-T450 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK450-10
- PLC Hex Register Address: DB21.DBW_0x5570
- Nominal Operating Range: 42 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-GLASS-RD-1710
- Overhaul Seal & Consumable Kit: KIT-MAK450-OPT-10
- Critical Fastener Torque Value: 20 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-OPT-10 at PLC register 0x5570.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-GLASS-RD-1710) with seal kit (KIT-MAK450-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 20 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-OPT-10 on Makino-Aero-T450 by installing PN-MAK450-GLASS-RD-1710 & kit KIT-MAK450-OPT-10. Fasteners torqued to 20 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-PLC-11

Target Machine Model: Makino-Aero-T450 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK450-11
- PLC Hex Register Address: DB21.DBW_0x5578
- Nominal Operating Range: 45 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-IGBT-MOD-1711
- Overhaul Seal & Consumable Kit: KIT-MAK450-PLC-11
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-PLC-11 at PLC register 0x5578.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-IGBT-MOD-1711) with seal kit (KIT-MAK450-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-PLC-11 on Makino-Aero-T450 by installing PN-MAK450-IGBT-MOD-1711 & kit KIT-MAK450-PLC-11. Fasteners torqued to 29 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-LUB-12

Target Machine Model: Makino-Aero-T450 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK450-12
- PLC Hex Register Address: DB21.DBW_0x5580
- Nominal Operating Range: 48 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-INJ-MET-1712
- Overhaul Seal & Consumable Kit: KIT-MAK450-LUB-12
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-LUB-12 at PLC register 0x5580.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-INJ-MET-1712) with seal kit (KIT-MAK450-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-LUB-12 on Makino-Aero-T450 by installing PN-MAK450-INJ-MET-1712 & kit KIT-MAK450-LUB-12. Fasteners torqued to 38 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-VAC-13

Target Machine Model: Makino-Aero-T450 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK450-13
- PLC Hex Register Address: DB21.DBW_0x5588
- Nominal Operating Range: 51 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-GASK-VIT-1713
- Overhaul Seal & Consumable Kit: KIT-MAK450-VAC-13
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-VAC-13 at PLC register 0x5588.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-GASK-VIT-1713) with seal kit (KIT-MAK450-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-VAC-13 on Makino-Aero-T450 by installing PN-MAK450-GASK-VIT-1713 & kit KIT-MAK450-VAC-13. Fasteners torqued to 47 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK450-THM-14

Target Machine Model: Makino-Aero-T450 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T450 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T450 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T450 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK450-14
- PLC Hex Register Address: DB21.DBW_0x5590
- Nominal Operating Range: 54 - 76 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 111 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK450-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK450-HEX-CORE-1714
- Overhaul Seal & Consumable Kit: KIT-MAK450-THM-14
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T450 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK450-THM-14 at PLC register 0x5590.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T450 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK450-HEX-CORE-1714) with seal kit (KIT-MAK450-THM-14).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK450-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T450, technician must log: "Resolved ERR-MAK450-THM-14 on Makino-Aero-T450 by installing PN-MAK450-HEX-CORE-1714 & kit KIT-MAK450-THM-14. Fasteners torqued to 56 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-SPN-1

Target Machine Model: Makino-Aero-T500 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK500-1
- PLC Hex Register Address: DB22.DBW_0x5848
- Nominal Operating Range: 15 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-BRG-CER-1801
- Overhaul Seal & Consumable Kit: KIT-MAK500-SPN-1
- Critical Fastener Torque Value: 79 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-SPN-1 at PLC register 0x5848.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-BRG-CER-1801) with seal kit (KIT-MAK500-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 79 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-SPN-1 on Makino-Aero-T500 by installing PN-MAK500-BRG-CER-1801 & kit KIT-MAK500-SPN-1. Fasteners torqued to 79 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-ATC-2

Target Machine Model: Makino-Aero-T500 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK500-2
- PLC Hex Register Address: DB22.DBW_0x5850
- Nominal Operating Range: 18 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-CAM-GRIP-1802
- Overhaul Seal & Consumable Kit: KIT-MAK500-ATC-2
- Critical Fastener Torque Value: 88 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-ATC-2 at PLC register 0x5850.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-CAM-GRIP-1802) with seal kit (KIT-MAK500-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 88 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-ATC-2 on Makino-Aero-T500 by installing PN-MAK500-CAM-GRIP-1802 & kit KIT-MAK500-ATC-2. Fasteners torqued to 88 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-HYD-3

Target Machine Model: Makino-Aero-T500 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK500-3
- PLC Hex Register Address: DB22.DBW_0x5858
- Nominal Operating Range: 21 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-VLV-PROP-1803
- Overhaul Seal & Consumable Kit: KIT-MAK500-HYD-3
- Critical Fastener Torque Value: 97 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-HYD-3 at PLC register 0x5858.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-VLV-PROP-1803) with seal kit (KIT-MAK500-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 97 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-HYD-3 on Makino-Aero-T500 by installing PN-MAK500-VLV-PROP-1803 & kit KIT-MAK500-HYD-3. Fasteners torqued to 97 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-CLN-4

Target Machine Model: Makino-Aero-T500 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK500-4
- PLC Hex Register Address: DB22.DBW_0x5860
- Nominal Operating Range: 24 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-PMP-PLG-1804
- Overhaul Seal & Consumable Kit: KIT-MAK500-CLN-4
- Critical Fastener Torque Value: 106 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-CLN-4 at PLC register 0x5860.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-PMP-PLG-1804) with seal kit (KIT-MAK500-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 106 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-CLN-4 on Makino-Aero-T500 by installing PN-MAK500-PMP-PLG-1804 & kit KIT-MAK500-CLN-4. Fasteners torqued to 106 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-XAX-5

Target Machine Model: Makino-Aero-T500 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK500-5
- PLC Hex Register Address: DB22.DBW_0x5868
- Nominal Operating Range: 27 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-LIN-COIL-1805
- Overhaul Seal & Consumable Kit: KIT-MAK500-XAX-5
- Critical Fastener Torque Value: 115 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-XAX-5 at PLC register 0x5868.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-LIN-COIL-1805) with seal kit (KIT-MAK500-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 115 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-XAX-5 on Makino-Aero-T500 by installing PN-MAK500-LIN-COIL-1805 & kit KIT-MAK500-XAX-5. Fasteners torqued to 115 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-YAX-6

Target Machine Model: Makino-Aero-T500 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK500-6
- PLC Hex Register Address: DB22.DBW_0x5870
- Nominal Operating Range: 30 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-BSW-NUT-1806
- Overhaul Seal & Consumable Kit: KIT-MAK500-YAX-6
- Critical Fastener Torque Value: 124 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-YAX-6 at PLC register 0x5870.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-BSW-NUT-1806) with seal kit (KIT-MAK500-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 124 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-YAX-6 on Makino-Aero-T500 by installing PN-MAK500-BSW-NUT-1806 & kit KIT-MAK500-YAX-6. Fasteners torqued to 124 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-ZAX-7

Target Machine Model: Makino-Aero-T500 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK500-7
- PLC Hex Register Address: DB22.DBW_0x5878
- Nominal Operating Range: 33 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-CYL-SEAL-1807
- Overhaul Seal & Consumable Kit: KIT-MAK500-ZAX-7
- Critical Fastener Torque Value: 133 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-ZAX-7 at PLC register 0x5878.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-CYL-SEAL-1807) with seal kit (KIT-MAK500-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 133 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-ZAX-7 on Makino-Aero-T500 by installing PN-MAK500-CYL-SEAL-1807 & kit KIT-MAK500-ZAX-7. Fasteners torqued to 133 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-BAX-8

Target Machine Model: Makino-Aero-T500 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK500-8
- PLC Hex Register Address: DB22.DBW_0x5880
- Nominal Operating Range: 36 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-TRQ-STAT-1808
- Overhaul Seal & Consumable Kit: KIT-MAK500-BAX-8
- Critical Fastener Torque Value: 142 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-BAX-8 at PLC register 0x5880.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-TRQ-STAT-1808) with seal kit (KIT-MAK500-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 142 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-BAX-8 on Makino-Aero-T500 by installing PN-MAK500-TRQ-STAT-1808 & kit KIT-MAK500-BAX-8. Fasteners torqued to 142 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-CAX-9

Target Machine Model: Makino-Aero-T500 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK500-9
- PLC Hex Register Address: DB22.DBW_0x5888
- Nominal Operating Range: 39 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-BRK-DISC-1809
- Overhaul Seal & Consumable Kit: KIT-MAK500-CAX-9
- Critical Fastener Torque Value: 151 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-CAX-9 at PLC register 0x5888.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-BRK-DISC-1809) with seal kit (KIT-MAK500-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 151 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-CAX-9 on Makino-Aero-T500 by installing PN-MAK500-BRK-DISC-1809 & kit KIT-MAK500-CAX-9. Fasteners torqued to 151 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-OPT-10

Target Machine Model: Makino-Aero-T500 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK500-10
- PLC Hex Register Address: DB22.DBW_0x5890
- Nominal Operating Range: 42 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-GLASS-RD-1810
- Overhaul Seal & Consumable Kit: KIT-MAK500-OPT-10
- Critical Fastener Torque Value: 160 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-OPT-10 at PLC register 0x5890.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-GLASS-RD-1810) with seal kit (KIT-MAK500-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 160 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-OPT-10 on Makino-Aero-T500 by installing PN-MAK500-GLASS-RD-1810 & kit KIT-MAK500-OPT-10. Fasteners torqued to 160 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-PLC-11

Target Machine Model: Makino-Aero-T500 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK500-11
- PLC Hex Register Address: DB22.DBW_0x5898
- Nominal Operating Range: 45 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-IGBT-MOD-1811
- Overhaul Seal & Consumable Kit: KIT-MAK500-PLC-11
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-PLC-11 at PLC register 0x5898.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-IGBT-MOD-1811) with seal kit (KIT-MAK500-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-PLC-11 on Makino-Aero-T500 by installing PN-MAK500-IGBT-MOD-1811 & kit KIT-MAK500-PLC-11. Fasteners torqued to 169 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-LUB-12

Target Machine Model: Makino-Aero-T500 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK500-12
- PLC Hex Register Address: DB22.DBW_0x58A0
- Nominal Operating Range: 48 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-INJ-MET-1812
- Overhaul Seal & Consumable Kit: KIT-MAK500-LUB-12
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-LUB-12 at PLC register 0x58A0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-INJ-MET-1812) with seal kit (KIT-MAK500-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-LUB-12 on Makino-Aero-T500 by installing PN-MAK500-INJ-MET-1812 & kit KIT-MAK500-LUB-12. Fasteners torqued to 178 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-VAC-13

Target Machine Model: Makino-Aero-T500 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK500-13
- PLC Hex Register Address: DB22.DBW_0x58A8
- Nominal Operating Range: 51 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-GASK-VIT-1813
- Overhaul Seal & Consumable Kit: KIT-MAK500-VAC-13
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-VAC-13 at PLC register 0x58A8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-GASK-VIT-1813) with seal kit (KIT-MAK500-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-VAC-13 on Makino-Aero-T500 by installing PN-MAK500-GASK-VIT-1813 & kit KIT-MAK500-VAC-13. Fasteners torqued to 187 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK500-THM-14

Target Machine Model: Makino-Aero-T500 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T500 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T500 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T500 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK500-14
- PLC Hex Register Address: DB22.DBW_0x58B0
- Nominal Operating Range: 54 - 77 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 112 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK500-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK500-HEX-CORE-1814
- Overhaul Seal & Consumable Kit: KIT-MAK500-THM-14
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T500 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK500-THM-14 at PLC register 0x58B0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T500 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK500-HEX-CORE-1814) with seal kit (KIT-MAK500-THM-14).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK500-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T500, technician must log: "Resolved ERR-MAK500-THM-14 on Makino-Aero-T500 by installing PN-MAK500-HEX-CORE-1814 & kit KIT-MAK500-THM-14. Fasteners torqued to 196 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-SPN-1

Target Machine Model: Makino-Aero-T550 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK550-1
- PLC Hex Register Address: DB23.DBW_0x5B68
- Nominal Operating Range: 15 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-BRG-CER-1901
- Overhaul Seal & Consumable Kit: KIT-MAK550-SPN-1
- Critical Fastener Torque Value: 29 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-SPN-1 at PLC register 0x5B68.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-BRG-CER-1901) with seal kit (KIT-MAK550-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 29 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-SPN-1 on Makino-Aero-T550 by installing PN-MAK550-BRG-CER-1901 & kit KIT-MAK550-SPN-1. Fasteners torqued to 29 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-ATC-2

Target Machine Model: Makino-Aero-T550 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK550-2
- PLC Hex Register Address: DB23.DBW_0x5B70
- Nominal Operating Range: 18 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-CAM-GRIP-1902
- Overhaul Seal & Consumable Kit: KIT-MAK550-ATC-2
- Critical Fastener Torque Value: 38 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-ATC-2 at PLC register 0x5B70.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-CAM-GRIP-1902) with seal kit (KIT-MAK550-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 38 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-ATC-2 on Makino-Aero-T550 by installing PN-MAK550-CAM-GRIP-1902 & kit KIT-MAK550-ATC-2. Fasteners torqued to 38 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-HYD-3

Target Machine Model: Makino-Aero-T550 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK550-3
- PLC Hex Register Address: DB23.DBW_0x5B78
- Nominal Operating Range: 21 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-VLV-PROP-1903
- Overhaul Seal & Consumable Kit: KIT-MAK550-HYD-3
- Critical Fastener Torque Value: 47 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-HYD-3 at PLC register 0x5B78.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-VLV-PROP-1903) with seal kit (KIT-MAK550-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 47 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-HYD-3 on Makino-Aero-T550 by installing PN-MAK550-VLV-PROP-1903 & kit KIT-MAK550-HYD-3. Fasteners torqued to 47 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-CLN-4

Target Machine Model: Makino-Aero-T550 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK550-4
- PLC Hex Register Address: DB23.DBW_0x5B80
- Nominal Operating Range: 24 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-PMP-PLG-1904
- Overhaul Seal & Consumable Kit: KIT-MAK550-CLN-4
- Critical Fastener Torque Value: 56 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-CLN-4 at PLC register 0x5B80.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-PMP-PLG-1904) with seal kit (KIT-MAK550-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 56 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-CLN-4 on Makino-Aero-T550 by installing PN-MAK550-PMP-PLG-1904 & kit KIT-MAK550-CLN-4. Fasteners torqued to 56 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-XAX-5

Target Machine Model: Makino-Aero-T550 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK550-5
- PLC Hex Register Address: DB23.DBW_0x5B88
- Nominal Operating Range: 27 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-LIN-COIL-1905
- Overhaul Seal & Consumable Kit: KIT-MAK550-XAX-5
- Critical Fastener Torque Value: 65 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-XAX-5 at PLC register 0x5B88.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-LIN-COIL-1905) with seal kit (KIT-MAK550-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 65 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-XAX-5 on Makino-Aero-T550 by installing PN-MAK550-LIN-COIL-1905 & kit KIT-MAK550-XAX-5. Fasteners torqued to 65 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-YAX-6

Target Machine Model: Makino-Aero-T550 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK550-6
- PLC Hex Register Address: DB23.DBW_0x5B90
- Nominal Operating Range: 30 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-BSW-NUT-1906
- Overhaul Seal & Consumable Kit: KIT-MAK550-YAX-6
- Critical Fastener Torque Value: 74 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-YAX-6 at PLC register 0x5B90.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-BSW-NUT-1906) with seal kit (KIT-MAK550-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 74 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-YAX-6 on Makino-Aero-T550 by installing PN-MAK550-BSW-NUT-1906 & kit KIT-MAK550-YAX-6. Fasteners torqued to 74 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-ZAX-7

Target Machine Model: Makino-Aero-T550 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK550-7
- PLC Hex Register Address: DB23.DBW_0x5B98
- Nominal Operating Range: 33 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-CYL-SEAL-1907
- Overhaul Seal & Consumable Kit: KIT-MAK550-ZAX-7
- Critical Fastener Torque Value: 83 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-ZAX-7 at PLC register 0x5B98.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-CYL-SEAL-1907) with seal kit (KIT-MAK550-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 83 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-ZAX-7 on Makino-Aero-T550 by installing PN-MAK550-CYL-SEAL-1907 & kit KIT-MAK550-ZAX-7. Fasteners torqued to 83 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-BAX-8

Target Machine Model: Makino-Aero-T550 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK550-8
- PLC Hex Register Address: DB23.DBW_0x5BA0
- Nominal Operating Range: 36 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-TRQ-STAT-1908
- Overhaul Seal & Consumable Kit: KIT-MAK550-BAX-8
- Critical Fastener Torque Value: 92 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-BAX-8 at PLC register 0x5BA0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-TRQ-STAT-1908) with seal kit (KIT-MAK550-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 92 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-BAX-8 on Makino-Aero-T550 by installing PN-MAK550-TRQ-STAT-1908 & kit KIT-MAK550-BAX-8. Fasteners torqued to 92 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-CAX-9

Target Machine Model: Makino-Aero-T550 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK550-9
- PLC Hex Register Address: DB23.DBW_0x5BA8
- Nominal Operating Range: 39 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-BRK-DISC-1909
- Overhaul Seal & Consumable Kit: KIT-MAK550-CAX-9
- Critical Fastener Torque Value: 101 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-CAX-9 at PLC register 0x5BA8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-BRK-DISC-1909) with seal kit (KIT-MAK550-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 101 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-CAX-9 on Makino-Aero-T550 by installing PN-MAK550-BRK-DISC-1909 & kit KIT-MAK550-CAX-9. Fasteners torqued to 101 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-OPT-10

Target Machine Model: Makino-Aero-T550 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK550-10
- PLC Hex Register Address: DB23.DBW_0x5BB0
- Nominal Operating Range: 42 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-GLASS-RD-1910
- Overhaul Seal & Consumable Kit: KIT-MAK550-OPT-10
- Critical Fastener Torque Value: 110 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-OPT-10 at PLC register 0x5BB0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-GLASS-RD-1910) with seal kit (KIT-MAK550-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 110 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-OPT-10 on Makino-Aero-T550 by installing PN-MAK550-GLASS-RD-1910 & kit KIT-MAK550-OPT-10. Fasteners torqued to 110 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-PLC-11

Target Machine Model: Makino-Aero-T550 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK550-11
- PLC Hex Register Address: DB23.DBW_0x5BB8
- Nominal Operating Range: 45 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-IGBT-MOD-1911
- Overhaul Seal & Consumable Kit: KIT-MAK550-PLC-11
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-PLC-11 at PLC register 0x5BB8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-IGBT-MOD-1911) with seal kit (KIT-MAK550-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-PLC-11 on Makino-Aero-T550 by installing PN-MAK550-IGBT-MOD-1911 & kit KIT-MAK550-PLC-11. Fasteners torqued to 119 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-LUB-12

Target Machine Model: Makino-Aero-T550 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK550-12
- PLC Hex Register Address: DB23.DBW_0x5BC0
- Nominal Operating Range: 48 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-INJ-MET-1912
- Overhaul Seal & Consumable Kit: KIT-MAK550-LUB-12
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-LUB-12 at PLC register 0x5BC0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-INJ-MET-1912) with seal kit (KIT-MAK550-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-LUB-12 on Makino-Aero-T550 by installing PN-MAK550-INJ-MET-1912 & kit KIT-MAK550-LUB-12. Fasteners torqued to 128 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-VAC-13

Target Machine Model: Makino-Aero-T550 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK550-13
- PLC Hex Register Address: DB23.DBW_0x5BC8
- Nominal Operating Range: 51 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-GASK-VIT-1913
- Overhaul Seal & Consumable Kit: KIT-MAK550-VAC-13
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-VAC-13 at PLC register 0x5BC8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-GASK-VIT-1913) with seal kit (KIT-MAK550-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-VAC-13 on Makino-Aero-T550 by installing PN-MAK550-GASK-VIT-1913 & kit KIT-MAK550-VAC-13. Fasteners torqued to 137 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK550-THM-14

Target Machine Model: Makino-Aero-T550 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T550 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T550 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T550 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK550-14
- PLC Hex Register Address: DB23.DBW_0x5BD0
- Nominal Operating Range: 54 - 78 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 113 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK550-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK550-HEX-CORE-1914
- Overhaul Seal & Consumable Kit: KIT-MAK550-THM-14
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T550 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK550-THM-14 at PLC register 0x5BD0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T550 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK550-HEX-CORE-1914) with seal kit (KIT-MAK550-THM-14).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK550-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T550, technician must log: "Resolved ERR-MAK550-THM-14 on Makino-Aero-T550 by installing PN-MAK550-HEX-CORE-1914 & kit KIT-MAK550-THM-14. Fasteners torqued to 146 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-SPN-1

Target Machine Model: Makino-Aero-T600 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: PT-SPN-MAK600-1
- PLC Hex Register Address: DB24.DBW_0x5E88
- Nominal Operating Range: 15 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-BRG-CER-2001
- Overhaul Seal & Consumable Kit: KIT-MAK600-SPN-1
- Critical Fastener Torque Value: 169 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-SPN-1 at PLC register 0x5E88.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-BRG-CER-2001) with seal kit (KIT-MAK600-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 169 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-SPN-1 on Makino-Aero-T600 by installing PN-MAK600-BRG-CER-2001 & kit KIT-MAK600-SPN-1. Fasteners torqued to 169 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-ATC-2

Target Machine Model: Makino-Aero-T600 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: PX-ATC-MAK600-2
- PLC Hex Register Address: DB24.DBW_0x5E90
- Nominal Operating Range: 18 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-CAM-GRIP-2002
- Overhaul Seal & Consumable Kit: KIT-MAK600-ATC-2
- Critical Fastener Torque Value: 178 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-ATC-2 at PLC register 0x5E90.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-CAM-GRIP-2002) with seal kit (KIT-MAK600-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 178 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-ATC-2 on Makino-Aero-T600 by installing PN-MAK600-CAM-GRIP-2002 & kit KIT-MAK600-ATC-2. Fasteners torqued to 178 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-HYD-3

Target Machine Model: Makino-Aero-T600 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: PR-HYD-MAK600-3
- PLC Hex Register Address: DB24.DBW_0x5E98
- Nominal Operating Range: 21 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-VLV-PROP-2003
- Overhaul Seal & Consumable Kit: KIT-MAK600-HYD-3
- Critical Fastener Torque Value: 187 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-HYD-3 at PLC register 0x5E98.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-VLV-PROP-2003) with seal kit (KIT-MAK600-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 187 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-HYD-3 on Makino-Aero-T600 by installing PN-MAK600-VLV-PROP-2003 & kit KIT-MAK600-HYD-3. Fasteners torqued to 187 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-CLN-4

Target Machine Model: Makino-Aero-T600 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: FL-CLN-MAK600-4
- PLC Hex Register Address: DB24.DBW_0x5EA0
- Nominal Operating Range: 24 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-PMP-PLG-2004
- Overhaul Seal & Consumable Kit: KIT-MAK600-CLN-4
- Critical Fastener Torque Value: 196 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-CLN-4 at PLC register 0x5EA0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-PMP-PLG-2004) with seal kit (KIT-MAK600-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 196 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-CLN-4 on Makino-Aero-T600 by installing PN-MAK600-PMP-PLG-2004 & kit KIT-MAK600-CLN-4. Fasteners torqued to 196 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-XAX-5

Target Machine Model: Makino-Aero-T600 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: EN-XAX-MAK600-5
- PLC Hex Register Address: DB24.DBW_0x5EA8
- Nominal Operating Range: 27 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-LIN-COIL-2005
- Overhaul Seal & Consumable Kit: KIT-MAK600-XAX-5
- Critical Fastener Torque Value: 205 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-XAX-5 at PLC register 0x5EA8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-LIN-COIL-2005) with seal kit (KIT-MAK600-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 205 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-XAX-5 on Makino-Aero-T600 by installing PN-MAK600-LIN-COIL-2005 & kit KIT-MAK600-XAX-5. Fasteners torqued to 205 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-YAX-6

Target Machine Model: Makino-Aero-T600 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: EN-YAX-MAK600-6
- PLC Hex Register Address: DB24.DBW_0x5EB0
- Nominal Operating Range: 30 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-BSW-NUT-2006
- Overhaul Seal & Consumable Kit: KIT-MAK600-YAX-6
- Critical Fastener Torque Value: 24 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-YAX-6 at PLC register 0x5EB0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-BSW-NUT-2006) with seal kit (KIT-MAK600-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 24 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-YAX-6 on Makino-Aero-T600 by installing PN-MAK600-BSW-NUT-2006 & kit KIT-MAK600-YAX-6. Fasteners torqued to 24 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-ZAX-7

Target Machine Model: Makino-Aero-T600 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: PR-ZAX-MAK600-7
- PLC Hex Register Address: DB24.DBW_0x5EB8
- Nominal Operating Range: 33 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-CYL-SEAL-2007
- Overhaul Seal & Consumable Kit: KIT-MAK600-ZAX-7
- Critical Fastener Torque Value: 33 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-ZAX-7 at PLC register 0x5EB8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-CYL-SEAL-2007) with seal kit (KIT-MAK600-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 33 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-ZAX-7 on Makino-Aero-T600 by installing PN-MAK600-CYL-SEAL-2007 & kit KIT-MAK600-ZAX-7. Fasteners torqued to 33 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-BAX-8

Target Machine Model: Makino-Aero-T600 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: EN-BAX-MAK600-8
- PLC Hex Register Address: DB24.DBW_0x5EC0
- Nominal Operating Range: 36 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-TRQ-STAT-2008
- Overhaul Seal & Consumable Kit: KIT-MAK600-BAX-8
- Critical Fastener Torque Value: 42 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-BAX-8 at PLC register 0x5EC0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-TRQ-STAT-2008) with seal kit (KIT-MAK600-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 42 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-BAX-8 on Makino-Aero-T600 by installing PN-MAK600-TRQ-STAT-2008 & kit KIT-MAK600-BAX-8. Fasteners torqued to 42 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-CAX-9

Target Machine Model: Makino-Aero-T600 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: PR-CAX-MAK600-9
- PLC Hex Register Address: DB24.DBW_0x5EC8
- Nominal Operating Range: 39 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-BRK-DISC-2009
- Overhaul Seal & Consumable Kit: KIT-MAK600-CAX-9
- Critical Fastener Torque Value: 51 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-CAX-9 at PLC register 0x5EC8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-BRK-DISC-2009) with seal kit (KIT-MAK600-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 51 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-CAX-9 on Makino-Aero-T600 by installing PN-MAK600-BRK-DISC-2009 & kit KIT-MAK600-CAX-9. Fasteners torqued to 51 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-OPT-10

Target Machine Model: Makino-Aero-T600 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: OP-SCL-MAK600-10
- PLC Hex Register Address: DB24.DBW_0x5ED0
- Nominal Operating Range: 42 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-GLASS-RD-2010
- Overhaul Seal & Consumable Kit: KIT-MAK600-OPT-10
- Critical Fastener Torque Value: 60 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-OPT-10 at PLC register 0x5ED0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-GLASS-RD-2010) with seal kit (KIT-MAK600-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 60 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-OPT-10 on Makino-Aero-T600 by installing PN-MAK600-GLASS-RD-2010 & kit KIT-MAK600-OPT-10. Fasteners torqued to 60 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-PLC-11

Target Machine Model: Makino-Aero-T600 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: VT-DRV-MAK600-11
- PLC Hex Register Address: DB24.DBW_0x5ED8
- Nominal Operating Range: 45 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-IGBT-MOD-2011
- Overhaul Seal & Consumable Kit: KIT-MAK600-PLC-11
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-PLC-11 at PLC register 0x5ED8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-IGBT-MOD-2011) with seal kit (KIT-MAK600-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-PLC-11 on Makino-Aero-T600 by installing PN-MAK600-IGBT-MOD-2011 & kit KIT-MAK600-PLC-11. Fasteners torqued to 69 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-LUB-12

Target Machine Model: Makino-Aero-T600 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: FL-LUB-MAK600-12
- PLC Hex Register Address: DB24.DBW_0x5EE0
- Nominal Operating Range: 48 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-INJ-MET-2012
- Overhaul Seal & Consumable Kit: KIT-MAK600-LUB-12
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-LUB-12 at PLC register 0x5EE0.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-INJ-MET-2012) with seal kit (KIT-MAK600-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-LUB-12 on Makino-Aero-T600 by installing PN-MAK600-INJ-MET-2012 & kit KIT-MAK600-LUB-12. Fasteners torqued to 78 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-VAC-13

Target Machine Model: Makino-Aero-T600 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: VC-MNF-MAK600-13
- PLC Hex Register Address: DB24.DBW_0x5EE8
- Nominal Operating Range: 51 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-GASK-VIT-2013
- Overhaul Seal & Consumable Kit: KIT-MAK600-VAC-13
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-VAC-13 at PLC register 0x5EE8.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-GASK-VIT-2013) with seal kit (KIT-MAK600-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-VAC-13 on Makino-Aero-T600 by installing PN-MAK600-GASK-VIT-2013 & kit KIT-MAK600-VAC-13. Fasteners torqued to 87 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAK600-THM-14

Target Machine Model: Makino-Aero-T600 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Makino-Aero-T600 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Makino-Aero-T600 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Makino-Aero-T600 (Makino Professional 6)
- Telemetry Sensor ID: TM-CHL-MAK600-14
- PLC Hex Register Address: DB24.DBW_0x5EF0
- Nominal Operating Range: 54 - 79 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 114 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAK600-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAK600-HEX-CORE-2014
- Overhaul Seal & Consumable Kit: KIT-MAK600-THM-14
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Makino-Aero-T600 (Makino Professional 6) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to Makino Professional 6 and verify fault latch ERR-MAK600-THM-14 at PLC register 0x5EF0.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Makino-Aero-T600 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAK600-HEX-CORE-2014) with seal kit (KIT-MAK600-THM-14).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAK600-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Makino-Aero-T600, technician must log: "Resolved ERR-MAK600-THM-14 on Makino-Aero-T600 by installing PN-MAK600-HEX-CORE-2014 & kit KIT-MAK600-THM-14. Fasteners torqued to 96 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-SPN-1

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ150-1
- PLC Hex Register Address: DB25.DBW_0x61A8
- Nominal Operating Range: 15 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-BRG-CER-2101
- Overhaul Seal & Consumable Kit: KIT-MAZ150-SPN-1
- Critical Fastener Torque Value: 119 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-SPN-1 at PLC register 0x61A8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-BRG-CER-2101) with seal kit (KIT-MAZ150-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 119 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-SPN-1 on Mazak-Integrex-i150 by installing PN-MAZ150-BRG-CER-2101 & kit KIT-MAZ150-SPN-1. Fasteners torqued to 119 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-ATC-2

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ150-2
- PLC Hex Register Address: DB25.DBW_0x61B0
- Nominal Operating Range: 18 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-CAM-GRIP-2102
- Overhaul Seal & Consumable Kit: KIT-MAZ150-ATC-2
- Critical Fastener Torque Value: 128 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-ATC-2 at PLC register 0x61B0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-CAM-GRIP-2102) with seal kit (KIT-MAZ150-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 128 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-ATC-2 on Mazak-Integrex-i150 by installing PN-MAZ150-CAM-GRIP-2102 & kit KIT-MAZ150-ATC-2. Fasteners torqued to 128 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-HYD-3

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ150-3
- PLC Hex Register Address: DB25.DBW_0x61B8
- Nominal Operating Range: 21 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-VLV-PROP-2103
- Overhaul Seal & Consumable Kit: KIT-MAZ150-HYD-3
- Critical Fastener Torque Value: 137 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-HYD-3 at PLC register 0x61B8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-VLV-PROP-2103) with seal kit (KIT-MAZ150-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 137 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-HYD-3 on Mazak-Integrex-i150 by installing PN-MAZ150-VLV-PROP-2103 & kit KIT-MAZ150-HYD-3. Fasteners torqued to 137 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-CLN-4

Target Machine Model: Mazak-Integrex-i150 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ150-4
- PLC Hex Register Address: DB25.DBW_0x61C0
- Nominal Operating Range: 24 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-PMP-PLG-2104
- Overhaul Seal & Consumable Kit: KIT-MAZ150-CLN-4
- Critical Fastener Torque Value: 146 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-CLN-4 at PLC register 0x61C0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-PMP-PLG-2104) with seal kit (KIT-MAZ150-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 146 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-CLN-4 on Mazak-Integrex-i150 by installing PN-MAZ150-PMP-PLG-2104 & kit KIT-MAZ150-CLN-4. Fasteners torqued to 146 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-XAX-5

Target Machine Model: Mazak-Integrex-i150 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ150-5
- PLC Hex Register Address: DB25.DBW_0x61C8
- Nominal Operating Range: 27 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-LIN-COIL-2105
- Overhaul Seal & Consumable Kit: KIT-MAZ150-XAX-5
- Critical Fastener Torque Value: 155 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-XAX-5 at PLC register 0x61C8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-LIN-COIL-2105) with seal kit (KIT-MAZ150-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 155 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-XAX-5 on Mazak-Integrex-i150 by installing PN-MAZ150-LIN-COIL-2105 & kit KIT-MAZ150-XAX-5. Fasteners torqued to 155 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-YAX-6

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ150-6
- PLC Hex Register Address: DB25.DBW_0x61D0
- Nominal Operating Range: 30 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-BSW-NUT-2106
- Overhaul Seal & Consumable Kit: KIT-MAZ150-YAX-6
- Critical Fastener Torque Value: 164 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-YAX-6 at PLC register 0x61D0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-BSW-NUT-2106) with seal kit (KIT-MAZ150-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 164 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-YAX-6 on Mazak-Integrex-i150 by installing PN-MAZ150-BSW-NUT-2106 & kit KIT-MAZ150-YAX-6. Fasteners torqued to 164 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-ZAX-7

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ150-7
- PLC Hex Register Address: DB25.DBW_0x61D8
- Nominal Operating Range: 33 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-CYL-SEAL-2107
- Overhaul Seal & Consumable Kit: KIT-MAZ150-ZAX-7
- Critical Fastener Torque Value: 173 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-ZAX-7 at PLC register 0x61D8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-CYL-SEAL-2107) with seal kit (KIT-MAZ150-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 173 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-ZAX-7 on Mazak-Integrex-i150 by installing PN-MAZ150-CYL-SEAL-2107 & kit KIT-MAZ150-ZAX-7. Fasteners torqued to 173 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-BAX-8

Target Machine Model: Mazak-Integrex-i150 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ150-8
- PLC Hex Register Address: DB25.DBW_0x61E0
- Nominal Operating Range: 36 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-TRQ-STAT-2108
- Overhaul Seal & Consumable Kit: KIT-MAZ150-BAX-8
- Critical Fastener Torque Value: 182 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-BAX-8 at PLC register 0x61E0.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-TRQ-STAT-2108) with seal kit (KIT-MAZ150-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 182 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-BAX-8 on Mazak-Integrex-i150 by installing PN-MAZ150-TRQ-STAT-2108 & kit KIT-MAZ150-BAX-8. Fasteners torqued to 182 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-CAX-9

Target Machine Model: Mazak-Integrex-i150 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ150-9
- PLC Hex Register Address: DB25.DBW_0x61E8
- Nominal Operating Range: 39 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-BRK-DISC-2109
- Overhaul Seal & Consumable Kit: KIT-MAZ150-CAX-9
- Critical Fastener Torque Value: 191 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-CAX-9 at PLC register 0x61E8.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-BRK-DISC-2109) with seal kit (KIT-MAZ150-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 191 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-CAX-9 on Mazak-Integrex-i150 by installing PN-MAZ150-BRK-DISC-2109 & kit KIT-MAZ150-CAX-9. Fasteners torqued to 191 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-OPT-10

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ150-10
- PLC Hex Register Address: DB25.DBW_0x61F0
- Nominal Operating Range: 42 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-GLASS-RD-2110
- Overhaul Seal & Consumable Kit: KIT-MAZ150-OPT-10
- Critical Fastener Torque Value: 200 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-OPT-10 at PLC register 0x61F0.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-GLASS-RD-2110) with seal kit (KIT-MAZ150-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 200 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-OPT-10 on Mazak-Integrex-i150 by installing PN-MAZ150-GLASS-RD-2110 & kit KIT-MAZ150-OPT-10. Fasteners torqued to 200 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-PLC-11

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ150-11
- PLC Hex Register Address: DB25.DBW_0x61F8
- Nominal Operating Range: 45 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-IGBT-MOD-2111
- Overhaul Seal & Consumable Kit: KIT-MAZ150-PLC-11
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-PLC-11 at PLC register 0x61F8.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-IGBT-MOD-2111) with seal kit (KIT-MAZ150-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-PLC-11 on Mazak-Integrex-i150 by installing PN-MAZ150-IGBT-MOD-2111 & kit KIT-MAZ150-PLC-11. Fasteners torqued to 209 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-LUB-12

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ150-12
- PLC Hex Register Address: DB25.DBW_0x6200
- Nominal Operating Range: 48 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-INJ-MET-2112
- Overhaul Seal & Consumable Kit: KIT-MAZ150-LUB-12
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-LUB-12 at PLC register 0x6200.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-INJ-MET-2112) with seal kit (KIT-MAZ150-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-LUB-12 on Mazak-Integrex-i150 by installing PN-MAZ150-INJ-MET-2112 & kit KIT-MAZ150-LUB-12. Fasteners torqued to 28 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-VAC-13

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ150-13
- PLC Hex Register Address: DB25.DBW_0x6208
- Nominal Operating Range: 51 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-GASK-VIT-2113
- Overhaul Seal & Consumable Kit: KIT-MAZ150-VAC-13
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-VAC-13 at PLC register 0x6208.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-GASK-VIT-2113) with seal kit (KIT-MAZ150-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-VAC-13 on Mazak-Integrex-i150 by installing PN-MAZ150-GASK-VIT-2113 & kit KIT-MAZ150-VAC-13. Fasteners torqued to 37 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ150-THM-14

Target Machine Model: Mazak-Integrex-i150 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i150 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i150 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i150 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ150-14
- PLC Hex Register Address: DB25.DBW_0x6210
- Nominal Operating Range: 54 - 80 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 115 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ150-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ150-HEX-CORE-2114
- Overhaul Seal & Consumable Kit: KIT-MAZ150-THM-14
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i150 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ150-THM-14 at PLC register 0x6210.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i150 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ150-HEX-CORE-2114) with seal kit (KIT-MAZ150-THM-14).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ150-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i150, technician must log: "Resolved ERR-MAZ150-THM-14 on Mazak-Integrex-i150 by installing PN-MAZ150-HEX-CORE-2114 & kit KIT-MAZ150-THM-14. Fasteners torqued to 46 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-SPN-1

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ200-1
- PLC Hex Register Address: DB26.DBW_0x64C8
- Nominal Operating Range: 15 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-BRG-CER-2201
- Overhaul Seal & Consumable Kit: KIT-MAZ200-SPN-1
- Critical Fastener Torque Value: 69 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-SPN-1 at PLC register 0x64C8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-BRG-CER-2201) with seal kit (KIT-MAZ200-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 69 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-SPN-1 on Mazak-Integrex-i200 by installing PN-MAZ200-BRG-CER-2201 & kit KIT-MAZ200-SPN-1. Fasteners torqued to 69 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-ATC-2

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ200-2
- PLC Hex Register Address: DB26.DBW_0x64D0
- Nominal Operating Range: 18 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-CAM-GRIP-2202
- Overhaul Seal & Consumable Kit: KIT-MAZ200-ATC-2
- Critical Fastener Torque Value: 78 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-ATC-2 at PLC register 0x64D0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-CAM-GRIP-2202) with seal kit (KIT-MAZ200-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 78 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-ATC-2 on Mazak-Integrex-i200 by installing PN-MAZ200-CAM-GRIP-2202 & kit KIT-MAZ200-ATC-2. Fasteners torqued to 78 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-HYD-3

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ200-3
- PLC Hex Register Address: DB26.DBW_0x64D8
- Nominal Operating Range: 21 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-VLV-PROP-2203
- Overhaul Seal & Consumable Kit: KIT-MAZ200-HYD-3
- Critical Fastener Torque Value: 87 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-HYD-3 at PLC register 0x64D8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-VLV-PROP-2203) with seal kit (KIT-MAZ200-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 87 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-HYD-3 on Mazak-Integrex-i200 by installing PN-MAZ200-VLV-PROP-2203 & kit KIT-MAZ200-HYD-3. Fasteners torqued to 87 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-CLN-4

Target Machine Model: Mazak-Integrex-i200 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ200-4
- PLC Hex Register Address: DB26.DBW_0x64E0
- Nominal Operating Range: 24 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-PMP-PLG-2204
- Overhaul Seal & Consumable Kit: KIT-MAZ200-CLN-4
- Critical Fastener Torque Value: 96 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-CLN-4 at PLC register 0x64E0.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-PMP-PLG-2204) with seal kit (KIT-MAZ200-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 96 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-CLN-4 on Mazak-Integrex-i200 by installing PN-MAZ200-PMP-PLG-2204 & kit KIT-MAZ200-CLN-4. Fasteners torqued to 96 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-XAX-5

Target Machine Model: Mazak-Integrex-i200 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ200-5
- PLC Hex Register Address: DB26.DBW_0x64E8
- Nominal Operating Range: 27 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-LIN-COIL-2205
- Overhaul Seal & Consumable Kit: KIT-MAZ200-XAX-5
- Critical Fastener Torque Value: 105 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-XAX-5 at PLC register 0x64E8.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-LIN-COIL-2205) with seal kit (KIT-MAZ200-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 105 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-XAX-5 on Mazak-Integrex-i200 by installing PN-MAZ200-LIN-COIL-2205 & kit KIT-MAZ200-XAX-5. Fasteners torqued to 105 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-YAX-6

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ200-6
- PLC Hex Register Address: DB26.DBW_0x64F0
- Nominal Operating Range: 30 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-BSW-NUT-2206
- Overhaul Seal & Consumable Kit: KIT-MAZ200-YAX-6
- Critical Fastener Torque Value: 114 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-YAX-6 at PLC register 0x64F0.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-BSW-NUT-2206) with seal kit (KIT-MAZ200-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 114 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-YAX-6 on Mazak-Integrex-i200 by installing PN-MAZ200-BSW-NUT-2206 & kit KIT-MAZ200-YAX-6. Fasteners torqued to 114 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-ZAX-7

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ200-7
- PLC Hex Register Address: DB26.DBW_0x64F8
- Nominal Operating Range: 33 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-CYL-SEAL-2207
- Overhaul Seal & Consumable Kit: KIT-MAZ200-ZAX-7
- Critical Fastener Torque Value: 123 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-ZAX-7 at PLC register 0x64F8.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-CYL-SEAL-2207) with seal kit (KIT-MAZ200-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 123 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-ZAX-7 on Mazak-Integrex-i200 by installing PN-MAZ200-CYL-SEAL-2207 & kit KIT-MAZ200-ZAX-7. Fasteners torqued to 123 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-BAX-8

Target Machine Model: Mazak-Integrex-i200 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ200-8
- PLC Hex Register Address: DB26.DBW_0x6500
- Nominal Operating Range: 36 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-TRQ-STAT-2208
- Overhaul Seal & Consumable Kit: KIT-MAZ200-BAX-8
- Critical Fastener Torque Value: 132 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-BAX-8 at PLC register 0x6500.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-TRQ-STAT-2208) with seal kit (KIT-MAZ200-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 132 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-BAX-8 on Mazak-Integrex-i200 by installing PN-MAZ200-TRQ-STAT-2208 & kit KIT-MAZ200-BAX-8. Fasteners torqued to 132 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-CAX-9

Target Machine Model: Mazak-Integrex-i200 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ200-9
- PLC Hex Register Address: DB26.DBW_0x6508
- Nominal Operating Range: 39 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-BRK-DISC-2209
- Overhaul Seal & Consumable Kit: KIT-MAZ200-CAX-9
- Critical Fastener Torque Value: 141 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-CAX-9 at PLC register 0x6508.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-BRK-DISC-2209) with seal kit (KIT-MAZ200-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 141 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-CAX-9 on Mazak-Integrex-i200 by installing PN-MAZ200-BRK-DISC-2209 & kit KIT-MAZ200-CAX-9. Fasteners torqued to 141 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-OPT-10

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ200-10
- PLC Hex Register Address: DB26.DBW_0x6510
- Nominal Operating Range: 42 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-GLASS-RD-2210
- Overhaul Seal & Consumable Kit: KIT-MAZ200-OPT-10
- Critical Fastener Torque Value: 150 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-OPT-10 at PLC register 0x6510.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-GLASS-RD-2210) with seal kit (KIT-MAZ200-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 150 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-OPT-10 on Mazak-Integrex-i200 by installing PN-MAZ200-GLASS-RD-2210 & kit KIT-MAZ200-OPT-10. Fasteners torqued to 150 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-PLC-11

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ200-11
- PLC Hex Register Address: DB26.DBW_0x6518
- Nominal Operating Range: 45 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-IGBT-MOD-2211
- Overhaul Seal & Consumable Kit: KIT-MAZ200-PLC-11
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-PLC-11 at PLC register 0x6518.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-IGBT-MOD-2211) with seal kit (KIT-MAZ200-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-PLC-11 on Mazak-Integrex-i200 by installing PN-MAZ200-IGBT-MOD-2211 & kit KIT-MAZ200-PLC-11. Fasteners torqued to 159 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-LUB-12

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ200-12
- PLC Hex Register Address: DB26.DBW_0x6520
- Nominal Operating Range: 48 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-INJ-MET-2212
- Overhaul Seal & Consumable Kit: KIT-MAZ200-LUB-12
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-LUB-12 at PLC register 0x6520.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-INJ-MET-2212) with seal kit (KIT-MAZ200-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-LUB-12 on Mazak-Integrex-i200 by installing PN-MAZ200-INJ-MET-2212 & kit KIT-MAZ200-LUB-12. Fasteners torqued to 168 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-VAC-13

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ200-13
- PLC Hex Register Address: DB26.DBW_0x6528
- Nominal Operating Range: 51 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-GASK-VIT-2213
- Overhaul Seal & Consumable Kit: KIT-MAZ200-VAC-13
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-VAC-13 at PLC register 0x6528.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-GASK-VIT-2213) with seal kit (KIT-MAZ200-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-VAC-13 on Mazak-Integrex-i200 by installing PN-MAZ200-GASK-VIT-2213 & kit KIT-MAZ200-VAC-13. Fasteners torqued to 177 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ200-THM-14

Target Machine Model: Mazak-Integrex-i200 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i200 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i200 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i200 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ200-14
- PLC Hex Register Address: DB26.DBW_0x6530
- Nominal Operating Range: 54 - 81 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 116 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ200-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ200-HEX-CORE-2214
- Overhaul Seal & Consumable Kit: KIT-MAZ200-THM-14
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i200 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ200-THM-14 at PLC register 0x6530.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i200 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ200-HEX-CORE-2214) with seal kit (KIT-MAZ200-THM-14).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ200-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i200, technician must log: "Resolved ERR-MAZ200-THM-14 on Mazak-Integrex-i200 by installing PN-MAZ200-HEX-CORE-2214 & kit KIT-MAZ200-THM-14. Fasteners torqued to 186 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-SPN-1

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ250-1
- PLC Hex Register Address: DB27.DBW_0x67E8
- Nominal Operating Range: 15 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-BRG-CER-2301
- Overhaul Seal & Consumable Kit: KIT-MAZ250-SPN-1
- Critical Fastener Torque Value: 209 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-SPN-1 at PLC register 0x67E8.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-BRG-CER-2301) with seal kit (KIT-MAZ250-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 209 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-SPN-1 on Mazak-Integrex-i250 by installing PN-MAZ250-BRG-CER-2301 & kit KIT-MAZ250-SPN-1. Fasteners torqued to 209 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-ATC-2

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ250-2
- PLC Hex Register Address: DB27.DBW_0x67F0
- Nominal Operating Range: 18 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-CAM-GRIP-2302
- Overhaul Seal & Consumable Kit: KIT-MAZ250-ATC-2
- Critical Fastener Torque Value: 28 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-ATC-2 at PLC register 0x67F0.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-CAM-GRIP-2302) with seal kit (KIT-MAZ250-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 28 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-ATC-2 on Mazak-Integrex-i250 by installing PN-MAZ250-CAM-GRIP-2302 & kit KIT-MAZ250-ATC-2. Fasteners torqued to 28 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-HYD-3

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ250-3
- PLC Hex Register Address: DB27.DBW_0x67F8
- Nominal Operating Range: 21 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-VLV-PROP-2303
- Overhaul Seal & Consumable Kit: KIT-MAZ250-HYD-3
- Critical Fastener Torque Value: 37 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-HYD-3 at PLC register 0x67F8.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-VLV-PROP-2303) with seal kit (KIT-MAZ250-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 37 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-HYD-3 on Mazak-Integrex-i250 by installing PN-MAZ250-VLV-PROP-2303 & kit KIT-MAZ250-HYD-3. Fasteners torqued to 37 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-CLN-4

Target Machine Model: Mazak-Integrex-i250 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ250-4
- PLC Hex Register Address: DB27.DBW_0x6800
- Nominal Operating Range: 24 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-PMP-PLG-2304
- Overhaul Seal & Consumable Kit: KIT-MAZ250-CLN-4
- Critical Fastener Torque Value: 46 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-CLN-4 at PLC register 0x6800.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-PMP-PLG-2304) with seal kit (KIT-MAZ250-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 46 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-CLN-4 on Mazak-Integrex-i250 by installing PN-MAZ250-PMP-PLG-2304 & kit KIT-MAZ250-CLN-4. Fasteners torqued to 46 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-XAX-5

Target Machine Model: Mazak-Integrex-i250 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ250-5
- PLC Hex Register Address: DB27.DBW_0x6808
- Nominal Operating Range: 27 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-LIN-COIL-2305
- Overhaul Seal & Consumable Kit: KIT-MAZ250-XAX-5
- Critical Fastener Torque Value: 55 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-XAX-5 at PLC register 0x6808.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-LIN-COIL-2305) with seal kit (KIT-MAZ250-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 55 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-XAX-5 on Mazak-Integrex-i250 by installing PN-MAZ250-LIN-COIL-2305 & kit KIT-MAZ250-XAX-5. Fasteners torqued to 55 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-YAX-6

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ250-6
- PLC Hex Register Address: DB27.DBW_0x6810
- Nominal Operating Range: 30 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-BSW-NUT-2306
- Overhaul Seal & Consumable Kit: KIT-MAZ250-YAX-6
- Critical Fastener Torque Value: 64 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-YAX-6 at PLC register 0x6810.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-BSW-NUT-2306) with seal kit (KIT-MAZ250-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 64 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-YAX-6 on Mazak-Integrex-i250 by installing PN-MAZ250-BSW-NUT-2306 & kit KIT-MAZ250-YAX-6. Fasteners torqued to 64 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-ZAX-7

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ250-7
- PLC Hex Register Address: DB27.DBW_0x6818
- Nominal Operating Range: 33 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-CYL-SEAL-2307
- Overhaul Seal & Consumable Kit: KIT-MAZ250-ZAX-7
- Critical Fastener Torque Value: 73 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-ZAX-7 at PLC register 0x6818.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-CYL-SEAL-2307) with seal kit (KIT-MAZ250-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 73 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-ZAX-7 on Mazak-Integrex-i250 by installing PN-MAZ250-CYL-SEAL-2307 & kit KIT-MAZ250-ZAX-7. Fasteners torqued to 73 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-BAX-8

Target Machine Model: Mazak-Integrex-i250 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ250-8
- PLC Hex Register Address: DB27.DBW_0x6820
- Nominal Operating Range: 36 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-TRQ-STAT-2308
- Overhaul Seal & Consumable Kit: KIT-MAZ250-BAX-8
- Critical Fastener Torque Value: 82 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-BAX-8 at PLC register 0x6820.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-TRQ-STAT-2308) with seal kit (KIT-MAZ250-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 82 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-BAX-8 on Mazak-Integrex-i250 by installing PN-MAZ250-TRQ-STAT-2308 & kit KIT-MAZ250-BAX-8. Fasteners torqued to 82 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-CAX-9

Target Machine Model: Mazak-Integrex-i250 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ250-9
- PLC Hex Register Address: DB27.DBW_0x6828
- Nominal Operating Range: 39 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-BRK-DISC-2309
- Overhaul Seal & Consumable Kit: KIT-MAZ250-CAX-9
- Critical Fastener Torque Value: 91 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-CAX-9 at PLC register 0x6828.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-BRK-DISC-2309) with seal kit (KIT-MAZ250-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 91 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-CAX-9 on Mazak-Integrex-i250 by installing PN-MAZ250-BRK-DISC-2309 & kit KIT-MAZ250-CAX-9. Fasteners torqued to 91 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-OPT-10

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ250-10
- PLC Hex Register Address: DB27.DBW_0x6830
- Nominal Operating Range: 42 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-GLASS-RD-2310
- Overhaul Seal & Consumable Kit: KIT-MAZ250-OPT-10
- Critical Fastener Torque Value: 100 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-OPT-10 at PLC register 0x6830.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-GLASS-RD-2310) with seal kit (KIT-MAZ250-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 100 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-OPT-10 on Mazak-Integrex-i250 by installing PN-MAZ250-GLASS-RD-2310 & kit KIT-MAZ250-OPT-10. Fasteners torqued to 100 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-PLC-11

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ250-11
- PLC Hex Register Address: DB27.DBW_0x6838
- Nominal Operating Range: 45 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-IGBT-MOD-2311
- Overhaul Seal & Consumable Kit: KIT-MAZ250-PLC-11
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-PLC-11 at PLC register 0x6838.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-IGBT-MOD-2311) with seal kit (KIT-MAZ250-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-PLC-11 on Mazak-Integrex-i250 by installing PN-MAZ250-IGBT-MOD-2311 & kit KIT-MAZ250-PLC-11. Fasteners torqued to 109 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-LUB-12

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ250-12
- PLC Hex Register Address: DB27.DBW_0x6840
- Nominal Operating Range: 48 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-INJ-MET-2312
- Overhaul Seal & Consumable Kit: KIT-MAZ250-LUB-12
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-LUB-12 at PLC register 0x6840.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-INJ-MET-2312) with seal kit (KIT-MAZ250-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-LUB-12 on Mazak-Integrex-i250 by installing PN-MAZ250-INJ-MET-2312 & kit KIT-MAZ250-LUB-12. Fasteners torqued to 118 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-VAC-13

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ250-13
- PLC Hex Register Address: DB27.DBW_0x6848
- Nominal Operating Range: 51 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-GASK-VIT-2313
- Overhaul Seal & Consumable Kit: KIT-MAZ250-VAC-13
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-VAC-13 at PLC register 0x6848.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-GASK-VIT-2313) with seal kit (KIT-MAZ250-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-VAC-13 on Mazak-Integrex-i250 by installing PN-MAZ250-GASK-VIT-2313 & kit KIT-MAZ250-VAC-13. Fasteners torqued to 127 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ250-THM-14

Target Machine Model: Mazak-Integrex-i250 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i250 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i250 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i250 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ250-14
- PLC Hex Register Address: DB27.DBW_0x6850
- Nominal Operating Range: 54 - 82 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 117 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ250-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ250-HEX-CORE-2314
- Overhaul Seal & Consumable Kit: KIT-MAZ250-THM-14
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i250 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ250-THM-14 at PLC register 0x6850.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i250 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ250-HEX-CORE-2314) with seal kit (KIT-MAZ250-THM-14).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ250-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i250, technician must log: "Resolved ERR-MAZ250-THM-14 on Mazak-Integrex-i250 by installing PN-MAZ250-HEX-CORE-2314 & kit KIT-MAZ250-THM-14. Fasteners torqued to 136 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-SPN-1

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ300-1
- PLC Hex Register Address: DB28.DBW_0x6B08
- Nominal Operating Range: 15 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-BRG-CER-2401
- Overhaul Seal & Consumable Kit: KIT-MAZ300-SPN-1
- Critical Fastener Torque Value: 159 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-SPN-1 at PLC register 0x6B08.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-BRG-CER-2401) with seal kit (KIT-MAZ300-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 159 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-SPN-1 on Mazak-Integrex-i300 by installing PN-MAZ300-BRG-CER-2401 & kit KIT-MAZ300-SPN-1. Fasteners torqued to 159 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-ATC-2

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ300-2
- PLC Hex Register Address: DB28.DBW_0x6B10
- Nominal Operating Range: 18 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-CAM-GRIP-2402
- Overhaul Seal & Consumable Kit: KIT-MAZ300-ATC-2
- Critical Fastener Torque Value: 168 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-ATC-2 at PLC register 0x6B10.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-CAM-GRIP-2402) with seal kit (KIT-MAZ300-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 168 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-ATC-2 on Mazak-Integrex-i300 by installing PN-MAZ300-CAM-GRIP-2402 & kit KIT-MAZ300-ATC-2. Fasteners torqued to 168 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-HYD-3

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ300-3
- PLC Hex Register Address: DB28.DBW_0x6B18
- Nominal Operating Range: 21 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-VLV-PROP-2403
- Overhaul Seal & Consumable Kit: KIT-MAZ300-HYD-3
- Critical Fastener Torque Value: 177 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-HYD-3 at PLC register 0x6B18.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-VLV-PROP-2403) with seal kit (KIT-MAZ300-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 177 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-HYD-3 on Mazak-Integrex-i300 by installing PN-MAZ300-VLV-PROP-2403 & kit KIT-MAZ300-HYD-3. Fasteners torqued to 177 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-CLN-4

Target Machine Model: Mazak-Integrex-i300 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ300-4
- PLC Hex Register Address: DB28.DBW_0x6B20
- Nominal Operating Range: 24 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-PMP-PLG-2404
- Overhaul Seal & Consumable Kit: KIT-MAZ300-CLN-4
- Critical Fastener Torque Value: 186 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-CLN-4 at PLC register 0x6B20.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-PMP-PLG-2404) with seal kit (KIT-MAZ300-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 186 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-CLN-4 on Mazak-Integrex-i300 by installing PN-MAZ300-PMP-PLG-2404 & kit KIT-MAZ300-CLN-4. Fasteners torqued to 186 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-XAX-5

Target Machine Model: Mazak-Integrex-i300 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ300-5
- PLC Hex Register Address: DB28.DBW_0x6B28
- Nominal Operating Range: 27 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-LIN-COIL-2405
- Overhaul Seal & Consumable Kit: KIT-MAZ300-XAX-5
- Critical Fastener Torque Value: 195 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-XAX-5 at PLC register 0x6B28.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-LIN-COIL-2405) with seal kit (KIT-MAZ300-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 195 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-XAX-5 on Mazak-Integrex-i300 by installing PN-MAZ300-LIN-COIL-2405 & kit KIT-MAZ300-XAX-5. Fasteners torqued to 195 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-YAX-6

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ300-6
- PLC Hex Register Address: DB28.DBW_0x6B30
- Nominal Operating Range: 30 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-BSW-NUT-2406
- Overhaul Seal & Consumable Kit: KIT-MAZ300-YAX-6
- Critical Fastener Torque Value: 204 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-YAX-6 at PLC register 0x6B30.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-BSW-NUT-2406) with seal kit (KIT-MAZ300-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 204 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-YAX-6 on Mazak-Integrex-i300 by installing PN-MAZ300-BSW-NUT-2406 & kit KIT-MAZ300-YAX-6. Fasteners torqued to 204 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-ZAX-7

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ300-7
- PLC Hex Register Address: DB28.DBW_0x6B38
- Nominal Operating Range: 33 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-CYL-SEAL-2407
- Overhaul Seal & Consumable Kit: KIT-MAZ300-ZAX-7
- Critical Fastener Torque Value: 23 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-ZAX-7 at PLC register 0x6B38.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-CYL-SEAL-2407) with seal kit (KIT-MAZ300-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 23 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-ZAX-7 on Mazak-Integrex-i300 by installing PN-MAZ300-CYL-SEAL-2407 & kit KIT-MAZ300-ZAX-7. Fasteners torqued to 23 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-BAX-8

Target Machine Model: Mazak-Integrex-i300 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ300-8
- PLC Hex Register Address: DB28.DBW_0x6B40
- Nominal Operating Range: 36 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-TRQ-STAT-2408
- Overhaul Seal & Consumable Kit: KIT-MAZ300-BAX-8
- Critical Fastener Torque Value: 32 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-BAX-8 at PLC register 0x6B40.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-TRQ-STAT-2408) with seal kit (KIT-MAZ300-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 32 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-BAX-8 on Mazak-Integrex-i300 by installing PN-MAZ300-TRQ-STAT-2408 & kit KIT-MAZ300-BAX-8. Fasteners torqued to 32 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-CAX-9

Target Machine Model: Mazak-Integrex-i300 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ300-9
- PLC Hex Register Address: DB28.DBW_0x6B48
- Nominal Operating Range: 39 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-BRK-DISC-2409
- Overhaul Seal & Consumable Kit: KIT-MAZ300-CAX-9
- Critical Fastener Torque Value: 41 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-CAX-9 at PLC register 0x6B48.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-BRK-DISC-2409) with seal kit (KIT-MAZ300-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 41 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-CAX-9 on Mazak-Integrex-i300 by installing PN-MAZ300-BRK-DISC-2409 & kit KIT-MAZ300-CAX-9. Fasteners torqued to 41 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-OPT-10

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ300-10
- PLC Hex Register Address: DB28.DBW_0x6B50
- Nominal Operating Range: 42 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-GLASS-RD-2410
- Overhaul Seal & Consumable Kit: KIT-MAZ300-OPT-10
- Critical Fastener Torque Value: 50 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-OPT-10 at PLC register 0x6B50.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-GLASS-RD-2410) with seal kit (KIT-MAZ300-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 50 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-OPT-10 on Mazak-Integrex-i300 by installing PN-MAZ300-GLASS-RD-2410 & kit KIT-MAZ300-OPT-10. Fasteners torqued to 50 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-PLC-11

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ300-11
- PLC Hex Register Address: DB28.DBW_0x6B58
- Nominal Operating Range: 45 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-IGBT-MOD-2411
- Overhaul Seal & Consumable Kit: KIT-MAZ300-PLC-11
- Critical Fastener Torque Value: 59 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-PLC-11 at PLC register 0x6B58.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-IGBT-MOD-2411) with seal kit (KIT-MAZ300-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 59 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-PLC-11 on Mazak-Integrex-i300 by installing PN-MAZ300-IGBT-MOD-2411 & kit KIT-MAZ300-PLC-11. Fasteners torqued to 59 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-LUB-12

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ300-12
- PLC Hex Register Address: DB28.DBW_0x6B60
- Nominal Operating Range: 48 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-INJ-MET-2412
- Overhaul Seal & Consumable Kit: KIT-MAZ300-LUB-12
- Critical Fastener Torque Value: 68 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-LUB-12 at PLC register 0x6B60.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-INJ-MET-2412) with seal kit (KIT-MAZ300-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 68 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-LUB-12 on Mazak-Integrex-i300 by installing PN-MAZ300-INJ-MET-2412 & kit KIT-MAZ300-LUB-12. Fasteners torqued to 68 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-VAC-13

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ300-13
- PLC Hex Register Address: DB28.DBW_0x6B68
- Nominal Operating Range: 51 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-GASK-VIT-2413
- Overhaul Seal & Consumable Kit: KIT-MAZ300-VAC-13
- Critical Fastener Torque Value: 77 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-VAC-13 at PLC register 0x6B68.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-GASK-VIT-2413) with seal kit (KIT-MAZ300-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 77 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-VAC-13 on Mazak-Integrex-i300 by installing PN-MAZ300-GASK-VIT-2413 & kit KIT-MAZ300-VAC-13. Fasteners torqued to 77 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ300-THM-14

Target Machine Model: Mazak-Integrex-i300 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i300 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i300 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i300 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ300-14
- PLC Hex Register Address: DB28.DBW_0x6B70
- Nominal Operating Range: 54 - 83 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 118 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ300-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ300-HEX-CORE-2414
- Overhaul Seal & Consumable Kit: KIT-MAZ300-THM-14
- Critical Fastener Torque Value: 86 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i300 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ300-THM-14 at PLC register 0x6B70.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i300 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ300-HEX-CORE-2414) with seal kit (KIT-MAZ300-THM-14).

Step 6: Torque retaining bolts in cross sequence to 86 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ300-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i300, technician must log: "Resolved ERR-MAZ300-THM-14 on Mazak-Integrex-i300 by installing PN-MAZ300-HEX-CORE-2414 & kit KIT-MAZ300-THM-14. Fasteners torqued to 86 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-SPN-1

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Electro-Spindle & Drawbar Assembly | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Electro-Spindle & Drawbar Assembly. Telemetry sensor PT-SPN-1 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PT-SPN-MAZ350-1
- PLC Hex Register Address: DB29.DBW_0x6E28
- Nominal Operating Range: 15 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-SPN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-SPN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-BRG-CER-2501
- Overhaul Seal & Consumable Kit: KIT-MAZ350-SPN-1
- Critical Fastener Torque Value: 109 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 6.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed SPN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-SPN-1 at PLC register 0x6E28.

Step 4: Remove Electro-Spindle & Drawbar Assembly housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-BRG-CER-2501) with seal kit (KIT-MAZ350-SPN-1).

Step 6: Torque retaining bolts in cross sequence to 109 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-SPN-1, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-SPN-1 on Mazak-Integrex-i350 by installing PN-MAZ350-BRG-CER-2501 & kit KIT-MAZ350-SPN-1. Fasteners torqued to 109 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-ATC-2

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Servo Tool Magazine & Gripper Arm | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Servo Tool Magazine & Gripper Arm. Telemetry sensor PX-ATC-2 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PX-ATC-MAZ350-2
- PLC Hex Register Address: DB29.DBW_0x6E30
- Nominal Operating Range: 18 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-ATC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ATC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-CAM-GRIP-2502
- Overhaul Seal & Consumable Kit: KIT-MAZ350-ATC-2
- Critical Fastener Torque Value: 118 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 3.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ATC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-ATC-2 at PLC register 0x6E30.

Step 4: Remove Servo Tool Magazine & Gripper Arm housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-CAM-GRIP-2502) with seal kit (KIT-MAZ350-ATC-2).

Step 6: Torque retaining bolts in cross sequence to 118 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-ATC-2, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-ATC-2 on Mazak-Integrex-i350 by installing PN-MAZ350-CAM-GRIP-2502 & kit KIT-MAZ350-ATC-2. Fasteners torqued to 118 Nm; alignment verified at ±0.0040 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-HYD-3

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Proportional Hydraulic Clamping Manifold | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Proportional Hydraulic Clamping Manifold. Telemetry sensor PR-HYD-3 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-HYD-MAZ350-3
- PLC Hex Register Address: DB29.DBW_0x6E38
- Nominal Operating Range: 21 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-HYD: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-HYD.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-VLV-PROP-2503
- Overhaul Seal & Consumable Kit: KIT-MAZ350-HYD-3
- Critical Fastener Torque Value: 127 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0045 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed HYD circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-HYD-3 at PLC register 0x6E38.

Step 4: Remove Proportional Hydraulic Clamping Manifold housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-VLV-PROP-2503) with seal kit (KIT-MAZ350-HYD-3).

Step 6: Torque retaining bolts in cross sequence to 127 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-HYD-3, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0045 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-HYD-3 on Mazak-Integrex-i350 by installing PN-MAZ350-VLV-PROP-2503 & kit KIT-MAZ350-HYD-3. Fasteners torqued to 127 Nm; alignment verified at ±0.0045 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-CLN-4

Target Machine Model: Mazak-Integrex-i350 | Subsystem: 120-Bar Through-Tool Cryo Coolant Loop | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in 120-Bar Through-Tool Cryo Coolant Loop. Telemetry sensor FL-CLN-4 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-CLN-MAZ350-4
- PLC Hex Register Address: DB29.DBW_0x6E40
- Nominal Operating Range: 24 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-CLN: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CLN.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-PMP-PLG-2504
- Overhaul Seal & Consumable Kit: KIT-MAZ350-CLN-4
- Critical Fastener Torque Value: 136 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0050 mm TIR
- Estimated Repair Duration (SLA MTTR): 2.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CLN circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-CLN-4 at PLC register 0x6E40.

Step 4: Remove 120-Bar Through-Tool Cryo Coolant Loop housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-PMP-PLG-2504) with seal kit (KIT-MAZ350-CLN-4).

Step 6: Torque retaining bolts in cross sequence to 136 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-CLN-4, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0050 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-CLN-4 on Mazak-Integrex-i350 by installing PN-MAZ350-PMP-PLG-2504 & kit KIT-MAZ350-CLN-4. Fasteners torqued to 136 Nm; alignment verified at ±0.0050 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-XAX-5

Target Machine Model: Mazak-Integrex-i350 | Subsystem: X-Axis Linear Motor & Hydrostatic Guideway | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in X-Axis Linear Motor & Hydrostatic Guideway. Telemetry sensor EN-XAX-5 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-XAX-MAZ350-5
- PLC Hex Register Address: DB29.DBW_0x6E48
- Nominal Operating Range: 27 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-XAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-XAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-LIN-COIL-2505
- Overhaul Seal & Consumable Kit: KIT-MAZ350-XAX-5
- Critical Fastener Torque Value: 145 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0055 mm TIR
- Estimated Repair Duration (SLA MTTR): 7 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed XAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-XAX-5 at PLC register 0x6E48.

Step 4: Remove X-Axis Linear Motor & Hydrostatic Guideway housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-LIN-COIL-2505) with seal kit (KIT-MAZ350-XAX-5).

Step 6: Torque retaining bolts in cross sequence to 145 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-XAX-5, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0055 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-XAX-5 on Mazak-Integrex-i350 by installing PN-MAZ350-LIN-COIL-2505 & kit KIT-MAZ350-XAX-5. Fasteners torqued to 145 Nm; alignment verified at ±0.0055 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-YAX-6

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Y-Axis Master-Slave Gantry Ballscrew | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Y-Axis Master-Slave Gantry Ballscrew. Telemetry sensor EN-YAX-6 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-YAX-MAZ350-6
- PLC Hex Register Address: DB29.DBW_0x6E50
- Nominal Operating Range: 30 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-YAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-YAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-BSW-NUT-2506
- Overhaul Seal & Consumable Kit: KIT-MAZ350-YAX-6
- Critical Fastener Torque Value: 154 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0060 mm TIR
- Estimated Repair Duration (SLA MTTR): 5.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed YAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-YAX-6 at PLC register 0x6E50.

Step 4: Remove Y-Axis Master-Slave Gantry Ballscrew housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-BSW-NUT-2506) with seal kit (KIT-MAZ350-YAX-6).

Step 6: Torque retaining bolts in cross sequence to 154 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-YAX-6, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0060 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-YAX-6 on Mazak-Integrex-i350 by installing PN-MAZ350-BSW-NUT-2506 & kit KIT-MAZ350-YAX-6. Fasteners torqued to 154 Nm; alignment verified at ±0.0060 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-ZAX-7

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Z-Axis Nitrogen Counterbalance & Ram | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Z-Axis Nitrogen Counterbalance & Ram. Telemetry sensor PR-ZAX-7 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-ZAX-MAZ350-7
- PLC Hex Register Address: DB29.DBW_0x6E58
- Nominal Operating Range: 33 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-ZAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-ZAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-CYL-SEAL-2507
- Overhaul Seal & Consumable Kit: KIT-MAZ350-ZAX-7
- Critical Fastener Torque Value: 163 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0065 mm TIR
- Estimated Repair Duration (SLA MTTR): 3 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed ZAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-ZAX-7 at PLC register 0x6E58.

Step 4: Remove Z-Axis Nitrogen Counterbalance & Ram housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-CYL-SEAL-2507) with seal kit (KIT-MAZ350-ZAX-7).

Step 6: Torque retaining bolts in cross sequence to 163 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-ZAX-7, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0065 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-ZAX-7 on Mazak-Integrex-i350 by installing PN-MAZ350-CYL-SEAL-2507 & kit KIT-MAZ350-ZAX-7. Fasteners torqued to 163 Nm; alignment verified at ±0.0065 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-BAX-8

Target Machine Model: Mazak-Integrex-i350 | Subsystem: B-Axis Direct-Drive Torque Swivel Head | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in B-Axis Direct-Drive Torque Swivel Head. Telemetry sensor EN-BAX-8 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: EN-BAX-MAZ350-8
- PLC Hex Register Address: DB29.DBW_0x6E60
- Nominal Operating Range: 36 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-BAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-BAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-TRQ-STAT-2508
- Overhaul Seal & Consumable Kit: KIT-MAZ350-BAX-8
- Critical Fastener Torque Value: 172 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0010 mm TIR
- Estimated Repair Duration (SLA MTTR): 8 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed BAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-BAX-8 at PLC register 0x6E60.

Step 4: Remove B-Axis Direct-Drive Torque Swivel Head housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-TRQ-STAT-2508) with seal kit (KIT-MAZ350-BAX-8).

Step 6: Torque retaining bolts in cross sequence to 172 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-BAX-8, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0010 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-BAX-8 on Mazak-Integrex-i350 by installing PN-MAZ350-TRQ-STAT-2508 & kit KIT-MAZ350-BAX-8. Fasteners torqued to 172 Nm; alignment verified at ±0.0010 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-CAX-9

Target Machine Model: Mazak-Integrex-i350 | Subsystem: C-Axis Hydro-Mechanical Rotary Table | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in C-Axis Hydro-Mechanical Rotary Table. Telemetry sensor PR-CAX-9 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: PR-CAX-MAZ350-9
- PLC Hex Register Address: DB29.DBW_0x6E68
- Nominal Operating Range: 39 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-CAX: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-CAX.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-BRK-DISC-2509
- Overhaul Seal & Consumable Kit: KIT-MAZ350-CAX-9
- Critical Fastener Torque Value: 181 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0015 mm TIR
- Estimated Repair Duration (SLA MTTR): 4.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed CAX circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-CAX-9 at PLC register 0x6E68.

Step 4: Remove C-Axis Hydro-Mechanical Rotary Table housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-BRK-DISC-2509) with seal kit (KIT-MAZ350-CAX-9).

Step 6: Torque retaining bolts in cross sequence to 181 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-CAX-9, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0015 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-CAX-9 on Mazak-Integrex-i350 by installing PN-MAZ350-BRK-DISC-2509 & kit KIT-MAZ350-CAX-9. Fasteners torqued to 181 Nm; alignment verified at ±0.0015 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-OPT-10

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Pressurized Optical Glass Linear Scales | Severity: MEDIUM

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Pressurized Optical Glass Linear Scales. Telemetry sensor OP-SCL-10 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Pressurized Optical Glass Linear Scales on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: OP-SCL-MAZ350-10
- PLC Hex Register Address: DB29.DBW_0x6E70
- Nominal Operating Range: 42 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-OPT: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-OPT.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-GLASS-RD-2510
- Overhaul Seal & Consumable Kit: KIT-MAZ350-OPT-10
- Critical Fastener Torque Value: 190 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0020 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed OPT circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-OPT-10 at PLC register 0x6E70.

Step 4: Remove Pressurized Optical Glass Linear Scales housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-GLASS-RD-2510) with seal kit (KIT-MAZ350-OPT-10).

Step 6: Torque retaining bolts in cross sequence to 190 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-OPT-10, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0020 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-OPT-10 on Mazak-Integrex-i350 by installing PN-MAZ350-GLASS-RD-2510 & kit KIT-MAZ350-OPT-10. Fasteners torqued to 190 Nm; alignment verified at ±0.0020 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-PLC-11

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Profinet Safety I/O & IGBT Servo Drive | Severity: CRITICAL

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Profinet Safety I/O & IGBT Servo Drive. Telemetry sensor VT-DRV-11 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VT-DRV-MAZ350-11
- PLC Hex Register Address: DB29.DBW_0x6E78
- Nominal Operating Range: 45 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-PLC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-PLC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-IGBT-MOD-2511
- Overhaul Seal & Consumable Kit: KIT-MAZ350-PLC-11
- Critical Fastener Torque Value: 199 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0025 mm TIR
- Estimated Repair Duration (SLA MTTR): 6 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed PLC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-PLC-11 at PLC register 0x6E78.

Step 4: Remove Profinet Safety I/O & IGBT Servo Drive housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-IGBT-MOD-2511) with seal kit (KIT-MAZ350-PLC-11).

Step 6: Torque retaining bolts in cross sequence to 199 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-PLC-11, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0025 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-PLC-11 on Mazak-Integrex-i350 by installing PN-MAZ350-IGBT-MOD-2511 & kit KIT-MAZ350-PLC-11. Fasteners torqued to 199 Nm; alignment verified at ±0.0025 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-LUB-12

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Air-Oil Spindle Micro-Dosage Lubrication | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Air-Oil Spindle Micro-Dosage Lubrication. Telemetry sensor FL-LUB-12 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: FL-LUB-MAZ350-12
- PLC Hex Register Address: DB29.DBW_0x6E80
- Nominal Operating Range: 48 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-LUB: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-LUB.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-INJ-MET-2512
- Overhaul Seal & Consumable Kit: KIT-MAZ350-LUB-12
- Critical Fastener Torque Value: 208 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0030 mm TIR
- Estimated Repair Duration (SLA MTTR): 1.5 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed LUB circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-LUB-12 at PLC register 0x6E80.

Step 4: Remove Air-Oil Spindle Micro-Dosage Lubrication housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-INJ-MET-2512) with seal kit (KIT-MAZ350-LUB-12).

Step 6: Torque retaining bolts in cross sequence to 208 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-LUB-12, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0030 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-LUB-12 on Mazak-Integrex-i350 by installing PN-MAZ350-INJ-MET-2512 & kit KIT-MAZ350-LUB-12. Fasteners torqued to 208 Nm; alignment verified at ±0.0030 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-VAC-13

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Workpiece Vacuum Fixture & Seal Manifold | Severity: LOW

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Workpiece Vacuum Fixture & Seal Manifold. Telemetry sensor VC-MNF-13 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: VC-MNF-MAZ350-13
- PLC Hex Register Address: DB29.DBW_0x6E88
- Nominal Operating Range: 51 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-VAC: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-VAC.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-GASK-VIT-2513
- Overhaul Seal & Consumable Kit: KIT-MAZ350-VAC-13
- Critical Fastener Torque Value: 27 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0035 mm TIR
- Estimated Repair Duration (SLA MTTR): 2 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed VAC circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-VAC-13 at PLC register 0x6E88.

Step 4: Remove Workpiece Vacuum Fixture & Seal Manifold housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-GASK-VIT-2513) with seal kit (KIT-MAZ350-VAC-13).

Step 6: Torque retaining bolts in cross sequence to 27 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-VAC-13, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0035 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-VAC-13 on Mazak-Integrex-i350 by installing PN-MAZ350-GASK-VIT-2513 & kit KIT-MAZ350-VAC-13. Fasteners torqued to 27 Nm; alignment verified at ±0.0035 mm."

STANDARD OPERATING PROCEDURE — FAULT CODE: ERR-MAZ350-THM-14

Target Machine Model: Mazak-Integrex-i350 | Subsystem: Active Glycol Chiller & Frame Compensation | Severity: HIGH

1. OBSERVED SHOP-FLOOR SYMPTOM & TICKET TRIGGER

Operator on Mazak-Integrex-i350 reported severe process deviation in Active Glycol Chiller & Frame Compensation. Telemetry sensor TM-CHL-14 breached threshold during aerospace Ti-6Al-4V / Inconel-718 cycle.

2. ENGINEERING ROOT CAUSE

Accelerated wear and thermal fatigue within Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i350 leading to seal bypass, preload loss, or closed-loop servo following error.

3. CONTROLLER TELEMETRY & PLC DIAGNOSTIC REGISTERS

- Applicable Machine Model: Mazak-Integrex-i350 (MAZATROL SmoothAi)
- Telemetry Sensor ID: TM-CHL-MAZ350-14
- PLC Hex Register Address: DB29.DBW_0x6E90
- Nominal Operating Range: 54 - 84 units (AS9100 Rev D Certified)
- Fault Trip Threshold: > 119 units sustained for > 120 ms

4. LOCKOUT / TAGOUT (LOTO) SAFETY ISOLATION

LOTO-MAZ350-THM: Lock 480VAC Main Breaker, Discharge DC Bus < 12V, Isolate Pneumatic Valve V-THM.

5. OEM REPLACEMENT PARTS, TORQUE SPECS & TOLERANCES

- Primary OEM Part Number: PN-MAZ350-HEX-CORE-2514
- Overhaul Seal & Consumable Kit: KIT-MAZ350-THM-14
- Critical Fastener Torque Value: 36 Nm ± 2%
- Post-Repair Geometric Tolerance: ±0.0040 mm TIR
- Estimated Repair Duration (SLA MTTR): 4 Hours

6. STEP-BY-STEP REMEDIATION INSTRUCTIONS

Step 1: Trigger controlled feed-hold on Mazak-Integrex-i350 (MAZATROL SmoothAi) and verify spindle/axis stop before opening interlock doors.

Step 2: Perform Lockout/Tagout (LOTO) at 480V disconnect and bleed THM circuit pressure to 0.0 Bar.

Step 3: Connect diagnostic terminal to MAZATROL SmoothAi and verify fault latch ERR-MAZ350-THM-14 at PLC register 0x6E90.

Step 4: Remove Active Glycol Chiller & Frame Compensation housing on Mazak-Integrex-i350 and extract damaged assembly; inspect datum surfaces for fretting corrosion.

Step 5: Install OEM replacement unit (P/N: PN-MAZ350-HEX-CORE-2514) with seal kit (KIT-MAZ350-THM-14).

Step 6: Torque retaining bolts in cross sequence to 36 Nm using calibrated digital torque wrench.

Step 7: Reset fault ERR-MAZ350-THM-14, run 30-minute laser interferometer / ballbar qualification, and verify runout within ±0.0040 mm.

7. MANDATORY WORK ORDER RESOLUTION SIGN-OFF

Upon completing remediation on Mazak-Integrex-i350, technician must log: "Resolved ERR-MAZ350-THM-14 on Mazak-Integrex-i350 by installing PN-MAZ350-HEX-CORE-2514 & kit KIT-MAZ350-THM-14. Fasteners torqued to 36 Nm; alignment verified at ±0.0040 mm."

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X150 (BANK 1)

Machine Model: Siemens-DMG-X150 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X150.

PRM_40121 [Siemens-DMG-X150] — ATC_LOOP_GAIN_1 | Default: 1.150 | Linked Fault: ERR-SIE150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X150.

PRM_40122 [Siemens-DMG-X150] — HYD_LOOP_GAIN_2 | Default: 2.300 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40123 [Siemens-DMG-X150] — CLN_LOOP_GAIN_3 | Default: 3.450 | Linked Fault: ERR-SIE150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X150.

PRM_40124 [Siemens-DMG-X150] — XAX_LOOP_GAIN_4 | Default: 4.600 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40125 [Siemens-DMG-X150] — YAX_LOOP_GAIN_5 | Default: 5.750 | Linked Fault: ERR-SIE150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X150.

PRM_40126 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_6 | Default: 6.900 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

PRM_40127 [Siemens-DMG-X150] — BAX_LOOP_GAIN_7 | Default: 8.050 | Linked Fault: ERR-SIE150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X150.

PRM_40128 [Siemens-DMG-X150] — CAX_LOOP_GAIN_8 | Default: 9.200 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

PRM_40129 [Siemens-DMG-X150] — OPT_LOOP_GAIN_9 | Default: 10.350 | Linked Fault: ERR-SIE150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X150.

PRM_40130 [Siemens-DMG-X150] — PLC_LOOP_GAIN_10 | Default: 11.500 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

PRM_40131 [Siemens-DMG-X150] — LUB_LOOP_GAIN_11 | Default: 12.650 | Linked Fault: ERR-SIE150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X150.

PRM_40132 [Siemens-DMG-X150] — VAC_LOOP_GAIN_12 | Default: 13.800 | Linked Fault: ERR-SIE150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150.

PRM_40133 [Siemens-DMG-X150] — THM_LOOP_GAIN_13 | Default: 14.950 | Linked Fault: ERR-SIE150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X150.

PRM_40134 [Siemens-DMG-X150] — SPN_LOOP_GAIN_14 | Default: 16.100 | Linked Fault: ERR-SIE150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150.

PRM_40135 [Siemens-DMG-X150] — ATC_LOOP_GAIN_15 | Default: 17.250 | Linked Fault: ERR-SIE150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X150.

PRM_40136 [Siemens-DMG-X150] — HYD_LOOP_GAIN_16 | Default: 18.400 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X150 (BANK 2)

Machine Model: Siemens-DMG-X150 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X150.

PRM_40141 [Siemens-DMG-X150] — HYD_LOOP_GAIN_1 | Default: 2.300 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40142 [Siemens-DMG-X150] — XAX_LOOP_GAIN_2 | Default: 4.600 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40143 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_3 | Default: 6.900 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

PRM_40144 [Siemens-DMG-X150] — CAX_LOOP_GAIN_4 | Default: 9.200 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

PRM_40145 [Siemens-DMG-X150] — PLC_LOOP_GAIN_5 | Default: 11.500 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

PRM_40146 [Siemens-DMG-X150] — VAC_LOOP_GAIN_6 | Default: 13.800 | Linked Fault: ERR-SIE150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150.

PRM_40147 [Siemens-DMG-X150] — SPN_LOOP_GAIN_7 | Default: 16.100 | Linked Fault: ERR-SIE150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150.

PRM_40148 [Siemens-DMG-X150] — HYD_LOOP_GAIN_8 | Default: 18.400 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40149 [Siemens-DMG-X150] — XAX_LOOP_GAIN_9 | Default: 20.700 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40150 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_10 | Default: 23.000 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

PRM_40151 [Siemens-DMG-X150] — CAX_LOOP_GAIN_11 | Default: 25.300 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

PRM_40152 [Siemens-DMG-X150] — PLC_LOOP_GAIN_12 | Default: 27.600 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

PRM_40153 [Siemens-DMG-X150] — VAC_LOOP_GAIN_13 | Default: 29.900 | Linked Fault: ERR-SIE150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150.

PRM_40154 [Siemens-DMG-X150] — SPN_LOOP_GAIN_14 | Default: 32.200 | Linked Fault: ERR-SIE150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150.

PRM_40155 [Siemens-DMG-X150] — HYD_LOOP_GAIN_15 | Default: 34.500 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40156 [Siemens-DMG-X150] — XAX_LOOP_GAIN_16 | Default: 36.800 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X150 (BANK 3)

Machine Model: Siemens-DMG-X150 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X150.

PRM_40161 [Siemens-DMG-X150] — CLN_LOOP_GAIN_1 | Default: 3.450 | Linked Fault: ERR-SIE150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X150.

PRM_40162 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_2 | Default: 6.900 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

PRM_40163 [Siemens-DMG-X150] — OPT_LOOP_GAIN_3 | Default: 10.350 | Linked Fault: ERR-SIE150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X150.

PRM_40164 [Siemens-DMG-X150] — VAC_LOOP_GAIN_4 | Default: 13.800 | Linked Fault: ERR-SIE150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150.

PRM_40165 [Siemens-DMG-X150] — ATC_LOOP_GAIN_5 | Default: 17.250 | Linked Fault: ERR-SIE150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X150.

PRM_40166 [Siemens-DMG-X150] — XAX_LOOP_GAIN_6 | Default: 20.700 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40167 [Siemens-DMG-X150] — BAX_LOOP_GAIN_7 | Default: 24.150 | Linked Fault: ERR-SIE150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X150.

PRM_40168 [Siemens-DMG-X150] — PLC_LOOP_GAIN_8 | Default: 27.600 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

PRM_40169 [Siemens-DMG-X150] — THM_LOOP_GAIN_9 | Default: 31.050 | Linked Fault: ERR-SIE150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X150.

PRM_40170 [Siemens-DMG-X150] — HYD_LOOP_GAIN_10 | Default: 34.500 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40171 [Siemens-DMG-X150] — YAX_LOOP_GAIN_11 | Default: 37.950 | Linked Fault: ERR-SIE150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X150.

PRM_40172 [Siemens-DMG-X150] — CAX_LOOP_GAIN_12 | Default: 41.400 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

PRM_40173 [Siemens-DMG-X150] — LUB_LOOP_GAIN_13 | Default: 44.850 | Linked Fault: ERR-SIE150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X150.

PRM_40174 [Siemens-DMG-X150] — SPN_LOOP_GAIN_14 | Default: 48.300 | Linked Fault: ERR-SIE150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150.

PRM_40175 [Siemens-DMG-X150] — CLN_LOOP_GAIN_15 | Default: 51.750 | Linked Fault: ERR-SIE150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X150.

PRM_40176 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_16 | Default: 55.200 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X150 (BANK 4)

Machine Model: Siemens-DMG-X150 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X150.

PRM_40181 [Siemens-DMG-X150] — XAX_LOOP_GAIN_1 | Default: 4.600 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40182 [Siemens-DMG-X150] — CAX_LOOP_GAIN_2 | Default: 9.200 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

PRM_40183 [Siemens-DMG-X150] — VAC_LOOP_GAIN_3 | Default: 13.800 | Linked Fault: ERR-SIE150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150.

PRM_40184 [Siemens-DMG-X150] — HYD_LOOP_GAIN_4 | Default: 18.400 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40185 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_5 | Default: 23.000 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

PRM_40186 [Siemens-DMG-X150] — PLC_LOOP_GAIN_6 | Default: 27.600 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

PRM_40187 [Siemens-DMG-X150] — SPN_LOOP_GAIN_7 | Default: 32.200 | Linked Fault: ERR-SIE150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150.

PRM_40188 [Siemens-DMG-X150] — XAX_LOOP_GAIN_8 | Default: 36.800 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40189 [Siemens-DMG-X150] — CAX_LOOP_GAIN_9 | Default: 41.400 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

PRM_40190 [Siemens-DMG-X150] — VAC_LOOP_GAIN_10 | Default: 46.000 | Linked Fault: ERR-SIE150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150.

PRM_40191 [Siemens-DMG-X150] — HYD_LOOP_GAIN_11 | Default: 50.600 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40192 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_12 | Default: 55.200 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

PRM_40193 [Siemens-DMG-X150] — PLC_LOOP_GAIN_13 | Default: 59.800 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

PRM_40194 [Siemens-DMG-X150] — SPN_LOOP_GAIN_14 | Default: 64.400 | Linked Fault: ERR-SIE150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150.

PRM_40195 [Siemens-DMG-X150] — XAX_LOOP_GAIN_15 | Default: 69.000 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40196 [Siemens-DMG-X150] — CAX_LOOP_GAIN_16 | Default: 73.600 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X150 (BANK 5)

Machine Model: Siemens-DMG-X150 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X150.

PRM_40201 [Siemens-DMG-X150] — YAX_LOOP_GAIN_1 | Default: 5.750 | Linked Fault: ERR-SIE150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X150.

PRM_40202 [Siemens-DMG-X150] — PLC_LOOP_GAIN_2 | Default: 11.500 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

PRM_40203 [Siemens-DMG-X150] — ATC_LOOP_GAIN_3 | Default: 17.250 | Linked Fault: ERR-SIE150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X150.

PRM_40204 [Siemens-DMG-X150] — ZAX_LOOP_GAIN_4 | Default: 23.000 | Linked Fault: ERR-SIE150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X150.

PRM_40205 [Siemens-DMG-X150] — LUB_LOOP_GAIN_5 | Default: 28.750 | Linked Fault: ERR-SIE150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X150.

PRM_40206 [Siemens-DMG-X150] — HYD_LOOP_GAIN_6 | Default: 34.500 | Linked Fault: ERR-SIE150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X150.

PRM_40207 [Siemens-DMG-X150] — BAX_LOOP_GAIN_7 | Default: 40.250 | Linked Fault: ERR-SIE150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X150.

PRM_40208 [Siemens-DMG-X150] — VAC_LOOP_GAIN_8 | Default: 46.000 | Linked Fault: ERR-SIE150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X150.

PRM_40209 [Siemens-DMG-X150] — CLN_LOOP_GAIN_9 | Default: 51.750 | Linked Fault: ERR-SIE150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X150.

PRM_40210 [Siemens-DMG-X150] — CAX_LOOP_GAIN_10 | Default: 57.500 | Linked Fault: ERR-SIE150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X150.

PRM_40211 [Siemens-DMG-X150] — THM_LOOP_GAIN_11 | Default: 63.250 | Linked Fault: ERR-SIE150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X150.

PRM_40212 [Siemens-DMG-X150] — XAX_LOOP_GAIN_12 | Default: 69.000 | Linked Fault: ERR-SIE150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X150.

PRM_40213 [Siemens-DMG-X150] — OPT_LOOP_GAIN_13 | Default: 74.750 | Linked Fault: ERR-SIE150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X150.

PRM_40214 [Siemens-DMG-X150] — SPN_LOOP_GAIN_14 | Default: 80.500 | Linked Fault: ERR-SIE150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X150.

PRM_40215 [Siemens-DMG-X150] — YAX_LOOP_GAIN_15 | Default: 86.250 | Linked Fault: ERR-SIE150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X150.

PRM_40216 [Siemens-DMG-X150] — PLC_LOOP_GAIN_16 | Default: 92.000 | Linked Fault: ERR-SIE150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X150.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X200 (BANK 1)

Machine Model: Siemens-DMG-X200 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X200.

PRM_40221 [Siemens-DMG-X200] — ATC_LOOP_GAIN_1 | Default: 2.300 | Linked Fault: ERR-SIE200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X200.

PRM_40222 [Siemens-DMG-X200] — HYD_LOOP_GAIN_2 | Default: 4.600 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40223 [Siemens-DMG-X200] — CLN_LOOP_GAIN_3 | Default: 6.900 | Linked Fault: ERR-SIE200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X200.

PRM_40224 [Siemens-DMG-X200] — XAX_LOOP_GAIN_4 | Default: 9.200 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40225 [Siemens-DMG-X200] — YAX_LOOP_GAIN_5 | Default: 11.500 | Linked Fault: ERR-SIE200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X200.

PRM_40226 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_6 | Default: 13.800 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

PRM_40227 [Siemens-DMG-X200] — BAX_LOOP_GAIN_7 | Default: 16.100 | Linked Fault: ERR-SIE200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X200.

PRM_40228 [Siemens-DMG-X200] — CAX_LOOP_GAIN_8 | Default: 18.400 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

PRM_40229 [Siemens-DMG-X200] — OPT_LOOP_GAIN_9 | Default: 20.700 | Linked Fault: ERR-SIE200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X200.

PRM_40230 [Siemens-DMG-X200] — PLC_LOOP_GAIN_10 | Default: 23.000 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

PRM_40231 [Siemens-DMG-X200] — LUB_LOOP_GAIN_11 | Default: 25.300 | Linked Fault: ERR-SIE200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X200.

PRM_40232 [Siemens-DMG-X200] — VAC_LOOP_GAIN_12 | Default: 27.600 | Linked Fault: ERR-SIE200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200.

PRM_40233 [Siemens-DMG-X200] — THM_LOOP_GAIN_13 | Default: 29.900 | Linked Fault: ERR-SIE200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X200.

PRM_40234 [Siemens-DMG-X200] — SPN_LOOP_GAIN_14 | Default: 32.200 | Linked Fault: ERR-SIE200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200.

PRM_40235 [Siemens-DMG-X200] — ATC_LOOP_GAIN_15 | Default: 34.500 | Linked Fault: ERR-SIE200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X200.

PRM_40236 [Siemens-DMG-X200] — HYD_LOOP_GAIN_16 | Default: 36.800 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X200 (BANK 2)

Machine Model: Siemens-DMG-X200 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X200.

PRM_40241 [Siemens-DMG-X200] — HYD_LOOP_GAIN_1 | Default: 4.600 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40242 [Siemens-DMG-X200] — XAX_LOOP_GAIN_2 | Default: 9.200 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40243 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_3 | Default: 13.800 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

PRM_40244 [Siemens-DMG-X200] — CAX_LOOP_GAIN_4 | Default: 18.400 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

PRM_40245 [Siemens-DMG-X200] — PLC_LOOP_GAIN_5 | Default: 23.000 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

PRM_40246 [Siemens-DMG-X200] — VAC_LOOP_GAIN_6 | Default: 27.600 | Linked Fault: ERR-SIE200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200.

PRM_40247 [Siemens-DMG-X200] — SPN_LOOP_GAIN_7 | Default: 32.200 | Linked Fault: ERR-SIE200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200.

PRM_40248 [Siemens-DMG-X200] — HYD_LOOP_GAIN_8 | Default: 36.800 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40249 [Siemens-DMG-X200] — XAX_LOOP_GAIN_9 | Default: 41.400 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40250 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_10 | Default: 46.000 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

PRM_40251 [Siemens-DMG-X200] — CAX_LOOP_GAIN_11 | Default: 50.600 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

PRM_40252 [Siemens-DMG-X200] — PLC_LOOP_GAIN_12 | Default: 55.200 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

PRM_40253 [Siemens-DMG-X200] — VAC_LOOP_GAIN_13 | Default: 59.800 | Linked Fault: ERR-SIE200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200.

PRM_40254 [Siemens-DMG-X200] — SPN_LOOP_GAIN_14 | Default: 64.400 | Linked Fault: ERR-SIE200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200.

PRM_40255 [Siemens-DMG-X200] — HYD_LOOP_GAIN_15 | Default: 69.000 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40256 [Siemens-DMG-X200] — XAX_LOOP_GAIN_16 | Default: 73.600 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X200 (BANK 3)

Machine Model: Siemens-DMG-X200 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X200.

PRM_40261 [Siemens-DMG-X200] — CLN_LOOP_GAIN_1 | Default: 6.900 | Linked Fault: ERR-SIE200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X200.

PRM_40262 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_2 | Default: 13.800 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

PRM_40263 [Siemens-DMG-X200] — OPT_LOOP_GAIN_3 | Default: 20.700 | Linked Fault: ERR-SIE200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X200.

PRM_40264 [Siemens-DMG-X200] — VAC_LOOP_GAIN_4 | Default: 27.600 | Linked Fault: ERR-SIE200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200.

PRM_40265 [Siemens-DMG-X200] — ATC_LOOP_GAIN_5 | Default: 34.500 | Linked Fault: ERR-SIE200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X200.

PRM_40266 [Siemens-DMG-X200] — XAX_LOOP_GAIN_6 | Default: 41.400 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40267 [Siemens-DMG-X200] — BAX_LOOP_GAIN_7 | Default: 48.300 | Linked Fault: ERR-SIE200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X200.

PRM_40268 [Siemens-DMG-X200] — PLC_LOOP_GAIN_8 | Default: 55.200 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

PRM_40269 [Siemens-DMG-X200] — THM_LOOP_GAIN_9 | Default: 62.100 | Linked Fault: ERR-SIE200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X200.

PRM_40270 [Siemens-DMG-X200] — HYD_LOOP_GAIN_10 | Default: 69.000 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40271 [Siemens-DMG-X200] — YAX_LOOP_GAIN_11 | Default: 75.900 | Linked Fault: ERR-SIE200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X200.

PRM_40272 [Siemens-DMG-X200] — CAX_LOOP_GAIN_12 | Default: 82.800 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

PRM_40273 [Siemens-DMG-X200] — LUB_LOOP_GAIN_13 | Default: 89.700 | Linked Fault: ERR-SIE200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X200.

PRM_40274 [Siemens-DMG-X200] — SPN_LOOP_GAIN_14 | Default: 96.600 | Linked Fault: ERR-SIE200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200.

PRM_40275 [Siemens-DMG-X200] — CLN_LOOP_GAIN_15 | Default: 103.500 | Linked Fault: ERR-SIE200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X200.

PRM_40276 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_16 | Default: 110.400 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X200 (BANK 4)

Machine Model: Siemens-DMG-X200 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X200.

PRM_40281 [Siemens-DMG-X200] — XAX_LOOP_GAIN_1 | Default: 9.200 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40282 [Siemens-DMG-X200] — CAX_LOOP_GAIN_2 | Default: 18.400 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

PRM_40283 [Siemens-DMG-X200] — VAC_LOOP_GAIN_3 | Default: 27.600 | Linked Fault: ERR-SIE200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200.

PRM_40284 [Siemens-DMG-X200] — HYD_LOOP_GAIN_4 | Default: 36.800 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40285 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_5 | Default: 46.000 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

PRM_40286 [Siemens-DMG-X200] — PLC_LOOP_GAIN_6 | Default: 55.200 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

PRM_40287 [Siemens-DMG-X200] — SPN_LOOP_GAIN_7 | Default: 64.400 | Linked Fault: ERR-SIE200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200.

PRM_40288 [Siemens-DMG-X200] — XAX_LOOP_GAIN_8 | Default: 73.600 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40289 [Siemens-DMG-X200] — CAX_LOOP_GAIN_9 | Default: 82.800 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

PRM_40290 [Siemens-DMG-X200] — VAC_LOOP_GAIN_10 | Default: 92.000 | Linked Fault: ERR-SIE200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200.

PRM_40291 [Siemens-DMG-X200] — HYD_LOOP_GAIN_11 | Default: 101.200 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40292 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_12 | Default: 110.400 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

PRM_40293 [Siemens-DMG-X200] — PLC_LOOP_GAIN_13 | Default: 119.600 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

PRM_40294 [Siemens-DMG-X200] — SPN_LOOP_GAIN_14 | Default: 128.800 | Linked Fault: ERR-SIE200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200.

PRM_40295 [Siemens-DMG-X200] — XAX_LOOP_GAIN_15 | Default: 138.000 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40296 [Siemens-DMG-X200] — CAX_LOOP_GAIN_16 | Default: 147.200 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X200 (BANK 5)

Machine Model: Siemens-DMG-X200 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X200.

PRM_40301 [Siemens-DMG-X200] — YAX_LOOP_GAIN_1 | Default: 11.500 | Linked Fault: ERR-SIE200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X200.

PRM_40302 [Siemens-DMG-X200] — PLC_LOOP_GAIN_2 | Default: 23.000 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

PRM_40303 [Siemens-DMG-X200] — ATC_LOOP_GAIN_3 | Default: 34.500 | Linked Fault: ERR-SIE200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X200.

PRM_40304 [Siemens-DMG-X200] — ZAX_LOOP_GAIN_4 | Default: 46.000 | Linked Fault: ERR-SIE200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X200.

PRM_40305 [Siemens-DMG-X200] — LUB_LOOP_GAIN_5 | Default: 57.500 | Linked Fault: ERR-SIE200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X200.

PRM_40306 [Siemens-DMG-X200] — HYD_LOOP_GAIN_6 | Default: 69.000 | Linked Fault: ERR-SIE200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X200.

PRM_40307 [Siemens-DMG-X200] — BAX_LOOP_GAIN_7 | Default: 80.500 | Linked Fault: ERR-SIE200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X200.

PRM_40308 [Siemens-DMG-X200] — VAC_LOOP_GAIN_8 | Default: 92.000 | Linked Fault: ERR-SIE200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X200.

PRM_40309 [Siemens-DMG-X200] — CLN_LOOP_GAIN_9 | Default: 103.500 | Linked Fault: ERR-SIE200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X200.

PRM_40310 [Siemens-DMG-X200] — CAX_LOOP_GAIN_10 | Default: 115.000 | Linked Fault: ERR-SIE200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X200.

PRM_40311 [Siemens-DMG-X200] — THM_LOOP_GAIN_11 | Default: 126.500 | Linked Fault: ERR-SIE200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X200.

PRM_40312 [Siemens-DMG-X200] — XAX_LOOP_GAIN_12 | Default: 138.000 | Linked Fault: ERR-SIE200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X200.

PRM_40313 [Siemens-DMG-X200] — OPT_LOOP_GAIN_13 | Default: 149.500 | Linked Fault: ERR-SIE200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X200.

PRM_40314 [Siemens-DMG-X200] — SPN_LOOP_GAIN_14 | Default: 161.000 | Linked Fault: ERR-SIE200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X200.

PRM_40315 [Siemens-DMG-X200] — YAX_LOOP_GAIN_15 | Default: 172.500 | Linked Fault: ERR-SIE200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X200.

PRM_40316 [Siemens-DMG-X200] — PLC_LOOP_GAIN_16 | Default: 184.000 | Linked Fault: ERR-SIE200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X200.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X250 (BANK 1)

Machine Model: Siemens-DMG-X250 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X250.

PRM_40321 [Siemens-DMG-X250] — ATC_LOOP_GAIN_1 | Default: 3.450 | Linked Fault: ERR-SIE250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X250.

PRM_40322 [Siemens-DMG-X250] — HYD_LOOP_GAIN_2 | Default: 6.900 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40323 [Siemens-DMG-X250] — CLN_LOOP_GAIN_3 | Default: 10.350 | Linked Fault: ERR-SIE250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X250.

PRM_40324 [Siemens-DMG-X250] — XAX_LOOP_GAIN_4 | Default: 13.800 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40325 [Siemens-DMG-X250] — YAX_LOOP_GAIN_5 | Default: 17.250 | Linked Fault: ERR-SIE250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X250.

PRM_40326 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_6 | Default: 20.700 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

PRM_40327 [Siemens-DMG-X250] — BAX_LOOP_GAIN_7 | Default: 24.150 | Linked Fault: ERR-SIE250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X250.

PRM_40328 [Siemens-DMG-X250] — CAX_LOOP_GAIN_8 | Default: 27.600 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

PRM_40329 [Siemens-DMG-X250] — OPT_LOOP_GAIN_9 | Default: 31.050 | Linked Fault: ERR-SIE250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X250.

PRM_40330 [Siemens-DMG-X250] — PLC_LOOP_GAIN_10 | Default: 34.500 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

PRM_40331 [Siemens-DMG-X250] — LUB_LOOP_GAIN_11 | Default: 37.950 | Linked Fault: ERR-SIE250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X250.

PRM_40332 [Siemens-DMG-X250] — VAC_LOOP_GAIN_12 | Default: 41.400 | Linked Fault: ERR-SIE250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250.

PRM_40333 [Siemens-DMG-X250] — THM_LOOP_GAIN_13 | Default: 44.850 | Linked Fault: ERR-SIE250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X250.

PRM_40334 [Siemens-DMG-X250] — SPN_LOOP_GAIN_14 | Default: 48.300 | Linked Fault: ERR-SIE250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250.

PRM_40335 [Siemens-DMG-X250] — ATC_LOOP_GAIN_15 | Default: 51.750 | Linked Fault: ERR-SIE250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X250.

PRM_40336 [Siemens-DMG-X250] — HYD_LOOP_GAIN_16 | Default: 55.200 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X250 (BANK 2)

Machine Model: Siemens-DMG-X250 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X250.

PRM_40341 [Siemens-DMG-X250] — HYD_LOOP_GAIN_1 | Default: 6.900 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40342 [Siemens-DMG-X250] — XAX_LOOP_GAIN_2 | Default: 13.800 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40343 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_3 | Default: 20.700 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

PRM_40344 [Siemens-DMG-X250] — CAX_LOOP_GAIN_4 | Default: 27.600 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

PRM_40345 [Siemens-DMG-X250] — PLC_LOOP_GAIN_5 | Default: 34.500 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

PRM_40346 [Siemens-DMG-X250] — VAC_LOOP_GAIN_6 | Default: 41.400 | Linked Fault: ERR-SIE250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250.

PRM_40347 [Siemens-DMG-X250] — SPN_LOOP_GAIN_7 | Default: 48.300 | Linked Fault: ERR-SIE250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250.

PRM_40348 [Siemens-DMG-X250] — HYD_LOOP_GAIN_8 | Default: 55.200 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40349 [Siemens-DMG-X250] — XAX_LOOP_GAIN_9 | Default: 62.100 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40350 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_10 | Default: 69.000 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

PRM_40351 [Siemens-DMG-X250] — CAX_LOOP_GAIN_11 | Default: 75.900 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

PRM_40352 [Siemens-DMG-X250] — PLC_LOOP_GAIN_12 | Default: 82.800 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

PRM_40353 [Siemens-DMG-X250] — VAC_LOOP_GAIN_13 | Default: 89.700 | Linked Fault: ERR-SIE250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250.

PRM_40354 [Siemens-DMG-X250] — SPN_LOOP_GAIN_14 | Default: 96.600 | Linked Fault: ERR-SIE250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250.

PRM_40355 [Siemens-DMG-X250] — HYD_LOOP_GAIN_15 | Default: 103.500 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40356 [Siemens-DMG-X250] — XAX_LOOP_GAIN_16 | Default: 110.400 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X250 (BANK 3)

Machine Model: Siemens-DMG-X250 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X250.

PRM_40361 [Siemens-DMG-X250] — CLN_LOOP_GAIN_1 | Default: 10.350 | Linked Fault: ERR-SIE250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X250.

PRM_40362 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_2 | Default: 20.700 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

PRM_40363 [Siemens-DMG-X250] — OPT_LOOP_GAIN_3 | Default: 31.050 | Linked Fault: ERR-SIE250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X250.

PRM_40364 [Siemens-DMG-X250] — VAC_LOOP_GAIN_4 | Default: 41.400 | Linked Fault: ERR-SIE250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250.

PRM_40365 [Siemens-DMG-X250] — ATC_LOOP_GAIN_5 | Default: 51.750 | Linked Fault: ERR-SIE250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X250.

PRM_40366 [Siemens-DMG-X250] — XAX_LOOP_GAIN_6 | Default: 62.100 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40367 [Siemens-DMG-X250] — BAX_LOOP_GAIN_7 | Default: 72.450 | Linked Fault: ERR-SIE250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X250.

PRM_40368 [Siemens-DMG-X250] — PLC_LOOP_GAIN_8 | Default: 82.800 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

PRM_40369 [Siemens-DMG-X250] — THM_LOOP_GAIN_9 | Default: 93.150 | Linked Fault: ERR-SIE250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X250.

PRM_40370 [Siemens-DMG-X250] — HYD_LOOP_GAIN_10 | Default: 103.500 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40371 [Siemens-DMG-X250] — YAX_LOOP_GAIN_11 | Default: 113.850 | Linked Fault: ERR-SIE250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X250.

PRM_40372 [Siemens-DMG-X250] — CAX_LOOP_GAIN_12 | Default: 124.200 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

PRM_40373 [Siemens-DMG-X250] — LUB_LOOP_GAIN_13 | Default: 134.550 | Linked Fault: ERR-SIE250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X250.

PRM_40374 [Siemens-DMG-X250] — SPN_LOOP_GAIN_14 | Default: 144.900 | Linked Fault: ERR-SIE250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250.

PRM_40375 [Siemens-DMG-X250] — CLN_LOOP_GAIN_15 | Default: 155.250 | Linked Fault: ERR-SIE250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X250.

PRM_40376 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_16 | Default: 165.600 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X250 (BANK 4)

Machine Model: Siemens-DMG-X250 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X250.

PRM_40381 [Siemens-DMG-X250] — XAX_LOOP_GAIN_1 | Default: 13.800 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40382 [Siemens-DMG-X250] — CAX_LOOP_GAIN_2 | Default: 27.600 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

PRM_40383 [Siemens-DMG-X250] — VAC_LOOP_GAIN_3 | Default: 41.400 | Linked Fault: ERR-SIE250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250.

PRM_40384 [Siemens-DMG-X250] — HYD_LOOP_GAIN_4 | Default: 55.200 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40385 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_5 | Default: 69.000 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

PRM_40386 [Siemens-DMG-X250] — PLC_LOOP_GAIN_6 | Default: 82.800 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

PRM_40387 [Siemens-DMG-X250] — SPN_LOOP_GAIN_7 | Default: 96.600 | Linked Fault: ERR-SIE250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250.

PRM_40388 [Siemens-DMG-X250] — XAX_LOOP_GAIN_8 | Default: 110.400 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40389 [Siemens-DMG-X250] — CAX_LOOP_GAIN_9 | Default: 124.200 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

PRM_40390 [Siemens-DMG-X250] — VAC_LOOP_GAIN_10 | Default: 138.000 | Linked Fault: ERR-SIE250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250.

PRM_40391 [Siemens-DMG-X250] — HYD_LOOP_GAIN_11 | Default: 151.800 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40392 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_12 | Default: 165.600 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

PRM_40393 [Siemens-DMG-X250] — PLC_LOOP_GAIN_13 | Default: 179.400 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

PRM_40394 [Siemens-DMG-X250] — SPN_LOOP_GAIN_14 | Default: 193.200 | Linked Fault: ERR-SIE250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250.

PRM_40395 [Siemens-DMG-X250] — XAX_LOOP_GAIN_15 | Default: 207.000 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40396 [Siemens-DMG-X250] — CAX_LOOP_GAIN_16 | Default: 220.800 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X250 (BANK 5)

Machine Model: Siemens-DMG-X250 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X250.

PRM_40401 [Siemens-DMG-X250] — YAX_LOOP_GAIN_1 | Default: 17.250 | Linked Fault: ERR-SIE250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X250.

PRM_40402 [Siemens-DMG-X250] — PLC_LOOP_GAIN_2 | Default: 34.500 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

PRM_40403 [Siemens-DMG-X250] — ATC_LOOP_GAIN_3 | Default: 51.750 | Linked Fault: ERR-SIE250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X250.

PRM_40404 [Siemens-DMG-X250] — ZAX_LOOP_GAIN_4 | Default: 69.000 | Linked Fault: ERR-SIE250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X250.

PRM_40405 [Siemens-DMG-X250] — LUB_LOOP_GAIN_5 | Default: 86.250 | Linked Fault: ERR-SIE250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X250.

PRM_40406 [Siemens-DMG-X250] — HYD_LOOP_GAIN_6 | Default: 103.500 | Linked Fault: ERR-SIE250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X250.

PRM_40407 [Siemens-DMG-X250] — BAX_LOOP_GAIN_7 | Default: 120.750 | Linked Fault: ERR-SIE250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X250.

PRM_40408 [Siemens-DMG-X250] — VAC_LOOP_GAIN_8 | Default: 138.000 | Linked Fault: ERR-SIE250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X250.

PRM_40409 [Siemens-DMG-X250] — CLN_LOOP_GAIN_9 | Default: 155.250 | Linked Fault: ERR-SIE250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X250.

PRM_40410 [Siemens-DMG-X250] — CAX_LOOP_GAIN_10 | Default: 172.500 | Linked Fault: ERR-SIE250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X250.

PRM_40411 [Siemens-DMG-X250] — THM_LOOP_GAIN_11 | Default: 189.750 | Linked Fault: ERR-SIE250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X250.

PRM_40412 [Siemens-DMG-X250] — XAX_LOOP_GAIN_12 | Default: 207.000 | Linked Fault: ERR-SIE250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X250.

PRM_40413 [Siemens-DMG-X250] — OPT_LOOP_GAIN_13 | Default: 224.250 | Linked Fault: ERR-SIE250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X250.

PRM_40414 [Siemens-DMG-X250] — SPN_LOOP_GAIN_14 | Default: 241.500 | Linked Fault: ERR-SIE250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X250.

PRM_40415 [Siemens-DMG-X250] — YAX_LOOP_GAIN_15 | Default: 8.750 | Linked Fault: ERR-SIE250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X250.

PRM_40416 [Siemens-DMG-X250] — PLC_LOOP_GAIN_16 | Default: 26.000 | Linked Fault: ERR-SIE250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X250.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X300 (BANK 1)

Machine Model: Siemens-DMG-X300 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X300.

PRM_40421 [Siemens-DMG-X300] — ATC_LOOP_GAIN_1 | Default: 4.600 | Linked Fault: ERR-SIE300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X300.

PRM_40422 [Siemens-DMG-X300] — HYD_LOOP_GAIN_2 | Default: 9.200 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40423 [Siemens-DMG-X300] — CLN_LOOP_GAIN_3 | Default: 13.800 | Linked Fault: ERR-SIE300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X300.

PRM_40424 [Siemens-DMG-X300] — XAX_LOOP_GAIN_4 | Default: 18.400 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40425 [Siemens-DMG-X300] — YAX_LOOP_GAIN_5 | Default: 23.000 | Linked Fault: ERR-SIE300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X300.

PRM_40426 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_6 | Default: 27.600 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

PRM_40427 [Siemens-DMG-X300] — BAX_LOOP_GAIN_7 | Default: 32.200 | Linked Fault: ERR-SIE300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X300.

PRM_40428 [Siemens-DMG-X300] — CAX_LOOP_GAIN_8 | Default: 36.800 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

PRM_40429 [Siemens-DMG-X300] — OPT_LOOP_GAIN_9 | Default: 41.400 | Linked Fault: ERR-SIE300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X300.

PRM_40430 [Siemens-DMG-X300] — PLC_LOOP_GAIN_10 | Default: 46.000 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

PRM_40431 [Siemens-DMG-X300] — LUB_LOOP_GAIN_11 | Default: 50.600 | Linked Fault: ERR-SIE300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X300.

PRM_40432 [Siemens-DMG-X300] — VAC_LOOP_GAIN_12 | Default: 55.200 | Linked Fault: ERR-SIE300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300.

PRM_40433 [Siemens-DMG-X300] — THM_LOOP_GAIN_13 | Default: 59.800 | Linked Fault: ERR-SIE300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X300.

PRM_40434 [Siemens-DMG-X300] — SPN_LOOP_GAIN_14 | Default: 64.400 | Linked Fault: ERR-SIE300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300.

PRM_40435 [Siemens-DMG-X300] — ATC_LOOP_GAIN_15 | Default: 69.000 | Linked Fault: ERR-SIE300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X300.

PRM_40436 [Siemens-DMG-X300] — HYD_LOOP_GAIN_16 | Default: 73.600 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X300 (BANK 2)

Machine Model: Siemens-DMG-X300 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X300.

PRM_40441 [Siemens-DMG-X300] — HYD_LOOP_GAIN_1 | Default: 9.200 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40442 [Siemens-DMG-X300] — XAX_LOOP_GAIN_2 | Default: 18.400 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40443 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_3 | Default: 27.600 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

PRM_40444 [Siemens-DMG-X300] — CAX_LOOP_GAIN_4 | Default: 36.800 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

PRM_40445 [Siemens-DMG-X300] — PLC_LOOP_GAIN_5 | Default: 46.000 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

PRM_40446 [Siemens-DMG-X300] — VAC_LOOP_GAIN_6 | Default: 55.200 | Linked Fault: ERR-SIE300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300.

PRM_40447 [Siemens-DMG-X300] — SPN_LOOP_GAIN_7 | Default: 64.400 | Linked Fault: ERR-SIE300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300.

PRM_40448 [Siemens-DMG-X300] — HYD_LOOP_GAIN_8 | Default: 73.600 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40449 [Siemens-DMG-X300] — XAX_LOOP_GAIN_9 | Default: 82.800 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40450 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_10 | Default: 92.000 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

PRM_40451 [Siemens-DMG-X300] — CAX_LOOP_GAIN_11 | Default: 101.200 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

PRM_40452 [Siemens-DMG-X300] — PLC_LOOP_GAIN_12 | Default: 110.400 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

PRM_40453 [Siemens-DMG-X300] — VAC_LOOP_GAIN_13 | Default: 119.600 | Linked Fault: ERR-SIE300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300.

PRM_40454 [Siemens-DMG-X300] — SPN_LOOP_GAIN_14 | Default: 128.800 | Linked Fault: ERR-SIE300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300.

PRM_40455 [Siemens-DMG-X300] — HYD_LOOP_GAIN_15 | Default: 138.000 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40456 [Siemens-DMG-X300] — XAX_LOOP_GAIN_16 | Default: 147.200 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X300 (BANK 3)

Machine Model: Siemens-DMG-X300 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X300.

PRM_40461 [Siemens-DMG-X300] — CLN_LOOP_GAIN_1 | Default: 13.800 | Linked Fault: ERR-SIE300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X300.

PRM_40462 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_2 | Default: 27.600 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

PRM_40463 [Siemens-DMG-X300] — OPT_LOOP_GAIN_3 | Default: 41.400 | Linked Fault: ERR-SIE300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X300.

PRM_40464 [Siemens-DMG-X300] — VAC_LOOP_GAIN_4 | Default: 55.200 | Linked Fault: ERR-SIE300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300.

PRM_40465 [Siemens-DMG-X300] — ATC_LOOP_GAIN_5 | Default: 69.000 | Linked Fault: ERR-SIE300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X300.

PRM_40466 [Siemens-DMG-X300] — XAX_LOOP_GAIN_6 | Default: 82.800 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40467 [Siemens-DMG-X300] — BAX_LOOP_GAIN_7 | Default: 96.600 | Linked Fault: ERR-SIE300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X300.

PRM_40468 [Siemens-DMG-X300] — PLC_LOOP_GAIN_8 | Default: 110.400 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

PRM_40469 [Siemens-DMG-X300] — THM_LOOP_GAIN_9 | Default: 124.200 | Linked Fault: ERR-SIE300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X300.

PRM_40470 [Siemens-DMG-X300] — HYD_LOOP_GAIN_10 | Default: 138.000 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40471 [Siemens-DMG-X300] — YAX_LOOP_GAIN_11 | Default: 151.800 | Linked Fault: ERR-SIE300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X300.

PRM_40472 [Siemens-DMG-X300] — CAX_LOOP_GAIN_12 | Default: 165.600 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

PRM_40473 [Siemens-DMG-X300] — LUB_LOOP_GAIN_13 | Default: 179.400 | Linked Fault: ERR-SIE300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X300.

PRM_40474 [Siemens-DMG-X300] — SPN_LOOP_GAIN_14 | Default: 193.200 | Linked Fault: ERR-SIE300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300.

PRM_40475 [Siemens-DMG-X300] — CLN_LOOP_GAIN_15 | Default: 207.000 | Linked Fault: ERR-SIE300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X300.

PRM_40476 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_16 | Default: 220.800 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X300 (BANK 4)

Machine Model: Siemens-DMG-X300 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X300.

PRM_40481 [Siemens-DMG-X300] — XAX_LOOP_GAIN_1 | Default: 18.400 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40482 [Siemens-DMG-X300] — CAX_LOOP_GAIN_2 | Default: 36.800 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

PRM_40483 [Siemens-DMG-X300] — VAC_LOOP_GAIN_3 | Default: 55.200 | Linked Fault: ERR-SIE300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300.

PRM_40484 [Siemens-DMG-X300] — HYD_LOOP_GAIN_4 | Default: 73.600 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40485 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_5 | Default: 92.000 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

PRM_40486 [Siemens-DMG-X300] — PLC_LOOP_GAIN_6 | Default: 110.400 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

PRM_40487 [Siemens-DMG-X300] — SPN_LOOP_GAIN_7 | Default: 128.800 | Linked Fault: ERR-SIE300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300.

PRM_40488 [Siemens-DMG-X300] — XAX_LOOP_GAIN_8 | Default: 147.200 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40489 [Siemens-DMG-X300] — CAX_LOOP_GAIN_9 | Default: 165.600 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

PRM_40490 [Siemens-DMG-X300] — VAC_LOOP_GAIN_10 | Default: 184.000 | Linked Fault: ERR-SIE300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300.

PRM_40491 [Siemens-DMG-X300] — HYD_LOOP_GAIN_11 | Default: 202.400 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40492 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_12 | Default: 220.800 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

PRM_40493 [Siemens-DMG-X300] — PLC_LOOP_GAIN_13 | Default: 239.200 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

PRM_40494 [Siemens-DMG-X300] — SPN_LOOP_GAIN_14 | Default: 7.600 | Linked Fault: ERR-SIE300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300.

PRM_40495 [Siemens-DMG-X300] — XAX_LOOP_GAIN_15 | Default: 26.000 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40496 [Siemens-DMG-X300] — CAX_LOOP_GAIN_16 | Default: 44.400 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X300 (BANK 5)

Machine Model: Siemens-DMG-X300 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X300.

PRM_40501 [Siemens-DMG-X300] — YAX_LOOP_GAIN_1 | Default: 23.000 | Linked Fault: ERR-SIE300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X300.

PRM_40502 [Siemens-DMG-X300] — PLC_LOOP_GAIN_2 | Default: 46.000 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

PRM_40503 [Siemens-DMG-X300] — ATC_LOOP_GAIN_3 | Default: 69.000 | Linked Fault: ERR-SIE300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X300.

PRM_40504 [Siemens-DMG-X300] — ZAX_LOOP_GAIN_4 | Default: 92.000 | Linked Fault: ERR-SIE300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X300.

PRM_40505 [Siemens-DMG-X300] — LUB_LOOP_GAIN_5 | Default: 115.000 | Linked Fault: ERR-SIE300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X300.

PRM_40506 [Siemens-DMG-X300] — HYD_LOOP_GAIN_6 | Default: 138.000 | Linked Fault: ERR-SIE300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X300.

PRM_40507 [Siemens-DMG-X300] — BAX_LOOP_GAIN_7 | Default: 161.000 | Linked Fault: ERR-SIE300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X300.

PRM_40508 [Siemens-DMG-X300] — VAC_LOOP_GAIN_8 | Default: 184.000 | Linked Fault: ERR-SIE300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X300.

PRM_40509 [Siemens-DMG-X300] — CLN_LOOP_GAIN_9 | Default: 207.000 | Linked Fault: ERR-SIE300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X300.

PRM_40510 [Siemens-DMG-X300] — CAX_LOOP_GAIN_10 | Default: 230.000 | Linked Fault: ERR-SIE300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X300.

PRM_40511 [Siemens-DMG-X300] — THM_LOOP_GAIN_11 | Default: 3.000 | Linked Fault: ERR-SIE300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X300.

PRM_40512 [Siemens-DMG-X300] — XAX_LOOP_GAIN_12 | Default: 26.000 | Linked Fault: ERR-SIE300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X300.

PRM_40513 [Siemens-DMG-X300] — OPT_LOOP_GAIN_13 | Default: 49.000 | Linked Fault: ERR-SIE300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X300.

PRM_40514 [Siemens-DMG-X300] — SPN_LOOP_GAIN_14 | Default: 72.000 | Linked Fault: ERR-SIE300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X300.

PRM_40515 [Siemens-DMG-X300] — YAX_LOOP_GAIN_15 | Default: 95.000 | Linked Fault: ERR-SIE300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X300.

PRM_40516 [Siemens-DMG-X300] — PLC_LOOP_GAIN_16 | Default: 118.000 | Linked Fault: ERR-SIE300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X300.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X350 (BANK 1)

Machine Model: Siemens-DMG-X350 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X350.

PRM_40521 [Siemens-DMG-X350] — ATC_LOOP_GAIN_1 | Default: 5.750 | Linked Fault: ERR-SIE350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X350.

PRM_40522 [Siemens-DMG-X350] — HYD_LOOP_GAIN_2 | Default: 11.500 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40523 [Siemens-DMG-X350] — CLN_LOOP_GAIN_3 | Default: 17.250 | Linked Fault: ERR-SIE350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X350.

PRM_40524 [Siemens-DMG-X350] — XAX_LOOP_GAIN_4 | Default: 23.000 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40525 [Siemens-DMG-X350] — YAX_LOOP_GAIN_5 | Default: 28.750 | Linked Fault: ERR-SIE350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X350.

PRM_40526 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_6 | Default: 34.500 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

PRM_40527 [Siemens-DMG-X350] — BAX_LOOP_GAIN_7 | Default: 40.250 | Linked Fault: ERR-SIE350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X350.

PRM_40528 [Siemens-DMG-X350] — CAX_LOOP_GAIN_8 | Default: 46.000 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

PRM_40529 [Siemens-DMG-X350] — OPT_LOOP_GAIN_9 | Default: 51.750 | Linked Fault: ERR-SIE350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X350.

PRM_40530 [Siemens-DMG-X350] — PLC_LOOP_GAIN_10 | Default: 57.500 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

PRM_40531 [Siemens-DMG-X350] — LUB_LOOP_GAIN_11 | Default: 63.250 | Linked Fault: ERR-SIE350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X350.

PRM_40532 [Siemens-DMG-X350] — VAC_LOOP_GAIN_12 | Default: 69.000 | Linked Fault: ERR-SIE350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350.

PRM_40533 [Siemens-DMG-X350] — THM_LOOP_GAIN_13 | Default: 74.750 | Linked Fault: ERR-SIE350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X350.

PRM_40534 [Siemens-DMG-X350] — SPN_LOOP_GAIN_14 | Default: 80.500 | Linked Fault: ERR-SIE350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350.

PRM_40535 [Siemens-DMG-X350] — ATC_LOOP_GAIN_15 | Default: 86.250 | Linked Fault: ERR-SIE350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X350.

PRM_40536 [Siemens-DMG-X350] — HYD_LOOP_GAIN_16 | Default: 92.000 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X350 (BANK 2)

Machine Model: Siemens-DMG-X350 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X350.

PRM_40541 [Siemens-DMG-X350] — HYD_LOOP_GAIN_1 | Default: 11.500 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40542 [Siemens-DMG-X350] — XAX_LOOP_GAIN_2 | Default: 23.000 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40543 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_3 | Default: 34.500 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

PRM_40544 [Siemens-DMG-X350] — CAX_LOOP_GAIN_4 | Default: 46.000 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

PRM_40545 [Siemens-DMG-X350] — PLC_LOOP_GAIN_5 | Default: 57.500 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

PRM_40546 [Siemens-DMG-X350] — VAC_LOOP_GAIN_6 | Default: 69.000 | Linked Fault: ERR-SIE350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350.

PRM_40547 [Siemens-DMG-X350] — SPN_LOOP_GAIN_7 | Default: 80.500 | Linked Fault: ERR-SIE350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350.

PRM_40548 [Siemens-DMG-X350] — HYD_LOOP_GAIN_8 | Default: 92.000 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40549 [Siemens-DMG-X350] — XAX_LOOP_GAIN_9 | Default: 103.500 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40550 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_10 | Default: 115.000 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

PRM_40551 [Siemens-DMG-X350] — CAX_LOOP_GAIN_11 | Default: 126.500 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

PRM_40552 [Siemens-DMG-X350] — PLC_LOOP_GAIN_12 | Default: 138.000 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

PRM_40553 [Siemens-DMG-X350] — VAC_LOOP_GAIN_13 | Default: 149.500 | Linked Fault: ERR-SIE350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350.

PRM_40554 [Siemens-DMG-X350] — SPN_LOOP_GAIN_14 | Default: 161.000 | Linked Fault: ERR-SIE350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350.

PRM_40555 [Siemens-DMG-X350] — HYD_LOOP_GAIN_15 | Default: 172.500 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40556 [Siemens-DMG-X350] — XAX_LOOP_GAIN_16 | Default: 184.000 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X350 (BANK 3)

Machine Model: Siemens-DMG-X350 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X350.

PRM_40561 [Siemens-DMG-X350] — CLN_LOOP_GAIN_1 | Default: 17.250 | Linked Fault: ERR-SIE350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X350.

PRM_40562 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_2 | Default: 34.500 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

PRM_40563 [Siemens-DMG-X350] — OPT_LOOP_GAIN_3 | Default: 51.750 | Linked Fault: ERR-SIE350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X350.

PRM_40564 [Siemens-DMG-X350] — VAC_LOOP_GAIN_4 | Default: 69.000 | Linked Fault: ERR-SIE350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350.

PRM_40565 [Siemens-DMG-X350] — ATC_LOOP_GAIN_5 | Default: 86.250 | Linked Fault: ERR-SIE350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X350.

PRM_40566 [Siemens-DMG-X350] — XAX_LOOP_GAIN_6 | Default: 103.500 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40567 [Siemens-DMG-X350] — BAX_LOOP_GAIN_7 | Default: 120.750 | Linked Fault: ERR-SIE350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X350.

PRM_40568 [Siemens-DMG-X350] — PLC_LOOP_GAIN_8 | Default: 138.000 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

PRM_40569 [Siemens-DMG-X350] — THM_LOOP_GAIN_9 | Default: 155.250 | Linked Fault: ERR-SIE350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X350.

PRM_40570 [Siemens-DMG-X350] — HYD_LOOP_GAIN_10 | Default: 172.500 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40571 [Siemens-DMG-X350] — YAX_LOOP_GAIN_11 | Default: 189.750 | Linked Fault: ERR-SIE350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X350.

PRM_40572 [Siemens-DMG-X350] — CAX_LOOP_GAIN_12 | Default: 207.000 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

PRM_40573 [Siemens-DMG-X350] — LUB_LOOP_GAIN_13 | Default: 224.250 | Linked Fault: ERR-SIE350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X350.

PRM_40574 [Siemens-DMG-X350] — SPN_LOOP_GAIN_14 | Default: 241.500 | Linked Fault: ERR-SIE350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350.

PRM_40575 [Siemens-DMG-X350] — CLN_LOOP_GAIN_15 | Default: 8.750 | Linked Fault: ERR-SIE350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X350.

PRM_40576 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_16 | Default: 26.000 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X350 (BANK 4)

Machine Model: Siemens-DMG-X350 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X350.

PRM_40581 [Siemens-DMG-X350] — XAX_LOOP_GAIN_1 | Default: 23.000 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40582 [Siemens-DMG-X350] — CAX_LOOP_GAIN_2 | Default: 46.000 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

PRM_40583 [Siemens-DMG-X350] — VAC_LOOP_GAIN_3 | Default: 69.000 | Linked Fault: ERR-SIE350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350.

PRM_40584 [Siemens-DMG-X350] — HYD_LOOP_GAIN_4 | Default: 92.000 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40585 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_5 | Default: 115.000 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

PRM_40586 [Siemens-DMG-X350] — PLC_LOOP_GAIN_6 | Default: 138.000 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

PRM_40587 [Siemens-DMG-X350] — SPN_LOOP_GAIN_7 | Default: 161.000 | Linked Fault: ERR-SIE350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350.

PRM_40588 [Siemens-DMG-X350] — XAX_LOOP_GAIN_8 | Default: 184.000 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40589 [Siemens-DMG-X350] — CAX_LOOP_GAIN_9 | Default: 207.000 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

PRM_40590 [Siemens-DMG-X350] — VAC_LOOP_GAIN_10 | Default: 230.000 | Linked Fault: ERR-SIE350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350.

PRM_40591 [Siemens-DMG-X350] — HYD_LOOP_GAIN_11 | Default: 3.000 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40592 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_12 | Default: 26.000 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

PRM_40593 [Siemens-DMG-X350] — PLC_LOOP_GAIN_13 | Default: 49.000 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

PRM_40594 [Siemens-DMG-X350] — SPN_LOOP_GAIN_14 | Default: 72.000 | Linked Fault: ERR-SIE350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350.

PRM_40595 [Siemens-DMG-X350] — XAX_LOOP_GAIN_15 | Default: 95.000 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40596 [Siemens-DMG-X350] — CAX_LOOP_GAIN_16 | Default: 118.000 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X350 (BANK 5)

Machine Model: Siemens-DMG-X350 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X350.

PRM_40601 [Siemens-DMG-X350] — YAX_LOOP_GAIN_1 | Default: 28.750 | Linked Fault: ERR-SIE350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X350.

PRM_40602 [Siemens-DMG-X350] — PLC_LOOP_GAIN_2 | Default: 57.500 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

PRM_40603 [Siemens-DMG-X350] — ATC_LOOP_GAIN_3 | Default: 86.250 | Linked Fault: ERR-SIE350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X350.

PRM_40604 [Siemens-DMG-X350] — ZAX_LOOP_GAIN_4 | Default: 115.000 | Linked Fault: ERR-SIE350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X350.

PRM_40605 [Siemens-DMG-X350] — LUB_LOOP_GAIN_5 | Default: 143.750 | Linked Fault: ERR-SIE350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X350.

PRM_40606 [Siemens-DMG-X350] — HYD_LOOP_GAIN_6 | Default: 172.500 | Linked Fault: ERR-SIE350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X350.

PRM_40607 [Siemens-DMG-X350] — BAX_LOOP_GAIN_7 | Default: 201.250 | Linked Fault: ERR-SIE350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X350.

PRM_40608 [Siemens-DMG-X350] — VAC_LOOP_GAIN_8 | Default: 230.000 | Linked Fault: ERR-SIE350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X350.

PRM_40609 [Siemens-DMG-X350] — CLN_LOOP_GAIN_9 | Default: 8.750 | Linked Fault: ERR-SIE350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X350.

PRM_40610 [Siemens-DMG-X350] — CAX_LOOP_GAIN_10 | Default: 37.500 | Linked Fault: ERR-SIE350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X350.

PRM_40611 [Siemens-DMG-X350] — THM_LOOP_GAIN_11 | Default: 66.250 | Linked Fault: ERR-SIE350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X350.

PRM_40612 [Siemens-DMG-X350] — XAX_LOOP_GAIN_12 | Default: 95.000 | Linked Fault: ERR-SIE350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X350.

PRM_40613 [Siemens-DMG-X350] — OPT_LOOP_GAIN_13 | Default: 123.750 | Linked Fault: ERR-SIE350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X350.

PRM_40614 [Siemens-DMG-X350] — SPN_LOOP_GAIN_14 | Default: 152.500 | Linked Fault: ERR-SIE350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X350.

PRM_40615 [Siemens-DMG-X350] — YAX_LOOP_GAIN_15 | Default: 181.250 | Linked Fault: ERR-SIE350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X350.

PRM_40616 [Siemens-DMG-X350] — PLC_LOOP_GAIN_16 | Default: 210.000 | Linked Fault: ERR-SIE350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X350.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X400 (BANK 1)

Machine Model: Siemens-DMG-X400 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X400.

PRM_40621 [Siemens-DMG-X400] — ATC_LOOP_GAIN_1 | Default: 6.900 | Linked Fault: ERR-SIE400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X400.

PRM_40622 [Siemens-DMG-X400] — HYD_LOOP_GAIN_2 | Default: 13.800 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40623 [Siemens-DMG-X400] — CLN_LOOP_GAIN_3 | Default: 20.700 | Linked Fault: ERR-SIE400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X400.

PRM_40624 [Siemens-DMG-X400] — XAX_LOOP_GAIN_4 | Default: 27.600 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40625 [Siemens-DMG-X400] — YAX_LOOP_GAIN_5 | Default: 34.500 | Linked Fault: ERR-SIE400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X400.

PRM_40626 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_6 | Default: 41.400 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

PRM_40627 [Siemens-DMG-X400] — BAX_LOOP_GAIN_7 | Default: 48.300 | Linked Fault: ERR-SIE400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X400.

PRM_40628 [Siemens-DMG-X400] — CAX_LOOP_GAIN_8 | Default: 55.200 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

PRM_40629 [Siemens-DMG-X400] — OPT_LOOP_GAIN_9 | Default: 62.100 | Linked Fault: ERR-SIE400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X400.

PRM_40630 [Siemens-DMG-X400] — PLC_LOOP_GAIN_10 | Default: 69.000 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

PRM_40631 [Siemens-DMG-X400] — LUB_LOOP_GAIN_11 | Default: 75.900 | Linked Fault: ERR-SIE400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X400.

PRM_40632 [Siemens-DMG-X400] — VAC_LOOP_GAIN_12 | Default: 82.800 | Linked Fault: ERR-SIE400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400.

PRM_40633 [Siemens-DMG-X400] — THM_LOOP_GAIN_13 | Default: 89.700 | Linked Fault: ERR-SIE400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X400.

PRM_40634 [Siemens-DMG-X400] — SPN_LOOP_GAIN_14 | Default: 96.600 | Linked Fault: ERR-SIE400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400.

PRM_40635 [Siemens-DMG-X400] — ATC_LOOP_GAIN_15 | Default: 103.500 | Linked Fault: ERR-SIE400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X400.

PRM_40636 [Siemens-DMG-X400] — HYD_LOOP_GAIN_16 | Default: 110.400 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X400 (BANK 2)

Machine Model: Siemens-DMG-X400 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X400.

PRM_40641 [Siemens-DMG-X400] — HYD_LOOP_GAIN_1 | Default: 13.800 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40642 [Siemens-DMG-X400] — XAX_LOOP_GAIN_2 | Default: 27.600 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40643 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_3 | Default: 41.400 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

PRM_40644 [Siemens-DMG-X400] — CAX_LOOP_GAIN_4 | Default: 55.200 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

PRM_40645 [Siemens-DMG-X400] — PLC_LOOP_GAIN_5 | Default: 69.000 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

PRM_40646 [Siemens-DMG-X400] — VAC_LOOP_GAIN_6 | Default: 82.800 | Linked Fault: ERR-SIE400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400.

PRM_40647 [Siemens-DMG-X400] — SPN_LOOP_GAIN_7 | Default: 96.600 | Linked Fault: ERR-SIE400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400.

PRM_40648 [Siemens-DMG-X400] — HYD_LOOP_GAIN_8 | Default: 110.400 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40649 [Siemens-DMG-X400] — XAX_LOOP_GAIN_9 | Default: 124.200 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40650 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_10 | Default: 138.000 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

PRM_40651 [Siemens-DMG-X400] — CAX_LOOP_GAIN_11 | Default: 151.800 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

PRM_40652 [Siemens-DMG-X400] — PLC_LOOP_GAIN_12 | Default: 165.600 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

PRM_40653 [Siemens-DMG-X400] — VAC_LOOP_GAIN_13 | Default: 179.400 | Linked Fault: ERR-SIE400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400.

PRM_40654 [Siemens-DMG-X400] — SPN_LOOP_GAIN_14 | Default: 193.200 | Linked Fault: ERR-SIE400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400.

PRM_40655 [Siemens-DMG-X400] — HYD_LOOP_GAIN_15 | Default: 207.000 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40656 [Siemens-DMG-X400] — XAX_LOOP_GAIN_16 | Default: 220.800 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X400 (BANK 3)

Machine Model: Siemens-DMG-X400 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X400.

PRM_40661 [Siemens-DMG-X400] — CLN_LOOP_GAIN_1 | Default: 20.700 | Linked Fault: ERR-SIE400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X400.

PRM_40662 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_2 | Default: 41.400 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

PRM_40663 [Siemens-DMG-X400] — OPT_LOOP_GAIN_3 | Default: 62.100 | Linked Fault: ERR-SIE400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X400.

PRM_40664 [Siemens-DMG-X400] — VAC_LOOP_GAIN_4 | Default: 82.800 | Linked Fault: ERR-SIE400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400.

PRM_40665 [Siemens-DMG-X400] — ATC_LOOP_GAIN_5 | Default: 103.500 | Linked Fault: ERR-SIE400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X400.

PRM_40666 [Siemens-DMG-X400] — XAX_LOOP_GAIN_6 | Default: 124.200 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40667 [Siemens-DMG-X400] — BAX_LOOP_GAIN_7 | Default: 144.900 | Linked Fault: ERR-SIE400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X400.

PRM_40668 [Siemens-DMG-X400] — PLC_LOOP_GAIN_8 | Default: 165.600 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

PRM_40669 [Siemens-DMG-X400] — THM_LOOP_GAIN_9 | Default: 186.300 | Linked Fault: ERR-SIE400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X400.

PRM_40670 [Siemens-DMG-X400] — HYD_LOOP_GAIN_10 | Default: 207.000 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40671 [Siemens-DMG-X400] — YAX_LOOP_GAIN_11 | Default: 227.700 | Linked Fault: ERR-SIE400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X400.

PRM_40672 [Siemens-DMG-X400] — CAX_LOOP_GAIN_12 | Default: 248.400 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

PRM_40673 [Siemens-DMG-X400] — LUB_LOOP_GAIN_13 | Default: 19.100 | Linked Fault: ERR-SIE400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X400.

PRM_40674 [Siemens-DMG-X400] — SPN_LOOP_GAIN_14 | Default: 39.800 | Linked Fault: ERR-SIE400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400.

PRM_40675 [Siemens-DMG-X400] — CLN_LOOP_GAIN_15 | Default: 60.500 | Linked Fault: ERR-SIE400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X400.

PRM_40676 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_16 | Default: 81.200 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X400 (BANK 4)

Machine Model: Siemens-DMG-X400 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X400.

PRM_40681 [Siemens-DMG-X400] — XAX_LOOP_GAIN_1 | Default: 27.600 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40682 [Siemens-DMG-X400] — CAX_LOOP_GAIN_2 | Default: 55.200 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

PRM_40683 [Siemens-DMG-X400] — VAC_LOOP_GAIN_3 | Default: 82.800 | Linked Fault: ERR-SIE400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400.

PRM_40684 [Siemens-DMG-X400] — HYD_LOOP_GAIN_4 | Default: 110.400 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40685 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_5 | Default: 138.000 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

PRM_40686 [Siemens-DMG-X400] — PLC_LOOP_GAIN_6 | Default: 165.600 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

PRM_40687 [Siemens-DMG-X400] — SPN_LOOP_GAIN_7 | Default: 193.200 | Linked Fault: ERR-SIE400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400.

PRM_40688 [Siemens-DMG-X400] — XAX_LOOP_GAIN_8 | Default: 220.800 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40689 [Siemens-DMG-X400] — CAX_LOOP_GAIN_9 | Default: 248.400 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

PRM_40690 [Siemens-DMG-X400] — VAC_LOOP_GAIN_10 | Default: 26.000 | Linked Fault: ERR-SIE400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400.

PRM_40691 [Siemens-DMG-X400] — HYD_LOOP_GAIN_11 | Default: 53.600 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40692 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_12 | Default: 81.200 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

PRM_40693 [Siemens-DMG-X400] — PLC_LOOP_GAIN_13 | Default: 108.800 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

PRM_40694 [Siemens-DMG-X400] — SPN_LOOP_GAIN_14 | Default: 136.400 | Linked Fault: ERR-SIE400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400.

PRM_40695 [Siemens-DMG-X400] — XAX_LOOP_GAIN_15 | Default: 164.000 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40696 [Siemens-DMG-X400] — CAX_LOOP_GAIN_16 | Default: 191.600 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X400 (BANK 5)

Machine Model: Siemens-DMG-X400 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X400.

PRM_40701 [Siemens-DMG-X400] — YAX_LOOP_GAIN_1 | Default: 34.500 | Linked Fault: ERR-SIE400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X400.

PRM_40702 [Siemens-DMG-X400] — PLC_LOOP_GAIN_2 | Default: 69.000 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

PRM_40703 [Siemens-DMG-X400] — ATC_LOOP_GAIN_3 | Default: 103.500 | Linked Fault: ERR-SIE400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X400.

PRM_40704 [Siemens-DMG-X400] — ZAX_LOOP_GAIN_4 | Default: 138.000 | Linked Fault: ERR-SIE400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X400.

PRM_40705 [Siemens-DMG-X400] — LUB_LOOP_GAIN_5 | Default: 172.500 | Linked Fault: ERR-SIE400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X400.

PRM_40706 [Siemens-DMG-X400] — HYD_LOOP_GAIN_6 | Default: 207.000 | Linked Fault: ERR-SIE400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X400.

PRM_40707 [Siemens-DMG-X400] — BAX_LOOP_GAIN_7 | Default: 241.500 | Linked Fault: ERR-SIE400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X400.

PRM_40708 [Siemens-DMG-X400] — VAC_LOOP_GAIN_8 | Default: 26.000 | Linked Fault: ERR-SIE400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X400.

PRM_40709 [Siemens-DMG-X400] — CLN_LOOP_GAIN_9 | Default: 60.500 | Linked Fault: ERR-SIE400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X400.

PRM_40710 [Siemens-DMG-X400] — CAX_LOOP_GAIN_10 | Default: 95.000 | Linked Fault: ERR-SIE400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X400.

PRM_40711 [Siemens-DMG-X400] — THM_LOOP_GAIN_11 | Default: 129.500 | Linked Fault: ERR-SIE400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X400.

PRM_40712 [Siemens-DMG-X400] — XAX_LOOP_GAIN_12 | Default: 164.000 | Linked Fault: ERR-SIE400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X400.

PRM_40713 [Siemens-DMG-X400] — OPT_LOOP_GAIN_13 | Default: 198.500 | Linked Fault: ERR-SIE400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X400.

PRM_40714 [Siemens-DMG-X400] — SPN_LOOP_GAIN_14 | Default: 233.000 | Linked Fault: ERR-SIE400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X400.

PRM_40715 [Siemens-DMG-X400] — YAX_LOOP_GAIN_15 | Default: 17.500 | Linked Fault: ERR-SIE400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X400.

PRM_40716 [Siemens-DMG-X400] — PLC_LOOP_GAIN_16 | Default: 52.000 | Linked Fault: ERR-SIE400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X400.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X450 (BANK 1)

Machine Model: Siemens-DMG-X450 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X450.

PRM_40721 [Siemens-DMG-X450] — ATC_LOOP_GAIN_1 | Default: 8.050 | Linked Fault: ERR-SIE450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X450.

PRM_40722 [Siemens-DMG-X450] — HYD_LOOP_GAIN_2 | Default: 16.100 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40723 [Siemens-DMG-X450] — CLN_LOOP_GAIN_3 | Default: 24.150 | Linked Fault: ERR-SIE450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X450.

PRM_40724 [Siemens-DMG-X450] — XAX_LOOP_GAIN_4 | Default: 32.200 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40725 [Siemens-DMG-X450] — YAX_LOOP_GAIN_5 | Default: 40.250 | Linked Fault: ERR-SIE450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X450.

PRM_40726 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_6 | Default: 48.300 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

PRM_40727 [Siemens-DMG-X450] — BAX_LOOP_GAIN_7 | Default: 56.350 | Linked Fault: ERR-SIE450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X450.

PRM_40728 [Siemens-DMG-X450] — CAX_LOOP_GAIN_8 | Default: 64.400 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

PRM_40729 [Siemens-DMG-X450] — OPT_LOOP_GAIN_9 | Default: 72.450 | Linked Fault: ERR-SIE450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X450.

PRM_40730 [Siemens-DMG-X450] — PLC_LOOP_GAIN_10 | Default: 80.500 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

PRM_40731 [Siemens-DMG-X450] — LUB_LOOP_GAIN_11 | Default: 88.550 | Linked Fault: ERR-SIE450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X450.

PRM_40732 [Siemens-DMG-X450] — VAC_LOOP_GAIN_12 | Default: 96.600 | Linked Fault: ERR-SIE450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450.

PRM_40733 [Siemens-DMG-X450] — THM_LOOP_GAIN_13 | Default: 104.650 | Linked Fault: ERR-SIE450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X450.

PRM_40734 [Siemens-DMG-X450] — SPN_LOOP_GAIN_14 | Default: 112.700 | Linked Fault: ERR-SIE450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450.

PRM_40735 [Siemens-DMG-X450] — ATC_LOOP_GAIN_15 | Default: 120.750 | Linked Fault: ERR-SIE450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X450.

PRM_40736 [Siemens-DMG-X450] — HYD_LOOP_GAIN_16 | Default: 128.800 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X450 (BANK 2)

Machine Model: Siemens-DMG-X450 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X450.

PRM_40741 [Siemens-DMG-X450] — HYD_LOOP_GAIN_1 | Default: 16.100 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40742 [Siemens-DMG-X450] — XAX_LOOP_GAIN_2 | Default: 32.200 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40743 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_3 | Default: 48.300 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

PRM_40744 [Siemens-DMG-X450] — CAX_LOOP_GAIN_4 | Default: 64.400 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

PRM_40745 [Siemens-DMG-X450] — PLC_LOOP_GAIN_5 | Default: 80.500 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

PRM_40746 [Siemens-DMG-X450] — VAC_LOOP_GAIN_6 | Default: 96.600 | Linked Fault: ERR-SIE450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450.

PRM_40747 [Siemens-DMG-X450] — SPN_LOOP_GAIN_7 | Default: 112.700 | Linked Fault: ERR-SIE450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450.

PRM_40748 [Siemens-DMG-X450] — HYD_LOOP_GAIN_8 | Default: 128.800 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40749 [Siemens-DMG-X450] — XAX_LOOP_GAIN_9 | Default: 144.900 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40750 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_10 | Default: 161.000 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

PRM_40751 [Siemens-DMG-X450] — CAX_LOOP_GAIN_11 | Default: 177.100 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

PRM_40752 [Siemens-DMG-X450] — PLC_LOOP_GAIN_12 | Default: 193.200 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

PRM_40753 [Siemens-DMG-X450] — VAC_LOOP_GAIN_13 | Default: 209.300 | Linked Fault: ERR-SIE450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450.

PRM_40754 [Siemens-DMG-X450] — SPN_LOOP_GAIN_14 | Default: 225.400 | Linked Fault: ERR-SIE450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450.

PRM_40755 [Siemens-DMG-X450] — HYD_LOOP_GAIN_15 | Default: 241.500 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40756 [Siemens-DMG-X450] — XAX_LOOP_GAIN_16 | Default: 7.600 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X450 (BANK 3)

Machine Model: Siemens-DMG-X450 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X450.

PRM_40761 [Siemens-DMG-X450] — CLN_LOOP_GAIN_1 | Default: 24.150 | Linked Fault: ERR-SIE450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X450.

PRM_40762 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_2 | Default: 48.300 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

PRM_40763 [Siemens-DMG-X450] — OPT_LOOP_GAIN_3 | Default: 72.450 | Linked Fault: ERR-SIE450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X450.

PRM_40764 [Siemens-DMG-X450] — VAC_LOOP_GAIN_4 | Default: 96.600 | Linked Fault: ERR-SIE450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450.

PRM_40765 [Siemens-DMG-X450] — ATC_LOOP_GAIN_5 | Default: 120.750 | Linked Fault: ERR-SIE450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X450.

PRM_40766 [Siemens-DMG-X450] — XAX_LOOP_GAIN_6 | Default: 144.900 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40767 [Siemens-DMG-X450] — BAX_LOOP_GAIN_7 | Default: 169.050 | Linked Fault: ERR-SIE450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X450.

PRM_40768 [Siemens-DMG-X450] — PLC_LOOP_GAIN_8 | Default: 193.200 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

PRM_40769 [Siemens-DMG-X450] — THM_LOOP_GAIN_9 | Default: 217.350 | Linked Fault: ERR-SIE450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X450.

PRM_40770 [Siemens-DMG-X450] — HYD_LOOP_GAIN_10 | Default: 241.500 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40771 [Siemens-DMG-X450] — YAX_LOOP_GAIN_11 | Default: 15.650 | Linked Fault: ERR-SIE450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X450.

PRM_40772 [Siemens-DMG-X450] — CAX_LOOP_GAIN_12 | Default: 39.800 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

PRM_40773 [Siemens-DMG-X450] — LUB_LOOP_GAIN_13 | Default: 63.950 | Linked Fault: ERR-SIE450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X450.

PRM_40774 [Siemens-DMG-X450] — SPN_LOOP_GAIN_14 | Default: 88.100 | Linked Fault: ERR-SIE450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450.

PRM_40775 [Siemens-DMG-X450] — CLN_LOOP_GAIN_15 | Default: 112.250 | Linked Fault: ERR-SIE450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X450.

PRM_40776 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_16 | Default: 136.400 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X450 (BANK 4)

Machine Model: Siemens-DMG-X450 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X450.

PRM_40781 [Siemens-DMG-X450] — XAX_LOOP_GAIN_1 | Default: 32.200 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40782 [Siemens-DMG-X450] — CAX_LOOP_GAIN_2 | Default: 64.400 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

PRM_40783 [Siemens-DMG-X450] — VAC_LOOP_GAIN_3 | Default: 96.600 | Linked Fault: ERR-SIE450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450.

PRM_40784 [Siemens-DMG-X450] — HYD_LOOP_GAIN_4 | Default: 128.800 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40785 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_5 | Default: 161.000 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

PRM_40786 [Siemens-DMG-X450] — PLC_LOOP_GAIN_6 | Default: 193.200 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

PRM_40787 [Siemens-DMG-X450] — SPN_LOOP_GAIN_7 | Default: 225.400 | Linked Fault: ERR-SIE450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450.

PRM_40788 [Siemens-DMG-X450] — XAX_LOOP_GAIN_8 | Default: 7.600 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40789 [Siemens-DMG-X450] — CAX_LOOP_GAIN_9 | Default: 39.800 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

PRM_40790 [Siemens-DMG-X450] — VAC_LOOP_GAIN_10 | Default: 72.000 | Linked Fault: ERR-SIE450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450.

PRM_40791 [Siemens-DMG-X450] — HYD_LOOP_GAIN_11 | Default: 104.200 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40792 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_12 | Default: 136.400 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

PRM_40793 [Siemens-DMG-X450] — PLC_LOOP_GAIN_13 | Default: 168.600 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

PRM_40794 [Siemens-DMG-X450] — SPN_LOOP_GAIN_14 | Default: 200.800 | Linked Fault: ERR-SIE450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450.

PRM_40795 [Siemens-DMG-X450] — XAX_LOOP_GAIN_15 | Default: 233.000 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40796 [Siemens-DMG-X450] — CAX_LOOP_GAIN_16 | Default: 15.200 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X450 (BANK 5)

Machine Model: Siemens-DMG-X450 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X450.

PRM_40801 [Siemens-DMG-X450] — YAX_LOOP_GAIN_1 | Default: 40.250 | Linked Fault: ERR-SIE450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X450.

PRM_40802 [Siemens-DMG-X450] — PLC_LOOP_GAIN_2 | Default: 80.500 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

PRM_40803 [Siemens-DMG-X450] — ATC_LOOP_GAIN_3 | Default: 120.750 | Linked Fault: ERR-SIE450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X450.

PRM_40804 [Siemens-DMG-X450] — ZAX_LOOP_GAIN_4 | Default: 161.000 | Linked Fault: ERR-SIE450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X450.

PRM_40805 [Siemens-DMG-X450] — LUB_LOOP_GAIN_5 | Default: 201.250 | Linked Fault: ERR-SIE450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X450.

PRM_40806 [Siemens-DMG-X450] — HYD_LOOP_GAIN_6 | Default: 241.500 | Linked Fault: ERR-SIE450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X450.

PRM_40807 [Siemens-DMG-X450] — BAX_LOOP_GAIN_7 | Default: 31.750 | Linked Fault: ERR-SIE450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X450.

PRM_40808 [Siemens-DMG-X450] — VAC_LOOP_GAIN_8 | Default: 72.000 | Linked Fault: ERR-SIE450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X450.

PRM_40809 [Siemens-DMG-X450] — CLN_LOOP_GAIN_9 | Default: 112.250 | Linked Fault: ERR-SIE450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X450.

PRM_40810 [Siemens-DMG-X450] — CAX_LOOP_GAIN_10 | Default: 152.500 | Linked Fault: ERR-SIE450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X450.

PRM_40811 [Siemens-DMG-X450] — THM_LOOP_GAIN_11 | Default: 192.750 | Linked Fault: ERR-SIE450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X450.

PRM_40812 [Siemens-DMG-X450] — XAX_LOOP_GAIN_12 | Default: 233.000 | Linked Fault: ERR-SIE450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X450.

PRM_40813 [Siemens-DMG-X450] — OPT_LOOP_GAIN_13 | Default: 23.250 | Linked Fault: ERR-SIE450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X450.

PRM_40814 [Siemens-DMG-X450] — SPN_LOOP_GAIN_14 | Default: 63.500 | Linked Fault: ERR-SIE450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X450.

PRM_40815 [Siemens-DMG-X450] — YAX_LOOP_GAIN_15 | Default: 103.750 | Linked Fault: ERR-SIE450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X450.

PRM_40816 [Siemens-DMG-X450] — PLC_LOOP_GAIN_16 | Default: 144.000 | Linked Fault: ERR-SIE450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X450.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X500 (BANK 1)

Machine Model: Siemens-DMG-X500 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X500.

PRM_40821 [Siemens-DMG-X500] — ATC_LOOP_GAIN_1 | Default: 9.200 | Linked Fault: ERR-SIE500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X500.

PRM_40822 [Siemens-DMG-X500] — HYD_LOOP_GAIN_2 | Default: 18.400 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40823 [Siemens-DMG-X500] — CLN_LOOP_GAIN_3 | Default: 27.600 | Linked Fault: ERR-SIE500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X500.

PRM_40824 [Siemens-DMG-X500] — XAX_LOOP_GAIN_4 | Default: 36.800 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40825 [Siemens-DMG-X500] — YAX_LOOP_GAIN_5 | Default: 46.000 | Linked Fault: ERR-SIE500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X500.

PRM_40826 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_6 | Default: 55.200 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

PRM_40827 [Siemens-DMG-X500] — BAX_LOOP_GAIN_7 | Default: 64.400 | Linked Fault: ERR-SIE500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X500.

PRM_40828 [Siemens-DMG-X500] — CAX_LOOP_GAIN_8 | Default: 73.600 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

PRM_40829 [Siemens-DMG-X500] — OPT_LOOP_GAIN_9 | Default: 82.800 | Linked Fault: ERR-SIE500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X500.

PRM_40830 [Siemens-DMG-X500] — PLC_LOOP_GAIN_10 | Default: 92.000 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

PRM_40831 [Siemens-DMG-X500] — LUB_LOOP_GAIN_11 | Default: 101.200 | Linked Fault: ERR-SIE500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X500.

PRM_40832 [Siemens-DMG-X500] — VAC_LOOP_GAIN_12 | Default: 110.400 | Linked Fault: ERR-SIE500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500.

PRM_40833 [Siemens-DMG-X500] — THM_LOOP_GAIN_13 | Default: 119.600 | Linked Fault: ERR-SIE500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X500.

PRM_40834 [Siemens-DMG-X500] — SPN_LOOP_GAIN_14 | Default: 128.800 | Linked Fault: ERR-SIE500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500.

PRM_40835 [Siemens-DMG-X500] — ATC_LOOP_GAIN_15 | Default: 138.000 | Linked Fault: ERR-SIE500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X500.

PRM_40836 [Siemens-DMG-X500] — HYD_LOOP_GAIN_16 | Default: 147.200 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X500 (BANK 2)

Machine Model: Siemens-DMG-X500 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X500.

PRM_40841 [Siemens-DMG-X500] — HYD_LOOP_GAIN_1 | Default: 18.400 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40842 [Siemens-DMG-X500] — XAX_LOOP_GAIN_2 | Default: 36.800 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40843 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_3 | Default: 55.200 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

PRM_40844 [Siemens-DMG-X500] — CAX_LOOP_GAIN_4 | Default: 73.600 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

PRM_40845 [Siemens-DMG-X500] — PLC_LOOP_GAIN_5 | Default: 92.000 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

PRM_40846 [Siemens-DMG-X500] — VAC_LOOP_GAIN_6 | Default: 110.400 | Linked Fault: ERR-SIE500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500.

PRM_40847 [Siemens-DMG-X500] — SPN_LOOP_GAIN_7 | Default: 128.800 | Linked Fault: ERR-SIE500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500.

PRM_40848 [Siemens-DMG-X500] — HYD_LOOP_GAIN_8 | Default: 147.200 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40849 [Siemens-DMG-X500] — XAX_LOOP_GAIN_9 | Default: 165.600 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40850 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_10 | Default: 184.000 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

PRM_40851 [Siemens-DMG-X500] — CAX_LOOP_GAIN_11 | Default: 202.400 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

PRM_40852 [Siemens-DMG-X500] — PLC_LOOP_GAIN_12 | Default: 220.800 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

PRM_40853 [Siemens-DMG-X500] — VAC_LOOP_GAIN_13 | Default: 239.200 | Linked Fault: ERR-SIE500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500.

PRM_40854 [Siemens-DMG-X500] — SPN_LOOP_GAIN_14 | Default: 7.600 | Linked Fault: ERR-SIE500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500.

PRM_40855 [Siemens-DMG-X500] — HYD_LOOP_GAIN_15 | Default: 26.000 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40856 [Siemens-DMG-X500] — XAX_LOOP_GAIN_16 | Default: 44.400 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X500 (BANK 3)

Machine Model: Siemens-DMG-X500 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X500.

PRM_40861 [Siemens-DMG-X500] — CLN_LOOP_GAIN_1 | Default: 27.600 | Linked Fault: ERR-SIE500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X500.

PRM_40862 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_2 | Default: 55.200 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

PRM_40863 [Siemens-DMG-X500] — OPT_LOOP_GAIN_3 | Default: 82.800 | Linked Fault: ERR-SIE500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X500.

PRM_40864 [Siemens-DMG-X500] — VAC_LOOP_GAIN_4 | Default: 110.400 | Linked Fault: ERR-SIE500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500.

PRM_40865 [Siemens-DMG-X500] — ATC_LOOP_GAIN_5 | Default: 138.000 | Linked Fault: ERR-SIE500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X500.

PRM_40866 [Siemens-DMG-X500] — XAX_LOOP_GAIN_6 | Default: 165.600 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40867 [Siemens-DMG-X500] — BAX_LOOP_GAIN_7 | Default: 193.200 | Linked Fault: ERR-SIE500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X500.

PRM_40868 [Siemens-DMG-X500] — PLC_LOOP_GAIN_8 | Default: 220.800 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

PRM_40869 [Siemens-DMG-X500] — THM_LOOP_GAIN_9 | Default: 248.400 | Linked Fault: ERR-SIE500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X500.

PRM_40870 [Siemens-DMG-X500] — HYD_LOOP_GAIN_10 | Default: 26.000 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40871 [Siemens-DMG-X500] — YAX_LOOP_GAIN_11 | Default: 53.600 | Linked Fault: ERR-SIE500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X500.

PRM_40872 [Siemens-DMG-X500] — CAX_LOOP_GAIN_12 | Default: 81.200 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

PRM_40873 [Siemens-DMG-X500] — LUB_LOOP_GAIN_13 | Default: 108.800 | Linked Fault: ERR-SIE500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X500.

PRM_40874 [Siemens-DMG-X500] — SPN_LOOP_GAIN_14 | Default: 136.400 | Linked Fault: ERR-SIE500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500.

PRM_40875 [Siemens-DMG-X500] — CLN_LOOP_GAIN_15 | Default: 164.000 | Linked Fault: ERR-SIE500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X500.

PRM_40876 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_16 | Default: 191.600 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X500 (BANK 4)

Machine Model: Siemens-DMG-X500 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X500.

PRM_40881 [Siemens-DMG-X500] — XAX_LOOP_GAIN_1 | Default: 36.800 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40882 [Siemens-DMG-X500] — CAX_LOOP_GAIN_2 | Default: 73.600 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

PRM_40883 [Siemens-DMG-X500] — VAC_LOOP_GAIN_3 | Default: 110.400 | Linked Fault: ERR-SIE500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500.

PRM_40884 [Siemens-DMG-X500] — HYD_LOOP_GAIN_4 | Default: 147.200 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40885 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_5 | Default: 184.000 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

PRM_40886 [Siemens-DMG-X500] — PLC_LOOP_GAIN_6 | Default: 220.800 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

PRM_40887 [Siemens-DMG-X500] — SPN_LOOP_GAIN_7 | Default: 7.600 | Linked Fault: ERR-SIE500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500.

PRM_40888 [Siemens-DMG-X500] — XAX_LOOP_GAIN_8 | Default: 44.400 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40889 [Siemens-DMG-X500] — CAX_LOOP_GAIN_9 | Default: 81.200 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

PRM_40890 [Siemens-DMG-X500] — VAC_LOOP_GAIN_10 | Default: 118.000 | Linked Fault: ERR-SIE500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500.

PRM_40891 [Siemens-DMG-X500] — HYD_LOOP_GAIN_11 | Default: 154.800 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40892 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_12 | Default: 191.600 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

PRM_40893 [Siemens-DMG-X500] — PLC_LOOP_GAIN_13 | Default: 228.400 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

PRM_40894 [Siemens-DMG-X500] — SPN_LOOP_GAIN_14 | Default: 15.200 | Linked Fault: ERR-SIE500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500.

PRM_40895 [Siemens-DMG-X500] — XAX_LOOP_GAIN_15 | Default: 52.000 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40896 [Siemens-DMG-X500] — CAX_LOOP_GAIN_16 | Default: 88.800 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X500 (BANK 5)

Machine Model: Siemens-DMG-X500 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X500.

PRM_40901 [Siemens-DMG-X500] — YAX_LOOP_GAIN_1 | Default: 46.000 | Linked Fault: ERR-SIE500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X500.

PRM_40902 [Siemens-DMG-X500] — PLC_LOOP_GAIN_2 | Default: 92.000 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

PRM_40903 [Siemens-DMG-X500] — ATC_LOOP_GAIN_3 | Default: 138.000 | Linked Fault: ERR-SIE500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X500.

PRM_40904 [Siemens-DMG-X500] — ZAX_LOOP_GAIN_4 | Default: 184.000 | Linked Fault: ERR-SIE500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X500.

PRM_40905 [Siemens-DMG-X500] — LUB_LOOP_GAIN_5 | Default: 230.000 | Linked Fault: ERR-SIE500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X500.

PRM_40906 [Siemens-DMG-X500] — HYD_LOOP_GAIN_6 | Default: 26.000 | Linked Fault: ERR-SIE500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X500.

PRM_40907 [Siemens-DMG-X500] — BAX_LOOP_GAIN_7 | Default: 72.000 | Linked Fault: ERR-SIE500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X500.

PRM_40908 [Siemens-DMG-X500] — VAC_LOOP_GAIN_8 | Default: 118.000 | Linked Fault: ERR-SIE500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X500.

PRM_40909 [Siemens-DMG-X500] — CLN_LOOP_GAIN_9 | Default: 164.000 | Linked Fault: ERR-SIE500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X500.

PRM_40910 [Siemens-DMG-X500] — CAX_LOOP_GAIN_10 | Default: 210.000 | Linked Fault: ERR-SIE500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X500.

PRM_40911 [Siemens-DMG-X500] — THM_LOOP_GAIN_11 | Default: 6.000 | Linked Fault: ERR-SIE500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X500.

PRM_40912 [Siemens-DMG-X500] — XAX_LOOP_GAIN_12 | Default: 52.000 | Linked Fault: ERR-SIE500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X500.

PRM_40913 [Siemens-DMG-X500] — OPT_LOOP_GAIN_13 | Default: 98.000 | Linked Fault: ERR-SIE500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X500.

PRM_40914 [Siemens-DMG-X500] — SPN_LOOP_GAIN_14 | Default: 144.000 | Linked Fault: ERR-SIE500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X500.

PRM_40915 [Siemens-DMG-X500] — YAX_LOOP_GAIN_15 | Default: 190.000 | Linked Fault: ERR-SIE500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X500.

PRM_40916 [Siemens-DMG-X500] — PLC_LOOP_GAIN_16 | Default: 236.000 | Linked Fault: ERR-SIE500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X500.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X550 (BANK 1)

Machine Model: Siemens-DMG-X550 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X550.

PRM_40921 [Siemens-DMG-X550] — ATC_LOOP_GAIN_1 | Default: 10.350 | Linked Fault: ERR-SIE550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X550.

PRM_40922 [Siemens-DMG-X550] — HYD_LOOP_GAIN_2 | Default: 20.700 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_40923 [Siemens-DMG-X550] — CLN_LOOP_GAIN_3 | Default: 31.050 | Linked Fault: ERR-SIE550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X550.

PRM_40924 [Siemens-DMG-X550] — XAX_LOOP_GAIN_4 | Default: 41.400 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_40925 [Siemens-DMG-X550] — YAX_LOOP_GAIN_5 | Default: 51.750 | Linked Fault: ERR-SIE550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X550.

PRM_40926 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_6 | Default: 62.100 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

PRM_40927 [Siemens-DMG-X550] — BAX_LOOP_GAIN_7 | Default: 72.450 | Linked Fault: ERR-SIE550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X550.

PRM_40928 [Siemens-DMG-X550] — CAX_LOOP_GAIN_8 | Default: 82.800 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

PRM_40929 [Siemens-DMG-X550] — OPT_LOOP_GAIN_9 | Default: 93.150 | Linked Fault: ERR-SIE550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X550.

PRM_40930 [Siemens-DMG-X550] — PLC_LOOP_GAIN_10 | Default: 103.500 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

PRM_40931 [Siemens-DMG-X550] — LUB_LOOP_GAIN_11 | Default: 113.850 | Linked Fault: ERR-SIE550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X550.

PRM_40932 [Siemens-DMG-X550] — VAC_LOOP_GAIN_12 | Default: 124.200 | Linked Fault: ERR-SIE550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550.

PRM_40933 [Siemens-DMG-X550] — THM_LOOP_GAIN_13 | Default: 134.550 | Linked Fault: ERR-SIE550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X550.

PRM_40934 [Siemens-DMG-X550] — SPN_LOOP_GAIN_14 | Default: 144.900 | Linked Fault: ERR-SIE550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550.

PRM_40935 [Siemens-DMG-X550] — ATC_LOOP_GAIN_15 | Default: 155.250 | Linked Fault: ERR-SIE550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X550.

PRM_40936 [Siemens-DMG-X550] — HYD_LOOP_GAIN_16 | Default: 165.600 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X550 (BANK 2)

Machine Model: Siemens-DMG-X550 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X550.

PRM_40941 [Siemens-DMG-X550] — HYD_LOOP_GAIN_1 | Default: 20.700 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_40942 [Siemens-DMG-X550] — XAX_LOOP_GAIN_2 | Default: 41.400 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_40943 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_3 | Default: 62.100 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

PRM_40944 [Siemens-DMG-X550] — CAX_LOOP_GAIN_4 | Default: 82.800 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

PRM_40945 [Siemens-DMG-X550] — PLC_LOOP_GAIN_5 | Default: 103.500 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

PRM_40946 [Siemens-DMG-X550] — VAC_LOOP_GAIN_6 | Default: 124.200 | Linked Fault: ERR-SIE550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550.

PRM_40947 [Siemens-DMG-X550] — SPN_LOOP_GAIN_7 | Default: 144.900 | Linked Fault: ERR-SIE550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550.

PRM_40948 [Siemens-DMG-X550] — HYD_LOOP_GAIN_8 | Default: 165.600 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_40949 [Siemens-DMG-X550] — XAX_LOOP_GAIN_9 | Default: 186.300 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_40950 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_10 | Default: 207.000 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

PRM_40951 [Siemens-DMG-X550] — CAX_LOOP_GAIN_11 | Default: 227.700 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

PRM_40952 [Siemens-DMG-X550] — PLC_LOOP_GAIN_12 | Default: 248.400 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

PRM_40953 [Siemens-DMG-X550] — VAC_LOOP_GAIN_13 | Default: 19.100 | Linked Fault: ERR-SIE550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550.

PRM_40954 [Siemens-DMG-X550] — SPN_LOOP_GAIN_14 | Default: 39.800 | Linked Fault: ERR-SIE550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550.

PRM_40955 [Siemens-DMG-X550] — HYD_LOOP_GAIN_15 | Default: 60.500 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_40956 [Siemens-DMG-X550] — XAX_LOOP_GAIN_16 | Default: 81.200 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X550 (BANK 3)

Machine Model: Siemens-DMG-X550 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X550.

PRM_40961 [Siemens-DMG-X550] — CLN_LOOP_GAIN_1 | Default: 31.050 | Linked Fault: ERR-SIE550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X550.

PRM_40962 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_2 | Default: 62.100 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

PRM_40963 [Siemens-DMG-X550] — OPT_LOOP_GAIN_3 | Default: 93.150 | Linked Fault: ERR-SIE550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X550.

PRM_40964 [Siemens-DMG-X550] — VAC_LOOP_GAIN_4 | Default: 124.200 | Linked Fault: ERR-SIE550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550.

PRM_40965 [Siemens-DMG-X550] — ATC_LOOP_GAIN_5 | Default: 155.250 | Linked Fault: ERR-SIE550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X550.

PRM_40966 [Siemens-DMG-X550] — XAX_LOOP_GAIN_6 | Default: 186.300 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_40967 [Siemens-DMG-X550] — BAX_LOOP_GAIN_7 | Default: 217.350 | Linked Fault: ERR-SIE550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X550.

PRM_40968 [Siemens-DMG-X550] — PLC_LOOP_GAIN_8 | Default: 248.400 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

PRM_40969 [Siemens-DMG-X550] — THM_LOOP_GAIN_9 | Default: 29.450 | Linked Fault: ERR-SIE550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X550.

PRM_40970 [Siemens-DMG-X550] — HYD_LOOP_GAIN_10 | Default: 60.500 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_40971 [Siemens-DMG-X550] — YAX_LOOP_GAIN_11 | Default: 91.550 | Linked Fault: ERR-SIE550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X550.

PRM_40972 [Siemens-DMG-X550] — CAX_LOOP_GAIN_12 | Default: 122.600 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

PRM_40973 [Siemens-DMG-X550] — LUB_LOOP_GAIN_13 | Default: 153.650 | Linked Fault: ERR-SIE550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X550.

PRM_40974 [Siemens-DMG-X550] — SPN_LOOP_GAIN_14 | Default: 184.700 | Linked Fault: ERR-SIE550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550.

PRM_40975 [Siemens-DMG-X550] — CLN_LOOP_GAIN_15 | Default: 215.750 | Linked Fault: ERR-SIE550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X550.

PRM_40976 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_16 | Default: 246.800 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X550 (BANK 4)

Machine Model: Siemens-DMG-X550 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X550.

PRM_40981 [Siemens-DMG-X550] — XAX_LOOP_GAIN_1 | Default: 41.400 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_40982 [Siemens-DMG-X550] — CAX_LOOP_GAIN_2 | Default: 82.800 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

PRM_40983 [Siemens-DMG-X550] — VAC_LOOP_GAIN_3 | Default: 124.200 | Linked Fault: ERR-SIE550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550.

PRM_40984 [Siemens-DMG-X550] — HYD_LOOP_GAIN_4 | Default: 165.600 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_40985 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_5 | Default: 207.000 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

PRM_40986 [Siemens-DMG-X550] — PLC_LOOP_GAIN_6 | Default: 248.400 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

PRM_40987 [Siemens-DMG-X550] — SPN_LOOP_GAIN_7 | Default: 39.800 | Linked Fault: ERR-SIE550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550.

PRM_40988 [Siemens-DMG-X550] — XAX_LOOP_GAIN_8 | Default: 81.200 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_40989 [Siemens-DMG-X550] — CAX_LOOP_GAIN_9 | Default: 122.600 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

PRM_40990 [Siemens-DMG-X550] — VAC_LOOP_GAIN_10 | Default: 164.000 | Linked Fault: ERR-SIE550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550.

PRM_40991 [Siemens-DMG-X550] — HYD_LOOP_GAIN_11 | Default: 205.400 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_40992 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_12 | Default: 246.800 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

PRM_40993 [Siemens-DMG-X550] — PLC_LOOP_GAIN_13 | Default: 38.200 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

PRM_40994 [Siemens-DMG-X550] — SPN_LOOP_GAIN_14 | Default: 79.600 | Linked Fault: ERR-SIE550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550.

PRM_40995 [Siemens-DMG-X550] — XAX_LOOP_GAIN_15 | Default: 121.000 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_40996 [Siemens-DMG-X550] — CAX_LOOP_GAIN_16 | Default: 162.400 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X550 (BANK 5)

Machine Model: Siemens-DMG-X550 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X550.

PRM_41001 [Siemens-DMG-X550] — YAX_LOOP_GAIN_1 | Default: 51.750 | Linked Fault: ERR-SIE550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X550.

PRM_41002 [Siemens-DMG-X550] — PLC_LOOP_GAIN_2 | Default: 103.500 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

PRM_41003 [Siemens-DMG-X550] — ATC_LOOP_GAIN_3 | Default: 155.250 | Linked Fault: ERR-SIE550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X550.

PRM_41004 [Siemens-DMG-X550] — ZAX_LOOP_GAIN_4 | Default: 207.000 | Linked Fault: ERR-SIE550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X550.

PRM_41005 [Siemens-DMG-X550] — LUB_LOOP_GAIN_5 | Default: 8.750 | Linked Fault: ERR-SIE550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X550.

PRM_41006 [Siemens-DMG-X550] — HYD_LOOP_GAIN_6 | Default: 60.500 | Linked Fault: ERR-SIE550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X550.

PRM_41007 [Siemens-DMG-X550] — BAX_LOOP_GAIN_7 | Default: 112.250 | Linked Fault: ERR-SIE550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X550.

PRM_41008 [Siemens-DMG-X550] — VAC_LOOP_GAIN_8 | Default: 164.000 | Linked Fault: ERR-SIE550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X550.

PRM_41009 [Siemens-DMG-X550] — CLN_LOOP_GAIN_9 | Default: 215.750 | Linked Fault: ERR-SIE550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X550.

PRM_41010 [Siemens-DMG-X550] — CAX_LOOP_GAIN_10 | Default: 17.500 | Linked Fault: ERR-SIE550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X550.

PRM_41011 [Siemens-DMG-X550] — THM_LOOP_GAIN_11 | Default: 69.250 | Linked Fault: ERR-SIE550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X550.

PRM_41012 [Siemens-DMG-X550] — XAX_LOOP_GAIN_12 | Default: 121.000 | Linked Fault: ERR-SIE550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X550.

PRM_41013 [Siemens-DMG-X550] — OPT_LOOP_GAIN_13 | Default: 172.750 | Linked Fault: ERR-SIE550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X550.

PRM_41014 [Siemens-DMG-X550] — SPN_LOOP_GAIN_14 | Default: 224.500 | Linked Fault: ERR-SIE550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X550.

PRM_41015 [Siemens-DMG-X550] — YAX_LOOP_GAIN_15 | Default: 26.250 | Linked Fault: ERR-SIE550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X550.

PRM_41016 [Siemens-DMG-X550] — PLC_LOOP_GAIN_16 | Default: 78.000 | Linked Fault: ERR-SIE550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X550.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X600 (BANK 1)

Machine Model: Siemens-DMG-X600 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X600.

PRM_41021 [Siemens-DMG-X600] — ATC_LOOP_GAIN_1 | Default: 11.500 | Linked Fault: ERR-SIE600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X600.

PRM_41022 [Siemens-DMG-X600] — HYD_LOOP_GAIN_2 | Default: 23.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41023 [Siemens-DMG-X600] — CLN_LOOP_GAIN_3 | Default: 34.500 | Linked Fault: ERR-SIE600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X600.

PRM_41024 [Siemens-DMG-X600] — XAX_LOOP_GAIN_4 | Default: 46.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41025 [Siemens-DMG-X600] — YAX_LOOP_GAIN_5 | Default: 57.500 | Linked Fault: ERR-SIE600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X600.

PRM_41026 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_6 | Default: 69.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

PRM_41027 [Siemens-DMG-X600] — BAX_LOOP_GAIN_7 | Default: 80.500 | Linked Fault: ERR-SIE600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X600.

PRM_41028 [Siemens-DMG-X600] — CAX_LOOP_GAIN_8 | Default: 92.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

PRM_41029 [Siemens-DMG-X600] — OPT_LOOP_GAIN_9 | Default: 103.500 | Linked Fault: ERR-SIE600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X600.

PRM_41030 [Siemens-DMG-X600] — PLC_LOOP_GAIN_10 | Default: 115.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

PRM_41031 [Siemens-DMG-X600] — LUB_LOOP_GAIN_11 | Default: 126.500 | Linked Fault: ERR-SIE600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X600.

PRM_41032 [Siemens-DMG-X600] — VAC_LOOP_GAIN_12 | Default: 138.000 | Linked Fault: ERR-SIE600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600.

PRM_41033 [Siemens-DMG-X600] — THM_LOOP_GAIN_13 | Default: 149.500 | Linked Fault: ERR-SIE600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X600.

PRM_41034 [Siemens-DMG-X600] — SPN_LOOP_GAIN_14 | Default: 161.000 | Linked Fault: ERR-SIE600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600.

PRM_41035 [Siemens-DMG-X600] — ATC_LOOP_GAIN_15 | Default: 172.500 | Linked Fault: ERR-SIE600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X600.

PRM_41036 [Siemens-DMG-X600] — HYD_LOOP_GAIN_16 | Default: 184.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X600 (BANK 2)

Machine Model: Siemens-DMG-X600 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X600.

PRM_41041 [Siemens-DMG-X600] — HYD_LOOP_GAIN_1 | Default: 23.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41042 [Siemens-DMG-X600] — XAX_LOOP_GAIN_2 | Default: 46.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41043 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_3 | Default: 69.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

PRM_41044 [Siemens-DMG-X600] — CAX_LOOP_GAIN_4 | Default: 92.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

PRM_41045 [Siemens-DMG-X600] — PLC_LOOP_GAIN_5 | Default: 115.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

PRM_41046 [Siemens-DMG-X600] — VAC_LOOP_GAIN_6 | Default: 138.000 | Linked Fault: ERR-SIE600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600.

PRM_41047 [Siemens-DMG-X600] — SPN_LOOP_GAIN_7 | Default: 161.000 | Linked Fault: ERR-SIE600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600.

PRM_41048 [Siemens-DMG-X600] — HYD_LOOP_GAIN_8 | Default: 184.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41049 [Siemens-DMG-X600] — XAX_LOOP_GAIN_9 | Default: 207.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41050 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_10 | Default: 230.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

PRM_41051 [Siemens-DMG-X600] — CAX_LOOP_GAIN_11 | Default: 3.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

PRM_41052 [Siemens-DMG-X600] — PLC_LOOP_GAIN_12 | Default: 26.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

PRM_41053 [Siemens-DMG-X600] — VAC_LOOP_GAIN_13 | Default: 49.000 | Linked Fault: ERR-SIE600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600.

PRM_41054 [Siemens-DMG-X600] — SPN_LOOP_GAIN_14 | Default: 72.000 | Linked Fault: ERR-SIE600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600.

PRM_41055 [Siemens-DMG-X600] — HYD_LOOP_GAIN_15 | Default: 95.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41056 [Siemens-DMG-X600] — XAX_LOOP_GAIN_16 | Default: 118.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X600 (BANK 3)

Machine Model: Siemens-DMG-X600 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X600.

PRM_41061 [Siemens-DMG-X600] — CLN_LOOP_GAIN_1 | Default: 34.500 | Linked Fault: ERR-SIE600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X600.

PRM_41062 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_2 | Default: 69.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

PRM_41063 [Siemens-DMG-X600] — OPT_LOOP_GAIN_3 | Default: 103.500 | Linked Fault: ERR-SIE600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X600.

PRM_41064 [Siemens-DMG-X600] — VAC_LOOP_GAIN_4 | Default: 138.000 | Linked Fault: ERR-SIE600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600.

PRM_41065 [Siemens-DMG-X600] — ATC_LOOP_GAIN_5 | Default: 172.500 | Linked Fault: ERR-SIE600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X600.

PRM_41066 [Siemens-DMG-X600] — XAX_LOOP_GAIN_6 | Default: 207.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41067 [Siemens-DMG-X600] — BAX_LOOP_GAIN_7 | Default: 241.500 | Linked Fault: ERR-SIE600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X600.

PRM_41068 [Siemens-DMG-X600] — PLC_LOOP_GAIN_8 | Default: 26.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

PRM_41069 [Siemens-DMG-X600] — THM_LOOP_GAIN_9 | Default: 60.500 | Linked Fault: ERR-SIE600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X600.

PRM_41070 [Siemens-DMG-X600] — HYD_LOOP_GAIN_10 | Default: 95.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41071 [Siemens-DMG-X600] — YAX_LOOP_GAIN_11 | Default: 129.500 | Linked Fault: ERR-SIE600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X600.

PRM_41072 [Siemens-DMG-X600] — CAX_LOOP_GAIN_12 | Default: 164.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

PRM_41073 [Siemens-DMG-X600] — LUB_LOOP_GAIN_13 | Default: 198.500 | Linked Fault: ERR-SIE600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X600.

PRM_41074 [Siemens-DMG-X600] — SPN_LOOP_GAIN_14 | Default: 233.000 | Linked Fault: ERR-SIE600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600.

PRM_41075 [Siemens-DMG-X600] — CLN_LOOP_GAIN_15 | Default: 17.500 | Linked Fault: ERR-SIE600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X600.

PRM_41076 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_16 | Default: 52.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X600 (BANK 4)

Machine Model: Siemens-DMG-X600 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X600.

PRM_41081 [Siemens-DMG-X600] — XAX_LOOP_GAIN_1 | Default: 46.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41082 [Siemens-DMG-X600] — CAX_LOOP_GAIN_2 | Default: 92.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

PRM_41083 [Siemens-DMG-X600] — VAC_LOOP_GAIN_3 | Default: 138.000 | Linked Fault: ERR-SIE600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600.

PRM_41084 [Siemens-DMG-X600] — HYD_LOOP_GAIN_4 | Default: 184.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41085 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_5 | Default: 230.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

PRM_41086 [Siemens-DMG-X600] — PLC_LOOP_GAIN_6 | Default: 26.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

PRM_41087 [Siemens-DMG-X600] — SPN_LOOP_GAIN_7 | Default: 72.000 | Linked Fault: ERR-SIE600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600.

PRM_41088 [Siemens-DMG-X600] — XAX_LOOP_GAIN_8 | Default: 118.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41089 [Siemens-DMG-X600] — CAX_LOOP_GAIN_9 | Default: 164.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

PRM_41090 [Siemens-DMG-X600] — VAC_LOOP_GAIN_10 | Default: 210.000 | Linked Fault: ERR-SIE600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600.

PRM_41091 [Siemens-DMG-X600] — HYD_LOOP_GAIN_11 | Default: 6.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41092 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_12 | Default: 52.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

PRM_41093 [Siemens-DMG-X600] — PLC_LOOP_GAIN_13 | Default: 98.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

PRM_41094 [Siemens-DMG-X600] — SPN_LOOP_GAIN_14 | Default: 144.000 | Linked Fault: ERR-SIE600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600.

PRM_41095 [Siemens-DMG-X600] — XAX_LOOP_GAIN_15 | Default: 190.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41096 [Siemens-DMG-X600] — CAX_LOOP_GAIN_16 | Default: 236.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

CNC/PLC SERVO PARAMETER DICTIONARY — Siemens-DMG-X600 (BANK 5)

Machine Model: Siemens-DMG-X600 | Controller: SINUMERIK ONE v6.15. Use these parameter registers when diagnosing servo following errors or thermal drift on Siemens-DMG-X600.

PRM_41101 [Siemens-DMG-X600] — YAX_LOOP_GAIN_1 | Default: 57.500 | Linked Fault: ERR-SIE600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X600.

PRM_41102 [Siemens-DMG-X600] — PLC_LOOP_GAIN_2 | Default: 115.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

PRM_41103 [Siemens-DMG-X600] — ATC_LOOP_GAIN_3 | Default: 172.500 | Linked Fault: ERR-SIE600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Siemens-DMG-X600.

PRM_41104 [Siemens-DMG-X600] — ZAX_LOOP_GAIN_4 | Default: 230.000 | Linked Fault: ERR-SIE600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Siemens-DMG-X600.

PRM_41105 [Siemens-DMG-X600] — LUB_LOOP_GAIN_5 | Default: 37.500 | Linked Fault: ERR-SIE600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Siemens-DMG-X600.

PRM_41106 [Siemens-DMG-X600] — HYD_LOOP_GAIN_6 | Default: 95.000 | Linked Fault: ERR-SIE600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Siemens-DMG-X600.

PRM_41107 [Siemens-DMG-X600] — BAX_LOOP_GAIN_7 | Default: 152.500 | Linked Fault: ERR-SIE600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Siemens-DMG-X600.

PRM_41108 [Siemens-DMG-X600] — VAC_LOOP_GAIN_8 | Default: 210.000 | Linked Fault: ERR-SIE600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Siemens-DMG-X600.

PRM_41109 [Siemens-DMG-X600] — CLN_LOOP_GAIN_9 | Default: 17.500 | Linked Fault: ERR-SIE600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Siemens-DMG-X600.

PRM_41110 [Siemens-DMG-X600] — CAX_LOOP_GAIN_10 | Default: 75.000 | Linked Fault: ERR-SIE600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Siemens-DMG-X600.

PRM_41111 [Siemens-DMG-X600] — THM_LOOP_GAIN_11 | Default: 132.500 | Linked Fault: ERR-SIE600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Siemens-DMG-X600.

PRM_41112 [Siemens-DMG-X600] — XAX_LOOP_GAIN_12 | Default: 190.000 | Linked Fault: ERR-SIE600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Siemens-DMG-X600.

PRM_41113 [Siemens-DMG-X600] — OPT_LOOP_GAIN_13 | Default: 247.500 | Linked Fault: ERR-SIE600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Siemens-DMG-X600.

PRM_41114 [Siemens-DMG-X600] — SPN_LOOP_GAIN_14 | Default: 55.000 | Linked Fault: ERR-SIE600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Siemens-DMG-X600.

PRM_41115 [Siemens-DMG-X600] — YAX_LOOP_GAIN_15 | Default: 112.500 | Linked Fault: ERR-SIE600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Siemens-DMG-X600.

PRM_41116 [Siemens-DMG-X600] — PLC_LOOP_GAIN_16 | Default: 170.000 | Linked Fault: ERR-SIE600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Siemens-DMG-X600.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T150 (BANK 1)

Machine Model: Makino-Aero-T150 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T150.

PRM_41121 [Makino-Aero-T150] — ATC_LOOP_GAIN_1 | Default: 12.650 | Linked Fault: ERR-MAK150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T150.

PRM_41122 [Makino-Aero-T150] — HYD_LOOP_GAIN_2 | Default: 25.300 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41123 [Makino-Aero-T150] — CLN_LOOP_GAIN_3 | Default: 37.950 | Linked Fault: ERR-MAK150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T150.

PRM_41124 [Makino-Aero-T150] — XAX_LOOP_GAIN_4 | Default: 50.600 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41125 [Makino-Aero-T150] — YAX_LOOP_GAIN_5 | Default: 63.250 | Linked Fault: ERR-MAK150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T150.

PRM_41126 [Makino-Aero-T150] — ZAX_LOOP_GAIN_6 | Default: 75.900 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

PRM_41127 [Makino-Aero-T150] — BAX_LOOP_GAIN_7 | Default: 88.550 | Linked Fault: ERR-MAK150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T150.

PRM_41128 [Makino-Aero-T150] — CAX_LOOP_GAIN_8 | Default: 101.200 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

PRM_41129 [Makino-Aero-T150] — OPT_LOOP_GAIN_9 | Default: 113.850 | Linked Fault: ERR-MAK150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T150.

PRM_41130 [Makino-Aero-T150] — PLC_LOOP_GAIN_10 | Default: 126.500 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

PRM_41131 [Makino-Aero-T150] — LUB_LOOP_GAIN_11 | Default: 139.150 | Linked Fault: ERR-MAK150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T150.

PRM_41132 [Makino-Aero-T150] — VAC_LOOP_GAIN_12 | Default: 151.800 | Linked Fault: ERR-MAK150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150.

PRM_41133 [Makino-Aero-T150] — THM_LOOP_GAIN_13 | Default: 164.450 | Linked Fault: ERR-MAK150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T150.

PRM_41134 [Makino-Aero-T150] — SPN_LOOP_GAIN_14 | Default: 177.100 | Linked Fault: ERR-MAK150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T150.

PRM_41135 [Makino-Aero-T150] — ATC_LOOP_GAIN_15 | Default: 189.750 | Linked Fault: ERR-MAK150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T150.

PRM_41136 [Makino-Aero-T150] — HYD_LOOP_GAIN_16 | Default: 202.400 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T150 (BANK 2)

Machine Model: Makino-Aero-T150 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T150.

PRM_41141 [Makino-Aero-T150] — HYD_LOOP_GAIN_1 | Default: 25.300 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41142 [Makino-Aero-T150] — XAX_LOOP_GAIN_2 | Default: 50.600 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41143 [Makino-Aero-T150] — ZAX_LOOP_GAIN_3 | Default: 75.900 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

PRM_41144 [Makino-Aero-T150] — CAX_LOOP_GAIN_4 | Default: 101.200 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

PRM_41145 [Makino-Aero-T150] — PLC_LOOP_GAIN_5 | Default: 126.500 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

PRM_41146 [Makino-Aero-T150] — VAC_LOOP_GAIN_6 | Default: 151.800 | Linked Fault: ERR-MAK150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150.

PRM_41147 [Makino-Aero-T150] — SPN_LOOP_GAIN_7 | Default: 177.100 | Linked Fault: ERR-MAK150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T150.

PRM_41148 [Makino-Aero-T150] — HYD_LOOP_GAIN_8 | Default: 202.400 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41149 [Makino-Aero-T150] — XAX_LOOP_GAIN_9 | Default: 227.700 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41150 [Makino-Aero-T150] — ZAX_LOOP_GAIN_10 | Default: 3.000 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

PRM_41151 [Makino-Aero-T150] — CAX_LOOP_GAIN_11 | Default: 28.300 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

PRM_41152 [Makino-Aero-T150] — PLC_LOOP_GAIN_12 | Default: 53.600 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

PRM_41153 [Makino-Aero-T150] — VAC_LOOP_GAIN_13 | Default: 78.900 | Linked Fault: ERR-MAK150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150.

PRM_41154 [Makino-Aero-T150] — SPN_LOOP_GAIN_14 | Default: 104.200 | Linked Fault: ERR-MAK150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T150.

PRM_41155 [Makino-Aero-T150] — HYD_LOOP_GAIN_15 | Default: 129.500 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41156 [Makino-Aero-T150] — XAX_LOOP_GAIN_16 | Default: 154.800 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T150 (BANK 3)

Machine Model: Makino-Aero-T150 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T150.

PRM_41161 [Makino-Aero-T150] — CLN_LOOP_GAIN_1 | Default: 37.950 | Linked Fault: ERR-MAK150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T150.

PRM_41162 [Makino-Aero-T150] — ZAX_LOOP_GAIN_2 | Default: 75.900 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

PRM_41163 [Makino-Aero-T150] — OPT_LOOP_GAIN_3 | Default: 113.850 | Linked Fault: ERR-MAK150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T150.

PRM_41164 [Makino-Aero-T150] — VAC_LOOP_GAIN_4 | Default: 151.800 | Linked Fault: ERR-MAK150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150.

PRM_41165 [Makino-Aero-T150] — ATC_LOOP_GAIN_5 | Default: 189.750 | Linked Fault: ERR-MAK150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T150.

PRM_41166 [Makino-Aero-T150] — XAX_LOOP_GAIN_6 | Default: 227.700 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41167 [Makino-Aero-T150] — BAX_LOOP_GAIN_7 | Default: 15.650 | Linked Fault: ERR-MAK150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T150.

PRM_41168 [Makino-Aero-T150] — PLC_LOOP_GAIN_8 | Default: 53.600 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

PRM_41169 [Makino-Aero-T150] — THM_LOOP_GAIN_9 | Default: 91.550 | Linked Fault: ERR-MAK150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T150.

PRM_41170 [Makino-Aero-T150] — HYD_LOOP_GAIN_10 | Default: 129.500 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41171 [Makino-Aero-T150] — YAX_LOOP_GAIN_11 | Default: 167.450 | Linked Fault: ERR-MAK150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T150.

PRM_41172 [Makino-Aero-T150] — CAX_LOOP_GAIN_12 | Default: 205.400 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

PRM_41173 [Makino-Aero-T150] — LUB_LOOP_GAIN_13 | Default: 243.350 | Linked Fault: ERR-MAK150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T150.

PRM_41174 [Makino-Aero-T150] — SPN_LOOP_GAIN_14 | Default: 31.300 | Linked Fault: ERR-MAK150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T150.

PRM_41175 [Makino-Aero-T150] — CLN_LOOP_GAIN_15 | Default: 69.250 | Linked Fault: ERR-MAK150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T150.

PRM_41176 [Makino-Aero-T150] — ZAX_LOOP_GAIN_16 | Default: 107.200 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T150 (BANK 4)

Machine Model: Makino-Aero-T150 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T150.

PRM_41181 [Makino-Aero-T150] — XAX_LOOP_GAIN_1 | Default: 50.600 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41182 [Makino-Aero-T150] — CAX_LOOP_GAIN_2 | Default: 101.200 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

PRM_41183 [Makino-Aero-T150] — VAC_LOOP_GAIN_3 | Default: 151.800 | Linked Fault: ERR-MAK150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150.

PRM_41184 [Makino-Aero-T150] — HYD_LOOP_GAIN_4 | Default: 202.400 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41185 [Makino-Aero-T150] — ZAX_LOOP_GAIN_5 | Default: 3.000 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

PRM_41186 [Makino-Aero-T150] — PLC_LOOP_GAIN_6 | Default: 53.600 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

PRM_41187 [Makino-Aero-T150] — SPN_LOOP_GAIN_7 | Default: 104.200 | Linked Fault: ERR-MAK150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T150.

PRM_41188 [Makino-Aero-T150] — XAX_LOOP_GAIN_8 | Default: 154.800 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41189 [Makino-Aero-T150] — CAX_LOOP_GAIN_9 | Default: 205.400 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

PRM_41190 [Makino-Aero-T150] — VAC_LOOP_GAIN_10 | Default: 6.000 | Linked Fault: ERR-MAK150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150.

PRM_41191 [Makino-Aero-T150] — HYD_LOOP_GAIN_11 | Default: 56.600 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41192 [Makino-Aero-T150] — ZAX_LOOP_GAIN_12 | Default: 107.200 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

PRM_41193 [Makino-Aero-T150] — PLC_LOOP_GAIN_13 | Default: 157.800 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

PRM_41194 [Makino-Aero-T150] — SPN_LOOP_GAIN_14 | Default: 208.400 | Linked Fault: ERR-MAK150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T150.

PRM_41195 [Makino-Aero-T150] — XAX_LOOP_GAIN_15 | Default: 9.000 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41196 [Makino-Aero-T150] — CAX_LOOP_GAIN_16 | Default: 59.600 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T150 (BANK 5)

Machine Model: Makino-Aero-T150 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T150.

PRM_41201 [Makino-Aero-T150] — YAX_LOOP_GAIN_1 | Default: 63.250 | Linked Fault: ERR-MAK150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T150.

PRM_41202 [Makino-Aero-T150] — PLC_LOOP_GAIN_2 | Default: 126.500 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

PRM_41203 [Makino-Aero-T150] — ATC_LOOP_GAIN_3 | Default: 189.750 | Linked Fault: ERR-MAK150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T150.

PRM_41204 [Makino-Aero-T150] — ZAX_LOOP_GAIN_4 | Default: 3.000 | Linked Fault: ERR-MAK150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T150.

PRM_41205 [Makino-Aero-T150] — LUB_LOOP_GAIN_5 | Default: 66.250 | Linked Fault: ERR-MAK150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T150.

PRM_41206 [Makino-Aero-T150] — HYD_LOOP_GAIN_6 | Default: 129.500 | Linked Fault: ERR-MAK150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T150.

PRM_41207 [Makino-Aero-T150] — BAX_LOOP_GAIN_7 | Default: 192.750 | Linked Fault: ERR-MAK150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T150.

PRM_41208 [Makino-Aero-T150] — VAC_LOOP_GAIN_8 | Default: 6.000 | Linked Fault: ERR-MAK150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T150.

PRM_41209 [Makino-Aero-T150] — CLN_LOOP_GAIN_9 | Default: 69.250 | Linked Fault: ERR-MAK150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T150.

PRM_41210 [Makino-Aero-T150] — CAX_LOOP_GAIN_10 | Default: 132.500 | Linked Fault: ERR-MAK150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T150.

PRM_41211 [Makino-Aero-T150] — THM_LOOP_GAIN_11 | Default: 195.750 | Linked Fault: ERR-MAK150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T150.

PRM_41212 [Makino-Aero-T150] — XAX_LOOP_GAIN_12 | Default: 9.000 | Linked Fault: ERR-MAK150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T150.

PRM_41213 [Makino-Aero-T150] — OPT_LOOP_GAIN_13 | Default: 72.250 | Linked Fault: ERR-MAK150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T150.

PRM_41214 [Makino-Aero-T150] — SPN_LOOP_GAIN_14 | Default: 135.500 | Linked Fault: ERR-MAK150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T150.

PRM_41215 [Makino-Aero-T150] — YAX_LOOP_GAIN_15 | Default: 198.750 | Linked Fault: ERR-MAK150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T150.

PRM_41216 [Makino-Aero-T150] — PLC_LOOP_GAIN_16 | Default: 12.000 | Linked Fault: ERR-MAK150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T150.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T200 (BANK 1)

Machine Model: Makino-Aero-T200 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T200.

PRM_41221 [Makino-Aero-T200] — ATC_LOOP_GAIN_1 | Default: 13.800 | Linked Fault: ERR-MAK200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T200.

PRM_41222 [Makino-Aero-T200] — HYD_LOOP_GAIN_2 | Default: 27.600 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41223 [Makino-Aero-T200] — CLN_LOOP_GAIN_3 | Default: 41.400 | Linked Fault: ERR-MAK200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T200.

PRM_41224 [Makino-Aero-T200] — XAX_LOOP_GAIN_4 | Default: 55.200 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41225 [Makino-Aero-T200] — YAX_LOOP_GAIN_5 | Default: 69.000 | Linked Fault: ERR-MAK200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T200.

PRM_41226 [Makino-Aero-T200] — ZAX_LOOP_GAIN_6 | Default: 82.800 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

PRM_41227 [Makino-Aero-T200] — BAX_LOOP_GAIN_7 | Default: 96.600 | Linked Fault: ERR-MAK200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T200.

PRM_41228 [Makino-Aero-T200] — CAX_LOOP_GAIN_8 | Default: 110.400 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

PRM_41229 [Makino-Aero-T200] — OPT_LOOP_GAIN_9 | Default: 124.200 | Linked Fault: ERR-MAK200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T200.

PRM_41230 [Makino-Aero-T200] — PLC_LOOP_GAIN_10 | Default: 138.000 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

PRM_41231 [Makino-Aero-T200] — LUB_LOOP_GAIN_11 | Default: 151.800 | Linked Fault: ERR-MAK200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T200.

PRM_41232 [Makino-Aero-T200] — VAC_LOOP_GAIN_12 | Default: 165.600 | Linked Fault: ERR-MAK200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200.

PRM_41233 [Makino-Aero-T200] — THM_LOOP_GAIN_13 | Default: 179.400 | Linked Fault: ERR-MAK200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T200.

PRM_41234 [Makino-Aero-T200] — SPN_LOOP_GAIN_14 | Default: 193.200 | Linked Fault: ERR-MAK200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T200.

PRM_41235 [Makino-Aero-T200] — ATC_LOOP_GAIN_15 | Default: 207.000 | Linked Fault: ERR-MAK200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T200.

PRM_41236 [Makino-Aero-T200] — HYD_LOOP_GAIN_16 | Default: 220.800 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T200 (BANK 2)

Machine Model: Makino-Aero-T200 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T200.

PRM_41241 [Makino-Aero-T200] — HYD_LOOP_GAIN_1 | Default: 27.600 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41242 [Makino-Aero-T200] — XAX_LOOP_GAIN_2 | Default: 55.200 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41243 [Makino-Aero-T200] — ZAX_LOOP_GAIN_3 | Default: 82.800 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

PRM_41244 [Makino-Aero-T200] — CAX_LOOP_GAIN_4 | Default: 110.400 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

PRM_41245 [Makino-Aero-T200] — PLC_LOOP_GAIN_5 | Default: 138.000 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

PRM_41246 [Makino-Aero-T200] — VAC_LOOP_GAIN_6 | Default: 165.600 | Linked Fault: ERR-MAK200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200.

PRM_41247 [Makino-Aero-T200] — SPN_LOOP_GAIN_7 | Default: 193.200 | Linked Fault: ERR-MAK200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T200.

PRM_41248 [Makino-Aero-T200] — HYD_LOOP_GAIN_8 | Default: 220.800 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41249 [Makino-Aero-T200] — XAX_LOOP_GAIN_9 | Default: 248.400 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41250 [Makino-Aero-T200] — ZAX_LOOP_GAIN_10 | Default: 26.000 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

PRM_41251 [Makino-Aero-T200] — CAX_LOOP_GAIN_11 | Default: 53.600 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

PRM_41252 [Makino-Aero-T200] — PLC_LOOP_GAIN_12 | Default: 81.200 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

PRM_41253 [Makino-Aero-T200] — VAC_LOOP_GAIN_13 | Default: 108.800 | Linked Fault: ERR-MAK200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200.

PRM_41254 [Makino-Aero-T200] — SPN_LOOP_GAIN_14 | Default: 136.400 | Linked Fault: ERR-MAK200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T200.

PRM_41255 [Makino-Aero-T200] — HYD_LOOP_GAIN_15 | Default: 164.000 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41256 [Makino-Aero-T200] — XAX_LOOP_GAIN_16 | Default: 191.600 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T200 (BANK 3)

Machine Model: Makino-Aero-T200 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T200.

PRM_41261 [Makino-Aero-T200] — CLN_LOOP_GAIN_1 | Default: 41.400 | Linked Fault: ERR-MAK200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T200.

PRM_41262 [Makino-Aero-T200] — ZAX_LOOP_GAIN_2 | Default: 82.800 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

PRM_41263 [Makino-Aero-T200] — OPT_LOOP_GAIN_3 | Default: 124.200 | Linked Fault: ERR-MAK200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T200.

PRM_41264 [Makino-Aero-T200] — VAC_LOOP_GAIN_4 | Default: 165.600 | Linked Fault: ERR-MAK200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200.

PRM_41265 [Makino-Aero-T200] — ATC_LOOP_GAIN_5 | Default: 207.000 | Linked Fault: ERR-MAK200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T200.

PRM_41266 [Makino-Aero-T200] — XAX_LOOP_GAIN_6 | Default: 248.400 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41267 [Makino-Aero-T200] — BAX_LOOP_GAIN_7 | Default: 39.800 | Linked Fault: ERR-MAK200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T200.

PRM_41268 [Makino-Aero-T200] — PLC_LOOP_GAIN_8 | Default: 81.200 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

PRM_41269 [Makino-Aero-T200] — THM_LOOP_GAIN_9 | Default: 122.600 | Linked Fault: ERR-MAK200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T200.

PRM_41270 [Makino-Aero-T200] — HYD_LOOP_GAIN_10 | Default: 164.000 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41271 [Makino-Aero-T200] — YAX_LOOP_GAIN_11 | Default: 205.400 | Linked Fault: ERR-MAK200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T200.

PRM_41272 [Makino-Aero-T200] — CAX_LOOP_GAIN_12 | Default: 246.800 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

PRM_41273 [Makino-Aero-T200] — LUB_LOOP_GAIN_13 | Default: 38.200 | Linked Fault: ERR-MAK200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T200.

PRM_41274 [Makino-Aero-T200] — SPN_LOOP_GAIN_14 | Default: 79.600 | Linked Fault: ERR-MAK200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T200.

PRM_41275 [Makino-Aero-T200] — CLN_LOOP_GAIN_15 | Default: 121.000 | Linked Fault: ERR-MAK200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T200.

PRM_41276 [Makino-Aero-T200] — ZAX_LOOP_GAIN_16 | Default: 162.400 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T200 (BANK 4)

Machine Model: Makino-Aero-T200 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T200.

PRM_41281 [Makino-Aero-T200] — XAX_LOOP_GAIN_1 | Default: 55.200 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41282 [Makino-Aero-T200] — CAX_LOOP_GAIN_2 | Default: 110.400 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

PRM_41283 [Makino-Aero-T200] — VAC_LOOP_GAIN_3 | Default: 165.600 | Linked Fault: ERR-MAK200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200.

PRM_41284 [Makino-Aero-T200] — HYD_LOOP_GAIN_4 | Default: 220.800 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41285 [Makino-Aero-T200] — ZAX_LOOP_GAIN_5 | Default: 26.000 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

PRM_41286 [Makino-Aero-T200] — PLC_LOOP_GAIN_6 | Default: 81.200 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

PRM_41287 [Makino-Aero-T200] — SPN_LOOP_GAIN_7 | Default: 136.400 | Linked Fault: ERR-MAK200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T200.

PRM_41288 [Makino-Aero-T200] — XAX_LOOP_GAIN_8 | Default: 191.600 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41289 [Makino-Aero-T200] — CAX_LOOP_GAIN_9 | Default: 246.800 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

PRM_41290 [Makino-Aero-T200] — VAC_LOOP_GAIN_10 | Default: 52.000 | Linked Fault: ERR-MAK200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200.

PRM_41291 [Makino-Aero-T200] — HYD_LOOP_GAIN_11 | Default: 107.200 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41292 [Makino-Aero-T200] — ZAX_LOOP_GAIN_12 | Default: 162.400 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

PRM_41293 [Makino-Aero-T200] — PLC_LOOP_GAIN_13 | Default: 217.600 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

PRM_41294 [Makino-Aero-T200] — SPN_LOOP_GAIN_14 | Default: 22.800 | Linked Fault: ERR-MAK200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T200.

PRM_41295 [Makino-Aero-T200] — XAX_LOOP_GAIN_15 | Default: 78.000 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41296 [Makino-Aero-T200] — CAX_LOOP_GAIN_16 | Default: 133.200 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T200 (BANK 5)

Machine Model: Makino-Aero-T200 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T200.

PRM_41301 [Makino-Aero-T200] — YAX_LOOP_GAIN_1 | Default: 69.000 | Linked Fault: ERR-MAK200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T200.

PRM_41302 [Makino-Aero-T200] — PLC_LOOP_GAIN_2 | Default: 138.000 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

PRM_41303 [Makino-Aero-T200] — ATC_LOOP_GAIN_3 | Default: 207.000 | Linked Fault: ERR-MAK200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T200.

PRM_41304 [Makino-Aero-T200] — ZAX_LOOP_GAIN_4 | Default: 26.000 | Linked Fault: ERR-MAK200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T200.

PRM_41305 [Makino-Aero-T200] — LUB_LOOP_GAIN_5 | Default: 95.000 | Linked Fault: ERR-MAK200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T200.

PRM_41306 [Makino-Aero-T200] — HYD_LOOP_GAIN_6 | Default: 164.000 | Linked Fault: ERR-MAK200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T200.

PRM_41307 [Makino-Aero-T200] — BAX_LOOP_GAIN_7 | Default: 233.000 | Linked Fault: ERR-MAK200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T200.

PRM_41308 [Makino-Aero-T200] — VAC_LOOP_GAIN_8 | Default: 52.000 | Linked Fault: ERR-MAK200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T200.

PRM_41309 [Makino-Aero-T200] — CLN_LOOP_GAIN_9 | Default: 121.000 | Linked Fault: ERR-MAK200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T200.

PRM_41310 [Makino-Aero-T200] — CAX_LOOP_GAIN_10 | Default: 190.000 | Linked Fault: ERR-MAK200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T200.

PRM_41311 [Makino-Aero-T200] — THM_LOOP_GAIN_11 | Default: 9.000 | Linked Fault: ERR-MAK200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T200.

PRM_41312 [Makino-Aero-T200] — XAX_LOOP_GAIN_12 | Default: 78.000 | Linked Fault: ERR-MAK200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T200.

PRM_41313 [Makino-Aero-T200] — OPT_LOOP_GAIN_13 | Default: 147.000 | Linked Fault: ERR-MAK200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T200.

PRM_41314 [Makino-Aero-T200] — SPN_LOOP_GAIN_14 | Default: 216.000 | Linked Fault: ERR-MAK200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T200.

PRM_41315 [Makino-Aero-T200] — YAX_LOOP_GAIN_15 | Default: 35.000 | Linked Fault: ERR-MAK200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T200.

PRM_41316 [Makino-Aero-T200] — PLC_LOOP_GAIN_16 | Default: 104.000 | Linked Fault: ERR-MAK200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T200.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T250 (BANK 1)

Machine Model: Makino-Aero-T250 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T250.

PRM_41321 [Makino-Aero-T250] — ATC_LOOP_GAIN_1 | Default: 14.950 | Linked Fault: ERR-MAK250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T250.

PRM_41322 [Makino-Aero-T250] — HYD_LOOP_GAIN_2 | Default: 29.900 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41323 [Makino-Aero-T250] — CLN_LOOP_GAIN_3 | Default: 44.850 | Linked Fault: ERR-MAK250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T250.

PRM_41324 [Makino-Aero-T250] — XAX_LOOP_GAIN_4 | Default: 59.800 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41325 [Makino-Aero-T250] — YAX_LOOP_GAIN_5 | Default: 74.750 | Linked Fault: ERR-MAK250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T250.

PRM_41326 [Makino-Aero-T250] — ZAX_LOOP_GAIN_6 | Default: 89.700 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

PRM_41327 [Makino-Aero-T250] — BAX_LOOP_GAIN_7 | Default: 104.650 | Linked Fault: ERR-MAK250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T250.

PRM_41328 [Makino-Aero-T250] — CAX_LOOP_GAIN_8 | Default: 119.600 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

PRM_41329 [Makino-Aero-T250] — OPT_LOOP_GAIN_9 | Default: 134.550 | Linked Fault: ERR-MAK250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T250.

PRM_41330 [Makino-Aero-T250] — PLC_LOOP_GAIN_10 | Default: 149.500 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

PRM_41331 [Makino-Aero-T250] — LUB_LOOP_GAIN_11 | Default: 164.450 | Linked Fault: ERR-MAK250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T250.

PRM_41332 [Makino-Aero-T250] — VAC_LOOP_GAIN_12 | Default: 179.400 | Linked Fault: ERR-MAK250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250.

PRM_41333 [Makino-Aero-T250] — THM_LOOP_GAIN_13 | Default: 194.350 | Linked Fault: ERR-MAK250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T250.

PRM_41334 [Makino-Aero-T250] — SPN_LOOP_GAIN_14 | Default: 209.300 | Linked Fault: ERR-MAK250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T250.

PRM_41335 [Makino-Aero-T250] — ATC_LOOP_GAIN_15 | Default: 224.250 | Linked Fault: ERR-MAK250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T250.

PRM_41336 [Makino-Aero-T250] — HYD_LOOP_GAIN_16 | Default: 239.200 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T250 (BANK 2)

Machine Model: Makino-Aero-T250 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T250.

PRM_41341 [Makino-Aero-T250] — HYD_LOOP_GAIN_1 | Default: 29.900 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41342 [Makino-Aero-T250] — XAX_LOOP_GAIN_2 | Default: 59.800 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41343 [Makino-Aero-T250] — ZAX_LOOP_GAIN_3 | Default: 89.700 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

PRM_41344 [Makino-Aero-T250] — CAX_LOOP_GAIN_4 | Default: 119.600 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

PRM_41345 [Makino-Aero-T250] — PLC_LOOP_GAIN_5 | Default: 149.500 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

PRM_41346 [Makino-Aero-T250] — VAC_LOOP_GAIN_6 | Default: 179.400 | Linked Fault: ERR-MAK250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250.

PRM_41347 [Makino-Aero-T250] — SPN_LOOP_GAIN_7 | Default: 209.300 | Linked Fault: ERR-MAK250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T250.

PRM_41348 [Makino-Aero-T250] — HYD_LOOP_GAIN_8 | Default: 239.200 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41349 [Makino-Aero-T250] — XAX_LOOP_GAIN_9 | Default: 19.100 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41350 [Makino-Aero-T250] — ZAX_LOOP_GAIN_10 | Default: 49.000 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

PRM_41351 [Makino-Aero-T250] — CAX_LOOP_GAIN_11 | Default: 78.900 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

PRM_41352 [Makino-Aero-T250] — PLC_LOOP_GAIN_12 | Default: 108.800 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

PRM_41353 [Makino-Aero-T250] — VAC_LOOP_GAIN_13 | Default: 138.700 | Linked Fault: ERR-MAK250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250.

PRM_41354 [Makino-Aero-T250] — SPN_LOOP_GAIN_14 | Default: 168.600 | Linked Fault: ERR-MAK250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T250.

PRM_41355 [Makino-Aero-T250] — HYD_LOOP_GAIN_15 | Default: 198.500 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41356 [Makino-Aero-T250] — XAX_LOOP_GAIN_16 | Default: 228.400 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T250 (BANK 3)

Machine Model: Makino-Aero-T250 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T250.

PRM_41361 [Makino-Aero-T250] — CLN_LOOP_GAIN_1 | Default: 44.850 | Linked Fault: ERR-MAK250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T250.

PRM_41362 [Makino-Aero-T250] — ZAX_LOOP_GAIN_2 | Default: 89.700 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

PRM_41363 [Makino-Aero-T250] — OPT_LOOP_GAIN_3 | Default: 134.550 | Linked Fault: ERR-MAK250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T250.

PRM_41364 [Makino-Aero-T250] — VAC_LOOP_GAIN_4 | Default: 179.400 | Linked Fault: ERR-MAK250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250.

PRM_41365 [Makino-Aero-T250] — ATC_LOOP_GAIN_5 | Default: 224.250 | Linked Fault: ERR-MAK250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T250.

PRM_41366 [Makino-Aero-T250] — XAX_LOOP_GAIN_6 | Default: 19.100 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41367 [Makino-Aero-T250] — BAX_LOOP_GAIN_7 | Default: 63.950 | Linked Fault: ERR-MAK250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T250.

PRM_41368 [Makino-Aero-T250] — PLC_LOOP_GAIN_8 | Default: 108.800 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

PRM_41369 [Makino-Aero-T250] — THM_LOOP_GAIN_9 | Default: 153.650 | Linked Fault: ERR-MAK250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T250.

PRM_41370 [Makino-Aero-T250] — HYD_LOOP_GAIN_10 | Default: 198.500 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41371 [Makino-Aero-T250] — YAX_LOOP_GAIN_11 | Default: 243.350 | Linked Fault: ERR-MAK250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T250.

PRM_41372 [Makino-Aero-T250] — CAX_LOOP_GAIN_12 | Default: 38.200 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

PRM_41373 [Makino-Aero-T250] — LUB_LOOP_GAIN_13 | Default: 83.050 | Linked Fault: ERR-MAK250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T250.

PRM_41374 [Makino-Aero-T250] — SPN_LOOP_GAIN_14 | Default: 127.900 | Linked Fault: ERR-MAK250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T250.

PRM_41375 [Makino-Aero-T250] — CLN_LOOP_GAIN_15 | Default: 172.750 | Linked Fault: ERR-MAK250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T250.

PRM_41376 [Makino-Aero-T250] — ZAX_LOOP_GAIN_16 | Default: 217.600 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T250 (BANK 4)

Machine Model: Makino-Aero-T250 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T250.

PRM_41381 [Makino-Aero-T250] — XAX_LOOP_GAIN_1 | Default: 59.800 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41382 [Makino-Aero-T250] — CAX_LOOP_GAIN_2 | Default: 119.600 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

PRM_41383 [Makino-Aero-T250] — VAC_LOOP_GAIN_3 | Default: 179.400 | Linked Fault: ERR-MAK250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250.

PRM_41384 [Makino-Aero-T250] — HYD_LOOP_GAIN_4 | Default: 239.200 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41385 [Makino-Aero-T250] — ZAX_LOOP_GAIN_5 | Default: 49.000 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

PRM_41386 [Makino-Aero-T250] — PLC_LOOP_GAIN_6 | Default: 108.800 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

PRM_41387 [Makino-Aero-T250] — SPN_LOOP_GAIN_7 | Default: 168.600 | Linked Fault: ERR-MAK250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T250.

PRM_41388 [Makino-Aero-T250] — XAX_LOOP_GAIN_8 | Default: 228.400 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41389 [Makino-Aero-T250] — CAX_LOOP_GAIN_9 | Default: 38.200 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

PRM_41390 [Makino-Aero-T250] — VAC_LOOP_GAIN_10 | Default: 98.000 | Linked Fault: ERR-MAK250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250.

PRM_41391 [Makino-Aero-T250] — HYD_LOOP_GAIN_11 | Default: 157.800 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41392 [Makino-Aero-T250] — ZAX_LOOP_GAIN_12 | Default: 217.600 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

PRM_41393 [Makino-Aero-T250] — PLC_LOOP_GAIN_13 | Default: 27.400 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

PRM_41394 [Makino-Aero-T250] — SPN_LOOP_GAIN_14 | Default: 87.200 | Linked Fault: ERR-MAK250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T250.

PRM_41395 [Makino-Aero-T250] — XAX_LOOP_GAIN_15 | Default: 147.000 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41396 [Makino-Aero-T250] — CAX_LOOP_GAIN_16 | Default: 206.800 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T250 (BANK 5)

Machine Model: Makino-Aero-T250 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T250.

PRM_41401 [Makino-Aero-T250] — YAX_LOOP_GAIN_1 | Default: 74.750 | Linked Fault: ERR-MAK250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T250.

PRM_41402 [Makino-Aero-T250] — PLC_LOOP_GAIN_2 | Default: 149.500 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

PRM_41403 [Makino-Aero-T250] — ATC_LOOP_GAIN_3 | Default: 224.250 | Linked Fault: ERR-MAK250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T250.

PRM_41404 [Makino-Aero-T250] — ZAX_LOOP_GAIN_4 | Default: 49.000 | Linked Fault: ERR-MAK250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T250.

PRM_41405 [Makino-Aero-T250] — LUB_LOOP_GAIN_5 | Default: 123.750 | Linked Fault: ERR-MAK250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T250.

PRM_41406 [Makino-Aero-T250] — HYD_LOOP_GAIN_6 | Default: 198.500 | Linked Fault: ERR-MAK250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T250.

PRM_41407 [Makino-Aero-T250] — BAX_LOOP_GAIN_7 | Default: 23.250 | Linked Fault: ERR-MAK250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T250.

PRM_41408 [Makino-Aero-T250] — VAC_LOOP_GAIN_8 | Default: 98.000 | Linked Fault: ERR-MAK250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T250.

PRM_41409 [Makino-Aero-T250] — CLN_LOOP_GAIN_9 | Default: 172.750 | Linked Fault: ERR-MAK250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T250.

PRM_41410 [Makino-Aero-T250] — CAX_LOOP_GAIN_10 | Default: 247.500 | Linked Fault: ERR-MAK250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T250.

PRM_41411 [Makino-Aero-T250] — THM_LOOP_GAIN_11 | Default: 72.250 | Linked Fault: ERR-MAK250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T250.

PRM_41412 [Makino-Aero-T250] — XAX_LOOP_GAIN_12 | Default: 147.000 | Linked Fault: ERR-MAK250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T250.

PRM_41413 [Makino-Aero-T250] — OPT_LOOP_GAIN_13 | Default: 221.750 | Linked Fault: ERR-MAK250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T250.

PRM_41414 [Makino-Aero-T250] — SPN_LOOP_GAIN_14 | Default: 46.500 | Linked Fault: ERR-MAK250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T250.

PRM_41415 [Makino-Aero-T250] — YAX_LOOP_GAIN_15 | Default: 121.250 | Linked Fault: ERR-MAK250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T250.

PRM_41416 [Makino-Aero-T250] — PLC_LOOP_GAIN_16 | Default: 196.000 | Linked Fault: ERR-MAK250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T250.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T300 (BANK 1)

Machine Model: Makino-Aero-T300 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T300.

PRM_41421 [Makino-Aero-T300] — ATC_LOOP_GAIN_1 | Default: 16.100 | Linked Fault: ERR-MAK300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T300.

PRM_41422 [Makino-Aero-T300] — HYD_LOOP_GAIN_2 | Default: 32.200 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41423 [Makino-Aero-T300] — CLN_LOOP_GAIN_3 | Default: 48.300 | Linked Fault: ERR-MAK300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T300.

PRM_41424 [Makino-Aero-T300] — XAX_LOOP_GAIN_4 | Default: 64.400 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41425 [Makino-Aero-T300] — YAX_LOOP_GAIN_5 | Default: 80.500 | Linked Fault: ERR-MAK300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T300.

PRM_41426 [Makino-Aero-T300] — ZAX_LOOP_GAIN_6 | Default: 96.600 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

PRM_41427 [Makino-Aero-T300] — BAX_LOOP_GAIN_7 | Default: 112.700 | Linked Fault: ERR-MAK300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T300.

PRM_41428 [Makino-Aero-T300] — CAX_LOOP_GAIN_8 | Default: 128.800 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

PRM_41429 [Makino-Aero-T300] — OPT_LOOP_GAIN_9 | Default: 144.900 | Linked Fault: ERR-MAK300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T300.

PRM_41430 [Makino-Aero-T300] — PLC_LOOP_GAIN_10 | Default: 161.000 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

PRM_41431 [Makino-Aero-T300] — LUB_LOOP_GAIN_11 | Default: 177.100 | Linked Fault: ERR-MAK300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T300.

PRM_41432 [Makino-Aero-T300] — VAC_LOOP_GAIN_12 | Default: 193.200 | Linked Fault: ERR-MAK300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300.

PRM_41433 [Makino-Aero-T300] — THM_LOOP_GAIN_13 | Default: 209.300 | Linked Fault: ERR-MAK300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T300.

PRM_41434 [Makino-Aero-T300] — SPN_LOOP_GAIN_14 | Default: 225.400 | Linked Fault: ERR-MAK300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T300.

PRM_41435 [Makino-Aero-T300] — ATC_LOOP_GAIN_15 | Default: 241.500 | Linked Fault: ERR-MAK300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T300.

PRM_41436 [Makino-Aero-T300] — HYD_LOOP_GAIN_16 | Default: 7.600 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T300 (BANK 2)

Machine Model: Makino-Aero-T300 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T300.

PRM_41441 [Makino-Aero-T300] — HYD_LOOP_GAIN_1 | Default: 32.200 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41442 [Makino-Aero-T300] — XAX_LOOP_GAIN_2 | Default: 64.400 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41443 [Makino-Aero-T300] — ZAX_LOOP_GAIN_3 | Default: 96.600 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

PRM_41444 [Makino-Aero-T300] — CAX_LOOP_GAIN_4 | Default: 128.800 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

PRM_41445 [Makino-Aero-T300] — PLC_LOOP_GAIN_5 | Default: 161.000 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

PRM_41446 [Makino-Aero-T300] — VAC_LOOP_GAIN_6 | Default: 193.200 | Linked Fault: ERR-MAK300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300.

PRM_41447 [Makino-Aero-T300] — SPN_LOOP_GAIN_7 | Default: 225.400 | Linked Fault: ERR-MAK300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T300.

PRM_41448 [Makino-Aero-T300] — HYD_LOOP_GAIN_8 | Default: 7.600 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41449 [Makino-Aero-T300] — XAX_LOOP_GAIN_9 | Default: 39.800 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41450 [Makino-Aero-T300] — ZAX_LOOP_GAIN_10 | Default: 72.000 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

PRM_41451 [Makino-Aero-T300] — CAX_LOOP_GAIN_11 | Default: 104.200 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

PRM_41452 [Makino-Aero-T300] — PLC_LOOP_GAIN_12 | Default: 136.400 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

PRM_41453 [Makino-Aero-T300] — VAC_LOOP_GAIN_13 | Default: 168.600 | Linked Fault: ERR-MAK300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300.

PRM_41454 [Makino-Aero-T300] — SPN_LOOP_GAIN_14 | Default: 200.800 | Linked Fault: ERR-MAK300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T300.

PRM_41455 [Makino-Aero-T300] — HYD_LOOP_GAIN_15 | Default: 233.000 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41456 [Makino-Aero-T300] — XAX_LOOP_GAIN_16 | Default: 15.200 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T300 (BANK 3)

Machine Model: Makino-Aero-T300 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T300.

PRM_41461 [Makino-Aero-T300] — CLN_LOOP_GAIN_1 | Default: 48.300 | Linked Fault: ERR-MAK300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T300.

PRM_41462 [Makino-Aero-T300] — ZAX_LOOP_GAIN_2 | Default: 96.600 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

PRM_41463 [Makino-Aero-T300] — OPT_LOOP_GAIN_3 | Default: 144.900 | Linked Fault: ERR-MAK300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T300.

PRM_41464 [Makino-Aero-T300] — VAC_LOOP_GAIN_4 | Default: 193.200 | Linked Fault: ERR-MAK300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300.

PRM_41465 [Makino-Aero-T300] — ATC_LOOP_GAIN_5 | Default: 241.500 | Linked Fault: ERR-MAK300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T300.

PRM_41466 [Makino-Aero-T300] — XAX_LOOP_GAIN_6 | Default: 39.800 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41467 [Makino-Aero-T300] — BAX_LOOP_GAIN_7 | Default: 88.100 | Linked Fault: ERR-MAK300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T300.

PRM_41468 [Makino-Aero-T300] — PLC_LOOP_GAIN_8 | Default: 136.400 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

PRM_41469 [Makino-Aero-T300] — THM_LOOP_GAIN_9 | Default: 184.700 | Linked Fault: ERR-MAK300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T300.

PRM_41470 [Makino-Aero-T300] — HYD_LOOP_GAIN_10 | Default: 233.000 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41471 [Makino-Aero-T300] — YAX_LOOP_GAIN_11 | Default: 31.300 | Linked Fault: ERR-MAK300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T300.

PRM_41472 [Makino-Aero-T300] — CAX_LOOP_GAIN_12 | Default: 79.600 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

PRM_41473 [Makino-Aero-T300] — LUB_LOOP_GAIN_13 | Default: 127.900 | Linked Fault: ERR-MAK300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T300.

PRM_41474 [Makino-Aero-T300] — SPN_LOOP_GAIN_14 | Default: 176.200 | Linked Fault: ERR-MAK300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T300.

PRM_41475 [Makino-Aero-T300] — CLN_LOOP_GAIN_15 | Default: 224.500 | Linked Fault: ERR-MAK300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T300.

PRM_41476 [Makino-Aero-T300] — ZAX_LOOP_GAIN_16 | Default: 22.800 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T300 (BANK 4)

Machine Model: Makino-Aero-T300 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T300.

PRM_41481 [Makino-Aero-T300] — XAX_LOOP_GAIN_1 | Default: 64.400 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41482 [Makino-Aero-T300] — CAX_LOOP_GAIN_2 | Default: 128.800 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

PRM_41483 [Makino-Aero-T300] — VAC_LOOP_GAIN_3 | Default: 193.200 | Linked Fault: ERR-MAK300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300.

PRM_41484 [Makino-Aero-T300] — HYD_LOOP_GAIN_4 | Default: 7.600 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41485 [Makino-Aero-T300] — ZAX_LOOP_GAIN_5 | Default: 72.000 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

PRM_41486 [Makino-Aero-T300] — PLC_LOOP_GAIN_6 | Default: 136.400 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

PRM_41487 [Makino-Aero-T300] — SPN_LOOP_GAIN_7 | Default: 200.800 | Linked Fault: ERR-MAK300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T300.

PRM_41488 [Makino-Aero-T300] — XAX_LOOP_GAIN_8 | Default: 15.200 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41489 [Makino-Aero-T300] — CAX_LOOP_GAIN_9 | Default: 79.600 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

PRM_41490 [Makino-Aero-T300] — VAC_LOOP_GAIN_10 | Default: 144.000 | Linked Fault: ERR-MAK300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300.

PRM_41491 [Makino-Aero-T300] — HYD_LOOP_GAIN_11 | Default: 208.400 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41492 [Makino-Aero-T300] — ZAX_LOOP_GAIN_12 | Default: 22.800 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

PRM_41493 [Makino-Aero-T300] — PLC_LOOP_GAIN_13 | Default: 87.200 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

PRM_41494 [Makino-Aero-T300] — SPN_LOOP_GAIN_14 | Default: 151.600 | Linked Fault: ERR-MAK300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T300.

PRM_41495 [Makino-Aero-T300] — XAX_LOOP_GAIN_15 | Default: 216.000 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41496 [Makino-Aero-T300] — CAX_LOOP_GAIN_16 | Default: 30.400 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T300 (BANK 5)

Machine Model: Makino-Aero-T300 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T300.

PRM_41501 [Makino-Aero-T300] — YAX_LOOP_GAIN_1 | Default: 80.500 | Linked Fault: ERR-MAK300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T300.

PRM_41502 [Makino-Aero-T300] — PLC_LOOP_GAIN_2 | Default: 161.000 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

PRM_41503 [Makino-Aero-T300] — ATC_LOOP_GAIN_3 | Default: 241.500 | Linked Fault: ERR-MAK300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T300.

PRM_41504 [Makino-Aero-T300] — ZAX_LOOP_GAIN_4 | Default: 72.000 | Linked Fault: ERR-MAK300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T300.

PRM_41505 [Makino-Aero-T300] — LUB_LOOP_GAIN_5 | Default: 152.500 | Linked Fault: ERR-MAK300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T300.

PRM_41506 [Makino-Aero-T300] — HYD_LOOP_GAIN_6 | Default: 233.000 | Linked Fault: ERR-MAK300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T300.

PRM_41507 [Makino-Aero-T300] — BAX_LOOP_GAIN_7 | Default: 63.500 | Linked Fault: ERR-MAK300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T300.

PRM_41508 [Makino-Aero-T300] — VAC_LOOP_GAIN_8 | Default: 144.000 | Linked Fault: ERR-MAK300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T300.

PRM_41509 [Makino-Aero-T300] — CLN_LOOP_GAIN_9 | Default: 224.500 | Linked Fault: ERR-MAK300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T300.

PRM_41510 [Makino-Aero-T300] — CAX_LOOP_GAIN_10 | Default: 55.000 | Linked Fault: ERR-MAK300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T300.

PRM_41511 [Makino-Aero-T300] — THM_LOOP_GAIN_11 | Default: 135.500 | Linked Fault: ERR-MAK300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T300.

PRM_41512 [Makino-Aero-T300] — XAX_LOOP_GAIN_12 | Default: 216.000 | Linked Fault: ERR-MAK300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T300.

PRM_41513 [Makino-Aero-T300] — OPT_LOOP_GAIN_13 | Default: 46.500 | Linked Fault: ERR-MAK300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T300.

PRM_41514 [Makino-Aero-T300] — SPN_LOOP_GAIN_14 | Default: 127.000 | Linked Fault: ERR-MAK300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T300.

PRM_41515 [Makino-Aero-T300] — YAX_LOOP_GAIN_15 | Default: 207.500 | Linked Fault: ERR-MAK300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T300.

PRM_41516 [Makino-Aero-T300] — PLC_LOOP_GAIN_16 | Default: 38.000 | Linked Fault: ERR-MAK300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T300.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T350 (BANK 1)

Machine Model: Makino-Aero-T350 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T350.

PRM_41521 [Makino-Aero-T350] — ATC_LOOP_GAIN_1 | Default: 17.250 | Linked Fault: ERR-MAK350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T350.

PRM_41522 [Makino-Aero-T350] — HYD_LOOP_GAIN_2 | Default: 34.500 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41523 [Makino-Aero-T350] — CLN_LOOP_GAIN_3 | Default: 51.750 | Linked Fault: ERR-MAK350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T350.

PRM_41524 [Makino-Aero-T350] — XAX_LOOP_GAIN_4 | Default: 69.000 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41525 [Makino-Aero-T350] — YAX_LOOP_GAIN_5 | Default: 86.250 | Linked Fault: ERR-MAK350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T350.

PRM_41526 [Makino-Aero-T350] — ZAX_LOOP_GAIN_6 | Default: 103.500 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

PRM_41527 [Makino-Aero-T350] — BAX_LOOP_GAIN_7 | Default: 120.750 | Linked Fault: ERR-MAK350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T350.

PRM_41528 [Makino-Aero-T350] — CAX_LOOP_GAIN_8 | Default: 138.000 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

PRM_41529 [Makino-Aero-T350] — OPT_LOOP_GAIN_9 | Default: 155.250 | Linked Fault: ERR-MAK350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T350.

PRM_41530 [Makino-Aero-T350] — PLC_LOOP_GAIN_10 | Default: 172.500 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

PRM_41531 [Makino-Aero-T350] — LUB_LOOP_GAIN_11 | Default: 189.750 | Linked Fault: ERR-MAK350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T350.

PRM_41532 [Makino-Aero-T350] — VAC_LOOP_GAIN_12 | Default: 207.000 | Linked Fault: ERR-MAK350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350.

PRM_41533 [Makino-Aero-T350] — THM_LOOP_GAIN_13 | Default: 224.250 | Linked Fault: ERR-MAK350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T350.

PRM_41534 [Makino-Aero-T350] — SPN_LOOP_GAIN_14 | Default: 241.500 | Linked Fault: ERR-MAK350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T350.

PRM_41535 [Makino-Aero-T350] — ATC_LOOP_GAIN_15 | Default: 8.750 | Linked Fault: ERR-MAK350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T350.

PRM_41536 [Makino-Aero-T350] — HYD_LOOP_GAIN_16 | Default: 26.000 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T350 (BANK 2)

Machine Model: Makino-Aero-T350 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T350.

PRM_41541 [Makino-Aero-T350] — HYD_LOOP_GAIN_1 | Default: 34.500 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41542 [Makino-Aero-T350] — XAX_LOOP_GAIN_2 | Default: 69.000 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41543 [Makino-Aero-T350] — ZAX_LOOP_GAIN_3 | Default: 103.500 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

PRM_41544 [Makino-Aero-T350] — CAX_LOOP_GAIN_4 | Default: 138.000 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

PRM_41545 [Makino-Aero-T350] — PLC_LOOP_GAIN_5 | Default: 172.500 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

PRM_41546 [Makino-Aero-T350] — VAC_LOOP_GAIN_6 | Default: 207.000 | Linked Fault: ERR-MAK350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350.

PRM_41547 [Makino-Aero-T350] — SPN_LOOP_GAIN_7 | Default: 241.500 | Linked Fault: ERR-MAK350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T350.

PRM_41548 [Makino-Aero-T350] — HYD_LOOP_GAIN_8 | Default: 26.000 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41549 [Makino-Aero-T350] — XAX_LOOP_GAIN_9 | Default: 60.500 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41550 [Makino-Aero-T350] — ZAX_LOOP_GAIN_10 | Default: 95.000 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

PRM_41551 [Makino-Aero-T350] — CAX_LOOP_GAIN_11 | Default: 129.500 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

PRM_41552 [Makino-Aero-T350] — PLC_LOOP_GAIN_12 | Default: 164.000 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

PRM_41553 [Makino-Aero-T350] — VAC_LOOP_GAIN_13 | Default: 198.500 | Linked Fault: ERR-MAK350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350.

PRM_41554 [Makino-Aero-T350] — SPN_LOOP_GAIN_14 | Default: 233.000 | Linked Fault: ERR-MAK350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T350.

PRM_41555 [Makino-Aero-T350] — HYD_LOOP_GAIN_15 | Default: 17.500 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41556 [Makino-Aero-T350] — XAX_LOOP_GAIN_16 | Default: 52.000 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T350 (BANK 3)

Machine Model: Makino-Aero-T350 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T350.

PRM_41561 [Makino-Aero-T350] — CLN_LOOP_GAIN_1 | Default: 51.750 | Linked Fault: ERR-MAK350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T350.

PRM_41562 [Makino-Aero-T350] — ZAX_LOOP_GAIN_2 | Default: 103.500 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

PRM_41563 [Makino-Aero-T350] — OPT_LOOP_GAIN_3 | Default: 155.250 | Linked Fault: ERR-MAK350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T350.

PRM_41564 [Makino-Aero-T350] — VAC_LOOP_GAIN_4 | Default: 207.000 | Linked Fault: ERR-MAK350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350.

PRM_41565 [Makino-Aero-T350] — ATC_LOOP_GAIN_5 | Default: 8.750 | Linked Fault: ERR-MAK350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T350.

PRM_41566 [Makino-Aero-T350] — XAX_LOOP_GAIN_6 | Default: 60.500 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41567 [Makino-Aero-T350] — BAX_LOOP_GAIN_7 | Default: 112.250 | Linked Fault: ERR-MAK350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T350.

PRM_41568 [Makino-Aero-T350] — PLC_LOOP_GAIN_8 | Default: 164.000 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

PRM_41569 [Makino-Aero-T350] — THM_LOOP_GAIN_9 | Default: 215.750 | Linked Fault: ERR-MAK350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T350.

PRM_41570 [Makino-Aero-T350] — HYD_LOOP_GAIN_10 | Default: 17.500 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41571 [Makino-Aero-T350] — YAX_LOOP_GAIN_11 | Default: 69.250 | Linked Fault: ERR-MAK350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T350.

PRM_41572 [Makino-Aero-T350] — CAX_LOOP_GAIN_12 | Default: 121.000 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

PRM_41573 [Makino-Aero-T350] — LUB_LOOP_GAIN_13 | Default: 172.750 | Linked Fault: ERR-MAK350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T350.

PRM_41574 [Makino-Aero-T350] — SPN_LOOP_GAIN_14 | Default: 224.500 | Linked Fault: ERR-MAK350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T350.

PRM_41575 [Makino-Aero-T350] — CLN_LOOP_GAIN_15 | Default: 26.250 | Linked Fault: ERR-MAK350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T350.

PRM_41576 [Makino-Aero-T350] — ZAX_LOOP_GAIN_16 | Default: 78.000 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T350 (BANK 4)

Machine Model: Makino-Aero-T350 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T350.

PRM_41581 [Makino-Aero-T350] — XAX_LOOP_GAIN_1 | Default: 69.000 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41582 [Makino-Aero-T350] — CAX_LOOP_GAIN_2 | Default: 138.000 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

PRM_41583 [Makino-Aero-T350] — VAC_LOOP_GAIN_3 | Default: 207.000 | Linked Fault: ERR-MAK350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350.

PRM_41584 [Makino-Aero-T350] — HYD_LOOP_GAIN_4 | Default: 26.000 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41585 [Makino-Aero-T350] — ZAX_LOOP_GAIN_5 | Default: 95.000 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

PRM_41586 [Makino-Aero-T350] — PLC_LOOP_GAIN_6 | Default: 164.000 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

PRM_41587 [Makino-Aero-T350] — SPN_LOOP_GAIN_7 | Default: 233.000 | Linked Fault: ERR-MAK350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T350.

PRM_41588 [Makino-Aero-T350] — XAX_LOOP_GAIN_8 | Default: 52.000 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41589 [Makino-Aero-T350] — CAX_LOOP_GAIN_9 | Default: 121.000 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

PRM_41590 [Makino-Aero-T350] — VAC_LOOP_GAIN_10 | Default: 190.000 | Linked Fault: ERR-MAK350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350.

PRM_41591 [Makino-Aero-T350] — HYD_LOOP_GAIN_11 | Default: 9.000 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41592 [Makino-Aero-T350] — ZAX_LOOP_GAIN_12 | Default: 78.000 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

PRM_41593 [Makino-Aero-T350] — PLC_LOOP_GAIN_13 | Default: 147.000 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

PRM_41594 [Makino-Aero-T350] — SPN_LOOP_GAIN_14 | Default: 216.000 | Linked Fault: ERR-MAK350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T350.

PRM_41595 [Makino-Aero-T350] — XAX_LOOP_GAIN_15 | Default: 35.000 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41596 [Makino-Aero-T350] — CAX_LOOP_GAIN_16 | Default: 104.000 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T350 (BANK 5)

Machine Model: Makino-Aero-T350 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T350.

PRM_41601 [Makino-Aero-T350] — YAX_LOOP_GAIN_1 | Default: 86.250 | Linked Fault: ERR-MAK350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T350.

PRM_41602 [Makino-Aero-T350] — PLC_LOOP_GAIN_2 | Default: 172.500 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

PRM_41603 [Makino-Aero-T350] — ATC_LOOP_GAIN_3 | Default: 8.750 | Linked Fault: ERR-MAK350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T350.

PRM_41604 [Makino-Aero-T350] — ZAX_LOOP_GAIN_4 | Default: 95.000 | Linked Fault: ERR-MAK350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T350.

PRM_41605 [Makino-Aero-T350] — LUB_LOOP_GAIN_5 | Default: 181.250 | Linked Fault: ERR-MAK350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T350.

PRM_41606 [Makino-Aero-T350] — HYD_LOOP_GAIN_6 | Default: 17.500 | Linked Fault: ERR-MAK350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T350.

PRM_41607 [Makino-Aero-T350] — BAX_LOOP_GAIN_7 | Default: 103.750 | Linked Fault: ERR-MAK350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T350.

PRM_41608 [Makino-Aero-T350] — VAC_LOOP_GAIN_8 | Default: 190.000 | Linked Fault: ERR-MAK350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T350.

PRM_41609 [Makino-Aero-T350] — CLN_LOOP_GAIN_9 | Default: 26.250 | Linked Fault: ERR-MAK350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T350.

PRM_41610 [Makino-Aero-T350] — CAX_LOOP_GAIN_10 | Default: 112.500 | Linked Fault: ERR-MAK350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T350.

PRM_41611 [Makino-Aero-T350] — THM_LOOP_GAIN_11 | Default: 198.750 | Linked Fault: ERR-MAK350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T350.

PRM_41612 [Makino-Aero-T350] — XAX_LOOP_GAIN_12 | Default: 35.000 | Linked Fault: ERR-MAK350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T350.

PRM_41613 [Makino-Aero-T350] — OPT_LOOP_GAIN_13 | Default: 121.250 | Linked Fault: ERR-MAK350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T350.

PRM_41614 [Makino-Aero-T350] — SPN_LOOP_GAIN_14 | Default: 207.500 | Linked Fault: ERR-MAK350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T350.

PRM_41615 [Makino-Aero-T350] — YAX_LOOP_GAIN_15 | Default: 43.750 | Linked Fault: ERR-MAK350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T350.

PRM_41616 [Makino-Aero-T350] — PLC_LOOP_GAIN_16 | Default: 130.000 | Linked Fault: ERR-MAK350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T350.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T400 (BANK 1)

Machine Model: Makino-Aero-T400 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T400.

PRM_41621 [Makino-Aero-T400] — ATC_LOOP_GAIN_1 | Default: 18.400 | Linked Fault: ERR-MAK400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T400.

PRM_41622 [Makino-Aero-T400] — HYD_LOOP_GAIN_2 | Default: 36.800 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41623 [Makino-Aero-T400] — CLN_LOOP_GAIN_3 | Default: 55.200 | Linked Fault: ERR-MAK400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T400.

PRM_41624 [Makino-Aero-T400] — XAX_LOOP_GAIN_4 | Default: 73.600 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41625 [Makino-Aero-T400] — YAX_LOOP_GAIN_5 | Default: 92.000 | Linked Fault: ERR-MAK400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T400.

PRM_41626 [Makino-Aero-T400] — ZAX_LOOP_GAIN_6 | Default: 110.400 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

PRM_41627 [Makino-Aero-T400] — BAX_LOOP_GAIN_7 | Default: 128.800 | Linked Fault: ERR-MAK400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T400.

PRM_41628 [Makino-Aero-T400] — CAX_LOOP_GAIN_8 | Default: 147.200 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

PRM_41629 [Makino-Aero-T400] — OPT_LOOP_GAIN_9 | Default: 165.600 | Linked Fault: ERR-MAK400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T400.

PRM_41630 [Makino-Aero-T400] — PLC_LOOP_GAIN_10 | Default: 184.000 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

PRM_41631 [Makino-Aero-T400] — LUB_LOOP_GAIN_11 | Default: 202.400 | Linked Fault: ERR-MAK400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T400.

PRM_41632 [Makino-Aero-T400] — VAC_LOOP_GAIN_12 | Default: 220.800 | Linked Fault: ERR-MAK400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400.

PRM_41633 [Makino-Aero-T400] — THM_LOOP_GAIN_13 | Default: 239.200 | Linked Fault: ERR-MAK400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T400.

PRM_41634 [Makino-Aero-T400] — SPN_LOOP_GAIN_14 | Default: 7.600 | Linked Fault: ERR-MAK400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T400.

PRM_41635 [Makino-Aero-T400] — ATC_LOOP_GAIN_15 | Default: 26.000 | Linked Fault: ERR-MAK400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T400.

PRM_41636 [Makino-Aero-T400] — HYD_LOOP_GAIN_16 | Default: 44.400 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T400 (BANK 2)

Machine Model: Makino-Aero-T400 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T400.

PRM_41641 [Makino-Aero-T400] — HYD_LOOP_GAIN_1 | Default: 36.800 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41642 [Makino-Aero-T400] — XAX_LOOP_GAIN_2 | Default: 73.600 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41643 [Makino-Aero-T400] — ZAX_LOOP_GAIN_3 | Default: 110.400 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

PRM_41644 [Makino-Aero-T400] — CAX_LOOP_GAIN_4 | Default: 147.200 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

PRM_41645 [Makino-Aero-T400] — PLC_LOOP_GAIN_5 | Default: 184.000 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

PRM_41646 [Makino-Aero-T400] — VAC_LOOP_GAIN_6 | Default: 220.800 | Linked Fault: ERR-MAK400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400.

PRM_41647 [Makino-Aero-T400] — SPN_LOOP_GAIN_7 | Default: 7.600 | Linked Fault: ERR-MAK400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T400.

PRM_41648 [Makino-Aero-T400] — HYD_LOOP_GAIN_8 | Default: 44.400 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41649 [Makino-Aero-T400] — XAX_LOOP_GAIN_9 | Default: 81.200 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41650 [Makino-Aero-T400] — ZAX_LOOP_GAIN_10 | Default: 118.000 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

PRM_41651 [Makino-Aero-T400] — CAX_LOOP_GAIN_11 | Default: 154.800 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

PRM_41652 [Makino-Aero-T400] — PLC_LOOP_GAIN_12 | Default: 191.600 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

PRM_41653 [Makino-Aero-T400] — VAC_LOOP_GAIN_13 | Default: 228.400 | Linked Fault: ERR-MAK400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400.

PRM_41654 [Makino-Aero-T400] — SPN_LOOP_GAIN_14 | Default: 15.200 | Linked Fault: ERR-MAK400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T400.

PRM_41655 [Makino-Aero-T400] — HYD_LOOP_GAIN_15 | Default: 52.000 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41656 [Makino-Aero-T400] — XAX_LOOP_GAIN_16 | Default: 88.800 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T400 (BANK 3)

Machine Model: Makino-Aero-T400 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T400.

PRM_41661 [Makino-Aero-T400] — CLN_LOOP_GAIN_1 | Default: 55.200 | Linked Fault: ERR-MAK400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T400.

PRM_41662 [Makino-Aero-T400] — ZAX_LOOP_GAIN_2 | Default: 110.400 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

PRM_41663 [Makino-Aero-T400] — OPT_LOOP_GAIN_3 | Default: 165.600 | Linked Fault: ERR-MAK400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T400.

PRM_41664 [Makino-Aero-T400] — VAC_LOOP_GAIN_4 | Default: 220.800 | Linked Fault: ERR-MAK400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400.

PRM_41665 [Makino-Aero-T400] — ATC_LOOP_GAIN_5 | Default: 26.000 | Linked Fault: ERR-MAK400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T400.

PRM_41666 [Makino-Aero-T400] — XAX_LOOP_GAIN_6 | Default: 81.200 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41667 [Makino-Aero-T400] — BAX_LOOP_GAIN_7 | Default: 136.400 | Linked Fault: ERR-MAK400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T400.

PRM_41668 [Makino-Aero-T400] — PLC_LOOP_GAIN_8 | Default: 191.600 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

PRM_41669 [Makino-Aero-T400] — THM_LOOP_GAIN_9 | Default: 246.800 | Linked Fault: ERR-MAK400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T400.

PRM_41670 [Makino-Aero-T400] — HYD_LOOP_GAIN_10 | Default: 52.000 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41671 [Makino-Aero-T400] — YAX_LOOP_GAIN_11 | Default: 107.200 | Linked Fault: ERR-MAK400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T400.

PRM_41672 [Makino-Aero-T400] — CAX_LOOP_GAIN_12 | Default: 162.400 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

PRM_41673 [Makino-Aero-T400] — LUB_LOOP_GAIN_13 | Default: 217.600 | Linked Fault: ERR-MAK400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T400.

PRM_41674 [Makino-Aero-T400] — SPN_LOOP_GAIN_14 | Default: 22.800 | Linked Fault: ERR-MAK400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T400.

PRM_41675 [Makino-Aero-T400] — CLN_LOOP_GAIN_15 | Default: 78.000 | Linked Fault: ERR-MAK400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T400.

PRM_41676 [Makino-Aero-T400] — ZAX_LOOP_GAIN_16 | Default: 133.200 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T400 (BANK 4)

Machine Model: Makino-Aero-T400 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T400.

PRM_41681 [Makino-Aero-T400] — XAX_LOOP_GAIN_1 | Default: 73.600 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41682 [Makino-Aero-T400] — CAX_LOOP_GAIN_2 | Default: 147.200 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

PRM_41683 [Makino-Aero-T400] — VAC_LOOP_GAIN_3 | Default: 220.800 | Linked Fault: ERR-MAK400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400.

PRM_41684 [Makino-Aero-T400] — HYD_LOOP_GAIN_4 | Default: 44.400 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41685 [Makino-Aero-T400] — ZAX_LOOP_GAIN_5 | Default: 118.000 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

PRM_41686 [Makino-Aero-T400] — PLC_LOOP_GAIN_6 | Default: 191.600 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

PRM_41687 [Makino-Aero-T400] — SPN_LOOP_GAIN_7 | Default: 15.200 | Linked Fault: ERR-MAK400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T400.

PRM_41688 [Makino-Aero-T400] — XAX_LOOP_GAIN_8 | Default: 88.800 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41689 [Makino-Aero-T400] — CAX_LOOP_GAIN_9 | Default: 162.400 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

PRM_41690 [Makino-Aero-T400] — VAC_LOOP_GAIN_10 | Default: 236.000 | Linked Fault: ERR-MAK400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400.

PRM_41691 [Makino-Aero-T400] — HYD_LOOP_GAIN_11 | Default: 59.600 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41692 [Makino-Aero-T400] — ZAX_LOOP_GAIN_12 | Default: 133.200 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

PRM_41693 [Makino-Aero-T400] — PLC_LOOP_GAIN_13 | Default: 206.800 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

PRM_41694 [Makino-Aero-T400] — SPN_LOOP_GAIN_14 | Default: 30.400 | Linked Fault: ERR-MAK400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T400.

PRM_41695 [Makino-Aero-T400] — XAX_LOOP_GAIN_15 | Default: 104.000 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41696 [Makino-Aero-T400] — CAX_LOOP_GAIN_16 | Default: 177.600 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T400 (BANK 5)

Machine Model: Makino-Aero-T400 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T400.

PRM_41701 [Makino-Aero-T400] — YAX_LOOP_GAIN_1 | Default: 92.000 | Linked Fault: ERR-MAK400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T400.

PRM_41702 [Makino-Aero-T400] — PLC_LOOP_GAIN_2 | Default: 184.000 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

PRM_41703 [Makino-Aero-T400] — ATC_LOOP_GAIN_3 | Default: 26.000 | Linked Fault: ERR-MAK400-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T400.

PRM_41704 [Makino-Aero-T400] — ZAX_LOOP_GAIN_4 | Default: 118.000 | Linked Fault: ERR-MAK400-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T400.

PRM_41705 [Makino-Aero-T400] — LUB_LOOP_GAIN_5 | Default: 210.000 | Linked Fault: ERR-MAK400-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T400.

PRM_41706 [Makino-Aero-T400] — HYD_LOOP_GAIN_6 | Default: 52.000 | Linked Fault: ERR-MAK400-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T400.

PRM_41707 [Makino-Aero-T400] — BAX_LOOP_GAIN_7 | Default: 144.000 | Linked Fault: ERR-MAK400-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T400.

PRM_41708 [Makino-Aero-T400] — VAC_LOOP_GAIN_8 | Default: 236.000 | Linked Fault: ERR-MAK400-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T400.

PRM_41709 [Makino-Aero-T400] — CLN_LOOP_GAIN_9 | Default: 78.000 | Linked Fault: ERR-MAK400-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T400.

PRM_41710 [Makino-Aero-T400] — CAX_LOOP_GAIN_10 | Default: 170.000 | Linked Fault: ERR-MAK400-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T400.

PRM_41711 [Makino-Aero-T400] — THM_LOOP_GAIN_11 | Default: 12.000 | Linked Fault: ERR-MAK400-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T400.

PRM_41712 [Makino-Aero-T400] — XAX_LOOP_GAIN_12 | Default: 104.000 | Linked Fault: ERR-MAK400-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T400.

PRM_41713 [Makino-Aero-T400] — OPT_LOOP_GAIN_13 | Default: 196.000 | Linked Fault: ERR-MAK400-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T400.

PRM_41714 [Makino-Aero-T400] — SPN_LOOP_GAIN_14 | Default: 38.000 | Linked Fault: ERR-MAK400-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T400.

PRM_41715 [Makino-Aero-T400] — YAX_LOOP_GAIN_15 | Default: 130.000 | Linked Fault: ERR-MAK400-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T400.

PRM_41716 [Makino-Aero-T400] — PLC_LOOP_GAIN_16 | Default: 222.000 | Linked Fault: ERR-MAK400-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T400.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T450 (BANK 1)

Machine Model: Makino-Aero-T450 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T450.

PRM_41721 [Makino-Aero-T450] — ATC_LOOP_GAIN_1 | Default: 19.550 | Linked Fault: ERR-MAK450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T450.

PRM_41722 [Makino-Aero-T450] — HYD_LOOP_GAIN_2 | Default: 39.100 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41723 [Makino-Aero-T450] — CLN_LOOP_GAIN_3 | Default: 58.650 | Linked Fault: ERR-MAK450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T450.

PRM_41724 [Makino-Aero-T450] — XAX_LOOP_GAIN_4 | Default: 78.200 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41725 [Makino-Aero-T450] — YAX_LOOP_GAIN_5 | Default: 97.750 | Linked Fault: ERR-MAK450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T450.

PRM_41726 [Makino-Aero-T450] — ZAX_LOOP_GAIN_6 | Default: 117.300 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

PRM_41727 [Makino-Aero-T450] — BAX_LOOP_GAIN_7 | Default: 136.850 | Linked Fault: ERR-MAK450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T450.

PRM_41728 [Makino-Aero-T450] — CAX_LOOP_GAIN_8 | Default: 156.400 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

PRM_41729 [Makino-Aero-T450] — OPT_LOOP_GAIN_9 | Default: 175.950 | Linked Fault: ERR-MAK450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T450.

PRM_41730 [Makino-Aero-T450] — PLC_LOOP_GAIN_10 | Default: 195.500 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

PRM_41731 [Makino-Aero-T450] — LUB_LOOP_GAIN_11 | Default: 215.050 | Linked Fault: ERR-MAK450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T450.

PRM_41732 [Makino-Aero-T450] — VAC_LOOP_GAIN_12 | Default: 234.600 | Linked Fault: ERR-MAK450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450.

PRM_41733 [Makino-Aero-T450] — THM_LOOP_GAIN_13 | Default: 4.150 | Linked Fault: ERR-MAK450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T450.

PRM_41734 [Makino-Aero-T450] — SPN_LOOP_GAIN_14 | Default: 23.700 | Linked Fault: ERR-MAK450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T450.

PRM_41735 [Makino-Aero-T450] — ATC_LOOP_GAIN_15 | Default: 43.250 | Linked Fault: ERR-MAK450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T450.

PRM_41736 [Makino-Aero-T450] — HYD_LOOP_GAIN_16 | Default: 62.800 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T450 (BANK 2)

Machine Model: Makino-Aero-T450 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T450.

PRM_41741 [Makino-Aero-T450] — HYD_LOOP_GAIN_1 | Default: 39.100 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41742 [Makino-Aero-T450] — XAX_LOOP_GAIN_2 | Default: 78.200 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41743 [Makino-Aero-T450] — ZAX_LOOP_GAIN_3 | Default: 117.300 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

PRM_41744 [Makino-Aero-T450] — CAX_LOOP_GAIN_4 | Default: 156.400 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

PRM_41745 [Makino-Aero-T450] — PLC_LOOP_GAIN_5 | Default: 195.500 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

PRM_41746 [Makino-Aero-T450] — VAC_LOOP_GAIN_6 | Default: 234.600 | Linked Fault: ERR-MAK450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450.

PRM_41747 [Makino-Aero-T450] — SPN_LOOP_GAIN_7 | Default: 23.700 | Linked Fault: ERR-MAK450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T450.

PRM_41748 [Makino-Aero-T450] — HYD_LOOP_GAIN_8 | Default: 62.800 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41749 [Makino-Aero-T450] — XAX_LOOP_GAIN_9 | Default: 101.900 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41750 [Makino-Aero-T450] — ZAX_LOOP_GAIN_10 | Default: 141.000 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

PRM_41751 [Makino-Aero-T450] — CAX_LOOP_GAIN_11 | Default: 180.100 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

PRM_41752 [Makino-Aero-T450] — PLC_LOOP_GAIN_12 | Default: 219.200 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

PRM_41753 [Makino-Aero-T450] — VAC_LOOP_GAIN_13 | Default: 8.300 | Linked Fault: ERR-MAK450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450.

PRM_41754 [Makino-Aero-T450] — SPN_LOOP_GAIN_14 | Default: 47.400 | Linked Fault: ERR-MAK450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T450.

PRM_41755 [Makino-Aero-T450] — HYD_LOOP_GAIN_15 | Default: 86.500 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41756 [Makino-Aero-T450] — XAX_LOOP_GAIN_16 | Default: 125.600 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T450 (BANK 3)

Machine Model: Makino-Aero-T450 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T450.

PRM_41761 [Makino-Aero-T450] — CLN_LOOP_GAIN_1 | Default: 58.650 | Linked Fault: ERR-MAK450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T450.

PRM_41762 [Makino-Aero-T450] — ZAX_LOOP_GAIN_2 | Default: 117.300 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

PRM_41763 [Makino-Aero-T450] — OPT_LOOP_GAIN_3 | Default: 175.950 | Linked Fault: ERR-MAK450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T450.

PRM_41764 [Makino-Aero-T450] — VAC_LOOP_GAIN_4 | Default: 234.600 | Linked Fault: ERR-MAK450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450.

PRM_41765 [Makino-Aero-T450] — ATC_LOOP_GAIN_5 | Default: 43.250 | Linked Fault: ERR-MAK450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T450.

PRM_41766 [Makino-Aero-T450] — XAX_LOOP_GAIN_6 | Default: 101.900 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41767 [Makino-Aero-T450] — BAX_LOOP_GAIN_7 | Default: 160.550 | Linked Fault: ERR-MAK450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T450.

PRM_41768 [Makino-Aero-T450] — PLC_LOOP_GAIN_8 | Default: 219.200 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

PRM_41769 [Makino-Aero-T450] — THM_LOOP_GAIN_9 | Default: 27.850 | Linked Fault: ERR-MAK450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T450.

PRM_41770 [Makino-Aero-T450] — HYD_LOOP_GAIN_10 | Default: 86.500 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41771 [Makino-Aero-T450] — YAX_LOOP_GAIN_11 | Default: 145.150 | Linked Fault: ERR-MAK450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T450.

PRM_41772 [Makino-Aero-T450] — CAX_LOOP_GAIN_12 | Default: 203.800 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

PRM_41773 [Makino-Aero-T450] — LUB_LOOP_GAIN_13 | Default: 12.450 | Linked Fault: ERR-MAK450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T450.

PRM_41774 [Makino-Aero-T450] — SPN_LOOP_GAIN_14 | Default: 71.100 | Linked Fault: ERR-MAK450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T450.

PRM_41775 [Makino-Aero-T450] — CLN_LOOP_GAIN_15 | Default: 129.750 | Linked Fault: ERR-MAK450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T450.

PRM_41776 [Makino-Aero-T450] — ZAX_LOOP_GAIN_16 | Default: 188.400 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T450 (BANK 4)

Machine Model: Makino-Aero-T450 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T450.

PRM_41781 [Makino-Aero-T450] — XAX_LOOP_GAIN_1 | Default: 78.200 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41782 [Makino-Aero-T450] — CAX_LOOP_GAIN_2 | Default: 156.400 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

PRM_41783 [Makino-Aero-T450] — VAC_LOOP_GAIN_3 | Default: 234.600 | Linked Fault: ERR-MAK450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450.

PRM_41784 [Makino-Aero-T450] — HYD_LOOP_GAIN_4 | Default: 62.800 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41785 [Makino-Aero-T450] — ZAX_LOOP_GAIN_5 | Default: 141.000 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

PRM_41786 [Makino-Aero-T450] — PLC_LOOP_GAIN_6 | Default: 219.200 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

PRM_41787 [Makino-Aero-T450] — SPN_LOOP_GAIN_7 | Default: 47.400 | Linked Fault: ERR-MAK450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T450.

PRM_41788 [Makino-Aero-T450] — XAX_LOOP_GAIN_8 | Default: 125.600 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41789 [Makino-Aero-T450] — CAX_LOOP_GAIN_9 | Default: 203.800 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

PRM_41790 [Makino-Aero-T450] — VAC_LOOP_GAIN_10 | Default: 32.000 | Linked Fault: ERR-MAK450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450.

PRM_41791 [Makino-Aero-T450] — HYD_LOOP_GAIN_11 | Default: 110.200 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41792 [Makino-Aero-T450] — ZAX_LOOP_GAIN_12 | Default: 188.400 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

PRM_41793 [Makino-Aero-T450] — PLC_LOOP_GAIN_13 | Default: 16.600 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

PRM_41794 [Makino-Aero-T450] — SPN_LOOP_GAIN_14 | Default: 94.800 | Linked Fault: ERR-MAK450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T450.

PRM_41795 [Makino-Aero-T450] — XAX_LOOP_GAIN_15 | Default: 173.000 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41796 [Makino-Aero-T450] — CAX_LOOP_GAIN_16 | Default: 1.200 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T450 (BANK 5)

Machine Model: Makino-Aero-T450 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T450.

PRM_41801 [Makino-Aero-T450] — YAX_LOOP_GAIN_1 | Default: 97.750 | Linked Fault: ERR-MAK450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T450.

PRM_41802 [Makino-Aero-T450] — PLC_LOOP_GAIN_2 | Default: 195.500 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

PRM_41803 [Makino-Aero-T450] — ATC_LOOP_GAIN_3 | Default: 43.250 | Linked Fault: ERR-MAK450-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T450.

PRM_41804 [Makino-Aero-T450] — ZAX_LOOP_GAIN_4 | Default: 141.000 | Linked Fault: ERR-MAK450-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T450.

PRM_41805 [Makino-Aero-T450] — LUB_LOOP_GAIN_5 | Default: 238.750 | Linked Fault: ERR-MAK450-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T450.

PRM_41806 [Makino-Aero-T450] — HYD_LOOP_GAIN_6 | Default: 86.500 | Linked Fault: ERR-MAK450-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T450.

PRM_41807 [Makino-Aero-T450] — BAX_LOOP_GAIN_7 | Default: 184.250 | Linked Fault: ERR-MAK450-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T450.

PRM_41808 [Makino-Aero-T450] — VAC_LOOP_GAIN_8 | Default: 32.000 | Linked Fault: ERR-MAK450-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T450.

PRM_41809 [Makino-Aero-T450] — CLN_LOOP_GAIN_9 | Default: 129.750 | Linked Fault: ERR-MAK450-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T450.

PRM_41810 [Makino-Aero-T450] — CAX_LOOP_GAIN_10 | Default: 227.500 | Linked Fault: ERR-MAK450-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T450.

PRM_41811 [Makino-Aero-T450] — THM_LOOP_GAIN_11 | Default: 75.250 | Linked Fault: ERR-MAK450-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T450.

PRM_41812 [Makino-Aero-T450] — XAX_LOOP_GAIN_12 | Default: 173.000 | Linked Fault: ERR-MAK450-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T450.

PRM_41813 [Makino-Aero-T450] — OPT_LOOP_GAIN_13 | Default: 20.750 | Linked Fault: ERR-MAK450-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T450.

PRM_41814 [Makino-Aero-T450] — SPN_LOOP_GAIN_14 | Default: 118.500 | Linked Fault: ERR-MAK450-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T450.

PRM_41815 [Makino-Aero-T450] — YAX_LOOP_GAIN_15 | Default: 216.250 | Linked Fault: ERR-MAK450-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T450.

PRM_41816 [Makino-Aero-T450] — PLC_LOOP_GAIN_16 | Default: 64.000 | Linked Fault: ERR-MAK450-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T450.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T500 (BANK 1)

Machine Model: Makino-Aero-T500 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T500.

PRM_41821 [Makino-Aero-T500] — ATC_LOOP_GAIN_1 | Default: 20.700 | Linked Fault: ERR-MAK500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T500.

PRM_41822 [Makino-Aero-T500] — HYD_LOOP_GAIN_2 | Default: 41.400 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41823 [Makino-Aero-T500] — CLN_LOOP_GAIN_3 | Default: 62.100 | Linked Fault: ERR-MAK500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T500.

PRM_41824 [Makino-Aero-T500] — XAX_LOOP_GAIN_4 | Default: 82.800 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41825 [Makino-Aero-T500] — YAX_LOOP_GAIN_5 | Default: 103.500 | Linked Fault: ERR-MAK500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T500.

PRM_41826 [Makino-Aero-T500] — ZAX_LOOP_GAIN_6 | Default: 124.200 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

PRM_41827 [Makino-Aero-T500] — BAX_LOOP_GAIN_7 | Default: 144.900 | Linked Fault: ERR-MAK500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T500.

PRM_41828 [Makino-Aero-T500] — CAX_LOOP_GAIN_8 | Default: 165.600 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

PRM_41829 [Makino-Aero-T500] — OPT_LOOP_GAIN_9 | Default: 186.300 | Linked Fault: ERR-MAK500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T500.

PRM_41830 [Makino-Aero-T500] — PLC_LOOP_GAIN_10 | Default: 207.000 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

PRM_41831 [Makino-Aero-T500] — LUB_LOOP_GAIN_11 | Default: 227.700 | Linked Fault: ERR-MAK500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T500.

PRM_41832 [Makino-Aero-T500] — VAC_LOOP_GAIN_12 | Default: 248.400 | Linked Fault: ERR-MAK500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500.

PRM_41833 [Makino-Aero-T500] — THM_LOOP_GAIN_13 | Default: 19.100 | Linked Fault: ERR-MAK500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T500.

PRM_41834 [Makino-Aero-T500] — SPN_LOOP_GAIN_14 | Default: 39.800 | Linked Fault: ERR-MAK500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T500.

PRM_41835 [Makino-Aero-T500] — ATC_LOOP_GAIN_15 | Default: 60.500 | Linked Fault: ERR-MAK500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T500.

PRM_41836 [Makino-Aero-T500] — HYD_LOOP_GAIN_16 | Default: 81.200 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T500 (BANK 2)

Machine Model: Makino-Aero-T500 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T500.

PRM_41841 [Makino-Aero-T500] — HYD_LOOP_GAIN_1 | Default: 41.400 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41842 [Makino-Aero-T500] — XAX_LOOP_GAIN_2 | Default: 82.800 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41843 [Makino-Aero-T500] — ZAX_LOOP_GAIN_3 | Default: 124.200 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

PRM_41844 [Makino-Aero-T500] — CAX_LOOP_GAIN_4 | Default: 165.600 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

PRM_41845 [Makino-Aero-T500] — PLC_LOOP_GAIN_5 | Default: 207.000 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

PRM_41846 [Makino-Aero-T500] — VAC_LOOP_GAIN_6 | Default: 248.400 | Linked Fault: ERR-MAK500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500.

PRM_41847 [Makino-Aero-T500] — SPN_LOOP_GAIN_7 | Default: 39.800 | Linked Fault: ERR-MAK500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T500.

PRM_41848 [Makino-Aero-T500] — HYD_LOOP_GAIN_8 | Default: 81.200 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41849 [Makino-Aero-T500] — XAX_LOOP_GAIN_9 | Default: 122.600 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41850 [Makino-Aero-T500] — ZAX_LOOP_GAIN_10 | Default: 164.000 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

PRM_41851 [Makino-Aero-T500] — CAX_LOOP_GAIN_11 | Default: 205.400 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

PRM_41852 [Makino-Aero-T500] — PLC_LOOP_GAIN_12 | Default: 246.800 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

PRM_41853 [Makino-Aero-T500] — VAC_LOOP_GAIN_13 | Default: 38.200 | Linked Fault: ERR-MAK500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500.

PRM_41854 [Makino-Aero-T500] — SPN_LOOP_GAIN_14 | Default: 79.600 | Linked Fault: ERR-MAK500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T500.

PRM_41855 [Makino-Aero-T500] — HYD_LOOP_GAIN_15 | Default: 121.000 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41856 [Makino-Aero-T500] — XAX_LOOP_GAIN_16 | Default: 162.400 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T500 (BANK 3)

Machine Model: Makino-Aero-T500 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T500.

PRM_41861 [Makino-Aero-T500] — CLN_LOOP_GAIN_1 | Default: 62.100 | Linked Fault: ERR-MAK500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T500.

PRM_41862 [Makino-Aero-T500] — ZAX_LOOP_GAIN_2 | Default: 124.200 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

PRM_41863 [Makino-Aero-T500] — OPT_LOOP_GAIN_3 | Default: 186.300 | Linked Fault: ERR-MAK500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T500.

PRM_41864 [Makino-Aero-T500] — VAC_LOOP_GAIN_4 | Default: 248.400 | Linked Fault: ERR-MAK500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500.

PRM_41865 [Makino-Aero-T500] — ATC_LOOP_GAIN_5 | Default: 60.500 | Linked Fault: ERR-MAK500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T500.

PRM_41866 [Makino-Aero-T500] — XAX_LOOP_GAIN_6 | Default: 122.600 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41867 [Makino-Aero-T500] — BAX_LOOP_GAIN_7 | Default: 184.700 | Linked Fault: ERR-MAK500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T500.

PRM_41868 [Makino-Aero-T500] — PLC_LOOP_GAIN_8 | Default: 246.800 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

PRM_41869 [Makino-Aero-T500] — THM_LOOP_GAIN_9 | Default: 58.900 | Linked Fault: ERR-MAK500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T500.

PRM_41870 [Makino-Aero-T500] — HYD_LOOP_GAIN_10 | Default: 121.000 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41871 [Makino-Aero-T500] — YAX_LOOP_GAIN_11 | Default: 183.100 | Linked Fault: ERR-MAK500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T500.

PRM_41872 [Makino-Aero-T500] — CAX_LOOP_GAIN_12 | Default: 245.200 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

PRM_41873 [Makino-Aero-T500] — LUB_LOOP_GAIN_13 | Default: 57.300 | Linked Fault: ERR-MAK500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T500.

PRM_41874 [Makino-Aero-T500] — SPN_LOOP_GAIN_14 | Default: 119.400 | Linked Fault: ERR-MAK500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T500.

PRM_41875 [Makino-Aero-T500] — CLN_LOOP_GAIN_15 | Default: 181.500 | Linked Fault: ERR-MAK500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T500.

PRM_41876 [Makino-Aero-T500] — ZAX_LOOP_GAIN_16 | Default: 243.600 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T500 (BANK 4)

Machine Model: Makino-Aero-T500 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T500.

PRM_41881 [Makino-Aero-T500] — XAX_LOOP_GAIN_1 | Default: 82.800 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41882 [Makino-Aero-T500] — CAX_LOOP_GAIN_2 | Default: 165.600 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

PRM_41883 [Makino-Aero-T500] — VAC_LOOP_GAIN_3 | Default: 248.400 | Linked Fault: ERR-MAK500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500.

PRM_41884 [Makino-Aero-T500] — HYD_LOOP_GAIN_4 | Default: 81.200 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41885 [Makino-Aero-T500] — ZAX_LOOP_GAIN_5 | Default: 164.000 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

PRM_41886 [Makino-Aero-T500] — PLC_LOOP_GAIN_6 | Default: 246.800 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

PRM_41887 [Makino-Aero-T500] — SPN_LOOP_GAIN_7 | Default: 79.600 | Linked Fault: ERR-MAK500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T500.

PRM_41888 [Makino-Aero-T500] — XAX_LOOP_GAIN_8 | Default: 162.400 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41889 [Makino-Aero-T500] — CAX_LOOP_GAIN_9 | Default: 245.200 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

PRM_41890 [Makino-Aero-T500] — VAC_LOOP_GAIN_10 | Default: 78.000 | Linked Fault: ERR-MAK500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500.

PRM_41891 [Makino-Aero-T500] — HYD_LOOP_GAIN_11 | Default: 160.800 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41892 [Makino-Aero-T500] — ZAX_LOOP_GAIN_12 | Default: 243.600 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

PRM_41893 [Makino-Aero-T500] — PLC_LOOP_GAIN_13 | Default: 76.400 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

PRM_41894 [Makino-Aero-T500] — SPN_LOOP_GAIN_14 | Default: 159.200 | Linked Fault: ERR-MAK500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T500.

PRM_41895 [Makino-Aero-T500] — XAX_LOOP_GAIN_15 | Default: 242.000 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41896 [Makino-Aero-T500] — CAX_LOOP_GAIN_16 | Default: 74.800 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T500 (BANK 5)

Machine Model: Makino-Aero-T500 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T500.

PRM_41901 [Makino-Aero-T500] — YAX_LOOP_GAIN_1 | Default: 103.500 | Linked Fault: ERR-MAK500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T500.

PRM_41902 [Makino-Aero-T500] — PLC_LOOP_GAIN_2 | Default: 207.000 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

PRM_41903 [Makino-Aero-T500] — ATC_LOOP_GAIN_3 | Default: 60.500 | Linked Fault: ERR-MAK500-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T500.

PRM_41904 [Makino-Aero-T500] — ZAX_LOOP_GAIN_4 | Default: 164.000 | Linked Fault: ERR-MAK500-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T500.

PRM_41905 [Makino-Aero-T500] — LUB_LOOP_GAIN_5 | Default: 17.500 | Linked Fault: ERR-MAK500-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T500.

PRM_41906 [Makino-Aero-T500] — HYD_LOOP_GAIN_6 | Default: 121.000 | Linked Fault: ERR-MAK500-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T500.

PRM_41907 [Makino-Aero-T500] — BAX_LOOP_GAIN_7 | Default: 224.500 | Linked Fault: ERR-MAK500-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T500.

PRM_41908 [Makino-Aero-T500] — VAC_LOOP_GAIN_8 | Default: 78.000 | Linked Fault: ERR-MAK500-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T500.

PRM_41909 [Makino-Aero-T500] — CLN_LOOP_GAIN_9 | Default: 181.500 | Linked Fault: ERR-MAK500-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T500.

PRM_41910 [Makino-Aero-T500] — CAX_LOOP_GAIN_10 | Default: 35.000 | Linked Fault: ERR-MAK500-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T500.

PRM_41911 [Makino-Aero-T500] — THM_LOOP_GAIN_11 | Default: 138.500 | Linked Fault: ERR-MAK500-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T500.

PRM_41912 [Makino-Aero-T500] — XAX_LOOP_GAIN_12 | Default: 242.000 | Linked Fault: ERR-MAK500-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T500.

PRM_41913 [Makino-Aero-T500] — OPT_LOOP_GAIN_13 | Default: 95.500 | Linked Fault: ERR-MAK500-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T500.

PRM_41914 [Makino-Aero-T500] — SPN_LOOP_GAIN_14 | Default: 199.000 | Linked Fault: ERR-MAK500-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T500.

PRM_41915 [Makino-Aero-T500] — YAX_LOOP_GAIN_15 | Default: 52.500 | Linked Fault: ERR-MAK500-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T500.

PRM_41916 [Makino-Aero-T500] — PLC_LOOP_GAIN_16 | Default: 156.000 | Linked Fault: ERR-MAK500-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T500.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T550 (BANK 1)

Machine Model: Makino-Aero-T550 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T550.

PRM_41921 [Makino-Aero-T550] — ATC_LOOP_GAIN_1 | Default: 21.850 | Linked Fault: ERR-MAK550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T550.

PRM_41922 [Makino-Aero-T550] — HYD_LOOP_GAIN_2 | Default: 43.700 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_41923 [Makino-Aero-T550] — CLN_LOOP_GAIN_3 | Default: 65.550 | Linked Fault: ERR-MAK550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T550.

PRM_41924 [Makino-Aero-T550] — XAX_LOOP_GAIN_4 | Default: 87.400 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_41925 [Makino-Aero-T550] — YAX_LOOP_GAIN_5 | Default: 109.250 | Linked Fault: ERR-MAK550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T550.

PRM_41926 [Makino-Aero-T550] — ZAX_LOOP_GAIN_6 | Default: 131.100 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

PRM_41927 [Makino-Aero-T550] — BAX_LOOP_GAIN_7 | Default: 152.950 | Linked Fault: ERR-MAK550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T550.

PRM_41928 [Makino-Aero-T550] — CAX_LOOP_GAIN_8 | Default: 174.800 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

PRM_41929 [Makino-Aero-T550] — OPT_LOOP_GAIN_9 | Default: 196.650 | Linked Fault: ERR-MAK550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T550.

PRM_41930 [Makino-Aero-T550] — PLC_LOOP_GAIN_10 | Default: 218.500 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

PRM_41931 [Makino-Aero-T550] — LUB_LOOP_GAIN_11 | Default: 240.350 | Linked Fault: ERR-MAK550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T550.

PRM_41932 [Makino-Aero-T550] — VAC_LOOP_GAIN_12 | Default: 12.200 | Linked Fault: ERR-MAK550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550.

PRM_41933 [Makino-Aero-T550] — THM_LOOP_GAIN_13 | Default: 34.050 | Linked Fault: ERR-MAK550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T550.

PRM_41934 [Makino-Aero-T550] — SPN_LOOP_GAIN_14 | Default: 55.900 | Linked Fault: ERR-MAK550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T550.

PRM_41935 [Makino-Aero-T550] — ATC_LOOP_GAIN_15 | Default: 77.750 | Linked Fault: ERR-MAK550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T550.

PRM_41936 [Makino-Aero-T550] — HYD_LOOP_GAIN_16 | Default: 99.600 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T550 (BANK 2)

Machine Model: Makino-Aero-T550 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T550.

PRM_41941 [Makino-Aero-T550] — HYD_LOOP_GAIN_1 | Default: 43.700 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_41942 [Makino-Aero-T550] — XAX_LOOP_GAIN_2 | Default: 87.400 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_41943 [Makino-Aero-T550] — ZAX_LOOP_GAIN_3 | Default: 131.100 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

PRM_41944 [Makino-Aero-T550] — CAX_LOOP_GAIN_4 | Default: 174.800 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

PRM_41945 [Makino-Aero-T550] — PLC_LOOP_GAIN_5 | Default: 218.500 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

PRM_41946 [Makino-Aero-T550] — VAC_LOOP_GAIN_6 | Default: 12.200 | Linked Fault: ERR-MAK550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550.

PRM_41947 [Makino-Aero-T550] — SPN_LOOP_GAIN_7 | Default: 55.900 | Linked Fault: ERR-MAK550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T550.

PRM_41948 [Makino-Aero-T550] — HYD_LOOP_GAIN_8 | Default: 99.600 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_41949 [Makino-Aero-T550] — XAX_LOOP_GAIN_9 | Default: 143.300 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_41950 [Makino-Aero-T550] — ZAX_LOOP_GAIN_10 | Default: 187.000 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

PRM_41951 [Makino-Aero-T550] — CAX_LOOP_GAIN_11 | Default: 230.700 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

PRM_41952 [Makino-Aero-T550] — PLC_LOOP_GAIN_12 | Default: 24.400 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

PRM_41953 [Makino-Aero-T550] — VAC_LOOP_GAIN_13 | Default: 68.100 | Linked Fault: ERR-MAK550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550.

PRM_41954 [Makino-Aero-T550] — SPN_LOOP_GAIN_14 | Default: 111.800 | Linked Fault: ERR-MAK550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T550.

PRM_41955 [Makino-Aero-T550] — HYD_LOOP_GAIN_15 | Default: 155.500 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_41956 [Makino-Aero-T550] — XAX_LOOP_GAIN_16 | Default: 199.200 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T550 (BANK 3)

Machine Model: Makino-Aero-T550 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T550.

PRM_41961 [Makino-Aero-T550] — CLN_LOOP_GAIN_1 | Default: 65.550 | Linked Fault: ERR-MAK550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T550.

PRM_41962 [Makino-Aero-T550] — ZAX_LOOP_GAIN_2 | Default: 131.100 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

PRM_41963 [Makino-Aero-T550] — OPT_LOOP_GAIN_3 | Default: 196.650 | Linked Fault: ERR-MAK550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T550.

PRM_41964 [Makino-Aero-T550] — VAC_LOOP_GAIN_4 | Default: 12.200 | Linked Fault: ERR-MAK550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550.

PRM_41965 [Makino-Aero-T550] — ATC_LOOP_GAIN_5 | Default: 77.750 | Linked Fault: ERR-MAK550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T550.

PRM_41966 [Makino-Aero-T550] — XAX_LOOP_GAIN_6 | Default: 143.300 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_41967 [Makino-Aero-T550] — BAX_LOOP_GAIN_7 | Default: 208.850 | Linked Fault: ERR-MAK550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T550.

PRM_41968 [Makino-Aero-T550] — PLC_LOOP_GAIN_8 | Default: 24.400 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

PRM_41969 [Makino-Aero-T550] — THM_LOOP_GAIN_9 | Default: 89.950 | Linked Fault: ERR-MAK550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T550.

PRM_41970 [Makino-Aero-T550] — HYD_LOOP_GAIN_10 | Default: 155.500 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_41971 [Makino-Aero-T550] — YAX_LOOP_GAIN_11 | Default: 221.050 | Linked Fault: ERR-MAK550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T550.

PRM_41972 [Makino-Aero-T550] — CAX_LOOP_GAIN_12 | Default: 36.600 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

PRM_41973 [Makino-Aero-T550] — LUB_LOOP_GAIN_13 | Default: 102.150 | Linked Fault: ERR-MAK550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T550.

PRM_41974 [Makino-Aero-T550] — SPN_LOOP_GAIN_14 | Default: 167.700 | Linked Fault: ERR-MAK550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T550.

PRM_41975 [Makino-Aero-T550] — CLN_LOOP_GAIN_15 | Default: 233.250 | Linked Fault: ERR-MAK550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T550.

PRM_41976 [Makino-Aero-T550] — ZAX_LOOP_GAIN_16 | Default: 48.800 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T550 (BANK 4)

Machine Model: Makino-Aero-T550 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T550.

PRM_41981 [Makino-Aero-T550] — XAX_LOOP_GAIN_1 | Default: 87.400 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_41982 [Makino-Aero-T550] — CAX_LOOP_GAIN_2 | Default: 174.800 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

PRM_41983 [Makino-Aero-T550] — VAC_LOOP_GAIN_3 | Default: 12.200 | Linked Fault: ERR-MAK550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550.

PRM_41984 [Makino-Aero-T550] — HYD_LOOP_GAIN_4 | Default: 99.600 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_41985 [Makino-Aero-T550] — ZAX_LOOP_GAIN_5 | Default: 187.000 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

PRM_41986 [Makino-Aero-T550] — PLC_LOOP_GAIN_6 | Default: 24.400 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

PRM_41987 [Makino-Aero-T550] — SPN_LOOP_GAIN_7 | Default: 111.800 | Linked Fault: ERR-MAK550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T550.

PRM_41988 [Makino-Aero-T550] — XAX_LOOP_GAIN_8 | Default: 199.200 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_41989 [Makino-Aero-T550] — CAX_LOOP_GAIN_9 | Default: 36.600 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

PRM_41990 [Makino-Aero-T550] — VAC_LOOP_GAIN_10 | Default: 124.000 | Linked Fault: ERR-MAK550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550.

PRM_41991 [Makino-Aero-T550] — HYD_LOOP_GAIN_11 | Default: 211.400 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_41992 [Makino-Aero-T550] — ZAX_LOOP_GAIN_12 | Default: 48.800 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

PRM_41993 [Makino-Aero-T550] — PLC_LOOP_GAIN_13 | Default: 136.200 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

PRM_41994 [Makino-Aero-T550] — SPN_LOOP_GAIN_14 | Default: 223.600 | Linked Fault: ERR-MAK550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T550.

PRM_41995 [Makino-Aero-T550] — XAX_LOOP_GAIN_15 | Default: 61.000 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_41996 [Makino-Aero-T550] — CAX_LOOP_GAIN_16 | Default: 148.400 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T550 (BANK 5)

Machine Model: Makino-Aero-T550 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T550.

PRM_42001 [Makino-Aero-T550] — YAX_LOOP_GAIN_1 | Default: 109.250 | Linked Fault: ERR-MAK550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T550.

PRM_42002 [Makino-Aero-T550] — PLC_LOOP_GAIN_2 | Default: 218.500 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

PRM_42003 [Makino-Aero-T550] — ATC_LOOP_GAIN_3 | Default: 77.750 | Linked Fault: ERR-MAK550-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T550.

PRM_42004 [Makino-Aero-T550] — ZAX_LOOP_GAIN_4 | Default: 187.000 | Linked Fault: ERR-MAK550-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T550.

PRM_42005 [Makino-Aero-T550] — LUB_LOOP_GAIN_5 | Default: 46.250 | Linked Fault: ERR-MAK550-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T550.

PRM_42006 [Makino-Aero-T550] — HYD_LOOP_GAIN_6 | Default: 155.500 | Linked Fault: ERR-MAK550-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T550.

PRM_42007 [Makino-Aero-T550] — BAX_LOOP_GAIN_7 | Default: 14.750 | Linked Fault: ERR-MAK550-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T550.

PRM_42008 [Makino-Aero-T550] — VAC_LOOP_GAIN_8 | Default: 124.000 | Linked Fault: ERR-MAK550-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T550.

PRM_42009 [Makino-Aero-T550] — CLN_LOOP_GAIN_9 | Default: 233.250 | Linked Fault: ERR-MAK550-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T550.

PRM_42010 [Makino-Aero-T550] — CAX_LOOP_GAIN_10 | Default: 92.500 | Linked Fault: ERR-MAK550-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T550.

PRM_42011 [Makino-Aero-T550] — THM_LOOP_GAIN_11 | Default: 201.750 | Linked Fault: ERR-MAK550-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T550.

PRM_42012 [Makino-Aero-T550] — XAX_LOOP_GAIN_12 | Default: 61.000 | Linked Fault: ERR-MAK550-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T550.

PRM_42013 [Makino-Aero-T550] — OPT_LOOP_GAIN_13 | Default: 170.250 | Linked Fault: ERR-MAK550-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T550.

PRM_42014 [Makino-Aero-T550] — SPN_LOOP_GAIN_14 | Default: 29.500 | Linked Fault: ERR-MAK550-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T550.

PRM_42015 [Makino-Aero-T550] — YAX_LOOP_GAIN_15 | Default: 138.750 | Linked Fault: ERR-MAK550-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T550.

PRM_42016 [Makino-Aero-T550] — PLC_LOOP_GAIN_16 | Default: 248.000 | Linked Fault: ERR-MAK550-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T550.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T600 (BANK 1)

Machine Model: Makino-Aero-T600 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T600.

PRM_42021 [Makino-Aero-T600] — ATC_LOOP_GAIN_1 | Default: 23.000 | Linked Fault: ERR-MAK600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T600.

PRM_42022 [Makino-Aero-T600] — HYD_LOOP_GAIN_2 | Default: 46.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42023 [Makino-Aero-T600] — CLN_LOOP_GAIN_3 | Default: 69.000 | Linked Fault: ERR-MAK600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T600.

PRM_42024 [Makino-Aero-T600] — XAX_LOOP_GAIN_4 | Default: 92.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42025 [Makino-Aero-T600] — YAX_LOOP_GAIN_5 | Default: 115.000 | Linked Fault: ERR-MAK600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T600.

PRM_42026 [Makino-Aero-T600] — ZAX_LOOP_GAIN_6 | Default: 138.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

PRM_42027 [Makino-Aero-T600] — BAX_LOOP_GAIN_7 | Default: 161.000 | Linked Fault: ERR-MAK600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T600.

PRM_42028 [Makino-Aero-T600] — CAX_LOOP_GAIN_8 | Default: 184.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

PRM_42029 [Makino-Aero-T600] — OPT_LOOP_GAIN_9 | Default: 207.000 | Linked Fault: ERR-MAK600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T600.

PRM_42030 [Makino-Aero-T600] — PLC_LOOP_GAIN_10 | Default: 230.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

PRM_42031 [Makino-Aero-T600] — LUB_LOOP_GAIN_11 | Default: 3.000 | Linked Fault: ERR-MAK600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T600.

PRM_42032 [Makino-Aero-T600] — VAC_LOOP_GAIN_12 | Default: 26.000 | Linked Fault: ERR-MAK600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600.

PRM_42033 [Makino-Aero-T600] — THM_LOOP_GAIN_13 | Default: 49.000 | Linked Fault: ERR-MAK600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T600.

PRM_42034 [Makino-Aero-T600] — SPN_LOOP_GAIN_14 | Default: 72.000 | Linked Fault: ERR-MAK600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T600.

PRM_42035 [Makino-Aero-T600] — ATC_LOOP_GAIN_15 | Default: 95.000 | Linked Fault: ERR-MAK600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T600.

PRM_42036 [Makino-Aero-T600] — HYD_LOOP_GAIN_16 | Default: 118.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T600 (BANK 2)

Machine Model: Makino-Aero-T600 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T600.

PRM_42041 [Makino-Aero-T600] — HYD_LOOP_GAIN_1 | Default: 46.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42042 [Makino-Aero-T600] — XAX_LOOP_GAIN_2 | Default: 92.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42043 [Makino-Aero-T600] — ZAX_LOOP_GAIN_3 | Default: 138.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

PRM_42044 [Makino-Aero-T600] — CAX_LOOP_GAIN_4 | Default: 184.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

PRM_42045 [Makino-Aero-T600] — PLC_LOOP_GAIN_5 | Default: 230.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

PRM_42046 [Makino-Aero-T600] — VAC_LOOP_GAIN_6 | Default: 26.000 | Linked Fault: ERR-MAK600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600.

PRM_42047 [Makino-Aero-T600] — SPN_LOOP_GAIN_7 | Default: 72.000 | Linked Fault: ERR-MAK600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T600.

PRM_42048 [Makino-Aero-T600] — HYD_LOOP_GAIN_8 | Default: 118.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42049 [Makino-Aero-T600] — XAX_LOOP_GAIN_9 | Default: 164.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42050 [Makino-Aero-T600] — ZAX_LOOP_GAIN_10 | Default: 210.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

PRM_42051 [Makino-Aero-T600] — CAX_LOOP_GAIN_11 | Default: 6.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

PRM_42052 [Makino-Aero-T600] — PLC_LOOP_GAIN_12 | Default: 52.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

PRM_42053 [Makino-Aero-T600] — VAC_LOOP_GAIN_13 | Default: 98.000 | Linked Fault: ERR-MAK600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600.

PRM_42054 [Makino-Aero-T600] — SPN_LOOP_GAIN_14 | Default: 144.000 | Linked Fault: ERR-MAK600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T600.

PRM_42055 [Makino-Aero-T600] — HYD_LOOP_GAIN_15 | Default: 190.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42056 [Makino-Aero-T600] — XAX_LOOP_GAIN_16 | Default: 236.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T600 (BANK 3)

Machine Model: Makino-Aero-T600 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T600.

PRM_42061 [Makino-Aero-T600] — CLN_LOOP_GAIN_1 | Default: 69.000 | Linked Fault: ERR-MAK600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T600.

PRM_42062 [Makino-Aero-T600] — ZAX_LOOP_GAIN_2 | Default: 138.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

PRM_42063 [Makino-Aero-T600] — OPT_LOOP_GAIN_3 | Default: 207.000 | Linked Fault: ERR-MAK600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T600.

PRM_42064 [Makino-Aero-T600] — VAC_LOOP_GAIN_4 | Default: 26.000 | Linked Fault: ERR-MAK600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600.

PRM_42065 [Makino-Aero-T600] — ATC_LOOP_GAIN_5 | Default: 95.000 | Linked Fault: ERR-MAK600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T600.

PRM_42066 [Makino-Aero-T600] — XAX_LOOP_GAIN_6 | Default: 164.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42067 [Makino-Aero-T600] — BAX_LOOP_GAIN_7 | Default: 233.000 | Linked Fault: ERR-MAK600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T600.

PRM_42068 [Makino-Aero-T600] — PLC_LOOP_GAIN_8 | Default: 52.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

PRM_42069 [Makino-Aero-T600] — THM_LOOP_GAIN_9 | Default: 121.000 | Linked Fault: ERR-MAK600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T600.

PRM_42070 [Makino-Aero-T600] — HYD_LOOP_GAIN_10 | Default: 190.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42071 [Makino-Aero-T600] — YAX_LOOP_GAIN_11 | Default: 9.000 | Linked Fault: ERR-MAK600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T600.

PRM_42072 [Makino-Aero-T600] — CAX_LOOP_GAIN_12 | Default: 78.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

PRM_42073 [Makino-Aero-T600] — LUB_LOOP_GAIN_13 | Default: 147.000 | Linked Fault: ERR-MAK600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T600.

PRM_42074 [Makino-Aero-T600] — SPN_LOOP_GAIN_14 | Default: 216.000 | Linked Fault: ERR-MAK600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T600.

PRM_42075 [Makino-Aero-T600] — CLN_LOOP_GAIN_15 | Default: 35.000 | Linked Fault: ERR-MAK600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T600.

PRM_42076 [Makino-Aero-T600] — ZAX_LOOP_GAIN_16 | Default: 104.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T600 (BANK 4)

Machine Model: Makino-Aero-T600 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T600.

PRM_42081 [Makino-Aero-T600] — XAX_LOOP_GAIN_1 | Default: 92.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42082 [Makino-Aero-T600] — CAX_LOOP_GAIN_2 | Default: 184.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

PRM_42083 [Makino-Aero-T600] — VAC_LOOP_GAIN_3 | Default: 26.000 | Linked Fault: ERR-MAK600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600.

PRM_42084 [Makino-Aero-T600] — HYD_LOOP_GAIN_4 | Default: 118.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42085 [Makino-Aero-T600] — ZAX_LOOP_GAIN_5 | Default: 210.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

PRM_42086 [Makino-Aero-T600] — PLC_LOOP_GAIN_6 | Default: 52.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

PRM_42087 [Makino-Aero-T600] — SPN_LOOP_GAIN_7 | Default: 144.000 | Linked Fault: ERR-MAK600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T600.

PRM_42088 [Makino-Aero-T600] — XAX_LOOP_GAIN_8 | Default: 236.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42089 [Makino-Aero-T600] — CAX_LOOP_GAIN_9 | Default: 78.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

PRM_42090 [Makino-Aero-T600] — VAC_LOOP_GAIN_10 | Default: 170.000 | Linked Fault: ERR-MAK600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600.

PRM_42091 [Makino-Aero-T600] — HYD_LOOP_GAIN_11 | Default: 12.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42092 [Makino-Aero-T600] — ZAX_LOOP_GAIN_12 | Default: 104.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

PRM_42093 [Makino-Aero-T600] — PLC_LOOP_GAIN_13 | Default: 196.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

PRM_42094 [Makino-Aero-T600] — SPN_LOOP_GAIN_14 | Default: 38.000 | Linked Fault: ERR-MAK600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T600.

PRM_42095 [Makino-Aero-T600] — XAX_LOOP_GAIN_15 | Default: 130.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42096 [Makino-Aero-T600] — CAX_LOOP_GAIN_16 | Default: 222.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

CNC/PLC SERVO PARAMETER DICTIONARY — Makino-Aero-T600 (BANK 5)

Machine Model: Makino-Aero-T600 | Controller: Makino Professional 6. Use these parameter registers when diagnosing servo following errors or thermal drift on Makino-Aero-T600.

PRM_42101 [Makino-Aero-T600] — YAX_LOOP_GAIN_1 | Default: 115.000 | Linked Fault: ERR-MAK600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T600.

PRM_42102 [Makino-Aero-T600] — PLC_LOOP_GAIN_2 | Default: 230.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

PRM_42103 [Makino-Aero-T600] — ATC_LOOP_GAIN_3 | Default: 95.000 | Linked Fault: ERR-MAK600-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Makino-Aero-T600.

PRM_42104 [Makino-Aero-T600] — ZAX_LOOP_GAIN_4 | Default: 210.000 | Linked Fault: ERR-MAK600-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Makino-Aero-T600.

PRM_42105 [Makino-Aero-T600] — LUB_LOOP_GAIN_5 | Default: 75.000 | Linked Fault: ERR-MAK600-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Makino-Aero-T600.

PRM_42106 [Makino-Aero-T600] — HYD_LOOP_GAIN_6 | Default: 190.000 | Linked Fault: ERR-MAK600-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Makino-Aero-T600.

PRM_42107 [Makino-Aero-T600] — BAX_LOOP_GAIN_7 | Default: 55.000 | Linked Fault: ERR-MAK600-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Makino-Aero-T600.

PRM_42108 [Makino-Aero-T600] — VAC_LOOP_GAIN_8 | Default: 170.000 | Linked Fault: ERR-MAK600-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Makino-Aero-T600.

PRM_42109 [Makino-Aero-T600] — CLN_LOOP_GAIN_9 | Default: 35.000 | Linked Fault: ERR-MAK600-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Makino-Aero-T600.

PRM_42110 [Makino-Aero-T600] — CAX_LOOP_GAIN_10 | Default: 150.000 | Linked Fault: ERR-MAK600-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Makino-Aero-T600.

PRM_42111 [Makino-Aero-T600] — THM_LOOP_GAIN_11 | Default: 15.000 | Linked Fault: ERR-MAK600-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Makino-Aero-T600.

PRM_42112 [Makino-Aero-T600] — XAX_LOOP_GAIN_12 | Default: 130.000 | Linked Fault: ERR-MAK600-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Makino-Aero-T600.

PRM_42113 [Makino-Aero-T600] — OPT_LOOP_GAIN_13 | Default: 245.000 | Linked Fault: ERR-MAK600-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Makino-Aero-T600.

PRM_42114 [Makino-Aero-T600] — SPN_LOOP_GAIN_14 | Default: 110.000 | Linked Fault: ERR-MAK600-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Makino-Aero-T600.

PRM_42115 [Makino-Aero-T600] — YAX_LOOP_GAIN_15 | Default: 225.000 | Linked Fault: ERR-MAK600-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Makino-Aero-T600.

PRM_42116 [Makino-Aero-T600] — PLC_LOOP_GAIN_16 | Default: 90.000 | Linked Fault: ERR-MAK600-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Makino-Aero-T600.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i150 (BANK 1)

Machine Model: Mazak-Integrex-i150 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i150.

PRM_42121 [Mazak-Integrex-i150] — ATC_LOOP_GAIN_1 | Default: 24.150 | Linked Fault: ERR-MAZ150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i150.

PRM_42122 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_2 | Default: 48.300 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42123 [Mazak-Integrex-i150] — CLN_LOOP_GAIN_3 | Default: 72.450 | Linked Fault: ERR-MAZ150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i150.

PRM_42124 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_4 | Default: 96.600 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42125 [Mazak-Integrex-i150] — YAX_LOOP_GAIN_5 | Default: 120.750 | Linked Fault: ERR-MAZ150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i150.

PRM_42126 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_6 | Default: 144.900 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

PRM_42127 [Mazak-Integrex-i150] — BAX_LOOP_GAIN_7 | Default: 169.050 | Linked Fault: ERR-MAZ150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i150.

PRM_42128 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_8 | Default: 193.200 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

PRM_42129 [Mazak-Integrex-i150] — OPT_LOOP_GAIN_9 | Default: 217.350 | Linked Fault: ERR-MAZ150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i150.

PRM_42130 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_10 | Default: 241.500 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

PRM_42131 [Mazak-Integrex-i150] — LUB_LOOP_GAIN_11 | Default: 15.650 | Linked Fault: ERR-MAZ150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i150.

PRM_42132 [Mazak-Integrex-i150] — VAC_LOOP_GAIN_12 | Default: 39.800 | Linked Fault: ERR-MAZ150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150.

PRM_42133 [Mazak-Integrex-i150] — THM_LOOP_GAIN_13 | Default: 63.950 | Linked Fault: ERR-MAZ150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i150.

PRM_42134 [Mazak-Integrex-i150] — SPN_LOOP_GAIN_14 | Default: 88.100 | Linked Fault: ERR-MAZ150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150.

PRM_42135 [Mazak-Integrex-i150] — ATC_LOOP_GAIN_15 | Default: 112.250 | Linked Fault: ERR-MAZ150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i150.

PRM_42136 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_16 | Default: 136.400 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i150 (BANK 2)

Machine Model: Mazak-Integrex-i150 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i150.

PRM_42141 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_1 | Default: 48.300 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42142 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_2 | Default: 96.600 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42143 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_3 | Default: 144.900 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

PRM_42144 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_4 | Default: 193.200 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

PRM_42145 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_5 | Default: 241.500 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

PRM_42146 [Mazak-Integrex-i150] — VAC_LOOP_GAIN_6 | Default: 39.800 | Linked Fault: ERR-MAZ150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150.

PRM_42147 [Mazak-Integrex-i150] — SPN_LOOP_GAIN_7 | Default: 88.100 | Linked Fault: ERR-MAZ150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150.

PRM_42148 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_8 | Default: 136.400 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42149 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_9 | Default: 184.700 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42150 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_10 | Default: 233.000 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

PRM_42151 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_11 | Default: 31.300 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

PRM_42152 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_12 | Default: 79.600 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

PRM_42153 [Mazak-Integrex-i150] — VAC_LOOP_GAIN_13 | Default: 127.900 | Linked Fault: ERR-MAZ150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150.

PRM_42154 [Mazak-Integrex-i150] — SPN_LOOP_GAIN_14 | Default: 176.200 | Linked Fault: ERR-MAZ150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150.

PRM_42155 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_15 | Default: 224.500 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42156 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_16 | Default: 22.800 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i150 (BANK 3)

Machine Model: Mazak-Integrex-i150 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i150.

PRM_42161 [Mazak-Integrex-i150] — CLN_LOOP_GAIN_1 | Default: 72.450 | Linked Fault: ERR-MAZ150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i150.

PRM_42162 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_2 | Default: 144.900 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

PRM_42163 [Mazak-Integrex-i150] — OPT_LOOP_GAIN_3 | Default: 217.350 | Linked Fault: ERR-MAZ150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i150.

PRM_42164 [Mazak-Integrex-i150] — VAC_LOOP_GAIN_4 | Default: 39.800 | Linked Fault: ERR-MAZ150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150.

PRM_42165 [Mazak-Integrex-i150] — ATC_LOOP_GAIN_5 | Default: 112.250 | Linked Fault: ERR-MAZ150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i150.

PRM_42166 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_6 | Default: 184.700 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42167 [Mazak-Integrex-i150] — BAX_LOOP_GAIN_7 | Default: 7.150 | Linked Fault: ERR-MAZ150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i150.

PRM_42168 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_8 | Default: 79.600 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

PRM_42169 [Mazak-Integrex-i150] — THM_LOOP_GAIN_9 | Default: 152.050 | Linked Fault: ERR-MAZ150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i150.

PRM_42170 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_10 | Default: 224.500 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42171 [Mazak-Integrex-i150] — YAX_LOOP_GAIN_11 | Default: 46.950 | Linked Fault: ERR-MAZ150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i150.

PRM_42172 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_12 | Default: 119.400 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

PRM_42173 [Mazak-Integrex-i150] — LUB_LOOP_GAIN_13 | Default: 191.850 | Linked Fault: ERR-MAZ150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i150.

PRM_42174 [Mazak-Integrex-i150] — SPN_LOOP_GAIN_14 | Default: 14.300 | Linked Fault: ERR-MAZ150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150.

PRM_42175 [Mazak-Integrex-i150] — CLN_LOOP_GAIN_15 | Default: 86.750 | Linked Fault: ERR-MAZ150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i150.

PRM_42176 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_16 | Default: 159.200 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i150 (BANK 4)

Machine Model: Mazak-Integrex-i150 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i150.

PRM_42181 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_1 | Default: 96.600 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42182 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_2 | Default: 193.200 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

PRM_42183 [Mazak-Integrex-i150] — VAC_LOOP_GAIN_3 | Default: 39.800 | Linked Fault: ERR-MAZ150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150.

PRM_42184 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_4 | Default: 136.400 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42185 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_5 | Default: 233.000 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

PRM_42186 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_6 | Default: 79.600 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

PRM_42187 [Mazak-Integrex-i150] — SPN_LOOP_GAIN_7 | Default: 176.200 | Linked Fault: ERR-MAZ150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150.

PRM_42188 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_8 | Default: 22.800 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42189 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_9 | Default: 119.400 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

PRM_42190 [Mazak-Integrex-i150] — VAC_LOOP_GAIN_10 | Default: 216.000 | Linked Fault: ERR-MAZ150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150.

PRM_42191 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_11 | Default: 62.600 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42192 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_12 | Default: 159.200 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

PRM_42193 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_13 | Default: 5.800 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

PRM_42194 [Mazak-Integrex-i150] — SPN_LOOP_GAIN_14 | Default: 102.400 | Linked Fault: ERR-MAZ150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150.

PRM_42195 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_15 | Default: 199.000 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42196 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_16 | Default: 45.600 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i150 (BANK 5)

Machine Model: Mazak-Integrex-i150 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i150.

PRM_42201 [Mazak-Integrex-i150] — YAX_LOOP_GAIN_1 | Default: 120.750 | Linked Fault: ERR-MAZ150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i150.

PRM_42202 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_2 | Default: 241.500 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

PRM_42203 [Mazak-Integrex-i150] — ATC_LOOP_GAIN_3 | Default: 112.250 | Linked Fault: ERR-MAZ150-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i150.

PRM_42204 [Mazak-Integrex-i150] — ZAX_LOOP_GAIN_4 | Default: 233.000 | Linked Fault: ERR-MAZ150-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i150.

PRM_42205 [Mazak-Integrex-i150] — LUB_LOOP_GAIN_5 | Default: 103.750 | Linked Fault: ERR-MAZ150-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i150.

PRM_42206 [Mazak-Integrex-i150] — HYD_LOOP_GAIN_6 | Default: 224.500 | Linked Fault: ERR-MAZ150-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i150.

PRM_42207 [Mazak-Integrex-i150] — BAX_LOOP_GAIN_7 | Default: 95.250 | Linked Fault: ERR-MAZ150-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i150.

PRM_42208 [Mazak-Integrex-i150] — VAC_LOOP_GAIN_8 | Default: 216.000 | Linked Fault: ERR-MAZ150-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i150.

PRM_42209 [Mazak-Integrex-i150] — CLN_LOOP_GAIN_9 | Default: 86.750 | Linked Fault: ERR-MAZ150-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i150.

PRM_42210 [Mazak-Integrex-i150] — CAX_LOOP_GAIN_10 | Default: 207.500 | Linked Fault: ERR-MAZ150-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i150.

PRM_42211 [Mazak-Integrex-i150] — THM_LOOP_GAIN_11 | Default: 78.250 | Linked Fault: ERR-MAZ150-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i150.

PRM_42212 [Mazak-Integrex-i150] — XAX_LOOP_GAIN_12 | Default: 199.000 | Linked Fault: ERR-MAZ150-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i150.

PRM_42213 [Mazak-Integrex-i150] — OPT_LOOP_GAIN_13 | Default: 69.750 | Linked Fault: ERR-MAZ150-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i150.

PRM_42214 [Mazak-Integrex-i150] — SPN_LOOP_GAIN_14 | Default: 190.500 | Linked Fault: ERR-MAZ150-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i150.

PRM_42215 [Mazak-Integrex-i150] — YAX_LOOP_GAIN_15 | Default: 61.250 | Linked Fault: ERR-MAZ150-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i150.

PRM_42216 [Mazak-Integrex-i150] — PLC_LOOP_GAIN_16 | Default: 182.000 | Linked Fault: ERR-MAZ150-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i150.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i200 (BANK 1)

Machine Model: Mazak-Integrex-i200 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i200.

PRM_42221 [Mazak-Integrex-i200] — ATC_LOOP_GAIN_1 | Default: 25.300 | Linked Fault: ERR-MAZ200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i200.

PRM_42222 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_2 | Default: 50.600 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42223 [Mazak-Integrex-i200] — CLN_LOOP_GAIN_3 | Default: 75.900 | Linked Fault: ERR-MAZ200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i200.

PRM_42224 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_4 | Default: 101.200 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42225 [Mazak-Integrex-i200] — YAX_LOOP_GAIN_5 | Default: 126.500 | Linked Fault: ERR-MAZ200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i200.

PRM_42226 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_6 | Default: 151.800 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

PRM_42227 [Mazak-Integrex-i200] — BAX_LOOP_GAIN_7 | Default: 177.100 | Linked Fault: ERR-MAZ200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i200.

PRM_42228 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_8 | Default: 202.400 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

PRM_42229 [Mazak-Integrex-i200] — OPT_LOOP_GAIN_9 | Default: 227.700 | Linked Fault: ERR-MAZ200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i200.

PRM_42230 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_10 | Default: 3.000 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

PRM_42231 [Mazak-Integrex-i200] — LUB_LOOP_GAIN_11 | Default: 28.300 | Linked Fault: ERR-MAZ200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i200.

PRM_42232 [Mazak-Integrex-i200] — VAC_LOOP_GAIN_12 | Default: 53.600 | Linked Fault: ERR-MAZ200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200.

PRM_42233 [Mazak-Integrex-i200] — THM_LOOP_GAIN_13 | Default: 78.900 | Linked Fault: ERR-MAZ200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i200.

PRM_42234 [Mazak-Integrex-i200] — SPN_LOOP_GAIN_14 | Default: 104.200 | Linked Fault: ERR-MAZ200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200.

PRM_42235 [Mazak-Integrex-i200] — ATC_LOOP_GAIN_15 | Default: 129.500 | Linked Fault: ERR-MAZ200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i200.

PRM_42236 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_16 | Default: 154.800 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i200 (BANK 2)

Machine Model: Mazak-Integrex-i200 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i200.

PRM_42241 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_1 | Default: 50.600 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42242 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_2 | Default: 101.200 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42243 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_3 | Default: 151.800 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

PRM_42244 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_4 | Default: 202.400 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

PRM_42245 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_5 | Default: 3.000 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

PRM_42246 [Mazak-Integrex-i200] — VAC_LOOP_GAIN_6 | Default: 53.600 | Linked Fault: ERR-MAZ200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200.

PRM_42247 [Mazak-Integrex-i200] — SPN_LOOP_GAIN_7 | Default: 104.200 | Linked Fault: ERR-MAZ200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200.

PRM_42248 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_8 | Default: 154.800 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42249 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_9 | Default: 205.400 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42250 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_10 | Default: 6.000 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

PRM_42251 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_11 | Default: 56.600 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

PRM_42252 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_12 | Default: 107.200 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

PRM_42253 [Mazak-Integrex-i200] — VAC_LOOP_GAIN_13 | Default: 157.800 | Linked Fault: ERR-MAZ200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200.

PRM_42254 [Mazak-Integrex-i200] — SPN_LOOP_GAIN_14 | Default: 208.400 | Linked Fault: ERR-MAZ200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200.

PRM_42255 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_15 | Default: 9.000 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42256 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_16 | Default: 59.600 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i200 (BANK 3)

Machine Model: Mazak-Integrex-i200 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i200.

PRM_42261 [Mazak-Integrex-i200] — CLN_LOOP_GAIN_1 | Default: 75.900 | Linked Fault: ERR-MAZ200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i200.

PRM_42262 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_2 | Default: 151.800 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

PRM_42263 [Mazak-Integrex-i200] — OPT_LOOP_GAIN_3 | Default: 227.700 | Linked Fault: ERR-MAZ200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i200.

PRM_42264 [Mazak-Integrex-i200] — VAC_LOOP_GAIN_4 | Default: 53.600 | Linked Fault: ERR-MAZ200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200.

PRM_42265 [Mazak-Integrex-i200] — ATC_LOOP_GAIN_5 | Default: 129.500 | Linked Fault: ERR-MAZ200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i200.

PRM_42266 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_6 | Default: 205.400 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42267 [Mazak-Integrex-i200] — BAX_LOOP_GAIN_7 | Default: 31.300 | Linked Fault: ERR-MAZ200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i200.

PRM_42268 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_8 | Default: 107.200 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

PRM_42269 [Mazak-Integrex-i200] — THM_LOOP_GAIN_9 | Default: 183.100 | Linked Fault: ERR-MAZ200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i200.

PRM_42270 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_10 | Default: 9.000 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42271 [Mazak-Integrex-i200] — YAX_LOOP_GAIN_11 | Default: 84.900 | Linked Fault: ERR-MAZ200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i200.

PRM_42272 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_12 | Default: 160.800 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

PRM_42273 [Mazak-Integrex-i200] — LUB_LOOP_GAIN_13 | Default: 236.700 | Linked Fault: ERR-MAZ200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i200.

PRM_42274 [Mazak-Integrex-i200] — SPN_LOOP_GAIN_14 | Default: 62.600 | Linked Fault: ERR-MAZ200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200.

PRM_42275 [Mazak-Integrex-i200] — CLN_LOOP_GAIN_15 | Default: 138.500 | Linked Fault: ERR-MAZ200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i200.

PRM_42276 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_16 | Default: 214.400 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i200 (BANK 4)

Machine Model: Mazak-Integrex-i200 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i200.

PRM_42281 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_1 | Default: 101.200 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42282 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_2 | Default: 202.400 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

PRM_42283 [Mazak-Integrex-i200] — VAC_LOOP_GAIN_3 | Default: 53.600 | Linked Fault: ERR-MAZ200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200.

PRM_42284 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_4 | Default: 154.800 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42285 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_5 | Default: 6.000 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

PRM_42286 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_6 | Default: 107.200 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

PRM_42287 [Mazak-Integrex-i200] — SPN_LOOP_GAIN_7 | Default: 208.400 | Linked Fault: ERR-MAZ200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200.

PRM_42288 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_8 | Default: 59.600 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42289 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_9 | Default: 160.800 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

PRM_42290 [Mazak-Integrex-i200] — VAC_LOOP_GAIN_10 | Default: 12.000 | Linked Fault: ERR-MAZ200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200.

PRM_42291 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_11 | Default: 113.200 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42292 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_12 | Default: 214.400 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

PRM_42293 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_13 | Default: 65.600 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

PRM_42294 [Mazak-Integrex-i200] — SPN_LOOP_GAIN_14 | Default: 166.800 | Linked Fault: ERR-MAZ200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200.

PRM_42295 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_15 | Default: 18.000 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42296 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_16 | Default: 119.200 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i200 (BANK 5)

Machine Model: Mazak-Integrex-i200 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i200.

PRM_42301 [Mazak-Integrex-i200] — YAX_LOOP_GAIN_1 | Default: 126.500 | Linked Fault: ERR-MAZ200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i200.

PRM_42302 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_2 | Default: 3.000 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

PRM_42303 [Mazak-Integrex-i200] — ATC_LOOP_GAIN_3 | Default: 129.500 | Linked Fault: ERR-MAZ200-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i200.

PRM_42304 [Mazak-Integrex-i200] — ZAX_LOOP_GAIN_4 | Default: 6.000 | Linked Fault: ERR-MAZ200-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i200.

PRM_42305 [Mazak-Integrex-i200] — LUB_LOOP_GAIN_5 | Default: 132.500 | Linked Fault: ERR-MAZ200-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i200.

PRM_42306 [Mazak-Integrex-i200] — HYD_LOOP_GAIN_6 | Default: 9.000 | Linked Fault: ERR-MAZ200-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i200.

PRM_42307 [Mazak-Integrex-i200] — BAX_LOOP_GAIN_7 | Default: 135.500 | Linked Fault: ERR-MAZ200-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i200.

PRM_42308 [Mazak-Integrex-i200] — VAC_LOOP_GAIN_8 | Default: 12.000 | Linked Fault: ERR-MAZ200-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i200.

PRM_42309 [Mazak-Integrex-i200] — CLN_LOOP_GAIN_9 | Default: 138.500 | Linked Fault: ERR-MAZ200-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i200.

PRM_42310 [Mazak-Integrex-i200] — CAX_LOOP_GAIN_10 | Default: 15.000 | Linked Fault: ERR-MAZ200-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i200.

PRM_42311 [Mazak-Integrex-i200] — THM_LOOP_GAIN_11 | Default: 141.500 | Linked Fault: ERR-MAZ200-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i200.

PRM_42312 [Mazak-Integrex-i200] — XAX_LOOP_GAIN_12 | Default: 18.000 | Linked Fault: ERR-MAZ200-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i200.

PRM_42313 [Mazak-Integrex-i200] — OPT_LOOP_GAIN_13 | Default: 144.500 | Linked Fault: ERR-MAZ200-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i200.

PRM_42314 [Mazak-Integrex-i200] — SPN_LOOP_GAIN_14 | Default: 21.000 | Linked Fault: ERR-MAZ200-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i200.

PRM_42315 [Mazak-Integrex-i200] — YAX_LOOP_GAIN_15 | Default: 147.500 | Linked Fault: ERR-MAZ200-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i200.

PRM_42316 [Mazak-Integrex-i200] — PLC_LOOP_GAIN_16 | Default: 24.000 | Linked Fault: ERR-MAZ200-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i200.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i250 (BANK 1)

Machine Model: Mazak-Integrex-i250 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i250.

PRM_42321 [Mazak-Integrex-i250] — ATC_LOOP_GAIN_1 | Default: 26.450 | Linked Fault: ERR-MAZ250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i250.

PRM_42322 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_2 | Default: 52.900 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42323 [Mazak-Integrex-i250] — CLN_LOOP_GAIN_3 | Default: 79.350 | Linked Fault: ERR-MAZ250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i250.

PRM_42324 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_4 | Default: 105.800 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42325 [Mazak-Integrex-i250] — YAX_LOOP_GAIN_5 | Default: 132.250 | Linked Fault: ERR-MAZ250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i250.

PRM_42326 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_6 | Default: 158.700 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

PRM_42327 [Mazak-Integrex-i250] — BAX_LOOP_GAIN_7 | Default: 185.150 | Linked Fault: ERR-MAZ250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i250.

PRM_42328 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_8 | Default: 211.600 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

PRM_42329 [Mazak-Integrex-i250] — OPT_LOOP_GAIN_9 | Default: 238.050 | Linked Fault: ERR-MAZ250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i250.

PRM_42330 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_10 | Default: 14.500 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

PRM_42331 [Mazak-Integrex-i250] — LUB_LOOP_GAIN_11 | Default: 40.950 | Linked Fault: ERR-MAZ250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i250.

PRM_42332 [Mazak-Integrex-i250] — VAC_LOOP_GAIN_12 | Default: 67.400 | Linked Fault: ERR-MAZ250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250.

PRM_42333 [Mazak-Integrex-i250] — THM_LOOP_GAIN_13 | Default: 93.850 | Linked Fault: ERR-MAZ250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i250.

PRM_42334 [Mazak-Integrex-i250] — SPN_LOOP_GAIN_14 | Default: 120.300 | Linked Fault: ERR-MAZ250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250.

PRM_42335 [Mazak-Integrex-i250] — ATC_LOOP_GAIN_15 | Default: 146.750 | Linked Fault: ERR-MAZ250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i250.

PRM_42336 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_16 | Default: 173.200 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i250 (BANK 2)

Machine Model: Mazak-Integrex-i250 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i250.

PRM_42341 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_1 | Default: 52.900 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42342 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_2 | Default: 105.800 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42343 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_3 | Default: 158.700 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

PRM_42344 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_4 | Default: 211.600 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

PRM_42345 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_5 | Default: 14.500 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

PRM_42346 [Mazak-Integrex-i250] — VAC_LOOP_GAIN_6 | Default: 67.400 | Linked Fault: ERR-MAZ250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250.

PRM_42347 [Mazak-Integrex-i250] — SPN_LOOP_GAIN_7 | Default: 120.300 | Linked Fault: ERR-MAZ250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250.

PRM_42348 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_8 | Default: 173.200 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42349 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_9 | Default: 226.100 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42350 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_10 | Default: 29.000 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

PRM_42351 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_11 | Default: 81.900 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

PRM_42352 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_12 | Default: 134.800 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

PRM_42353 [Mazak-Integrex-i250] — VAC_LOOP_GAIN_13 | Default: 187.700 | Linked Fault: ERR-MAZ250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250.

PRM_42354 [Mazak-Integrex-i250] — SPN_LOOP_GAIN_14 | Default: 240.600 | Linked Fault: ERR-MAZ250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250.

PRM_42355 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_15 | Default: 43.500 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42356 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_16 | Default: 96.400 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i250 (BANK 3)

Machine Model: Mazak-Integrex-i250 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i250.

PRM_42361 [Mazak-Integrex-i250] — CLN_LOOP_GAIN_1 | Default: 79.350 | Linked Fault: ERR-MAZ250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i250.

PRM_42362 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_2 | Default: 158.700 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

PRM_42363 [Mazak-Integrex-i250] — OPT_LOOP_GAIN_3 | Default: 238.050 | Linked Fault: ERR-MAZ250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i250.

PRM_42364 [Mazak-Integrex-i250] — VAC_LOOP_GAIN_4 | Default: 67.400 | Linked Fault: ERR-MAZ250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250.

PRM_42365 [Mazak-Integrex-i250] — ATC_LOOP_GAIN_5 | Default: 146.750 | Linked Fault: ERR-MAZ250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i250.

PRM_42366 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_6 | Default: 226.100 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42367 [Mazak-Integrex-i250] — BAX_LOOP_GAIN_7 | Default: 55.450 | Linked Fault: ERR-MAZ250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i250.

PRM_42368 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_8 | Default: 134.800 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

PRM_42369 [Mazak-Integrex-i250] — THM_LOOP_GAIN_9 | Default: 214.150 | Linked Fault: ERR-MAZ250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i250.

PRM_42370 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_10 | Default: 43.500 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42371 [Mazak-Integrex-i250] — YAX_LOOP_GAIN_11 | Default: 122.850 | Linked Fault: ERR-MAZ250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i250.

PRM_42372 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_12 | Default: 202.200 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

PRM_42373 [Mazak-Integrex-i250] — LUB_LOOP_GAIN_13 | Default: 31.550 | Linked Fault: ERR-MAZ250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i250.

PRM_42374 [Mazak-Integrex-i250] — SPN_LOOP_GAIN_14 | Default: 110.900 | Linked Fault: ERR-MAZ250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250.

PRM_42375 [Mazak-Integrex-i250] — CLN_LOOP_GAIN_15 | Default: 190.250 | Linked Fault: ERR-MAZ250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i250.

PRM_42376 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_16 | Default: 19.600 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i250 (BANK 4)

Machine Model: Mazak-Integrex-i250 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i250.

PRM_42381 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_1 | Default: 105.800 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42382 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_2 | Default: 211.600 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

PRM_42383 [Mazak-Integrex-i250] — VAC_LOOP_GAIN_3 | Default: 67.400 | Linked Fault: ERR-MAZ250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250.

PRM_42384 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_4 | Default: 173.200 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42385 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_5 | Default: 29.000 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

PRM_42386 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_6 | Default: 134.800 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

PRM_42387 [Mazak-Integrex-i250] — SPN_LOOP_GAIN_7 | Default: 240.600 | Linked Fault: ERR-MAZ250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250.

PRM_42388 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_8 | Default: 96.400 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42389 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_9 | Default: 202.200 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

PRM_42390 [Mazak-Integrex-i250] — VAC_LOOP_GAIN_10 | Default: 58.000 | Linked Fault: ERR-MAZ250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250.

PRM_42391 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_11 | Default: 163.800 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42392 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_12 | Default: 19.600 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

PRM_42393 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_13 | Default: 125.400 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

PRM_42394 [Mazak-Integrex-i250] — SPN_LOOP_GAIN_14 | Default: 231.200 | Linked Fault: ERR-MAZ250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250.

PRM_42395 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_15 | Default: 87.000 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42396 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_16 | Default: 192.800 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i250 (BANK 5)

Machine Model: Mazak-Integrex-i250 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i250.

PRM_42401 [Mazak-Integrex-i250] — YAX_LOOP_GAIN_1 | Default: 132.250 | Linked Fault: ERR-MAZ250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i250.

PRM_42402 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_2 | Default: 14.500 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

PRM_42403 [Mazak-Integrex-i250] — ATC_LOOP_GAIN_3 | Default: 146.750 | Linked Fault: ERR-MAZ250-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i250.

PRM_42404 [Mazak-Integrex-i250] — ZAX_LOOP_GAIN_4 | Default: 29.000 | Linked Fault: ERR-MAZ250-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i250.

PRM_42405 [Mazak-Integrex-i250] — LUB_LOOP_GAIN_5 | Default: 161.250 | Linked Fault: ERR-MAZ250-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i250.

PRM_42406 [Mazak-Integrex-i250] — HYD_LOOP_GAIN_6 | Default: 43.500 | Linked Fault: ERR-MAZ250-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i250.

PRM_42407 [Mazak-Integrex-i250] — BAX_LOOP_GAIN_7 | Default: 175.750 | Linked Fault: ERR-MAZ250-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i250.

PRM_42408 [Mazak-Integrex-i250] — VAC_LOOP_GAIN_8 | Default: 58.000 | Linked Fault: ERR-MAZ250-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i250.

PRM_42409 [Mazak-Integrex-i250] — CLN_LOOP_GAIN_9 | Default: 190.250 | Linked Fault: ERR-MAZ250-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i250.

PRM_42410 [Mazak-Integrex-i250] — CAX_LOOP_GAIN_10 | Default: 72.500 | Linked Fault: ERR-MAZ250-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i250.

PRM_42411 [Mazak-Integrex-i250] — THM_LOOP_GAIN_11 | Default: 204.750 | Linked Fault: ERR-MAZ250-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i250.

PRM_42412 [Mazak-Integrex-i250] — XAX_LOOP_GAIN_12 | Default: 87.000 | Linked Fault: ERR-MAZ250-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i250.

PRM_42413 [Mazak-Integrex-i250] — OPT_LOOP_GAIN_13 | Default: 219.250 | Linked Fault: ERR-MAZ250-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i250.

PRM_42414 [Mazak-Integrex-i250] — SPN_LOOP_GAIN_14 | Default: 101.500 | Linked Fault: ERR-MAZ250-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i250.

PRM_42415 [Mazak-Integrex-i250] — YAX_LOOP_GAIN_15 | Default: 233.750 | Linked Fault: ERR-MAZ250-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i250.

PRM_42416 [Mazak-Integrex-i250] — PLC_LOOP_GAIN_16 | Default: 116.000 | Linked Fault: ERR-MAZ250-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i250.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i300 (BANK 1)

Machine Model: Mazak-Integrex-i300 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i300.

PRM_42421 [Mazak-Integrex-i300] — ATC_LOOP_GAIN_1 | Default: 27.600 | Linked Fault: ERR-MAZ300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i300.

PRM_42422 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_2 | Default: 55.200 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42423 [Mazak-Integrex-i300] — CLN_LOOP_GAIN_3 | Default: 82.800 | Linked Fault: ERR-MAZ300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i300.

PRM_42424 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_4 | Default: 110.400 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42425 [Mazak-Integrex-i300] — YAX_LOOP_GAIN_5 | Default: 138.000 | Linked Fault: ERR-MAZ300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i300.

PRM_42426 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_6 | Default: 165.600 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

PRM_42427 [Mazak-Integrex-i300] — BAX_LOOP_GAIN_7 | Default: 193.200 | Linked Fault: ERR-MAZ300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i300.

PRM_42428 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_8 | Default: 220.800 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

PRM_42429 [Mazak-Integrex-i300] — OPT_LOOP_GAIN_9 | Default: 248.400 | Linked Fault: ERR-MAZ300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i300.

PRM_42430 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_10 | Default: 26.000 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

PRM_42431 [Mazak-Integrex-i300] — LUB_LOOP_GAIN_11 | Default: 53.600 | Linked Fault: ERR-MAZ300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i300.

PRM_42432 [Mazak-Integrex-i300] — VAC_LOOP_GAIN_12 | Default: 81.200 | Linked Fault: ERR-MAZ300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300.

PRM_42433 [Mazak-Integrex-i300] — THM_LOOP_GAIN_13 | Default: 108.800 | Linked Fault: ERR-MAZ300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i300.

PRM_42434 [Mazak-Integrex-i300] — SPN_LOOP_GAIN_14 | Default: 136.400 | Linked Fault: ERR-MAZ300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300.

PRM_42435 [Mazak-Integrex-i300] — ATC_LOOP_GAIN_15 | Default: 164.000 | Linked Fault: ERR-MAZ300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i300.

PRM_42436 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_16 | Default: 191.600 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i300 (BANK 2)

Machine Model: Mazak-Integrex-i300 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i300.

PRM_42441 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_1 | Default: 55.200 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42442 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_2 | Default: 110.400 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42443 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_3 | Default: 165.600 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

PRM_42444 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_4 | Default: 220.800 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

PRM_42445 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_5 | Default: 26.000 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

PRM_42446 [Mazak-Integrex-i300] — VAC_LOOP_GAIN_6 | Default: 81.200 | Linked Fault: ERR-MAZ300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300.

PRM_42447 [Mazak-Integrex-i300] — SPN_LOOP_GAIN_7 | Default: 136.400 | Linked Fault: ERR-MAZ300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300.

PRM_42448 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_8 | Default: 191.600 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42449 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_9 | Default: 246.800 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42450 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_10 | Default: 52.000 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

PRM_42451 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_11 | Default: 107.200 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

PRM_42452 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_12 | Default: 162.400 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

PRM_42453 [Mazak-Integrex-i300] — VAC_LOOP_GAIN_13 | Default: 217.600 | Linked Fault: ERR-MAZ300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300.

PRM_42454 [Mazak-Integrex-i300] — SPN_LOOP_GAIN_14 | Default: 22.800 | Linked Fault: ERR-MAZ300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300.

PRM_42455 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_15 | Default: 78.000 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42456 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_16 | Default: 133.200 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i300 (BANK 3)

Machine Model: Mazak-Integrex-i300 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i300.

PRM_42461 [Mazak-Integrex-i300] — CLN_LOOP_GAIN_1 | Default: 82.800 | Linked Fault: ERR-MAZ300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i300.

PRM_42462 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_2 | Default: 165.600 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

PRM_42463 [Mazak-Integrex-i300] — OPT_LOOP_GAIN_3 | Default: 248.400 | Linked Fault: ERR-MAZ300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i300.

PRM_42464 [Mazak-Integrex-i300] — VAC_LOOP_GAIN_4 | Default: 81.200 | Linked Fault: ERR-MAZ300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300.

PRM_42465 [Mazak-Integrex-i300] — ATC_LOOP_GAIN_5 | Default: 164.000 | Linked Fault: ERR-MAZ300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i300.

PRM_42466 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_6 | Default: 246.800 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42467 [Mazak-Integrex-i300] — BAX_LOOP_GAIN_7 | Default: 79.600 | Linked Fault: ERR-MAZ300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i300.

PRM_42468 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_8 | Default: 162.400 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

PRM_42469 [Mazak-Integrex-i300] — THM_LOOP_GAIN_9 | Default: 245.200 | Linked Fault: ERR-MAZ300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i300.

PRM_42470 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_10 | Default: 78.000 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42471 [Mazak-Integrex-i300] — YAX_LOOP_GAIN_11 | Default: 160.800 | Linked Fault: ERR-MAZ300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i300.

PRM_42472 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_12 | Default: 243.600 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

PRM_42473 [Mazak-Integrex-i300] — LUB_LOOP_GAIN_13 | Default: 76.400 | Linked Fault: ERR-MAZ300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i300.

PRM_42474 [Mazak-Integrex-i300] — SPN_LOOP_GAIN_14 | Default: 159.200 | Linked Fault: ERR-MAZ300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300.

PRM_42475 [Mazak-Integrex-i300] — CLN_LOOP_GAIN_15 | Default: 242.000 | Linked Fault: ERR-MAZ300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i300.

PRM_42476 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_16 | Default: 74.800 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i300 (BANK 4)

Machine Model: Mazak-Integrex-i300 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i300.

PRM_42481 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_1 | Default: 110.400 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42482 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_2 | Default: 220.800 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

PRM_42483 [Mazak-Integrex-i300] — VAC_LOOP_GAIN_3 | Default: 81.200 | Linked Fault: ERR-MAZ300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300.

PRM_42484 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_4 | Default: 191.600 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42485 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_5 | Default: 52.000 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

PRM_42486 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_6 | Default: 162.400 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

PRM_42487 [Mazak-Integrex-i300] — SPN_LOOP_GAIN_7 | Default: 22.800 | Linked Fault: ERR-MAZ300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300.

PRM_42488 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_8 | Default: 133.200 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42489 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_9 | Default: 243.600 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

PRM_42490 [Mazak-Integrex-i300] — VAC_LOOP_GAIN_10 | Default: 104.000 | Linked Fault: ERR-MAZ300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300.

PRM_42491 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_11 | Default: 214.400 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42492 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_12 | Default: 74.800 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

PRM_42493 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_13 | Default: 185.200 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

PRM_42494 [Mazak-Integrex-i300] — SPN_LOOP_GAIN_14 | Default: 45.600 | Linked Fault: ERR-MAZ300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300.

PRM_42495 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_15 | Default: 156.000 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42496 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_16 | Default: 16.400 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i300 (BANK 5)

Machine Model: Mazak-Integrex-i300 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i300.

PRM_42501 [Mazak-Integrex-i300] — YAX_LOOP_GAIN_1 | Default: 138.000 | Linked Fault: ERR-MAZ300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i300.

PRM_42502 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_2 | Default: 26.000 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

PRM_42503 [Mazak-Integrex-i300] — ATC_LOOP_GAIN_3 | Default: 164.000 | Linked Fault: ERR-MAZ300-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i300.

PRM_42504 [Mazak-Integrex-i300] — ZAX_LOOP_GAIN_4 | Default: 52.000 | Linked Fault: ERR-MAZ300-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i300.

PRM_42505 [Mazak-Integrex-i300] — LUB_LOOP_GAIN_5 | Default: 190.000 | Linked Fault: ERR-MAZ300-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i300.

PRM_42506 [Mazak-Integrex-i300] — HYD_LOOP_GAIN_6 | Default: 78.000 | Linked Fault: ERR-MAZ300-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i300.

PRM_42507 [Mazak-Integrex-i300] — BAX_LOOP_GAIN_7 | Default: 216.000 | Linked Fault: ERR-MAZ300-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i300.

PRM_42508 [Mazak-Integrex-i300] — VAC_LOOP_GAIN_8 | Default: 104.000 | Linked Fault: ERR-MAZ300-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i300.

PRM_42509 [Mazak-Integrex-i300] — CLN_LOOP_GAIN_9 | Default: 242.000 | Linked Fault: ERR-MAZ300-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i300.

PRM_42510 [Mazak-Integrex-i300] — CAX_LOOP_GAIN_10 | Default: 130.000 | Linked Fault: ERR-MAZ300-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i300.

PRM_42511 [Mazak-Integrex-i300] — THM_LOOP_GAIN_11 | Default: 18.000 | Linked Fault: ERR-MAZ300-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i300.

PRM_42512 [Mazak-Integrex-i300] — XAX_LOOP_GAIN_12 | Default: 156.000 | Linked Fault: ERR-MAZ300-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i300.

PRM_42513 [Mazak-Integrex-i300] — OPT_LOOP_GAIN_13 | Default: 44.000 | Linked Fault: ERR-MAZ300-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i300.

PRM_42514 [Mazak-Integrex-i300] — SPN_LOOP_GAIN_14 | Default: 182.000 | Linked Fault: ERR-MAZ300-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i300.

PRM_42515 [Mazak-Integrex-i300] — YAX_LOOP_GAIN_15 | Default: 70.000 | Linked Fault: ERR-MAZ300-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i300.

PRM_42516 [Mazak-Integrex-i300] — PLC_LOOP_GAIN_16 | Default: 208.000 | Linked Fault: ERR-MAZ300-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i300.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i350 (BANK 1)

Machine Model: Mazak-Integrex-i350 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i350.

PRM_42521 [Mazak-Integrex-i350] — ATC_LOOP_GAIN_1 | Default: 28.750 | Linked Fault: ERR-MAZ350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i350.

PRM_42522 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_2 | Default: 57.500 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42523 [Mazak-Integrex-i350] — CLN_LOOP_GAIN_3 | Default: 86.250 | Linked Fault: ERR-MAZ350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i350.

PRM_42524 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_4 | Default: 115.000 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42525 [Mazak-Integrex-i350] — YAX_LOOP_GAIN_5 | Default: 143.750 | Linked Fault: ERR-MAZ350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i350.

PRM_42526 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_6 | Default: 172.500 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

PRM_42527 [Mazak-Integrex-i350] — BAX_LOOP_GAIN_7 | Default: 201.250 | Linked Fault: ERR-MAZ350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i350.

PRM_42528 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_8 | Default: 230.000 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

PRM_42529 [Mazak-Integrex-i350] — OPT_LOOP_GAIN_9 | Default: 8.750 | Linked Fault: ERR-MAZ350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i350.

PRM_42530 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_10 | Default: 37.500 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

PRM_42531 [Mazak-Integrex-i350] — LUB_LOOP_GAIN_11 | Default: 66.250 | Linked Fault: ERR-MAZ350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i350.

PRM_42532 [Mazak-Integrex-i350] — VAC_LOOP_GAIN_12 | Default: 95.000 | Linked Fault: ERR-MAZ350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350.

PRM_42533 [Mazak-Integrex-i350] — THM_LOOP_GAIN_13 | Default: 123.750 | Linked Fault: ERR-MAZ350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i350.

PRM_42534 [Mazak-Integrex-i350] — SPN_LOOP_GAIN_14 | Default: 152.500 | Linked Fault: ERR-MAZ350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350.

PRM_42535 [Mazak-Integrex-i350] — ATC_LOOP_GAIN_15 | Default: 181.250 | Linked Fault: ERR-MAZ350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i350.

PRM_42536 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_16 | Default: 210.000 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i350 (BANK 2)

Machine Model: Mazak-Integrex-i350 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i350.

PRM_42541 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_1 | Default: 57.500 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42542 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_2 | Default: 115.000 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42543 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_3 | Default: 172.500 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

PRM_42544 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_4 | Default: 230.000 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

PRM_42545 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_5 | Default: 37.500 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

PRM_42546 [Mazak-Integrex-i350] — VAC_LOOP_GAIN_6 | Default: 95.000 | Linked Fault: ERR-MAZ350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350.

PRM_42547 [Mazak-Integrex-i350] — SPN_LOOP_GAIN_7 | Default: 152.500 | Linked Fault: ERR-MAZ350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350.

PRM_42548 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_8 | Default: 210.000 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42549 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_9 | Default: 17.500 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42550 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_10 | Default: 75.000 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

PRM_42551 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_11 | Default: 132.500 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

PRM_42552 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_12 | Default: 190.000 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

PRM_42553 [Mazak-Integrex-i350] — VAC_LOOP_GAIN_13 | Default: 247.500 | Linked Fault: ERR-MAZ350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350.

PRM_42554 [Mazak-Integrex-i350] — SPN_LOOP_GAIN_14 | Default: 55.000 | Linked Fault: ERR-MAZ350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350.

PRM_42555 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_15 | Default: 112.500 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42556 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_16 | Default: 170.000 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i350 (BANK 3)

Machine Model: Mazak-Integrex-i350 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i350.

PRM_42561 [Mazak-Integrex-i350] — CLN_LOOP_GAIN_1 | Default: 86.250 | Linked Fault: ERR-MAZ350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i350.

PRM_42562 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_2 | Default: 172.500 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

PRM_42563 [Mazak-Integrex-i350] — OPT_LOOP_GAIN_3 | Default: 8.750 | Linked Fault: ERR-MAZ350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i350.

PRM_42564 [Mazak-Integrex-i350] — VAC_LOOP_GAIN_4 | Default: 95.000 | Linked Fault: ERR-MAZ350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350.

PRM_42565 [Mazak-Integrex-i350] — ATC_LOOP_GAIN_5 | Default: 181.250 | Linked Fault: ERR-MAZ350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i350.

PRM_42566 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_6 | Default: 17.500 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42567 [Mazak-Integrex-i350] — BAX_LOOP_GAIN_7 | Default: 103.750 | Linked Fault: ERR-MAZ350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i350.

PRM_42568 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_8 | Default: 190.000 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

PRM_42569 [Mazak-Integrex-i350] — THM_LOOP_GAIN_9 | Default: 26.250 | Linked Fault: ERR-MAZ350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i350.

PRM_42570 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_10 | Default: 112.500 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42571 [Mazak-Integrex-i350] — YAX_LOOP_GAIN_11 | Default: 198.750 | Linked Fault: ERR-MAZ350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i350.

PRM_42572 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_12 | Default: 35.000 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

PRM_42573 [Mazak-Integrex-i350] — LUB_LOOP_GAIN_13 | Default: 121.250 | Linked Fault: ERR-MAZ350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i350.

PRM_42574 [Mazak-Integrex-i350] — SPN_LOOP_GAIN_14 | Default: 207.500 | Linked Fault: ERR-MAZ350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350.

PRM_42575 [Mazak-Integrex-i350] — CLN_LOOP_GAIN_15 | Default: 43.750 | Linked Fault: ERR-MAZ350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i350.

PRM_42576 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_16 | Default: 130.000 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i350 (BANK 4)

Machine Model: Mazak-Integrex-i350 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i350.

PRM_42581 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_1 | Default: 115.000 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42582 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_2 | Default: 230.000 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

PRM_42583 [Mazak-Integrex-i350] — VAC_LOOP_GAIN_3 | Default: 95.000 | Linked Fault: ERR-MAZ350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350.

PRM_42584 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_4 | Default: 210.000 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42585 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_5 | Default: 75.000 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

PRM_42586 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_6 | Default: 190.000 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

PRM_42587 [Mazak-Integrex-i350] — SPN_LOOP_GAIN_7 | Default: 55.000 | Linked Fault: ERR-MAZ350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350.

PRM_42588 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_8 | Default: 170.000 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42589 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_9 | Default: 35.000 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

PRM_42590 [Mazak-Integrex-i350] — VAC_LOOP_GAIN_10 | Default: 150.000 | Linked Fault: ERR-MAZ350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350.

PRM_42591 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_11 | Default: 15.000 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42592 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_12 | Default: 130.000 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

PRM_42593 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_13 | Default: 245.000 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

PRM_42594 [Mazak-Integrex-i350] — SPN_LOOP_GAIN_14 | Default: 110.000 | Linked Fault: ERR-MAZ350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350.

PRM_42595 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_15 | Default: 225.000 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42596 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_16 | Default: 90.000 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

CNC/PLC SERVO PARAMETER DICTIONARY — Mazak-Integrex-i350 (BANK 5)

Machine Model: Mazak-Integrex-i350 | Controller: MAZATROL SmoothAi. Use these parameter registers when diagnosing servo following errors or thermal drift on Mazak-Integrex-i350.

PRM_42601 [Mazak-Integrex-i350] — YAX_LOOP_GAIN_1 | Default: 143.750 | Linked Fault: ERR-MAZ350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i350.

PRM_42602 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_2 | Default: 37.500 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

PRM_42603 [Mazak-Integrex-i350] — ATC_LOOP_GAIN_3 | Default: 181.250 | Linked Fault: ERR-MAZ350-ATC-2
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Servo Tool Magazine & Gripper Arm on Mazak-Integrex-i350.

PRM_42604 [Mazak-Integrex-i350] — ZAX_LOOP_GAIN_4 | Default: 75.000 | Linked Fault: ERR-MAZ350-ZAX-7
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Z-Axis Nitrogen Counterbalance & Ram on Mazak-Integrex-i350.

PRM_42605 [Mazak-Integrex-i350] — LUB_LOOP_GAIN_5 | Default: 218.750 | Linked Fault: ERR-MAZ350-LUB-12
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Air-Oil Spindle Micro-Dosage Lubrication on Mazak-Integrex-i350.

PRM_42606 [Mazak-Integrex-i350] — HYD_LOOP_GAIN_6 | Default: 112.500 | Linked Fault: ERR-MAZ350-HYD-3
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Proportional Hydraulic Clamping Manifold on Mazak-Integrex-i350.

PRM_42607 [Mazak-Integrex-i350] — BAX_LOOP_GAIN_7 | Default: 6.250 | Linked Fault: ERR-MAZ350-BAX-8
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for B-Axis Direct-Drive Torque Swivel Head on Mazak-Integrex-i350.

PRM_42608 [Mazak-Integrex-i350] — VAC_LOOP_GAIN_8 | Default: 150.000 | Linked Fault: ERR-MAZ350-VAC-13
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Workpiece Vacuum Fixture & Seal Manifold on Mazak-Integrex-i350.

PRM_42609 [Mazak-Integrex-i350] — CLN_LOOP_GAIN_9 | Default: 43.750 | Linked Fault: ERR-MAZ350-CLN-4
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for 120-Bar Through-Tool Cryo Coolant Loop on Mazak-Integrex-i350.

PRM_42610 [Mazak-Integrex-i350] — CAX_LOOP_GAIN_10 | Default: 187.500 | Linked Fault: ERR-MAZ350-CAX-9
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for C-Axis Hydro-Mechanical Rotary Table on Mazak-Integrex-i350.

PRM_42611 [Mazak-Integrex-i350] — THM_LOOP_GAIN_11 | Default: 81.250 | Linked Fault: ERR-MAZ350-THM-14
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Active Glycol Chiller & Frame Compensation on Mazak-Integrex-i350.

PRM_42612 [Mazak-Integrex-i350] — XAX_LOOP_GAIN_12 | Default: 225.000 | Linked Fault: ERR-MAZ350-XAX-5
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for X-Axis Linear Motor & Hydrostatic Guideway on Mazak-Integrex-i350.

PRM_42613 [Mazak-Integrex-i350] — OPT_LOOP_GAIN_13 | Default: 118.750 | Linked Fault: ERR-MAZ350-OPT-10
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Pressurized Optical Glass Linear Scales on Mazak-Integrex-i350.

PRM_42614 [Mazak-Integrex-i350] — SPN_LOOP_GAIN_14 | Default: 12.500 | Linked Fault: ERR-MAZ350-SPN-1
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Electro-Spindle & Drawbar Assembly on Mazak-Integrex-i350.

PRM_42615 [Mazak-Integrex-i350] — YAX_LOOP_GAIN_15 | Default: 156.250 | Linked Fault: ERR-MAZ350-YAX-6
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Y-Axis Master-Slave Gantry Ballscrew on Mazak-Integrex-i350.

PRM_42616 [Mazak-Integrex-i350] — PLC_LOOP_GAIN_16 | Default: 50.000 | Linked Fault: ERR-MAZ350-PLC-11
Description: Closed-loop velocity feedforward and harmonic notch filter parameter for Profinet Safety I/O & IGBT Servo Drive on Mazak-Integrex-i350.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X150 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X150. Verify part numbers before closing Work Orders.

BOM-1-1.1: PN-SIE150-CAM-GRIP-102 / Kit KIT-SIE150-ATC-2 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-1-1.2: PN-SIE150-VLV-PROP-103 / Kit KIT-SIE150-HYD-3 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-1-1.3: PN-SIE150-PMP-PLG-104 / Kit KIT-SIE150-CLN-4 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-1-1.4: PN-SIE150-LIN-COIL-105 / Kit KIT-SIE150-XAX-5 (Siemens-DMG-X150) | PM Interval: 1250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-1-1.5: PN-SIE150-BSW-NUT-106 / Kit KIT-SIE150-YAX-6 (Siemens-DMG-X150) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-1-1.6: PN-SIE150-CYL-SEAL-107 / Kit KIT-SIE150-ZAX-7 (Siemens-DMG-X150) | PM Interval: 250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-1-1.7: PN-SIE150-TRQ-STAT-108 / Kit KIT-SIE150-BAX-8 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-1-1.8: PN-SIE150-BRK-DISC-109 / Kit KIT-SIE150-CAX-9 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-1-1.9: PN-SIE150-GLASS-RD-110 / Kit KIT-SIE150-OPT-10 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-1-1.10: PN-SIE150-IGBT-MOD-111 / Kit KIT-SIE150-PLC-11 (Siemens-DMG-X150) | PM Interval: 1250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-1-1.11: PN-SIE150-INJ-MET-112 / Kit KIT-SIE150-LUB-12 (Siemens-DMG-X150) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-1-1.12: PN-SIE150-GASK-VIT-113 / Kit KIT-SIE150-VAC-13 (Siemens-DMG-X150) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-1-1.13: PN-SIE150-HEX-CORE-114 / Kit KIT-SIE150-THM-14 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-1-1.14: PN-SIE150-BRG-CER-101 / Kit KIT-SIE150-SPN-1 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-1-1.15: PN-SIE150-CAM-GRIP-102 / Kit KIT-SIE150-ATC-2 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE150-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X150 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X150. Verify part numbers before closing Work Orders.

BOM-1-2.1: PN-SIE150-VLV-PROP-103 / Kit KIT-SIE150-HYD-3 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-1-2.2: PN-SIE150-LIN-COIL-105 / Kit KIT-SIE150-XAX-5 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-1-2.3: PN-SIE150-CYL-SEAL-107 / Kit KIT-SIE150-ZAX-7 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-1-2.4: PN-SIE150-BRK-DISC-109 / Kit KIT-SIE150-CAX-9 (Siemens-DMG-X150) | PM Interval: 1250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-1-2.5: PN-SIE150-IGBT-MOD-111 / Kit KIT-SIE150-PLC-11 (Siemens-DMG-X150) | PM Interval: 1500 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-1-2.6: PN-SIE150-GASK-VIT-113 / Kit KIT-SIE150-VAC-13 (Siemens-DMG-X150) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-1-2.7: PN-SIE150-BRG-CER-101 / Kit KIT-SIE150-SPN-1 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-1-2.8: PN-SIE150-VLV-PROP-103 / Kit KIT-SIE150-HYD-3 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-1-2.9: PN-SIE150-LIN-COIL-105 / Kit KIT-SIE150-XAX-5 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-1-2.10: PN-SIE150-CYL-SEAL-107 / Kit KIT-SIE150-ZAX-7 (Siemens-DMG-X150) | PM Interval: 1250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-1-2.11: PN-SIE150-BRK-DISC-109 / Kit KIT-SIE150-CAX-9 (Siemens-DMG-X150) | PM Interval: 1500 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-1-2.12: PN-SIE150-IGBT-MOD-111 / Kit KIT-SIE150-PLC-11 (Siemens-DMG-X150) | PM Interval: 250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-1-2.13: PN-SIE150-GASK-VIT-113 / Kit KIT-SIE150-VAC-13 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-1-2.14: PN-SIE150-BRG-CER-101 / Kit KIT-SIE150-SPN-1 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-1-2.15: PN-SIE150-VLV-PROP-103 / Kit KIT-SIE150-HYD-3 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE150-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X150 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X150. Verify part numbers before closing Work Orders.

BOM-1-3.1: PN-SIE150-PMP-PLG-104 / Kit KIT-SIE150-CLN-4 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE150-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-1-3.2: PN-SIE150-CYL-SEAL-107 / Kit KIT-SIE150-ZAX-7 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE150-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-1-3.3: PN-SIE150-GLASS-RD-110 / Kit KIT-SIE150-OPT-10 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE150-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-1-3.4: PN-SIE150-GASK-VIT-113 / Kit KIT-SIE150-VAC-13 (Siemens-DMG-X150) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE150-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-1-3.5: PN-SIE150-CAM-GRIP-102 / Kit KIT-SIE150-ATC-2 (Siemens-DMG-X150) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE150-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-1-3.6: PN-SIE150-LIN-COIL-105 / Kit KIT-SIE150-XAX-5 (Siemens-DMG-X150) | PM Interval: 250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE150-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-1-3.7: PN-SIE150-TRQ-STAT-108 / Kit KIT-SIE150-BAX-8 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE150-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-1-3.8: PN-SIE150-IGBT-MOD-111 / Kit KIT-SIE150-PLC-11 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE150-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-1-3.9: PN-SIE150-HEX-CORE-114 / Kit KIT-SIE150-THM-14 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE150-THM-14 within tolerance ±0.0040 mm TIR.

BOM-1-3.10: PN-SIE150-VLV-PROP-103 / Kit KIT-SIE150-HYD-3 (Siemens-DMG-X150) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE150-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-1-3.11: PN-SIE150-BSW-NUT-106 / Kit KIT-SIE150-YAX-6 (Siemens-DMG-X150) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE150-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-1-3.12: PN-SIE150-BRK-DISC-109 / Kit KIT-SIE150-CAX-9 (Siemens-DMG-X150) | PM Interval: 250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE150-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-1-3.13: PN-SIE150-INJ-MET-112 / Kit KIT-SIE150-LUB-12 (Siemens-DMG-X150) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE150-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-1-3.14: PN-SIE150-BRG-CER-101 / Kit KIT-SIE150-SPN-1 (Siemens-DMG-X150) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE150-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-1-3.15: PN-SIE150-PMP-PLG-104 / Kit KIT-SIE150-CLN-4 (Siemens-DMG-X150) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE150-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X200 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X200. Verify part numbers before closing Work Orders.

BOM-2-1.1: PN-SIE200-CAM-GRIP-202 / Kit KIT-SIE200-ATC-2 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-2-1.2: PN-SIE200-VLV-PROP-203 / Kit KIT-SIE200-HYD-3 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-2-1.3: PN-SIE200-PMP-PLG-204 / Kit KIT-SIE200-CLN-4 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-2-1.4: PN-SIE200-LIN-COIL-205 / Kit KIT-SIE200-XAX-5 (Siemens-DMG-X200) | PM Interval: 1250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-2-1.5: PN-SIE200-BSW-NUT-206 / Kit KIT-SIE200-YAX-6 (Siemens-DMG-X200) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-2-1.6: PN-SIE200-CYL-SEAL-207 / Kit KIT-SIE200-ZAX-7 (Siemens-DMG-X200) | PM Interval: 250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-2-1.7: PN-SIE200-TRQ-STAT-208 / Kit KIT-SIE200-BAX-8 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-2-1.8: PN-SIE200-BRK-DISC-209 / Kit KIT-SIE200-CAX-9 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-2-1.9: PN-SIE200-GLASS-RD-210 / Kit KIT-SIE200-OPT-10 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-2-1.10: PN-SIE200-IGBT-MOD-211 / Kit KIT-SIE200-PLC-11 (Siemens-DMG-X200) | PM Interval: 1250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-2-1.11: PN-SIE200-INJ-MET-212 / Kit KIT-SIE200-LUB-12 (Siemens-DMG-X200) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-2-1.12: PN-SIE200-GASK-VIT-213 / Kit KIT-SIE200-VAC-13 (Siemens-DMG-X200) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-2-1.13: PN-SIE200-HEX-CORE-214 / Kit KIT-SIE200-THM-14 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-2-1.14: PN-SIE200-BRG-CER-201 / Kit KIT-SIE200-SPN-1 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-2-1.15: PN-SIE200-CAM-GRIP-202 / Kit KIT-SIE200-ATC-2 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE200-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X200 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X200. Verify part numbers before closing Work Orders.

BOM-2-2.1: PN-SIE200-VLV-PROP-203 / Kit KIT-SIE200-HYD-3 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-2-2.2: PN-SIE200-LIN-COIL-205 / Kit KIT-SIE200-XAX-5 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-2-2.3: PN-SIE200-CYL-SEAL-207 / Kit KIT-SIE200-ZAX-7 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-2-2.4: PN-SIE200-BRK-DISC-209 / Kit KIT-SIE200-CAX-9 (Siemens-DMG-X200) | PM Interval: 1250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-2-2.5: PN-SIE200-IGBT-MOD-211 / Kit KIT-SIE200-PLC-11 (Siemens-DMG-X200) | PM Interval: 1500 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-2-2.6: PN-SIE200-GASK-VIT-213 / Kit KIT-SIE200-VAC-13 (Siemens-DMG-X200) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-2-2.7: PN-SIE200-BRG-CER-201 / Kit KIT-SIE200-SPN-1 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-2-2.8: PN-SIE200-VLV-PROP-203 / Kit KIT-SIE200-HYD-3 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-2-2.9: PN-SIE200-LIN-COIL-205 / Kit KIT-SIE200-XAX-5 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-2-2.10: PN-SIE200-CYL-SEAL-207 / Kit KIT-SIE200-ZAX-7 (Siemens-DMG-X200) | PM Interval: 1250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-2-2.11: PN-SIE200-BRK-DISC-209 / Kit KIT-SIE200-CAX-9 (Siemens-DMG-X200) | PM Interval: 1500 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-2-2.12: PN-SIE200-IGBT-MOD-211 / Kit KIT-SIE200-PLC-11 (Siemens-DMG-X200) | PM Interval: 250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-2-2.13: PN-SIE200-GASK-VIT-213 / Kit KIT-SIE200-VAC-13 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-2-2.14: PN-SIE200-BRG-CER-201 / Kit KIT-SIE200-SPN-1 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-2-2.15: PN-SIE200-VLV-PROP-203 / Kit KIT-SIE200-HYD-3 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE200-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X200 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X200. Verify part numbers before closing Work Orders.

BOM-2-3.1: PN-SIE200-PMP-PLG-204 / Kit KIT-SIE200-CLN-4 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE200-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-2-3.2: PN-SIE200-CYL-SEAL-207 / Kit KIT-SIE200-ZAX-7 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE200-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-2-3.3: PN-SIE200-GLASS-RD-210 / Kit KIT-SIE200-OPT-10 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE200-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-2-3.4: PN-SIE200-GASK-VIT-213 / Kit KIT-SIE200-VAC-13 (Siemens-DMG-X200) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE200-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-2-3.5: PN-SIE200-CAM-GRIP-202 / Kit KIT-SIE200-ATC-2 (Siemens-DMG-X200) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE200-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-2-3.6: PN-SIE200-LIN-COIL-205 / Kit KIT-SIE200-XAX-5 (Siemens-DMG-X200) | PM Interval: 250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE200-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-2-3.7: PN-SIE200-TRQ-STAT-208 / Kit KIT-SIE200-BAX-8 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE200-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-2-3.8: PN-SIE200-IGBT-MOD-211 / Kit KIT-SIE200-PLC-11 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE200-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-2-3.9: PN-SIE200-HEX-CORE-214 / Kit KIT-SIE200-THM-14 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE200-THM-14 within tolerance ±0.0060 mm TIR.

BOM-2-3.10: PN-SIE200-VLV-PROP-203 / Kit KIT-SIE200-HYD-3 (Siemens-DMG-X200) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE200-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-2-3.11: PN-SIE200-BSW-NUT-206 / Kit KIT-SIE200-YAX-6 (Siemens-DMG-X200) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE200-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-2-3.12: PN-SIE200-BRK-DISC-209 / Kit KIT-SIE200-CAX-9 (Siemens-DMG-X200) | PM Interval: 250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE200-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-2-3.13: PN-SIE200-INJ-MET-212 / Kit KIT-SIE200-LUB-12 (Siemens-DMG-X200) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE200-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-2-3.14: PN-SIE200-BRG-CER-201 / Kit KIT-SIE200-SPN-1 (Siemens-DMG-X200) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE200-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-2-3.15: PN-SIE200-PMP-PLG-204 / Kit KIT-SIE200-CLN-4 (Siemens-DMG-X200) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE200-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X250 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X250. Verify part numbers before closing Work Orders.

BOM-3-1.1: PN-SIE250-CAM-GRIP-302 / Kit KIT-SIE250-ATC-2 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-3-1.2: PN-SIE250-VLV-PROP-303 / Kit KIT-SIE250-HYD-3 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-3-1.3: PN-SIE250-PMP-PLG-304 / Kit KIT-SIE250-CLN-4 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-3-1.4: PN-SIE250-LIN-COIL-305 / Kit KIT-SIE250-XAX-5 (Siemens-DMG-X250) | PM Interval: 1250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-3-1.5: PN-SIE250-BSW-NUT-306 / Kit KIT-SIE250-YAX-6 (Siemens-DMG-X250) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-3-1.6: PN-SIE250-CYL-SEAL-307 / Kit KIT-SIE250-ZAX-7 (Siemens-DMG-X250) | PM Interval: 250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-3-1.7: PN-SIE250-TRQ-STAT-308 / Kit KIT-SIE250-BAX-8 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-3-1.8: PN-SIE250-BRK-DISC-309 / Kit KIT-SIE250-CAX-9 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-3-1.9: PN-SIE250-GLASS-RD-310 / Kit KIT-SIE250-OPT-10 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-3-1.10: PN-SIE250-IGBT-MOD-311 / Kit KIT-SIE250-PLC-11 (Siemens-DMG-X250) | PM Interval: 1250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-3-1.11: PN-SIE250-INJ-MET-312 / Kit KIT-SIE250-LUB-12 (Siemens-DMG-X250) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-3-1.12: PN-SIE250-GASK-VIT-313 / Kit KIT-SIE250-VAC-13 (Siemens-DMG-X250) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-3-1.13: PN-SIE250-HEX-CORE-314 / Kit KIT-SIE250-THM-14 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-3-1.14: PN-SIE250-BRG-CER-301 / Kit KIT-SIE250-SPN-1 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-3-1.15: PN-SIE250-CAM-GRIP-302 / Kit KIT-SIE250-ATC-2 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE250-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X250 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X250. Verify part numbers before closing Work Orders.

BOM-3-2.1: PN-SIE250-VLV-PROP-303 / Kit KIT-SIE250-HYD-3 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-3-2.2: PN-SIE250-LIN-COIL-305 / Kit KIT-SIE250-XAX-5 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-3-2.3: PN-SIE250-CYL-SEAL-307 / Kit KIT-SIE250-ZAX-7 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-3-2.4: PN-SIE250-BRK-DISC-309 / Kit KIT-SIE250-CAX-9 (Siemens-DMG-X250) | PM Interval: 1250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-3-2.5: PN-SIE250-IGBT-MOD-311 / Kit KIT-SIE250-PLC-11 (Siemens-DMG-X250) | PM Interval: 1500 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-3-2.6: PN-SIE250-GASK-VIT-313 / Kit KIT-SIE250-VAC-13 (Siemens-DMG-X250) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-3-2.7: PN-SIE250-BRG-CER-301 / Kit KIT-SIE250-SPN-1 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-3-2.8: PN-SIE250-VLV-PROP-303 / Kit KIT-SIE250-HYD-3 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-3-2.9: PN-SIE250-LIN-COIL-305 / Kit KIT-SIE250-XAX-5 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-3-2.10: PN-SIE250-CYL-SEAL-307 / Kit KIT-SIE250-ZAX-7 (Siemens-DMG-X250) | PM Interval: 1250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-3-2.11: PN-SIE250-BRK-DISC-309 / Kit KIT-SIE250-CAX-9 (Siemens-DMG-X250) | PM Interval: 1500 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-3-2.12: PN-SIE250-IGBT-MOD-311 / Kit KIT-SIE250-PLC-11 (Siemens-DMG-X250) | PM Interval: 250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-3-2.13: PN-SIE250-GASK-VIT-313 / Kit KIT-SIE250-VAC-13 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-3-2.14: PN-SIE250-BRG-CER-301 / Kit KIT-SIE250-SPN-1 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-3-2.15: PN-SIE250-VLV-PROP-303 / Kit KIT-SIE250-HYD-3 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE250-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X250 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X250. Verify part numbers before closing Work Orders.

BOM-3-3.1: PN-SIE250-PMP-PLG-304 / Kit KIT-SIE250-CLN-4 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE250-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-3-3.2: PN-SIE250-CYL-SEAL-307 / Kit KIT-SIE250-ZAX-7 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE250-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-3-3.3: PN-SIE250-GLASS-RD-310 / Kit KIT-SIE250-OPT-10 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE250-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-3-3.4: PN-SIE250-GASK-VIT-313 / Kit KIT-SIE250-VAC-13 (Siemens-DMG-X250) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE250-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-3-3.5: PN-SIE250-CAM-GRIP-302 / Kit KIT-SIE250-ATC-2 (Siemens-DMG-X250) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE250-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-3-3.6: PN-SIE250-LIN-COIL-305 / Kit KIT-SIE250-XAX-5 (Siemens-DMG-X250) | PM Interval: 250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE250-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-3-3.7: PN-SIE250-TRQ-STAT-308 / Kit KIT-SIE250-BAX-8 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE250-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-3-3.8: PN-SIE250-IGBT-MOD-311 / Kit KIT-SIE250-PLC-11 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE250-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-3-3.9: PN-SIE250-HEX-CORE-314 / Kit KIT-SIE250-THM-14 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE250-THM-14 within tolerance ±0.0020 mm TIR.

BOM-3-3.10: PN-SIE250-VLV-PROP-303 / Kit KIT-SIE250-HYD-3 (Siemens-DMG-X250) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE250-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-3-3.11: PN-SIE250-BSW-NUT-306 / Kit KIT-SIE250-YAX-6 (Siemens-DMG-X250) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE250-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-3-3.12: PN-SIE250-BRK-DISC-309 / Kit KIT-SIE250-CAX-9 (Siemens-DMG-X250) | PM Interval: 250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE250-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-3-3.13: PN-SIE250-INJ-MET-312 / Kit KIT-SIE250-LUB-12 (Siemens-DMG-X250) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE250-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-3-3.14: PN-SIE250-BRG-CER-301 / Kit KIT-SIE250-SPN-1 (Siemens-DMG-X250) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE250-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-3-3.15: PN-SIE250-PMP-PLG-304 / Kit KIT-SIE250-CLN-4 (Siemens-DMG-X250) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE250-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X300 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X300. Verify part numbers before closing Work Orders.

BOM-4-1.1: PN-SIE300-CAM-GRIP-402 / Kit KIT-SIE300-ATC-2 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-4-1.2: PN-SIE300-VLV-PROP-403 / Kit KIT-SIE300-HYD-3 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-4-1.3: PN-SIE300-PMP-PLG-404 / Kit KIT-SIE300-CLN-4 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-4-1.4: PN-SIE300-LIN-COIL-405 / Kit KIT-SIE300-XAX-5 (Siemens-DMG-X300) | PM Interval: 1250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-4-1.5: PN-SIE300-BSW-NUT-406 / Kit KIT-SIE300-YAX-6 (Siemens-DMG-X300) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-4-1.6: PN-SIE300-CYL-SEAL-407 / Kit KIT-SIE300-ZAX-7 (Siemens-DMG-X300) | PM Interval: 250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-4-1.7: PN-SIE300-TRQ-STAT-408 / Kit KIT-SIE300-BAX-8 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-4-1.8: PN-SIE300-BRK-DISC-409 / Kit KIT-SIE300-CAX-9 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-4-1.9: PN-SIE300-GLASS-RD-410 / Kit KIT-SIE300-OPT-10 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-4-1.10: PN-SIE300-IGBT-MOD-411 / Kit KIT-SIE300-PLC-11 (Siemens-DMG-X300) | PM Interval: 1250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-4-1.11: PN-SIE300-INJ-MET-412 / Kit KIT-SIE300-LUB-12 (Siemens-DMG-X300) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-4-1.12: PN-SIE300-GASK-VIT-413 / Kit KIT-SIE300-VAC-13 (Siemens-DMG-X300) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-4-1.13: PN-SIE300-HEX-CORE-414 / Kit KIT-SIE300-THM-14 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-4-1.14: PN-SIE300-BRG-CER-401 / Kit KIT-SIE300-SPN-1 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-4-1.15: PN-SIE300-CAM-GRIP-402 / Kit KIT-SIE300-ATC-2 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE300-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X300 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X300. Verify part numbers before closing Work Orders.

BOM-4-2.1: PN-SIE300-VLV-PROP-403 / Kit KIT-SIE300-HYD-3 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-4-2.2: PN-SIE300-LIN-COIL-405 / Kit KIT-SIE300-XAX-5 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-4-2.3: PN-SIE300-CYL-SEAL-407 / Kit KIT-SIE300-ZAX-7 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-4-2.4: PN-SIE300-BRK-DISC-409 / Kit KIT-SIE300-CAX-9 (Siemens-DMG-X300) | PM Interval: 1250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-4-2.5: PN-SIE300-IGBT-MOD-411 / Kit KIT-SIE300-PLC-11 (Siemens-DMG-X300) | PM Interval: 1500 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-4-2.6: PN-SIE300-GASK-VIT-413 / Kit KIT-SIE300-VAC-13 (Siemens-DMG-X300) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-4-2.7: PN-SIE300-BRG-CER-401 / Kit KIT-SIE300-SPN-1 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-4-2.8: PN-SIE300-VLV-PROP-403 / Kit KIT-SIE300-HYD-3 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-4-2.9: PN-SIE300-LIN-COIL-405 / Kit KIT-SIE300-XAX-5 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-4-2.10: PN-SIE300-CYL-SEAL-407 / Kit KIT-SIE300-ZAX-7 (Siemens-DMG-X300) | PM Interval: 1250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-4-2.11: PN-SIE300-BRK-DISC-409 / Kit KIT-SIE300-CAX-9 (Siemens-DMG-X300) | PM Interval: 1500 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-4-2.12: PN-SIE300-IGBT-MOD-411 / Kit KIT-SIE300-PLC-11 (Siemens-DMG-X300) | PM Interval: 250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-4-2.13: PN-SIE300-GASK-VIT-413 / Kit KIT-SIE300-VAC-13 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-4-2.14: PN-SIE300-BRG-CER-401 / Kit KIT-SIE300-SPN-1 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-4-2.15: PN-SIE300-VLV-PROP-403 / Kit KIT-SIE300-HYD-3 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE300-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X300 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X300. Verify part numbers before closing Work Orders.

BOM-4-3.1: PN-SIE300-PMP-PLG-404 / Kit KIT-SIE300-CLN-4 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE300-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-4-3.2: PN-SIE300-CYL-SEAL-407 / Kit KIT-SIE300-ZAX-7 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE300-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-4-3.3: PN-SIE300-GLASS-RD-410 / Kit KIT-SIE300-OPT-10 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE300-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-4-3.4: PN-SIE300-GASK-VIT-413 / Kit KIT-SIE300-VAC-13 (Siemens-DMG-X300) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE300-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-4-3.5: PN-SIE300-CAM-GRIP-402 / Kit KIT-SIE300-ATC-2 (Siemens-DMG-X300) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE300-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-4-3.6: PN-SIE300-LIN-COIL-405 / Kit KIT-SIE300-XAX-5 (Siemens-DMG-X300) | PM Interval: 250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE300-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-4-3.7: PN-SIE300-TRQ-STAT-408 / Kit KIT-SIE300-BAX-8 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE300-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-4-3.8: PN-SIE300-IGBT-MOD-411 / Kit KIT-SIE300-PLC-11 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE300-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-4-3.9: PN-SIE300-HEX-CORE-414 / Kit KIT-SIE300-THM-14 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE300-THM-14 within tolerance ±0.0040 mm TIR.

BOM-4-3.10: PN-SIE300-VLV-PROP-403 / Kit KIT-SIE300-HYD-3 (Siemens-DMG-X300) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE300-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-4-3.11: PN-SIE300-BSW-NUT-406 / Kit KIT-SIE300-YAX-6 (Siemens-DMG-X300) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE300-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-4-3.12: PN-SIE300-BRK-DISC-409 / Kit KIT-SIE300-CAX-9 (Siemens-DMG-X300) | PM Interval: 250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE300-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-4-3.13: PN-SIE300-INJ-MET-412 / Kit KIT-SIE300-LUB-12 (Siemens-DMG-X300) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE300-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-4-3.14: PN-SIE300-BRG-CER-401 / Kit KIT-SIE300-SPN-1 (Siemens-DMG-X300) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE300-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-4-3.15: PN-SIE300-PMP-PLG-404 / Kit KIT-SIE300-CLN-4 (Siemens-DMG-X300) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE300-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X350 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X350. Verify part numbers before closing Work Orders.

BOM-5-1.1: PN-SIE350-CAM-GRIP-502 / Kit KIT-SIE350-ATC-2 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-5-1.2: PN-SIE350-VLV-PROP-503 / Kit KIT-SIE350-HYD-3 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-5-1.3: PN-SIE350-PMP-PLG-504 / Kit KIT-SIE350-CLN-4 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-5-1.4: PN-SIE350-LIN-COIL-505 / Kit KIT-SIE350-XAX-5 (Siemens-DMG-X350) | PM Interval: 1250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-5-1.5: PN-SIE350-BSW-NUT-506 / Kit KIT-SIE350-YAX-6 (Siemens-DMG-X350) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-5-1.6: PN-SIE350-CYL-SEAL-507 / Kit KIT-SIE350-ZAX-7 (Siemens-DMG-X350) | PM Interval: 250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-5-1.7: PN-SIE350-TRQ-STAT-508 / Kit KIT-SIE350-BAX-8 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-5-1.8: PN-SIE350-BRK-DISC-509 / Kit KIT-SIE350-CAX-9 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-5-1.9: PN-SIE350-GLASS-RD-510 / Kit KIT-SIE350-OPT-10 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-5-1.10: PN-SIE350-IGBT-MOD-511 / Kit KIT-SIE350-PLC-11 (Siemens-DMG-X350) | PM Interval: 1250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-5-1.11: PN-SIE350-INJ-MET-512 / Kit KIT-SIE350-LUB-12 (Siemens-DMG-X350) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-5-1.12: PN-SIE350-GASK-VIT-513 / Kit KIT-SIE350-VAC-13 (Siemens-DMG-X350) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-5-1.13: PN-SIE350-HEX-CORE-514 / Kit KIT-SIE350-THM-14 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-5-1.14: PN-SIE350-BRG-CER-501 / Kit KIT-SIE350-SPN-1 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-5-1.15: PN-SIE350-CAM-GRIP-502 / Kit KIT-SIE350-ATC-2 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE350-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X350 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X350. Verify part numbers before closing Work Orders.

BOM-5-2.1: PN-SIE350-VLV-PROP-503 / Kit KIT-SIE350-HYD-3 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-5-2.2: PN-SIE350-LIN-COIL-505 / Kit KIT-SIE350-XAX-5 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-5-2.3: PN-SIE350-CYL-SEAL-507 / Kit KIT-SIE350-ZAX-7 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-5-2.4: PN-SIE350-BRK-DISC-509 / Kit KIT-SIE350-CAX-9 (Siemens-DMG-X350) | PM Interval: 1250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-5-2.5: PN-SIE350-IGBT-MOD-511 / Kit KIT-SIE350-PLC-11 (Siemens-DMG-X350) | PM Interval: 1500 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-5-2.6: PN-SIE350-GASK-VIT-513 / Kit KIT-SIE350-VAC-13 (Siemens-DMG-X350) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-5-2.7: PN-SIE350-BRG-CER-501 / Kit KIT-SIE350-SPN-1 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-5-2.8: PN-SIE350-VLV-PROP-503 / Kit KIT-SIE350-HYD-3 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-5-2.9: PN-SIE350-LIN-COIL-505 / Kit KIT-SIE350-XAX-5 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-5-2.10: PN-SIE350-CYL-SEAL-507 / Kit KIT-SIE350-ZAX-7 (Siemens-DMG-X350) | PM Interval: 1250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-5-2.11: PN-SIE350-BRK-DISC-509 / Kit KIT-SIE350-CAX-9 (Siemens-DMG-X350) | PM Interval: 1500 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-5-2.12: PN-SIE350-IGBT-MOD-511 / Kit KIT-SIE350-PLC-11 (Siemens-DMG-X350) | PM Interval: 250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-5-2.13: PN-SIE350-GASK-VIT-513 / Kit KIT-SIE350-VAC-13 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-5-2.14: PN-SIE350-BRG-CER-501 / Kit KIT-SIE350-SPN-1 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-5-2.15: PN-SIE350-VLV-PROP-503 / Kit KIT-SIE350-HYD-3 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE350-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X350 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X350. Verify part numbers before closing Work Orders.

BOM-5-3.1: PN-SIE350-PMP-PLG-504 / Kit KIT-SIE350-CLN-4 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE350-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-5-3.2: PN-SIE350-CYL-SEAL-507 / Kit KIT-SIE350-ZAX-7 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE350-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-5-3.3: PN-SIE350-GLASS-RD-510 / Kit KIT-SIE350-OPT-10 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE350-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-5-3.4: PN-SIE350-GASK-VIT-513 / Kit KIT-SIE350-VAC-13 (Siemens-DMG-X350) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE350-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-5-3.5: PN-SIE350-CAM-GRIP-502 / Kit KIT-SIE350-ATC-2 (Siemens-DMG-X350) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE350-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-5-3.6: PN-SIE350-LIN-COIL-505 / Kit KIT-SIE350-XAX-5 (Siemens-DMG-X350) | PM Interval: 250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE350-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-5-3.7: PN-SIE350-TRQ-STAT-508 / Kit KIT-SIE350-BAX-8 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE350-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-5-3.8: PN-SIE350-IGBT-MOD-511 / Kit KIT-SIE350-PLC-11 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE350-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-5-3.9: PN-SIE350-HEX-CORE-514 / Kit KIT-SIE350-THM-14 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE350-THM-14 within tolerance ±0.0060 mm TIR.

BOM-5-3.10: PN-SIE350-VLV-PROP-503 / Kit KIT-SIE350-HYD-3 (Siemens-DMG-X350) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE350-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-5-3.11: PN-SIE350-BSW-NUT-506 / Kit KIT-SIE350-YAX-6 (Siemens-DMG-X350) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE350-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-5-3.12: PN-SIE350-BRK-DISC-509 / Kit KIT-SIE350-CAX-9 (Siemens-DMG-X350) | PM Interval: 250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE350-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-5-3.13: PN-SIE350-INJ-MET-512 / Kit KIT-SIE350-LUB-12 (Siemens-DMG-X350) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE350-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-5-3.14: PN-SIE350-BRG-CER-501 / Kit KIT-SIE350-SPN-1 (Siemens-DMG-X350) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE350-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-5-3.15: PN-SIE350-PMP-PLG-504 / Kit KIT-SIE350-CLN-4 (Siemens-DMG-X350) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE350-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X400 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X400. Verify part numbers before closing Work Orders.

BOM-6-1.1: PN-SIE400-CAM-GRIP-602 / Kit KIT-SIE400-ATC-2 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-6-1.2: PN-SIE400-VLV-PROP-603 / Kit KIT-SIE400-HYD-3 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-6-1.3: PN-SIE400-PMP-PLG-604 / Kit KIT-SIE400-CLN-4 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-6-1.4: PN-SIE400-LIN-COIL-605 / Kit KIT-SIE400-XAX-5 (Siemens-DMG-X400) | PM Interval: 1250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-6-1.5: PN-SIE400-BSW-NUT-606 / Kit KIT-SIE400-YAX-6 (Siemens-DMG-X400) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-6-1.6: PN-SIE400-CYL-SEAL-607 / Kit KIT-SIE400-ZAX-7 (Siemens-DMG-X400) | PM Interval: 250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-6-1.7: PN-SIE400-TRQ-STAT-608 / Kit KIT-SIE400-BAX-8 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-6-1.8: PN-SIE400-BRK-DISC-609 / Kit KIT-SIE400-CAX-9 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-6-1.9: PN-SIE400-GLASS-RD-610 / Kit KIT-SIE400-OPT-10 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-6-1.10: PN-SIE400-IGBT-MOD-611 / Kit KIT-SIE400-PLC-11 (Siemens-DMG-X400) | PM Interval: 1250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-6-1.11: PN-SIE400-INJ-MET-612 / Kit KIT-SIE400-LUB-12 (Siemens-DMG-X400) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-6-1.12: PN-SIE400-GASK-VIT-613 / Kit KIT-SIE400-VAC-13 (Siemens-DMG-X400) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-6-1.13: PN-SIE400-HEX-CORE-614 / Kit KIT-SIE400-THM-14 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-6-1.14: PN-SIE400-BRG-CER-601 / Kit KIT-SIE400-SPN-1 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-6-1.15: PN-SIE400-CAM-GRIP-602 / Kit KIT-SIE400-ATC-2 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE400-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X400 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X400. Verify part numbers before closing Work Orders.

BOM-6-2.1: PN-SIE400-VLV-PROP-603 / Kit KIT-SIE400-HYD-3 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-6-2.2: PN-SIE400-LIN-COIL-605 / Kit KIT-SIE400-XAX-5 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-6-2.3: PN-SIE400-CYL-SEAL-607 / Kit KIT-SIE400-ZAX-7 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-6-2.4: PN-SIE400-BRK-DISC-609 / Kit KIT-SIE400-CAX-9 (Siemens-DMG-X400) | PM Interval: 1250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-6-2.5: PN-SIE400-IGBT-MOD-611 / Kit KIT-SIE400-PLC-11 (Siemens-DMG-X400) | PM Interval: 1500 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-6-2.6: PN-SIE400-GASK-VIT-613 / Kit KIT-SIE400-VAC-13 (Siemens-DMG-X400) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-6-2.7: PN-SIE400-BRG-CER-601 / Kit KIT-SIE400-SPN-1 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-6-2.8: PN-SIE400-VLV-PROP-603 / Kit KIT-SIE400-HYD-3 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-6-2.9: PN-SIE400-LIN-COIL-605 / Kit KIT-SIE400-XAX-5 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-6-2.10: PN-SIE400-CYL-SEAL-607 / Kit KIT-SIE400-ZAX-7 (Siemens-DMG-X400) | PM Interval: 1250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-6-2.11: PN-SIE400-BRK-DISC-609 / Kit KIT-SIE400-CAX-9 (Siemens-DMG-X400) | PM Interval: 1500 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-6-2.12: PN-SIE400-IGBT-MOD-611 / Kit KIT-SIE400-PLC-11 (Siemens-DMG-X400) | PM Interval: 250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-6-2.13: PN-SIE400-GASK-VIT-613 / Kit KIT-SIE400-VAC-13 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-6-2.14: PN-SIE400-BRG-CER-601 / Kit KIT-SIE400-SPN-1 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-6-2.15: PN-SIE400-VLV-PROP-603 / Kit KIT-SIE400-HYD-3 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE400-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X400 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X400. Verify part numbers before closing Work Orders.

BOM-6-3.1: PN-SIE400-PMP-PLG-604 / Kit KIT-SIE400-CLN-4 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE400-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-6-3.2: PN-SIE400-CYL-SEAL-607 / Kit KIT-SIE400-ZAX-7 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE400-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-6-3.3: PN-SIE400-GLASS-RD-610 / Kit KIT-SIE400-OPT-10 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE400-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-6-3.4: PN-SIE400-GASK-VIT-613 / Kit KIT-SIE400-VAC-13 (Siemens-DMG-X400) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE400-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-6-3.5: PN-SIE400-CAM-GRIP-602 / Kit KIT-SIE400-ATC-2 (Siemens-DMG-X400) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE400-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-6-3.6: PN-SIE400-LIN-COIL-605 / Kit KIT-SIE400-XAX-5 (Siemens-DMG-X400) | PM Interval: 250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE400-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-6-3.7: PN-SIE400-TRQ-STAT-608 / Kit KIT-SIE400-BAX-8 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE400-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-6-3.8: PN-SIE400-IGBT-MOD-611 / Kit KIT-SIE400-PLC-11 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE400-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-6-3.9: PN-SIE400-HEX-CORE-614 / Kit KIT-SIE400-THM-14 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE400-THM-14 within tolerance ±0.0020 mm TIR.

BOM-6-3.10: PN-SIE400-VLV-PROP-603 / Kit KIT-SIE400-HYD-3 (Siemens-DMG-X400) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE400-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-6-3.11: PN-SIE400-BSW-NUT-606 / Kit KIT-SIE400-YAX-6 (Siemens-DMG-X400) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE400-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-6-3.12: PN-SIE400-BRK-DISC-609 / Kit KIT-SIE400-CAX-9 (Siemens-DMG-X400) | PM Interval: 250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE400-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-6-3.13: PN-SIE400-INJ-MET-612 / Kit KIT-SIE400-LUB-12 (Siemens-DMG-X400) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE400-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-6-3.14: PN-SIE400-BRG-CER-601 / Kit KIT-SIE400-SPN-1 (Siemens-DMG-X400) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE400-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-6-3.15: PN-SIE400-PMP-PLG-604 / Kit KIT-SIE400-CLN-4 (Siemens-DMG-X400) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE400-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X450 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X450. Verify part numbers before closing Work Orders.

BOM-7-1.1: PN-SIE450-CAM-GRIP-702 / Kit KIT-SIE450-ATC-2 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-7-1.2: PN-SIE450-VLV-PROP-703 / Kit KIT-SIE450-HYD-3 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-7-1.3: PN-SIE450-PMP-PLG-704 / Kit KIT-SIE450-CLN-4 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-7-1.4: PN-SIE450-LIN-COIL-705 / Kit KIT-SIE450-XAX-5 (Siemens-DMG-X450) | PM Interval: 1250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-7-1.5: PN-SIE450-BSW-NUT-706 / Kit KIT-SIE450-YAX-6 (Siemens-DMG-X450) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-7-1.6: PN-SIE450-CYL-SEAL-707 / Kit KIT-SIE450-ZAX-7 (Siemens-DMG-X450) | PM Interval: 250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-7-1.7: PN-SIE450-TRQ-STAT-708 / Kit KIT-SIE450-BAX-8 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-7-1.8: PN-SIE450-BRK-DISC-709 / Kit KIT-SIE450-CAX-9 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-7-1.9: PN-SIE450-GLASS-RD-710 / Kit KIT-SIE450-OPT-10 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-7-1.10: PN-SIE450-IGBT-MOD-711 / Kit KIT-SIE450-PLC-11 (Siemens-DMG-X450) | PM Interval: 1250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-7-1.11: PN-SIE450-INJ-MET-712 / Kit KIT-SIE450-LUB-12 (Siemens-DMG-X450) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-7-1.12: PN-SIE450-GASK-VIT-713 / Kit KIT-SIE450-VAC-13 (Siemens-DMG-X450) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-7-1.13: PN-SIE450-HEX-CORE-714 / Kit KIT-SIE450-THM-14 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-7-1.14: PN-SIE450-BRG-CER-701 / Kit KIT-SIE450-SPN-1 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-7-1.15: PN-SIE450-CAM-GRIP-702 / Kit KIT-SIE450-ATC-2 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE450-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X450 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X450. Verify part numbers before closing Work Orders.

BOM-7-2.1: PN-SIE450-VLV-PROP-703 / Kit KIT-SIE450-HYD-3 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-7-2.2: PN-SIE450-LIN-COIL-705 / Kit KIT-SIE450-XAX-5 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-7-2.3: PN-SIE450-CYL-SEAL-707 / Kit KIT-SIE450-ZAX-7 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-7-2.4: PN-SIE450-BRK-DISC-709 / Kit KIT-SIE450-CAX-9 (Siemens-DMG-X450) | PM Interval: 1250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-7-2.5: PN-SIE450-IGBT-MOD-711 / Kit KIT-SIE450-PLC-11 (Siemens-DMG-X450) | PM Interval: 1500 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-7-2.6: PN-SIE450-GASK-VIT-713 / Kit KIT-SIE450-VAC-13 (Siemens-DMG-X450) | PM Interval: 250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-7-2.7: PN-SIE450-BRG-CER-701 / Kit KIT-SIE450-SPN-1 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-7-2.8: PN-SIE450-VLV-PROP-703 / Kit KIT-SIE450-HYD-3 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-7-2.9: PN-SIE450-LIN-COIL-705 / Kit KIT-SIE450-XAX-5 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-7-2.10: PN-SIE450-CYL-SEAL-707 / Kit KIT-SIE450-ZAX-7 (Siemens-DMG-X450) | PM Interval: 1250 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-7-2.11: PN-SIE450-BRK-DISC-709 / Kit KIT-SIE450-CAX-9 (Siemens-DMG-X450) | PM Interval: 1500 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-7-2.12: PN-SIE450-IGBT-MOD-711 / Kit KIT-SIE450-PLC-11 (Siemens-DMG-X450) | PM Interval: 250 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-7-2.13: PN-SIE450-GASK-VIT-713 / Kit KIT-SIE450-VAC-13 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-7-2.14: PN-SIE450-BRG-CER-701 / Kit KIT-SIE450-SPN-1 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-7-2.15: PN-SIE450-VLV-PROP-703 / Kit KIT-SIE450-HYD-3 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE450-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X450 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X450. Verify part numbers before closing Work Orders.

BOM-7-3.1: PN-SIE450-PMP-PLG-704 / Kit KIT-SIE450-CLN-4 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE450-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-7-3.2: PN-SIE450-CYL-SEAL-707 / Kit KIT-SIE450-ZAX-7 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 113 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE450-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-7-3.3: PN-SIE450-GLASS-RD-710 / Kit KIT-SIE450-OPT-10 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 140 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE450-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-7-3.4: PN-SIE450-GASK-VIT-713 / Kit KIT-SIE450-VAC-13 (Siemens-DMG-X450) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE450-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-7-3.5: PN-SIE450-CAM-GRIP-702 / Kit KIT-SIE450-ATC-2 (Siemens-DMG-X450) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE450-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-7-3.6: PN-SIE450-LIN-COIL-705 / Kit KIT-SIE450-XAX-5 (Siemens-DMG-X450) | PM Interval: 250 Hrs | Torque: 95 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE450-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-7-3.7: PN-SIE450-TRQ-STAT-708 / Kit KIT-SIE450-BAX-8 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 122 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE450-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-7-3.8: PN-SIE450-IGBT-MOD-711 / Kit KIT-SIE450-PLC-11 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE450-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-7-3.9: PN-SIE450-HEX-CORE-714 / Kit KIT-SIE450-THM-14 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE450-THM-14 within tolerance ±0.0040 mm TIR.

BOM-7-3.10: PN-SIE450-VLV-PROP-703 / Kit KIT-SIE450-HYD-3 (Siemens-DMG-X450) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE450-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-7-3.11: PN-SIE450-BSW-NUT-706 / Kit KIT-SIE450-YAX-6 (Siemens-DMG-X450) | PM Interval: 1500 Hrs | Torque: 104 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE450-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-7-3.12: PN-SIE450-BRK-DISC-709 / Kit KIT-SIE450-CAX-9 (Siemens-DMG-X450) | PM Interval: 250 Hrs | Torque: 131 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE450-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-7-3.13: PN-SIE450-INJ-MET-712 / Kit KIT-SIE450-LUB-12 (Siemens-DMG-X450) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE450-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-7-3.14: PN-SIE450-BRG-CER-701 / Kit KIT-SIE450-SPN-1 (Siemens-DMG-X450) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE450-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-7-3.15: PN-SIE450-PMP-PLG-704 / Kit KIT-SIE450-CLN-4 (Siemens-DMG-X450) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE450-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X500 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X500. Verify part numbers before closing Work Orders.

BOM-8-1.1: PN-SIE500-CAM-GRIP-802 / Kit KIT-SIE500-ATC-2 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-8-1.2: PN-SIE500-VLV-PROP-803 / Kit KIT-SIE500-HYD-3 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-8-1.3: PN-SIE500-PMP-PLG-804 / Kit KIT-SIE500-CLN-4 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-8-1.4: PN-SIE500-LIN-COIL-805 / Kit KIT-SIE500-XAX-5 (Siemens-DMG-X500) | PM Interval: 1250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-8-1.5: PN-SIE500-BSW-NUT-806 / Kit KIT-SIE500-YAX-6 (Siemens-DMG-X500) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-8-1.6: PN-SIE500-CYL-SEAL-807 / Kit KIT-SIE500-ZAX-7 (Siemens-DMG-X500) | PM Interval: 250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-8-1.7: PN-SIE500-TRQ-STAT-808 / Kit KIT-SIE500-BAX-8 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-8-1.8: PN-SIE500-BRK-DISC-809 / Kit KIT-SIE500-CAX-9 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-8-1.9: PN-SIE500-GLASS-RD-810 / Kit KIT-SIE500-OPT-10 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-8-1.10: PN-SIE500-IGBT-MOD-811 / Kit KIT-SIE500-PLC-11 (Siemens-DMG-X500) | PM Interval: 1250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-8-1.11: PN-SIE500-INJ-MET-812 / Kit KIT-SIE500-LUB-12 (Siemens-DMG-X500) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-8-1.12: PN-SIE500-GASK-VIT-813 / Kit KIT-SIE500-VAC-13 (Siemens-DMG-X500) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-8-1.13: PN-SIE500-HEX-CORE-814 / Kit KIT-SIE500-THM-14 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-8-1.14: PN-SIE500-BRG-CER-801 / Kit KIT-SIE500-SPN-1 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-8-1.15: PN-SIE500-CAM-GRIP-802 / Kit KIT-SIE500-ATC-2 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE500-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X500 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X500. Verify part numbers before closing Work Orders.

BOM-8-2.1: PN-SIE500-VLV-PROP-803 / Kit KIT-SIE500-HYD-3 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-8-2.2: PN-SIE500-LIN-COIL-805 / Kit KIT-SIE500-XAX-5 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-8-2.3: PN-SIE500-CYL-SEAL-807 / Kit KIT-SIE500-ZAX-7 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-8-2.4: PN-SIE500-BRK-DISC-809 / Kit KIT-SIE500-CAX-9 (Siemens-DMG-X500) | PM Interval: 1250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-8-2.5: PN-SIE500-IGBT-MOD-811 / Kit KIT-SIE500-PLC-11 (Siemens-DMG-X500) | PM Interval: 1500 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-8-2.6: PN-SIE500-GASK-VIT-813 / Kit KIT-SIE500-VAC-13 (Siemens-DMG-X500) | PM Interval: 250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-8-2.7: PN-SIE500-BRG-CER-801 / Kit KIT-SIE500-SPN-1 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-8-2.8: PN-SIE500-VLV-PROP-803 / Kit KIT-SIE500-HYD-3 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-8-2.9: PN-SIE500-LIN-COIL-805 / Kit KIT-SIE500-XAX-5 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-8-2.10: PN-SIE500-CYL-SEAL-807 / Kit KIT-SIE500-ZAX-7 (Siemens-DMG-X500) | PM Interval: 1250 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-8-2.11: PN-SIE500-BRK-DISC-809 / Kit KIT-SIE500-CAX-9 (Siemens-DMG-X500) | PM Interval: 1500 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-8-2.12: PN-SIE500-IGBT-MOD-811 / Kit KIT-SIE500-PLC-11 (Siemens-DMG-X500) | PM Interval: 250 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-8-2.13: PN-SIE500-GASK-VIT-813 / Kit KIT-SIE500-VAC-13 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-8-2.14: PN-SIE500-BRG-CER-801 / Kit KIT-SIE500-SPN-1 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-8-2.15: PN-SIE500-VLV-PROP-803 / Kit KIT-SIE500-HYD-3 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE500-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X500 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X500. Verify part numbers before closing Work Orders.

BOM-8-3.1: PN-SIE500-PMP-PLG-804 / Kit KIT-SIE500-CLN-4 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE500-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-8-3.2: PN-SIE500-CYL-SEAL-807 / Kit KIT-SIE500-ZAX-7 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 63 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE500-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-8-3.3: PN-SIE500-GLASS-RD-810 / Kit KIT-SIE500-OPT-10 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 90 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE500-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-8-3.4: PN-SIE500-GASK-VIT-813 / Kit KIT-SIE500-VAC-13 (Siemens-DMG-X500) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE500-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-8-3.5: PN-SIE500-CAM-GRIP-802 / Kit KIT-SIE500-ATC-2 (Siemens-DMG-X500) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE500-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-8-3.6: PN-SIE500-LIN-COIL-805 / Kit KIT-SIE500-XAX-5 (Siemens-DMG-X500) | PM Interval: 250 Hrs | Torque: 45 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE500-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-8-3.7: PN-SIE500-TRQ-STAT-808 / Kit KIT-SIE500-BAX-8 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 72 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE500-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-8-3.8: PN-SIE500-IGBT-MOD-811 / Kit KIT-SIE500-PLC-11 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE500-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-8-3.9: PN-SIE500-HEX-CORE-814 / Kit KIT-SIE500-THM-14 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE500-THM-14 within tolerance ±0.0060 mm TIR.

BOM-8-3.10: PN-SIE500-VLV-PROP-803 / Kit KIT-SIE500-HYD-3 (Siemens-DMG-X500) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE500-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-8-3.11: PN-SIE500-BSW-NUT-806 / Kit KIT-SIE500-YAX-6 (Siemens-DMG-X500) | PM Interval: 1500 Hrs | Torque: 54 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE500-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-8-3.12: PN-SIE500-BRK-DISC-809 / Kit KIT-SIE500-CAX-9 (Siemens-DMG-X500) | PM Interval: 250 Hrs | Torque: 81 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE500-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-8-3.13: PN-SIE500-INJ-MET-812 / Kit KIT-SIE500-LUB-12 (Siemens-DMG-X500) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE500-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-8-3.14: PN-SIE500-BRG-CER-801 / Kit KIT-SIE500-SPN-1 (Siemens-DMG-X500) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE500-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-8-3.15: PN-SIE500-PMP-PLG-804 / Kit KIT-SIE500-CLN-4 (Siemens-DMG-X500) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE500-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X550 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X550. Verify part numbers before closing Work Orders.

BOM-9-1.1: PN-SIE550-CAM-GRIP-902 / Kit KIT-SIE550-ATC-2 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-9-1.2: PN-SIE550-VLV-PROP-903 / Kit KIT-SIE550-HYD-3 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-9-1.3: PN-SIE550-PMP-PLG-904 / Kit KIT-SIE550-CLN-4 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-9-1.4: PN-SIE550-LIN-COIL-905 / Kit KIT-SIE550-XAX-5 (Siemens-DMG-X550) | PM Interval: 1250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-9-1.5: PN-SIE550-BSW-NUT-906 / Kit KIT-SIE550-YAX-6 (Siemens-DMG-X550) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-9-1.6: PN-SIE550-CYL-SEAL-907 / Kit KIT-SIE550-ZAX-7 (Siemens-DMG-X550) | PM Interval: 250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-9-1.7: PN-SIE550-TRQ-STAT-908 / Kit KIT-SIE550-BAX-8 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-9-1.8: PN-SIE550-BRK-DISC-909 / Kit KIT-SIE550-CAX-9 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-9-1.9: PN-SIE550-GLASS-RD-910 / Kit KIT-SIE550-OPT-10 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-9-1.10: PN-SIE550-IGBT-MOD-911 / Kit KIT-SIE550-PLC-11 (Siemens-DMG-X550) | PM Interval: 1250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-9-1.11: PN-SIE550-INJ-MET-912 / Kit KIT-SIE550-LUB-12 (Siemens-DMG-X550) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-9-1.12: PN-SIE550-GASK-VIT-913 / Kit KIT-SIE550-VAC-13 (Siemens-DMG-X550) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-9-1.13: PN-SIE550-HEX-CORE-914 / Kit KIT-SIE550-THM-14 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-9-1.14: PN-SIE550-BRG-CER-901 / Kit KIT-SIE550-SPN-1 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-9-1.15: PN-SIE550-CAM-GRIP-902 / Kit KIT-SIE550-ATC-2 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE550-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X550 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X550. Verify part numbers before closing Work Orders.

BOM-9-2.1: PN-SIE550-VLV-PROP-903 / Kit KIT-SIE550-HYD-3 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-9-2.2: PN-SIE550-LIN-COIL-905 / Kit KIT-SIE550-XAX-5 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-9-2.3: PN-SIE550-CYL-SEAL-907 / Kit KIT-SIE550-ZAX-7 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-9-2.4: PN-SIE550-BRK-DISC-909 / Kit KIT-SIE550-CAX-9 (Siemens-DMG-X550) | PM Interval: 1250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-9-2.5: PN-SIE550-IGBT-MOD-911 / Kit KIT-SIE550-PLC-11 (Siemens-DMG-X550) | PM Interval: 1500 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-9-2.6: PN-SIE550-GASK-VIT-913 / Kit KIT-SIE550-VAC-13 (Siemens-DMG-X550) | PM Interval: 250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-9-2.7: PN-SIE550-BRG-CER-901 / Kit KIT-SIE550-SPN-1 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-9-2.8: PN-SIE550-VLV-PROP-903 / Kit KIT-SIE550-HYD-3 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-9-2.9: PN-SIE550-LIN-COIL-905 / Kit KIT-SIE550-XAX-5 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-9-2.10: PN-SIE550-CYL-SEAL-907 / Kit KIT-SIE550-ZAX-7 (Siemens-DMG-X550) | PM Interval: 1250 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-9-2.11: PN-SIE550-BRK-DISC-909 / Kit KIT-SIE550-CAX-9 (Siemens-DMG-X550) | PM Interval: 1500 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-9-2.12: PN-SIE550-IGBT-MOD-911 / Kit KIT-SIE550-PLC-11 (Siemens-DMG-X550) | PM Interval: 250 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-9-2.13: PN-SIE550-GASK-VIT-913 / Kit KIT-SIE550-VAC-13 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-9-2.14: PN-SIE550-BRG-CER-901 / Kit KIT-SIE550-SPN-1 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-9-2.15: PN-SIE550-VLV-PROP-903 / Kit KIT-SIE550-HYD-3 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE550-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X550 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X550. Verify part numbers before closing Work Orders.

BOM-9-3.1: PN-SIE550-PMP-PLG-904 / Kit KIT-SIE550-CLN-4 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE550-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-9-3.2: PN-SIE550-CYL-SEAL-907 / Kit KIT-SIE550-ZAX-7 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 203 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE550-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-9-3.3: PN-SIE550-GLASS-RD-910 / Kit KIT-SIE550-OPT-10 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 40 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE550-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-9-3.4: PN-SIE550-GASK-VIT-913 / Kit KIT-SIE550-VAC-13 (Siemens-DMG-X550) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE550-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-9-3.5: PN-SIE550-CAM-GRIP-902 / Kit KIT-SIE550-ATC-2 (Siemens-DMG-X550) | PM Interval: 1500 Hrs | Torque: 158 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE550-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-9-3.6: PN-SIE550-LIN-COIL-905 / Kit KIT-SIE550-XAX-5 (Siemens-DMG-X550) | PM Interval: 250 Hrs | Torque: 185 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE550-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-9-3.7: PN-SIE550-TRQ-STAT-908 / Kit KIT-SIE550-BAX-8 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 22 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE550-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-9-3.8: PN-SIE550-IGBT-MOD-911 / Kit KIT-SIE550-PLC-11 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE550-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-9-3.9: PN-SIE550-HEX-CORE-914 / Kit KIT-SIE550-THM-14 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE550-THM-14 within tolerance ±0.0020 mm TIR.

BOM-9-3.10: PN-SIE550-VLV-PROP-903 / Kit KIT-SIE550-HYD-3 (Siemens-DMG-X550) | PM Interval: 1250 Hrs | Torque: 167 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE550-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-9-3.11: PN-SIE550-BSW-NUT-906 / Kit KIT-SIE550-YAX-6 (Siemens-DMG-X550) | PM Interval: 1500 Hrs | Torque: 194 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE550-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-9-3.12: PN-SIE550-BRK-DISC-909 / Kit KIT-SIE550-CAX-9 (Siemens-DMG-X550) | PM Interval: 250 Hrs | Torque: 31 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE550-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-9-3.13: PN-SIE550-INJ-MET-912 / Kit KIT-SIE550-LUB-12 (Siemens-DMG-X550) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE550-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-9-3.14: PN-SIE550-BRG-CER-901 / Kit KIT-SIE550-SPN-1 (Siemens-DMG-X550) | PM Interval: 750 Hrs | Torque: 149 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE550-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-9-3.15: PN-SIE550-PMP-PLG-904 / Kit KIT-SIE550-CLN-4 (Siemens-DMG-X550) | PM Interval: 1000 Hrs | Torque: 176 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE550-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X600 (GROUP 1)

Certified Aerospace Bill of Materials for Siemens-DMG-X600. Verify part numbers before closing Work Orders.

BOM-10-1.1: PN-SIE600-CAM-GRIP-1002 / Kit KIT-SIE600-ATC-2 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-10-1.2: PN-SIE600-VLV-PROP-1003 / Kit KIT-SIE600-HYD-3 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-10-1.3: PN-SIE600-PMP-PLG-1004 / Kit KIT-SIE600-CLN-4 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-10-1.4: PN-SIE600-LIN-COIL-1005 / Kit KIT-SIE600-XAX-5 (Siemens-DMG-X600) | PM Interval: 1250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-10-1.5: PN-SIE600-BSW-NUT-1006 / Kit KIT-SIE600-YAX-6 (Siemens-DMG-X600) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-10-1.6: PN-SIE600-CYL-SEAL-1007 / Kit KIT-SIE600-ZAX-7 (Siemens-DMG-X600) | PM Interval: 250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-10-1.7: PN-SIE600-TRQ-STAT-1008 / Kit KIT-SIE600-BAX-8 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-10-1.8: PN-SIE600-BRK-DISC-1009 / Kit KIT-SIE600-CAX-9 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-10-1.9: PN-SIE600-GLASS-RD-1010 / Kit KIT-SIE600-OPT-10 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-10-1.10: PN-SIE600-IGBT-MOD-1011 / Kit KIT-SIE600-PLC-11 (Siemens-DMG-X600) | PM Interval: 1250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-10-1.11: PN-SIE600-INJ-MET-1012 / Kit KIT-SIE600-LUB-12 (Siemens-DMG-X600) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-10-1.12: PN-SIE600-GASK-VIT-1013 / Kit KIT-SIE600-VAC-13 (Siemens-DMG-X600) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-10-1.13: PN-SIE600-HEX-CORE-1014 / Kit KIT-SIE600-THM-14 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-10-1.14: PN-SIE600-BRG-CER-1001 / Kit KIT-SIE600-SPN-1 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-10-1.15: PN-SIE600-CAM-GRIP-1002 / Kit KIT-SIE600-ATC-2 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE600-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X600 (GROUP 2)

Certified Aerospace Bill of Materials for Siemens-DMG-X600. Verify part numbers before closing Work Orders.

BOM-10-2.1: PN-SIE600-VLV-PROP-1003 / Kit KIT-SIE600-HYD-3 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-10-2.2: PN-SIE600-LIN-COIL-1005 / Kit KIT-SIE600-XAX-5 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-10-2.3: PN-SIE600-CYL-SEAL-1007 / Kit KIT-SIE600-ZAX-7 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-10-2.4: PN-SIE600-BRK-DISC-1009 / Kit KIT-SIE600-CAX-9 (Siemens-DMG-X600) | PM Interval: 1250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-10-2.5: PN-SIE600-IGBT-MOD-1011 / Kit KIT-SIE600-PLC-11 (Siemens-DMG-X600) | PM Interval: 1500 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-10-2.6: PN-SIE600-GASK-VIT-1013 / Kit KIT-SIE600-VAC-13 (Siemens-DMG-X600) | PM Interval: 250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-10-2.7: PN-SIE600-BRG-CER-1001 / Kit KIT-SIE600-SPN-1 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-10-2.8: PN-SIE600-VLV-PROP-1003 / Kit KIT-SIE600-HYD-3 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-10-2.9: PN-SIE600-LIN-COIL-1005 / Kit KIT-SIE600-XAX-5 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-10-2.10: PN-SIE600-CYL-SEAL-1007 / Kit KIT-SIE600-ZAX-7 (Siemens-DMG-X600) | PM Interval: 1250 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-10-2.11: PN-SIE600-BRK-DISC-1009 / Kit KIT-SIE600-CAX-9 (Siemens-DMG-X600) | PM Interval: 1500 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-10-2.12: PN-SIE600-IGBT-MOD-1011 / Kit KIT-SIE600-PLC-11 (Siemens-DMG-X600) | PM Interval: 250 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-10-2.13: PN-SIE600-GASK-VIT-1013 / Kit KIT-SIE600-VAC-13 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-10-2.14: PN-SIE600-BRG-CER-1001 / Kit KIT-SIE600-SPN-1 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-10-2.15: PN-SIE600-VLV-PROP-1003 / Kit KIT-SIE600-HYD-3 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE600-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Siemens-DMG-X600 (GROUP 3)

Certified Aerospace Bill of Materials for Siemens-DMG-X600. Verify part numbers before closing Work Orders.

BOM-10-3.1: PN-SIE600-PMP-PLG-1004 / Kit KIT-SIE600-CLN-4 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE600-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-10-3.2: PN-SIE600-CYL-SEAL-1007 / Kit KIT-SIE600-ZAX-7 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 153 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-SIE600-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-10-3.3: PN-SIE600-GLASS-RD-1010 / Kit KIT-SIE600-OPT-10 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 180 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-SIE600-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-10-3.4: PN-SIE600-GASK-VIT-1013 / Kit KIT-SIE600-VAC-13 (Siemens-DMG-X600) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-SIE600-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-10-3.5: PN-SIE600-CAM-GRIP-1002 / Kit KIT-SIE600-ATC-2 (Siemens-DMG-X600) | PM Interval: 1500 Hrs | Torque: 108 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-SIE600-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-10-3.6: PN-SIE600-LIN-COIL-1005 / Kit KIT-SIE600-XAX-5 (Siemens-DMG-X600) | PM Interval: 250 Hrs | Torque: 135 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-SIE600-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-10-3.7: PN-SIE600-TRQ-STAT-1008 / Kit KIT-SIE600-BAX-8 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 162 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-SIE600-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-10-3.8: PN-SIE600-IGBT-MOD-1011 / Kit KIT-SIE600-PLC-11 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-SIE600-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-10-3.9: PN-SIE600-HEX-CORE-1014 / Kit KIT-SIE600-THM-14 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-SIE600-THM-14 within tolerance ±0.0040 mm TIR.

BOM-10-3.10: PN-SIE600-VLV-PROP-1003 / Kit KIT-SIE600-HYD-3 (Siemens-DMG-X600) | PM Interval: 1250 Hrs | Torque: 117 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-SIE600-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-10-3.11: PN-SIE600-BSW-NUT-1006 / Kit KIT-SIE600-YAX-6 (Siemens-DMG-X600) | PM Interval: 1500 Hrs | Torque: 144 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-SIE600-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-10-3.12: PN-SIE600-BRK-DISC-1009 / Kit KIT-SIE600-CAX-9 (Siemens-DMG-X600) | PM Interval: 250 Hrs | Torque: 171 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-SIE600-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-10-3.13: PN-SIE600-INJ-MET-1012 / Kit KIT-SIE600-LUB-12 (Siemens-DMG-X600) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-SIE600-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-10-3.14: PN-SIE600-BRG-CER-1001 / Kit KIT-SIE600-SPN-1 (Siemens-DMG-X600) | PM Interval: 750 Hrs | Torque: 99 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-SIE600-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-10-3.15: PN-SIE600-PMP-PLG-1004 / Kit KIT-SIE600-CLN-4 (Siemens-DMG-X600) | PM Interval: 1000 Hrs | Torque: 126 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-SIE600-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T150 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T150. Verify part numbers before closing Work Orders.

BOM-11-1.1: PN-MAK150-CAM-GRIP-1102 / Kit KIT-MAK150-ATC-2 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK150-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-11-1.2: PN-MAK150-VLV-PROP-1103 / Kit KIT-MAK150-HYD-3 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-11-1.3: PN-MAK150-PMP-PLG-1104 / Kit KIT-MAK150-CLN-4 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK150-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-11-1.4: PN-MAK150-LIN-COIL-1105 / Kit KIT-MAK150-XAX-5 (Makino-Aero-T150) | PM Interval: 1250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-11-1.5: PN-MAK150-BSW-NUT-1106 / Kit KIT-MAK150-YAX-6 (Makino-Aero-T150) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK150-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-11-1.6: PN-MAK150-CYL-SEAL-1107 / Kit KIT-MAK150-ZAX-7 (Makino-Aero-T150) | PM Interval: 250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-11-1.7: PN-MAK150-TRQ-STAT-1108 / Kit KIT-MAK150-BAX-8 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK150-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-11-1.8: PN-MAK150-BRK-DISC-1109 / Kit KIT-MAK150-CAX-9 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-11-1.9: PN-MAK150-GLASS-RD-1110 / Kit KIT-MAK150-OPT-10 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK150-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-11-1.10: PN-MAK150-IGBT-MOD-1111 / Kit KIT-MAK150-PLC-11 (Makino-Aero-T150) | PM Interval: 1250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-11-1.11: PN-MAK150-INJ-MET-1112 / Kit KIT-MAK150-LUB-12 (Makino-Aero-T150) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK150-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-11-1.12: PN-MAK150-GASK-VIT-1113 / Kit KIT-MAK150-VAC-13 (Makino-Aero-T150) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-11-1.13: PN-MAK150-HEX-CORE-1114 / Kit KIT-MAK150-THM-14 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK150-THM-14 within tolerance ±0.0060 mm TIR.

BOM-11-1.14: PN-MAK150-BRG-CER-1101 / Kit KIT-MAK150-SPN-1 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-11-1.15: PN-MAK150-CAM-GRIP-1102 / Kit KIT-MAK150-ATC-2 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK150-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T150 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T150. Verify part numbers before closing Work Orders.

BOM-11-2.1: PN-MAK150-VLV-PROP-1103 / Kit KIT-MAK150-HYD-3 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-11-2.2: PN-MAK150-LIN-COIL-1105 / Kit KIT-MAK150-XAX-5 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-11-2.3: PN-MAK150-CYL-SEAL-1107 / Kit KIT-MAK150-ZAX-7 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-11-2.4: PN-MAK150-BRK-DISC-1109 / Kit KIT-MAK150-CAX-9 (Makino-Aero-T150) | PM Interval: 1250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-11-2.5: PN-MAK150-IGBT-MOD-1111 / Kit KIT-MAK150-PLC-11 (Makino-Aero-T150) | PM Interval: 1500 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-11-2.6: PN-MAK150-GASK-VIT-1113 / Kit KIT-MAK150-VAC-13 (Makino-Aero-T150) | PM Interval: 250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-11-2.7: PN-MAK150-BRG-CER-1101 / Kit KIT-MAK150-SPN-1 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-11-2.8: PN-MAK150-VLV-PROP-1103 / Kit KIT-MAK150-HYD-3 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-11-2.9: PN-MAK150-LIN-COIL-1105 / Kit KIT-MAK150-XAX-5 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-11-2.10: PN-MAK150-CYL-SEAL-1107 / Kit KIT-MAK150-ZAX-7 (Makino-Aero-T150) | PM Interval: 1250 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-11-2.11: PN-MAK150-BRK-DISC-1109 / Kit KIT-MAK150-CAX-9 (Makino-Aero-T150) | PM Interval: 1500 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-11-2.12: PN-MAK150-IGBT-MOD-1111 / Kit KIT-MAK150-PLC-11 (Makino-Aero-T150) | PM Interval: 250 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-11-2.13: PN-MAK150-GASK-VIT-1113 / Kit KIT-MAK150-VAC-13 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-11-2.14: PN-MAK150-BRG-CER-1101 / Kit KIT-MAK150-SPN-1 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-11-2.15: PN-MAK150-VLV-PROP-1103 / Kit KIT-MAK150-HYD-3 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK150-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T150 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T150. Verify part numbers before closing Work Orders.

BOM-11-3.1: PN-MAK150-PMP-PLG-1104 / Kit KIT-MAK150-CLN-4 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK150-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-11-3.2: PN-MAK150-CYL-SEAL-1107 / Kit KIT-MAK150-ZAX-7 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 103 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK150-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-11-3.3: PN-MAK150-GLASS-RD-1110 / Kit KIT-MAK150-OPT-10 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 130 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK150-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-11-3.4: PN-MAK150-GASK-VIT-1113 / Kit KIT-MAK150-VAC-13 (Makino-Aero-T150) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK150-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-11-3.5: PN-MAK150-CAM-GRIP-1102 / Kit KIT-MAK150-ATC-2 (Makino-Aero-T150) | PM Interval: 1500 Hrs | Torque: 58 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK150-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-11-3.6: PN-MAK150-LIN-COIL-1105 / Kit KIT-MAK150-XAX-5 (Makino-Aero-T150) | PM Interval: 250 Hrs | Torque: 85 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK150-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-11-3.7: PN-MAK150-TRQ-STAT-1108 / Kit KIT-MAK150-BAX-8 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 112 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK150-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-11-3.8: PN-MAK150-IGBT-MOD-1111 / Kit KIT-MAK150-PLC-11 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK150-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-11-3.9: PN-MAK150-HEX-CORE-1114 / Kit KIT-MAK150-THM-14 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK150-THM-14 within tolerance ±0.0060 mm TIR.

BOM-11-3.10: PN-MAK150-VLV-PROP-1103 / Kit KIT-MAK150-HYD-3 (Makino-Aero-T150) | PM Interval: 1250 Hrs | Torque: 67 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK150-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-11-3.11: PN-MAK150-BSW-NUT-1106 / Kit KIT-MAK150-YAX-6 (Makino-Aero-T150) | PM Interval: 1500 Hrs | Torque: 94 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK150-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-11-3.12: PN-MAK150-BRK-DISC-1109 / Kit KIT-MAK150-CAX-9 (Makino-Aero-T150) | PM Interval: 250 Hrs | Torque: 121 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK150-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-11-3.13: PN-MAK150-INJ-MET-1112 / Kit KIT-MAK150-LUB-12 (Makino-Aero-T150) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK150-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-11-3.14: PN-MAK150-BRG-CER-1101 / Kit KIT-MAK150-SPN-1 (Makino-Aero-T150) | PM Interval: 750 Hrs | Torque: 49 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK150-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-11-3.15: PN-MAK150-PMP-PLG-1104 / Kit KIT-MAK150-CLN-4 (Makino-Aero-T150) | PM Interval: 1000 Hrs | Torque: 76 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK150-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T200 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T200. Verify part numbers before closing Work Orders.

BOM-12-1.1: PN-MAK200-CAM-GRIP-1202 / Kit KIT-MAK200-ATC-2 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK200-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-12-1.2: PN-MAK200-VLV-PROP-1203 / Kit KIT-MAK200-HYD-3 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-12-1.3: PN-MAK200-PMP-PLG-1204 / Kit KIT-MAK200-CLN-4 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK200-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-12-1.4: PN-MAK200-LIN-COIL-1205 / Kit KIT-MAK200-XAX-5 (Makino-Aero-T200) | PM Interval: 1250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-12-1.5: PN-MAK200-BSW-NUT-1206 / Kit KIT-MAK200-YAX-6 (Makino-Aero-T200) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK200-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-12-1.6: PN-MAK200-CYL-SEAL-1207 / Kit KIT-MAK200-ZAX-7 (Makino-Aero-T200) | PM Interval: 250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-12-1.7: PN-MAK200-TRQ-STAT-1208 / Kit KIT-MAK200-BAX-8 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK200-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-12-1.8: PN-MAK200-BRK-DISC-1209 / Kit KIT-MAK200-CAX-9 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-12-1.9: PN-MAK200-GLASS-RD-1210 / Kit KIT-MAK200-OPT-10 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK200-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-12-1.10: PN-MAK200-IGBT-MOD-1211 / Kit KIT-MAK200-PLC-11 (Makino-Aero-T200) | PM Interval: 1250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-12-1.11: PN-MAK200-INJ-MET-1212 / Kit KIT-MAK200-LUB-12 (Makino-Aero-T200) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK200-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-12-1.12: PN-MAK200-GASK-VIT-1213 / Kit KIT-MAK200-VAC-13 (Makino-Aero-T200) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-12-1.13: PN-MAK200-HEX-CORE-1214 / Kit KIT-MAK200-THM-14 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK200-THM-14 within tolerance ±0.0020 mm TIR.

BOM-12-1.14: PN-MAK200-BRG-CER-1201 / Kit KIT-MAK200-SPN-1 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-12-1.15: PN-MAK200-CAM-GRIP-1202 / Kit KIT-MAK200-ATC-2 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK200-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T200 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T200. Verify part numbers before closing Work Orders.

BOM-12-2.1: PN-MAK200-VLV-PROP-1203 / Kit KIT-MAK200-HYD-3 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-12-2.2: PN-MAK200-LIN-COIL-1205 / Kit KIT-MAK200-XAX-5 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-12-2.3: PN-MAK200-CYL-SEAL-1207 / Kit KIT-MAK200-ZAX-7 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-12-2.4: PN-MAK200-BRK-DISC-1209 / Kit KIT-MAK200-CAX-9 (Makino-Aero-T200) | PM Interval: 1250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-12-2.5: PN-MAK200-IGBT-MOD-1211 / Kit KIT-MAK200-PLC-11 (Makino-Aero-T200) | PM Interval: 1500 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-12-2.6: PN-MAK200-GASK-VIT-1213 / Kit KIT-MAK200-VAC-13 (Makino-Aero-T200) | PM Interval: 250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-12-2.7: PN-MAK200-BRG-CER-1201 / Kit KIT-MAK200-SPN-1 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-12-2.8: PN-MAK200-VLV-PROP-1203 / Kit KIT-MAK200-HYD-3 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-12-2.9: PN-MAK200-LIN-COIL-1205 / Kit KIT-MAK200-XAX-5 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-12-2.10: PN-MAK200-CYL-SEAL-1207 / Kit KIT-MAK200-ZAX-7 (Makino-Aero-T200) | PM Interval: 1250 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-12-2.11: PN-MAK200-BRK-DISC-1209 / Kit KIT-MAK200-CAX-9 (Makino-Aero-T200) | PM Interval: 1500 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-12-2.12: PN-MAK200-IGBT-MOD-1211 / Kit KIT-MAK200-PLC-11 (Makino-Aero-T200) | PM Interval: 250 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-12-2.13: PN-MAK200-GASK-VIT-1213 / Kit KIT-MAK200-VAC-13 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-12-2.14: PN-MAK200-BRG-CER-1201 / Kit KIT-MAK200-SPN-1 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-12-2.15: PN-MAK200-VLV-PROP-1203 / Kit KIT-MAK200-HYD-3 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK200-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T200 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T200. Verify part numbers before closing Work Orders.

BOM-12-3.1: PN-MAK200-PMP-PLG-1204 / Kit KIT-MAK200-CLN-4 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK200-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-12-3.2: PN-MAK200-CYL-SEAL-1207 / Kit KIT-MAK200-ZAX-7 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 53 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK200-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-12-3.3: PN-MAK200-GLASS-RD-1210 / Kit KIT-MAK200-OPT-10 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 80 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK200-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-12-3.4: PN-MAK200-GASK-VIT-1213 / Kit KIT-MAK200-VAC-13 (Makino-Aero-T200) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK200-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-12-3.5: PN-MAK200-CAM-GRIP-1202 / Kit KIT-MAK200-ATC-2 (Makino-Aero-T200) | PM Interval: 1500 Hrs | Torque: 198 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK200-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-12-3.6: PN-MAK200-LIN-COIL-1205 / Kit KIT-MAK200-XAX-5 (Makino-Aero-T200) | PM Interval: 250 Hrs | Torque: 35 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK200-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-12-3.7: PN-MAK200-TRQ-STAT-1208 / Kit KIT-MAK200-BAX-8 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 62 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK200-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-12-3.8: PN-MAK200-IGBT-MOD-1211 / Kit KIT-MAK200-PLC-11 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK200-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-12-3.9: PN-MAK200-HEX-CORE-1214 / Kit KIT-MAK200-THM-14 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK200-THM-14 within tolerance ±0.0020 mm TIR.

BOM-12-3.10: PN-MAK200-VLV-PROP-1203 / Kit KIT-MAK200-HYD-3 (Makino-Aero-T200) | PM Interval: 1250 Hrs | Torque: 207 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK200-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-12-3.11: PN-MAK200-BSW-NUT-1206 / Kit KIT-MAK200-YAX-6 (Makino-Aero-T200) | PM Interval: 1500 Hrs | Torque: 44 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK200-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-12-3.12: PN-MAK200-BRK-DISC-1209 / Kit KIT-MAK200-CAX-9 (Makino-Aero-T200) | PM Interval: 250 Hrs | Torque: 71 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK200-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-12-3.13: PN-MAK200-INJ-MET-1212 / Kit KIT-MAK200-LUB-12 (Makino-Aero-T200) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK200-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-12-3.14: PN-MAK200-BRG-CER-1201 / Kit KIT-MAK200-SPN-1 (Makino-Aero-T200) | PM Interval: 750 Hrs | Torque: 189 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK200-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-12-3.15: PN-MAK200-PMP-PLG-1204 / Kit KIT-MAK200-CLN-4 (Makino-Aero-T200) | PM Interval: 1000 Hrs | Torque: 26 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK200-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T250 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T250. Verify part numbers before closing Work Orders.

BOM-13-1.1: PN-MAK250-CAM-GRIP-1302 / Kit KIT-MAK250-ATC-2 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK250-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-13-1.2: PN-MAK250-VLV-PROP-1303 / Kit KIT-MAK250-HYD-3 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-13-1.3: PN-MAK250-PMP-PLG-1304 / Kit KIT-MAK250-CLN-4 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK250-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-13-1.4: PN-MAK250-LIN-COIL-1305 / Kit KIT-MAK250-XAX-5 (Makino-Aero-T250) | PM Interval: 1250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-13-1.5: PN-MAK250-BSW-NUT-1306 / Kit KIT-MAK250-YAX-6 (Makino-Aero-T250) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK250-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-13-1.6: PN-MAK250-CYL-SEAL-1307 / Kit KIT-MAK250-ZAX-7 (Makino-Aero-T250) | PM Interval: 250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-13-1.7: PN-MAK250-TRQ-STAT-1308 / Kit KIT-MAK250-BAX-8 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK250-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-13-1.8: PN-MAK250-BRK-DISC-1309 / Kit KIT-MAK250-CAX-9 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-13-1.9: PN-MAK250-GLASS-RD-1310 / Kit KIT-MAK250-OPT-10 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK250-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-13-1.10: PN-MAK250-IGBT-MOD-1311 / Kit KIT-MAK250-PLC-11 (Makino-Aero-T250) | PM Interval: 1250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-13-1.11: PN-MAK250-INJ-MET-1312 / Kit KIT-MAK250-LUB-12 (Makino-Aero-T250) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK250-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-13-1.12: PN-MAK250-GASK-VIT-1313 / Kit KIT-MAK250-VAC-13 (Makino-Aero-T250) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-13-1.13: PN-MAK250-HEX-CORE-1314 / Kit KIT-MAK250-THM-14 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK250-THM-14 within tolerance ±0.0040 mm TIR.

BOM-13-1.14: PN-MAK250-BRG-CER-1301 / Kit KIT-MAK250-SPN-1 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-13-1.15: PN-MAK250-CAM-GRIP-1302 / Kit KIT-MAK250-ATC-2 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK250-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T250 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T250. Verify part numbers before closing Work Orders.

BOM-13-2.1: PN-MAK250-VLV-PROP-1303 / Kit KIT-MAK250-HYD-3 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-13-2.2: PN-MAK250-LIN-COIL-1305 / Kit KIT-MAK250-XAX-5 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-13-2.3: PN-MAK250-CYL-SEAL-1307 / Kit KIT-MAK250-ZAX-7 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-13-2.4: PN-MAK250-BRK-DISC-1309 / Kit KIT-MAK250-CAX-9 (Makino-Aero-T250) | PM Interval: 1250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-13-2.5: PN-MAK250-IGBT-MOD-1311 / Kit KIT-MAK250-PLC-11 (Makino-Aero-T250) | PM Interval: 1500 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-13-2.6: PN-MAK250-GASK-VIT-1313 / Kit KIT-MAK250-VAC-13 (Makino-Aero-T250) | PM Interval: 250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-13-2.7: PN-MAK250-BRG-CER-1301 / Kit KIT-MAK250-SPN-1 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-13-2.8: PN-MAK250-VLV-PROP-1303 / Kit KIT-MAK250-HYD-3 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-13-2.9: PN-MAK250-LIN-COIL-1305 / Kit KIT-MAK250-XAX-5 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-13-2.10: PN-MAK250-CYL-SEAL-1307 / Kit KIT-MAK250-ZAX-7 (Makino-Aero-T250) | PM Interval: 1250 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-13-2.11: PN-MAK250-BRK-DISC-1309 / Kit KIT-MAK250-CAX-9 (Makino-Aero-T250) | PM Interval: 1500 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-13-2.12: PN-MAK250-IGBT-MOD-1311 / Kit KIT-MAK250-PLC-11 (Makino-Aero-T250) | PM Interval: 250 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-13-2.13: PN-MAK250-GASK-VIT-1313 / Kit KIT-MAK250-VAC-13 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-13-2.14: PN-MAK250-BRG-CER-1301 / Kit KIT-MAK250-SPN-1 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-13-2.15: PN-MAK250-VLV-PROP-1303 / Kit KIT-MAK250-HYD-3 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK250-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T250 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T250. Verify part numbers before closing Work Orders.

BOM-13-3.1: PN-MAK250-PMP-PLG-1304 / Kit KIT-MAK250-CLN-4 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK250-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-13-3.2: PN-MAK250-CYL-SEAL-1307 / Kit KIT-MAK250-ZAX-7 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 193 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK250-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-13-3.3: PN-MAK250-GLASS-RD-1310 / Kit KIT-MAK250-OPT-10 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 30 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK250-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-13-3.4: PN-MAK250-GASK-VIT-1313 / Kit KIT-MAK250-VAC-13 (Makino-Aero-T250) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK250-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-13-3.5: PN-MAK250-CAM-GRIP-1302 / Kit KIT-MAK250-ATC-2 (Makino-Aero-T250) | PM Interval: 1500 Hrs | Torque: 148 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK250-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-13-3.6: PN-MAK250-LIN-COIL-1305 / Kit KIT-MAK250-XAX-5 (Makino-Aero-T250) | PM Interval: 250 Hrs | Torque: 175 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK250-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-13-3.7: PN-MAK250-TRQ-STAT-1308 / Kit KIT-MAK250-BAX-8 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 202 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK250-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-13-3.8: PN-MAK250-IGBT-MOD-1311 / Kit KIT-MAK250-PLC-11 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK250-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-13-3.9: PN-MAK250-HEX-CORE-1314 / Kit KIT-MAK250-THM-14 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK250-THM-14 within tolerance ±0.0040 mm TIR.

BOM-13-3.10: PN-MAK250-VLV-PROP-1303 / Kit KIT-MAK250-HYD-3 (Makino-Aero-T250) | PM Interval: 1250 Hrs | Torque: 157 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK250-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-13-3.11: PN-MAK250-BSW-NUT-1306 / Kit KIT-MAK250-YAX-6 (Makino-Aero-T250) | PM Interval: 1500 Hrs | Torque: 184 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK250-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-13-3.12: PN-MAK250-BRK-DISC-1309 / Kit KIT-MAK250-CAX-9 (Makino-Aero-T250) | PM Interval: 250 Hrs | Torque: 21 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK250-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-13-3.13: PN-MAK250-INJ-MET-1312 / Kit KIT-MAK250-LUB-12 (Makino-Aero-T250) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK250-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-13-3.14: PN-MAK250-BRG-CER-1301 / Kit KIT-MAK250-SPN-1 (Makino-Aero-T250) | PM Interval: 750 Hrs | Torque: 139 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK250-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-13-3.15: PN-MAK250-PMP-PLG-1304 / Kit KIT-MAK250-CLN-4 (Makino-Aero-T250) | PM Interval: 1000 Hrs | Torque: 166 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK250-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T300 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T300. Verify part numbers before closing Work Orders.

BOM-14-1.1: PN-MAK300-CAM-GRIP-1402 / Kit KIT-MAK300-ATC-2 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK300-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-14-1.2: PN-MAK300-VLV-PROP-1403 / Kit KIT-MAK300-HYD-3 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-14-1.3: PN-MAK300-PMP-PLG-1404 / Kit KIT-MAK300-CLN-4 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK300-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-14-1.4: PN-MAK300-LIN-COIL-1405 / Kit KIT-MAK300-XAX-5 (Makino-Aero-T300) | PM Interval: 1250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-14-1.5: PN-MAK300-BSW-NUT-1406 / Kit KIT-MAK300-YAX-6 (Makino-Aero-T300) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK300-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-14-1.6: PN-MAK300-CYL-SEAL-1407 / Kit KIT-MAK300-ZAX-7 (Makino-Aero-T300) | PM Interval: 250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-14-1.7: PN-MAK300-TRQ-STAT-1408 / Kit KIT-MAK300-BAX-8 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK300-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-14-1.8: PN-MAK300-BRK-DISC-1409 / Kit KIT-MAK300-CAX-9 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-14-1.9: PN-MAK300-GLASS-RD-1410 / Kit KIT-MAK300-OPT-10 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK300-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-14-1.10: PN-MAK300-IGBT-MOD-1411 / Kit KIT-MAK300-PLC-11 (Makino-Aero-T300) | PM Interval: 1250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-14-1.11: PN-MAK300-INJ-MET-1412 / Kit KIT-MAK300-LUB-12 (Makino-Aero-T300) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK300-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-14-1.12: PN-MAK300-GASK-VIT-1413 / Kit KIT-MAK300-VAC-13 (Makino-Aero-T300) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-14-1.13: PN-MAK300-HEX-CORE-1414 / Kit KIT-MAK300-THM-14 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK300-THM-14 within tolerance ±0.0060 mm TIR.

BOM-14-1.14: PN-MAK300-BRG-CER-1401 / Kit KIT-MAK300-SPN-1 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-14-1.15: PN-MAK300-CAM-GRIP-1402 / Kit KIT-MAK300-ATC-2 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK300-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T300 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T300. Verify part numbers before closing Work Orders.

BOM-14-2.1: PN-MAK300-VLV-PROP-1403 / Kit KIT-MAK300-HYD-3 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-14-2.2: PN-MAK300-LIN-COIL-1405 / Kit KIT-MAK300-XAX-5 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-14-2.3: PN-MAK300-CYL-SEAL-1407 / Kit KIT-MAK300-ZAX-7 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-14-2.4: PN-MAK300-BRK-DISC-1409 / Kit KIT-MAK300-CAX-9 (Makino-Aero-T300) | PM Interval: 1250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-14-2.5: PN-MAK300-IGBT-MOD-1411 / Kit KIT-MAK300-PLC-11 (Makino-Aero-T300) | PM Interval: 1500 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-14-2.6: PN-MAK300-GASK-VIT-1413 / Kit KIT-MAK300-VAC-13 (Makino-Aero-T300) | PM Interval: 250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-14-2.7: PN-MAK300-BRG-CER-1401 / Kit KIT-MAK300-SPN-1 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-14-2.8: PN-MAK300-VLV-PROP-1403 / Kit KIT-MAK300-HYD-3 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-14-2.9: PN-MAK300-LIN-COIL-1405 / Kit KIT-MAK300-XAX-5 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-14-2.10: PN-MAK300-CYL-SEAL-1407 / Kit KIT-MAK300-ZAX-7 (Makino-Aero-T300) | PM Interval: 1250 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-14-2.11: PN-MAK300-BRK-DISC-1409 / Kit KIT-MAK300-CAX-9 (Makino-Aero-T300) | PM Interval: 1500 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-14-2.12: PN-MAK300-IGBT-MOD-1411 / Kit KIT-MAK300-PLC-11 (Makino-Aero-T300) | PM Interval: 250 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-14-2.13: PN-MAK300-GASK-VIT-1413 / Kit KIT-MAK300-VAC-13 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-14-2.14: PN-MAK300-BRG-CER-1401 / Kit KIT-MAK300-SPN-1 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-14-2.15: PN-MAK300-VLV-PROP-1403 / Kit KIT-MAK300-HYD-3 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK300-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T300 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T300. Verify part numbers before closing Work Orders.

BOM-14-3.1: PN-MAK300-PMP-PLG-1404 / Kit KIT-MAK300-CLN-4 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK300-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-14-3.2: PN-MAK300-CYL-SEAL-1407 / Kit KIT-MAK300-ZAX-7 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 143 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK300-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-14-3.3: PN-MAK300-GLASS-RD-1410 / Kit KIT-MAK300-OPT-10 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 170 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK300-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-14-3.4: PN-MAK300-GASK-VIT-1413 / Kit KIT-MAK300-VAC-13 (Makino-Aero-T300) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK300-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-14-3.5: PN-MAK300-CAM-GRIP-1402 / Kit KIT-MAK300-ATC-2 (Makino-Aero-T300) | PM Interval: 1500 Hrs | Torque: 98 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK300-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-14-3.6: PN-MAK300-LIN-COIL-1405 / Kit KIT-MAK300-XAX-5 (Makino-Aero-T300) | PM Interval: 250 Hrs | Torque: 125 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK300-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-14-3.7: PN-MAK300-TRQ-STAT-1408 / Kit KIT-MAK300-BAX-8 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 152 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK300-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-14-3.8: PN-MAK300-IGBT-MOD-1411 / Kit KIT-MAK300-PLC-11 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK300-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-14-3.9: PN-MAK300-HEX-CORE-1414 / Kit KIT-MAK300-THM-14 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK300-THM-14 within tolerance ±0.0060 mm TIR.

BOM-14-3.10: PN-MAK300-VLV-PROP-1403 / Kit KIT-MAK300-HYD-3 (Makino-Aero-T300) | PM Interval: 1250 Hrs | Torque: 107 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK300-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-14-3.11: PN-MAK300-BSW-NUT-1406 / Kit KIT-MAK300-YAX-6 (Makino-Aero-T300) | PM Interval: 1500 Hrs | Torque: 134 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK300-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-14-3.12: PN-MAK300-BRK-DISC-1409 / Kit KIT-MAK300-CAX-9 (Makino-Aero-T300) | PM Interval: 250 Hrs | Torque: 161 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK300-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-14-3.13: PN-MAK300-INJ-MET-1412 / Kit KIT-MAK300-LUB-12 (Makino-Aero-T300) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK300-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-14-3.14: PN-MAK300-BRG-CER-1401 / Kit KIT-MAK300-SPN-1 (Makino-Aero-T300) | PM Interval: 750 Hrs | Torque: 89 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK300-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-14-3.15: PN-MAK300-PMP-PLG-1404 / Kit KIT-MAK300-CLN-4 (Makino-Aero-T300) | PM Interval: 1000 Hrs | Torque: 116 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK300-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T350 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T350. Verify part numbers before closing Work Orders.

BOM-15-1.1: PN-MAK350-CAM-GRIP-1502 / Kit KIT-MAK350-ATC-2 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK350-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-15-1.2: PN-MAK350-VLV-PROP-1503 / Kit KIT-MAK350-HYD-3 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-15-1.3: PN-MAK350-PMP-PLG-1504 / Kit KIT-MAK350-CLN-4 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK350-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-15-1.4: PN-MAK350-LIN-COIL-1505 / Kit KIT-MAK350-XAX-5 (Makino-Aero-T350) | PM Interval: 1250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-15-1.5: PN-MAK350-BSW-NUT-1506 / Kit KIT-MAK350-YAX-6 (Makino-Aero-T350) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK350-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-15-1.6: PN-MAK350-CYL-SEAL-1507 / Kit KIT-MAK350-ZAX-7 (Makino-Aero-T350) | PM Interval: 250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-15-1.7: PN-MAK350-TRQ-STAT-1508 / Kit KIT-MAK350-BAX-8 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK350-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-15-1.8: PN-MAK350-BRK-DISC-1509 / Kit KIT-MAK350-CAX-9 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-15-1.9: PN-MAK350-GLASS-RD-1510 / Kit KIT-MAK350-OPT-10 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK350-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-15-1.10: PN-MAK350-IGBT-MOD-1511 / Kit KIT-MAK350-PLC-11 (Makino-Aero-T350) | PM Interval: 1250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-15-1.11: PN-MAK350-INJ-MET-1512 / Kit KIT-MAK350-LUB-12 (Makino-Aero-T350) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK350-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-15-1.12: PN-MAK350-GASK-VIT-1513 / Kit KIT-MAK350-VAC-13 (Makino-Aero-T350) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-15-1.13: PN-MAK350-HEX-CORE-1514 / Kit KIT-MAK350-THM-14 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK350-THM-14 within tolerance ±0.0020 mm TIR.

BOM-15-1.14: PN-MAK350-BRG-CER-1501 / Kit KIT-MAK350-SPN-1 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-15-1.15: PN-MAK350-CAM-GRIP-1502 / Kit KIT-MAK350-ATC-2 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK350-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T350 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T350. Verify part numbers before closing Work Orders.

BOM-15-2.1: PN-MAK350-VLV-PROP-1503 / Kit KIT-MAK350-HYD-3 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-15-2.2: PN-MAK350-LIN-COIL-1505 / Kit KIT-MAK350-XAX-5 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-15-2.3: PN-MAK350-CYL-SEAL-1507 / Kit KIT-MAK350-ZAX-7 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-15-2.4: PN-MAK350-BRK-DISC-1509 / Kit KIT-MAK350-CAX-9 (Makino-Aero-T350) | PM Interval: 1250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-15-2.5: PN-MAK350-IGBT-MOD-1511 / Kit KIT-MAK350-PLC-11 (Makino-Aero-T350) | PM Interval: 1500 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-15-2.6: PN-MAK350-GASK-VIT-1513 / Kit KIT-MAK350-VAC-13 (Makino-Aero-T350) | PM Interval: 250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-15-2.7: PN-MAK350-BRG-CER-1501 / Kit KIT-MAK350-SPN-1 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-15-2.8: PN-MAK350-VLV-PROP-1503 / Kit KIT-MAK350-HYD-3 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-15-2.9: PN-MAK350-LIN-COIL-1505 / Kit KIT-MAK350-XAX-5 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-15-2.10: PN-MAK350-CYL-SEAL-1507 / Kit KIT-MAK350-ZAX-7 (Makino-Aero-T350) | PM Interval: 1250 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-15-2.11: PN-MAK350-BRK-DISC-1509 / Kit KIT-MAK350-CAX-9 (Makino-Aero-T350) | PM Interval: 1500 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-15-2.12: PN-MAK350-IGBT-MOD-1511 / Kit KIT-MAK350-PLC-11 (Makino-Aero-T350) | PM Interval: 250 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-15-2.13: PN-MAK350-GASK-VIT-1513 / Kit KIT-MAK350-VAC-13 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-15-2.14: PN-MAK350-BRG-CER-1501 / Kit KIT-MAK350-SPN-1 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-15-2.15: PN-MAK350-VLV-PROP-1503 / Kit KIT-MAK350-HYD-3 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK350-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T350 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T350. Verify part numbers before closing Work Orders.

BOM-15-3.1: PN-MAK350-PMP-PLG-1504 / Kit KIT-MAK350-CLN-4 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK350-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-15-3.2: PN-MAK350-CYL-SEAL-1507 / Kit KIT-MAK350-ZAX-7 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 93 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK350-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-15-3.3: PN-MAK350-GLASS-RD-1510 / Kit KIT-MAK350-OPT-10 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 120 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK350-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-15-3.4: PN-MAK350-GASK-VIT-1513 / Kit KIT-MAK350-VAC-13 (Makino-Aero-T350) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK350-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-15-3.5: PN-MAK350-CAM-GRIP-1502 / Kit KIT-MAK350-ATC-2 (Makino-Aero-T350) | PM Interval: 1500 Hrs | Torque: 48 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK350-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-15-3.6: PN-MAK350-LIN-COIL-1505 / Kit KIT-MAK350-XAX-5 (Makino-Aero-T350) | PM Interval: 250 Hrs | Torque: 75 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK350-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-15-3.7: PN-MAK350-TRQ-STAT-1508 / Kit KIT-MAK350-BAX-8 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 102 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK350-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-15-3.8: PN-MAK350-IGBT-MOD-1511 / Kit KIT-MAK350-PLC-11 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK350-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-15-3.9: PN-MAK350-HEX-CORE-1514 / Kit KIT-MAK350-THM-14 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK350-THM-14 within tolerance ±0.0020 mm TIR.

BOM-15-3.10: PN-MAK350-VLV-PROP-1503 / Kit KIT-MAK350-HYD-3 (Makino-Aero-T350) | PM Interval: 1250 Hrs | Torque: 57 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK350-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-15-3.11: PN-MAK350-BSW-NUT-1506 / Kit KIT-MAK350-YAX-6 (Makino-Aero-T350) | PM Interval: 1500 Hrs | Torque: 84 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK350-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-15-3.12: PN-MAK350-BRK-DISC-1509 / Kit KIT-MAK350-CAX-9 (Makino-Aero-T350) | PM Interval: 250 Hrs | Torque: 111 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK350-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-15-3.13: PN-MAK350-INJ-MET-1512 / Kit KIT-MAK350-LUB-12 (Makino-Aero-T350) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK350-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-15-3.14: PN-MAK350-BRG-CER-1501 / Kit KIT-MAK350-SPN-1 (Makino-Aero-T350) | PM Interval: 750 Hrs | Torque: 39 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK350-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-15-3.15: PN-MAK350-PMP-PLG-1504 / Kit KIT-MAK350-CLN-4 (Makino-Aero-T350) | PM Interval: 1000 Hrs | Torque: 66 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK350-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T400 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T400. Verify part numbers before closing Work Orders.

BOM-16-1.1: PN-MAK400-CAM-GRIP-1602 / Kit KIT-MAK400-ATC-2 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK400-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-16-1.2: PN-MAK400-VLV-PROP-1603 / Kit KIT-MAK400-HYD-3 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-16-1.3: PN-MAK400-PMP-PLG-1604 / Kit KIT-MAK400-CLN-4 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK400-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-16-1.4: PN-MAK400-LIN-COIL-1605 / Kit KIT-MAK400-XAX-5 (Makino-Aero-T400) | PM Interval: 1250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-16-1.5: PN-MAK400-BSW-NUT-1606 / Kit KIT-MAK400-YAX-6 (Makino-Aero-T400) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK400-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-16-1.6: PN-MAK400-CYL-SEAL-1607 / Kit KIT-MAK400-ZAX-7 (Makino-Aero-T400) | PM Interval: 250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-16-1.7: PN-MAK400-TRQ-STAT-1608 / Kit KIT-MAK400-BAX-8 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK400-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-16-1.8: PN-MAK400-BRK-DISC-1609 / Kit KIT-MAK400-CAX-9 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-16-1.9: PN-MAK400-GLASS-RD-1610 / Kit KIT-MAK400-OPT-10 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK400-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-16-1.10: PN-MAK400-IGBT-MOD-1611 / Kit KIT-MAK400-PLC-11 (Makino-Aero-T400) | PM Interval: 1250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-16-1.11: PN-MAK400-INJ-MET-1612 / Kit KIT-MAK400-LUB-12 (Makino-Aero-T400) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK400-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-16-1.12: PN-MAK400-GASK-VIT-1613 / Kit KIT-MAK400-VAC-13 (Makino-Aero-T400) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-16-1.13: PN-MAK400-HEX-CORE-1614 / Kit KIT-MAK400-THM-14 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK400-THM-14 within tolerance ±0.0040 mm TIR.

BOM-16-1.14: PN-MAK400-BRG-CER-1601 / Kit KIT-MAK400-SPN-1 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-16-1.15: PN-MAK400-CAM-GRIP-1602 / Kit KIT-MAK400-ATC-2 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK400-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T400 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T400. Verify part numbers before closing Work Orders.

BOM-16-2.1: PN-MAK400-VLV-PROP-1603 / Kit KIT-MAK400-HYD-3 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-16-2.2: PN-MAK400-LIN-COIL-1605 / Kit KIT-MAK400-XAX-5 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-16-2.3: PN-MAK400-CYL-SEAL-1607 / Kit KIT-MAK400-ZAX-7 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-16-2.4: PN-MAK400-BRK-DISC-1609 / Kit KIT-MAK400-CAX-9 (Makino-Aero-T400) | PM Interval: 1250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-16-2.5: PN-MAK400-IGBT-MOD-1611 / Kit KIT-MAK400-PLC-11 (Makino-Aero-T400) | PM Interval: 1500 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-16-2.6: PN-MAK400-GASK-VIT-1613 / Kit KIT-MAK400-VAC-13 (Makino-Aero-T400) | PM Interval: 250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-16-2.7: PN-MAK400-BRG-CER-1601 / Kit KIT-MAK400-SPN-1 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-16-2.8: PN-MAK400-VLV-PROP-1603 / Kit KIT-MAK400-HYD-3 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-16-2.9: PN-MAK400-LIN-COIL-1605 / Kit KIT-MAK400-XAX-5 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-16-2.10: PN-MAK400-CYL-SEAL-1607 / Kit KIT-MAK400-ZAX-7 (Makino-Aero-T400) | PM Interval: 1250 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-16-2.11: PN-MAK400-BRK-DISC-1609 / Kit KIT-MAK400-CAX-9 (Makino-Aero-T400) | PM Interval: 1500 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-16-2.12: PN-MAK400-IGBT-MOD-1611 / Kit KIT-MAK400-PLC-11 (Makino-Aero-T400) | PM Interval: 250 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-16-2.13: PN-MAK400-GASK-VIT-1613 / Kit KIT-MAK400-VAC-13 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-16-2.14: PN-MAK400-BRG-CER-1601 / Kit KIT-MAK400-SPN-1 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-16-2.15: PN-MAK400-VLV-PROP-1603 / Kit KIT-MAK400-HYD-3 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK400-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T400 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T400. Verify part numbers before closing Work Orders.

BOM-16-3.1: PN-MAK400-PMP-PLG-1604 / Kit KIT-MAK400-CLN-4 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK400-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-16-3.2: PN-MAK400-CYL-SEAL-1607 / Kit KIT-MAK400-ZAX-7 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 43 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK400-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-16-3.3: PN-MAK400-GLASS-RD-1610 / Kit KIT-MAK400-OPT-10 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 70 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK400-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-16-3.4: PN-MAK400-GASK-VIT-1613 / Kit KIT-MAK400-VAC-13 (Makino-Aero-T400) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK400-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-16-3.5: PN-MAK400-CAM-GRIP-1602 / Kit KIT-MAK400-ATC-2 (Makino-Aero-T400) | PM Interval: 1500 Hrs | Torque: 188 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK400-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-16-3.6: PN-MAK400-LIN-COIL-1605 / Kit KIT-MAK400-XAX-5 (Makino-Aero-T400) | PM Interval: 250 Hrs | Torque: 25 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK400-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-16-3.7: PN-MAK400-TRQ-STAT-1608 / Kit KIT-MAK400-BAX-8 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 52 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK400-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-16-3.8: PN-MAK400-IGBT-MOD-1611 / Kit KIT-MAK400-PLC-11 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK400-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-16-3.9: PN-MAK400-HEX-CORE-1614 / Kit KIT-MAK400-THM-14 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK400-THM-14 within tolerance ±0.0040 mm TIR.

BOM-16-3.10: PN-MAK400-VLV-PROP-1603 / Kit KIT-MAK400-HYD-3 (Makino-Aero-T400) | PM Interval: 1250 Hrs | Torque: 197 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK400-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-16-3.11: PN-MAK400-BSW-NUT-1606 / Kit KIT-MAK400-YAX-6 (Makino-Aero-T400) | PM Interval: 1500 Hrs | Torque: 34 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK400-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-16-3.12: PN-MAK400-BRK-DISC-1609 / Kit KIT-MAK400-CAX-9 (Makino-Aero-T400) | PM Interval: 250 Hrs | Torque: 61 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK400-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-16-3.13: PN-MAK400-INJ-MET-1612 / Kit KIT-MAK400-LUB-12 (Makino-Aero-T400) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK400-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-16-3.14: PN-MAK400-BRG-CER-1601 / Kit KIT-MAK400-SPN-1 (Makino-Aero-T400) | PM Interval: 750 Hrs | Torque: 179 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK400-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-16-3.15: PN-MAK400-PMP-PLG-1604 / Kit KIT-MAK400-CLN-4 (Makino-Aero-T400) | PM Interval: 1000 Hrs | Torque: 206 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK400-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T450 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T450. Verify part numbers before closing Work Orders.

BOM-17-1.1: PN-MAK450-CAM-GRIP-1702 / Kit KIT-MAK450-ATC-2 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK450-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-17-1.2: PN-MAK450-VLV-PROP-1703 / Kit KIT-MAK450-HYD-3 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-17-1.3: PN-MAK450-PMP-PLG-1704 / Kit KIT-MAK450-CLN-4 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK450-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-17-1.4: PN-MAK450-LIN-COIL-1705 / Kit KIT-MAK450-XAX-5 (Makino-Aero-T450) | PM Interval: 1250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-17-1.5: PN-MAK450-BSW-NUT-1706 / Kit KIT-MAK450-YAX-6 (Makino-Aero-T450) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK450-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-17-1.6: PN-MAK450-CYL-SEAL-1707 / Kit KIT-MAK450-ZAX-7 (Makino-Aero-T450) | PM Interval: 250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-17-1.7: PN-MAK450-TRQ-STAT-1708 / Kit KIT-MAK450-BAX-8 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK450-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-17-1.8: PN-MAK450-BRK-DISC-1709 / Kit KIT-MAK450-CAX-9 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-17-1.9: PN-MAK450-GLASS-RD-1710 / Kit KIT-MAK450-OPT-10 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK450-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-17-1.10: PN-MAK450-IGBT-MOD-1711 / Kit KIT-MAK450-PLC-11 (Makino-Aero-T450) | PM Interval: 1250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-17-1.11: PN-MAK450-INJ-MET-1712 / Kit KIT-MAK450-LUB-12 (Makino-Aero-T450) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK450-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-17-1.12: PN-MAK450-GASK-VIT-1713 / Kit KIT-MAK450-VAC-13 (Makino-Aero-T450) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-17-1.13: PN-MAK450-HEX-CORE-1714 / Kit KIT-MAK450-THM-14 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK450-THM-14 within tolerance ±0.0060 mm TIR.

BOM-17-1.14: PN-MAK450-BRG-CER-1701 / Kit KIT-MAK450-SPN-1 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-17-1.15: PN-MAK450-CAM-GRIP-1702 / Kit KIT-MAK450-ATC-2 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK450-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T450 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T450. Verify part numbers before closing Work Orders.

BOM-17-2.1: PN-MAK450-VLV-PROP-1703 / Kit KIT-MAK450-HYD-3 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-17-2.2: PN-MAK450-LIN-COIL-1705 / Kit KIT-MAK450-XAX-5 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-17-2.3: PN-MAK450-CYL-SEAL-1707 / Kit KIT-MAK450-ZAX-7 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-17-2.4: PN-MAK450-BRK-DISC-1709 / Kit KIT-MAK450-CAX-9 (Makino-Aero-T450) | PM Interval: 1250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-17-2.5: PN-MAK450-IGBT-MOD-1711 / Kit KIT-MAK450-PLC-11 (Makino-Aero-T450) | PM Interval: 1500 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-17-2.6: PN-MAK450-GASK-VIT-1713 / Kit KIT-MAK450-VAC-13 (Makino-Aero-T450) | PM Interval: 250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-17-2.7: PN-MAK450-BRG-CER-1701 / Kit KIT-MAK450-SPN-1 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-17-2.8: PN-MAK450-VLV-PROP-1703 / Kit KIT-MAK450-HYD-3 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-17-2.9: PN-MAK450-LIN-COIL-1705 / Kit KIT-MAK450-XAX-5 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-17-2.10: PN-MAK450-CYL-SEAL-1707 / Kit KIT-MAK450-ZAX-7 (Makino-Aero-T450) | PM Interval: 1250 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-17-2.11: PN-MAK450-BRK-DISC-1709 / Kit KIT-MAK450-CAX-9 (Makino-Aero-T450) | PM Interval: 1500 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-17-2.12: PN-MAK450-IGBT-MOD-1711 / Kit KIT-MAK450-PLC-11 (Makino-Aero-T450) | PM Interval: 250 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-17-2.13: PN-MAK450-GASK-VIT-1713 / Kit KIT-MAK450-VAC-13 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-17-2.14: PN-MAK450-BRG-CER-1701 / Kit KIT-MAK450-SPN-1 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-17-2.15: PN-MAK450-VLV-PROP-1703 / Kit KIT-MAK450-HYD-3 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK450-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T450 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T450. Verify part numbers before closing Work Orders.

BOM-17-3.1: PN-MAK450-PMP-PLG-1704 / Kit KIT-MAK450-CLN-4 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK450-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-17-3.2: PN-MAK450-CYL-SEAL-1707 / Kit KIT-MAK450-ZAX-7 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 183 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK450-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-17-3.3: PN-MAK450-GLASS-RD-1710 / Kit KIT-MAK450-OPT-10 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 20 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK450-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-17-3.4: PN-MAK450-GASK-VIT-1713 / Kit KIT-MAK450-VAC-13 (Makino-Aero-T450) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK450-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-17-3.5: PN-MAK450-CAM-GRIP-1702 / Kit KIT-MAK450-ATC-2 (Makino-Aero-T450) | PM Interval: 1500 Hrs | Torque: 138 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK450-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-17-3.6: PN-MAK450-LIN-COIL-1705 / Kit KIT-MAK450-XAX-5 (Makino-Aero-T450) | PM Interval: 250 Hrs | Torque: 165 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK450-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-17-3.7: PN-MAK450-TRQ-STAT-1708 / Kit KIT-MAK450-BAX-8 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 192 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK450-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-17-3.8: PN-MAK450-IGBT-MOD-1711 / Kit KIT-MAK450-PLC-11 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK450-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-17-3.9: PN-MAK450-HEX-CORE-1714 / Kit KIT-MAK450-THM-14 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK450-THM-14 within tolerance ±0.0060 mm TIR.

BOM-17-3.10: PN-MAK450-VLV-PROP-1703 / Kit KIT-MAK450-HYD-3 (Makino-Aero-T450) | PM Interval: 1250 Hrs | Torque: 147 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK450-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-17-3.11: PN-MAK450-BSW-NUT-1706 / Kit KIT-MAK450-YAX-6 (Makino-Aero-T450) | PM Interval: 1500 Hrs | Torque: 174 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK450-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-17-3.12: PN-MAK450-BRK-DISC-1709 / Kit KIT-MAK450-CAX-9 (Makino-Aero-T450) | PM Interval: 250 Hrs | Torque: 201 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK450-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-17-3.13: PN-MAK450-INJ-MET-1712 / Kit KIT-MAK450-LUB-12 (Makino-Aero-T450) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK450-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-17-3.14: PN-MAK450-BRG-CER-1701 / Kit KIT-MAK450-SPN-1 (Makino-Aero-T450) | PM Interval: 750 Hrs | Torque: 129 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK450-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-17-3.15: PN-MAK450-PMP-PLG-1704 / Kit KIT-MAK450-CLN-4 (Makino-Aero-T450) | PM Interval: 1000 Hrs | Torque: 156 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK450-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T500 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T500. Verify part numbers before closing Work Orders.

BOM-18-1.1: PN-MAK500-CAM-GRIP-1802 / Kit KIT-MAK500-ATC-2 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK500-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-18-1.2: PN-MAK500-VLV-PROP-1803 / Kit KIT-MAK500-HYD-3 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-18-1.3: PN-MAK500-PMP-PLG-1804 / Kit KIT-MAK500-CLN-4 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK500-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-18-1.4: PN-MAK500-LIN-COIL-1805 / Kit KIT-MAK500-XAX-5 (Makino-Aero-T500) | PM Interval: 1250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-18-1.5: PN-MAK500-BSW-NUT-1806 / Kit KIT-MAK500-YAX-6 (Makino-Aero-T500) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK500-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-18-1.6: PN-MAK500-CYL-SEAL-1807 / Kit KIT-MAK500-ZAX-7 (Makino-Aero-T500) | PM Interval: 250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-18-1.7: PN-MAK500-TRQ-STAT-1808 / Kit KIT-MAK500-BAX-8 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK500-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-18-1.8: PN-MAK500-BRK-DISC-1809 / Kit KIT-MAK500-CAX-9 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-18-1.9: PN-MAK500-GLASS-RD-1810 / Kit KIT-MAK500-OPT-10 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK500-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-18-1.10: PN-MAK500-IGBT-MOD-1811 / Kit KIT-MAK500-PLC-11 (Makino-Aero-T500) | PM Interval: 1250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-18-1.11: PN-MAK500-INJ-MET-1812 / Kit KIT-MAK500-LUB-12 (Makino-Aero-T500) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK500-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-18-1.12: PN-MAK500-GASK-VIT-1813 / Kit KIT-MAK500-VAC-13 (Makino-Aero-T500) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-18-1.13: PN-MAK500-HEX-CORE-1814 / Kit KIT-MAK500-THM-14 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK500-THM-14 within tolerance ±0.0020 mm TIR.

BOM-18-1.14: PN-MAK500-BRG-CER-1801 / Kit KIT-MAK500-SPN-1 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-18-1.15: PN-MAK500-CAM-GRIP-1802 / Kit KIT-MAK500-ATC-2 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK500-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T500 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T500. Verify part numbers before closing Work Orders.

BOM-18-2.1: PN-MAK500-VLV-PROP-1803 / Kit KIT-MAK500-HYD-3 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-18-2.2: PN-MAK500-LIN-COIL-1805 / Kit KIT-MAK500-XAX-5 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-18-2.3: PN-MAK500-CYL-SEAL-1807 / Kit KIT-MAK500-ZAX-7 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-18-2.4: PN-MAK500-BRK-DISC-1809 / Kit KIT-MAK500-CAX-9 (Makino-Aero-T500) | PM Interval: 1250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-18-2.5: PN-MAK500-IGBT-MOD-1811 / Kit KIT-MAK500-PLC-11 (Makino-Aero-T500) | PM Interval: 1500 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-18-2.6: PN-MAK500-GASK-VIT-1813 / Kit KIT-MAK500-VAC-13 (Makino-Aero-T500) | PM Interval: 250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-18-2.7: PN-MAK500-BRG-CER-1801 / Kit KIT-MAK500-SPN-1 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-18-2.8: PN-MAK500-VLV-PROP-1803 / Kit KIT-MAK500-HYD-3 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-18-2.9: PN-MAK500-LIN-COIL-1805 / Kit KIT-MAK500-XAX-5 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-18-2.10: PN-MAK500-CYL-SEAL-1807 / Kit KIT-MAK500-ZAX-7 (Makino-Aero-T500) | PM Interval: 1250 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-18-2.11: PN-MAK500-BRK-DISC-1809 / Kit KIT-MAK500-CAX-9 (Makino-Aero-T500) | PM Interval: 1500 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-18-2.12: PN-MAK500-IGBT-MOD-1811 / Kit KIT-MAK500-PLC-11 (Makino-Aero-T500) | PM Interval: 250 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-18-2.13: PN-MAK500-GASK-VIT-1813 / Kit KIT-MAK500-VAC-13 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-18-2.14: PN-MAK500-BRG-CER-1801 / Kit KIT-MAK500-SPN-1 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-18-2.15: PN-MAK500-VLV-PROP-1803 / Kit KIT-MAK500-HYD-3 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK500-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T500 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T500. Verify part numbers before closing Work Orders.

BOM-18-3.1: PN-MAK500-PMP-PLG-1804 / Kit KIT-MAK500-CLN-4 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK500-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-18-3.2: PN-MAK500-CYL-SEAL-1807 / Kit KIT-MAK500-ZAX-7 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 133 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK500-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-18-3.3: PN-MAK500-GLASS-RD-1810 / Kit KIT-MAK500-OPT-10 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 160 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK500-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-18-3.4: PN-MAK500-GASK-VIT-1813 / Kit KIT-MAK500-VAC-13 (Makino-Aero-T500) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK500-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-18-3.5: PN-MAK500-CAM-GRIP-1802 / Kit KIT-MAK500-ATC-2 (Makino-Aero-T500) | PM Interval: 1500 Hrs | Torque: 88 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK500-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-18-3.6: PN-MAK500-LIN-COIL-1805 / Kit KIT-MAK500-XAX-5 (Makino-Aero-T500) | PM Interval: 250 Hrs | Torque: 115 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK500-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-18-3.7: PN-MAK500-TRQ-STAT-1808 / Kit KIT-MAK500-BAX-8 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 142 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK500-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-18-3.8: PN-MAK500-IGBT-MOD-1811 / Kit KIT-MAK500-PLC-11 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK500-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-18-3.9: PN-MAK500-HEX-CORE-1814 / Kit KIT-MAK500-THM-14 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK500-THM-14 within tolerance ±0.0020 mm TIR.

BOM-18-3.10: PN-MAK500-VLV-PROP-1803 / Kit KIT-MAK500-HYD-3 (Makino-Aero-T500) | PM Interval: 1250 Hrs | Torque: 97 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK500-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-18-3.11: PN-MAK500-BSW-NUT-1806 / Kit KIT-MAK500-YAX-6 (Makino-Aero-T500) | PM Interval: 1500 Hrs | Torque: 124 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK500-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-18-3.12: PN-MAK500-BRK-DISC-1809 / Kit KIT-MAK500-CAX-9 (Makino-Aero-T500) | PM Interval: 250 Hrs | Torque: 151 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK500-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-18-3.13: PN-MAK500-INJ-MET-1812 / Kit KIT-MAK500-LUB-12 (Makino-Aero-T500) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK500-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-18-3.14: PN-MAK500-BRG-CER-1801 / Kit KIT-MAK500-SPN-1 (Makino-Aero-T500) | PM Interval: 750 Hrs | Torque: 79 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK500-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-18-3.15: PN-MAK500-PMP-PLG-1804 / Kit KIT-MAK500-CLN-4 (Makino-Aero-T500) | PM Interval: 1000 Hrs | Torque: 106 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK500-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T550 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T550. Verify part numbers before closing Work Orders.

BOM-19-1.1: PN-MAK550-CAM-GRIP-1902 / Kit KIT-MAK550-ATC-2 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK550-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-19-1.2: PN-MAK550-VLV-PROP-1903 / Kit KIT-MAK550-HYD-3 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-19-1.3: PN-MAK550-PMP-PLG-1904 / Kit KIT-MAK550-CLN-4 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK550-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-19-1.4: PN-MAK550-LIN-COIL-1905 / Kit KIT-MAK550-XAX-5 (Makino-Aero-T550) | PM Interval: 1250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-19-1.5: PN-MAK550-BSW-NUT-1906 / Kit KIT-MAK550-YAX-6 (Makino-Aero-T550) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK550-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-19-1.6: PN-MAK550-CYL-SEAL-1907 / Kit KIT-MAK550-ZAX-7 (Makino-Aero-T550) | PM Interval: 250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-19-1.7: PN-MAK550-TRQ-STAT-1908 / Kit KIT-MAK550-BAX-8 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK550-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-19-1.8: PN-MAK550-BRK-DISC-1909 / Kit KIT-MAK550-CAX-9 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-19-1.9: PN-MAK550-GLASS-RD-1910 / Kit KIT-MAK550-OPT-10 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK550-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-19-1.10: PN-MAK550-IGBT-MOD-1911 / Kit KIT-MAK550-PLC-11 (Makino-Aero-T550) | PM Interval: 1250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-19-1.11: PN-MAK550-INJ-MET-1912 / Kit KIT-MAK550-LUB-12 (Makino-Aero-T550) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK550-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-19-1.12: PN-MAK550-GASK-VIT-1913 / Kit KIT-MAK550-VAC-13 (Makino-Aero-T550) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-19-1.13: PN-MAK550-HEX-CORE-1914 / Kit KIT-MAK550-THM-14 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK550-THM-14 within tolerance ±0.0040 mm TIR.

BOM-19-1.14: PN-MAK550-BRG-CER-1901 / Kit KIT-MAK550-SPN-1 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-19-1.15: PN-MAK550-CAM-GRIP-1902 / Kit KIT-MAK550-ATC-2 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK550-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T550 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T550. Verify part numbers before closing Work Orders.

BOM-19-2.1: PN-MAK550-VLV-PROP-1903 / Kit KIT-MAK550-HYD-3 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-19-2.2: PN-MAK550-LIN-COIL-1905 / Kit KIT-MAK550-XAX-5 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-19-2.3: PN-MAK550-CYL-SEAL-1907 / Kit KIT-MAK550-ZAX-7 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-19-2.4: PN-MAK550-BRK-DISC-1909 / Kit KIT-MAK550-CAX-9 (Makino-Aero-T550) | PM Interval: 1250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-19-2.5: PN-MAK550-IGBT-MOD-1911 / Kit KIT-MAK550-PLC-11 (Makino-Aero-T550) | PM Interval: 1500 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-19-2.6: PN-MAK550-GASK-VIT-1913 / Kit KIT-MAK550-VAC-13 (Makino-Aero-T550) | PM Interval: 250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-19-2.7: PN-MAK550-BRG-CER-1901 / Kit KIT-MAK550-SPN-1 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-19-2.8: PN-MAK550-VLV-PROP-1903 / Kit KIT-MAK550-HYD-3 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-19-2.9: PN-MAK550-LIN-COIL-1905 / Kit KIT-MAK550-XAX-5 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-19-2.10: PN-MAK550-CYL-SEAL-1907 / Kit KIT-MAK550-ZAX-7 (Makino-Aero-T550) | PM Interval: 1250 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-19-2.11: PN-MAK550-BRK-DISC-1909 / Kit KIT-MAK550-CAX-9 (Makino-Aero-T550) | PM Interval: 1500 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-19-2.12: PN-MAK550-IGBT-MOD-1911 / Kit KIT-MAK550-PLC-11 (Makino-Aero-T550) | PM Interval: 250 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-19-2.13: PN-MAK550-GASK-VIT-1913 / Kit KIT-MAK550-VAC-13 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-19-2.14: PN-MAK550-BRG-CER-1901 / Kit KIT-MAK550-SPN-1 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-19-2.15: PN-MAK550-VLV-PROP-1903 / Kit KIT-MAK550-HYD-3 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK550-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T550 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T550. Verify part numbers before closing Work Orders.

BOM-19-3.1: PN-MAK550-PMP-PLG-1904 / Kit KIT-MAK550-CLN-4 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK550-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-19-3.2: PN-MAK550-CYL-SEAL-1907 / Kit KIT-MAK550-ZAX-7 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 83 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK550-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-19-3.3: PN-MAK550-GLASS-RD-1910 / Kit KIT-MAK550-OPT-10 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 110 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK550-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-19-3.4: PN-MAK550-GASK-VIT-1913 / Kit KIT-MAK550-VAC-13 (Makino-Aero-T550) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK550-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-19-3.5: PN-MAK550-CAM-GRIP-1902 / Kit KIT-MAK550-ATC-2 (Makino-Aero-T550) | PM Interval: 1500 Hrs | Torque: 38 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK550-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-19-3.6: PN-MAK550-LIN-COIL-1905 / Kit KIT-MAK550-XAX-5 (Makino-Aero-T550) | PM Interval: 250 Hrs | Torque: 65 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK550-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-19-3.7: PN-MAK550-TRQ-STAT-1908 / Kit KIT-MAK550-BAX-8 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 92 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK550-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-19-3.8: PN-MAK550-IGBT-MOD-1911 / Kit KIT-MAK550-PLC-11 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK550-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-19-3.9: PN-MAK550-HEX-CORE-1914 / Kit KIT-MAK550-THM-14 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK550-THM-14 within tolerance ±0.0040 mm TIR.

BOM-19-3.10: PN-MAK550-VLV-PROP-1903 / Kit KIT-MAK550-HYD-3 (Makino-Aero-T550) | PM Interval: 1250 Hrs | Torque: 47 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK550-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-19-3.11: PN-MAK550-BSW-NUT-1906 / Kit KIT-MAK550-YAX-6 (Makino-Aero-T550) | PM Interval: 1500 Hrs | Torque: 74 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK550-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-19-3.12: PN-MAK550-BRK-DISC-1909 / Kit KIT-MAK550-CAX-9 (Makino-Aero-T550) | PM Interval: 250 Hrs | Torque: 101 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK550-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-19-3.13: PN-MAK550-INJ-MET-1912 / Kit KIT-MAK550-LUB-12 (Makino-Aero-T550) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK550-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-19-3.14: PN-MAK550-BRG-CER-1901 / Kit KIT-MAK550-SPN-1 (Makino-Aero-T550) | PM Interval: 750 Hrs | Torque: 29 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK550-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-19-3.15: PN-MAK550-PMP-PLG-1904 / Kit KIT-MAK550-CLN-4 (Makino-Aero-T550) | PM Interval: 1000 Hrs | Torque: 56 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK550-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T600 (GROUP 1)

Certified Aerospace Bill of Materials for Makino-Aero-T600. Verify part numbers before closing Work Orders.

BOM-20-1.1: PN-MAK600-CAM-GRIP-2002 / Kit KIT-MAK600-ATC-2 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK600-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-20-1.2: PN-MAK600-VLV-PROP-2003 / Kit KIT-MAK600-HYD-3 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-20-1.3: PN-MAK600-PMP-PLG-2004 / Kit KIT-MAK600-CLN-4 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK600-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-20-1.4: PN-MAK600-LIN-COIL-2005 / Kit KIT-MAK600-XAX-5 (Makino-Aero-T600) | PM Interval: 1250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-20-1.5: PN-MAK600-BSW-NUT-2006 / Kit KIT-MAK600-YAX-6 (Makino-Aero-T600) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK600-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-20-1.6: PN-MAK600-CYL-SEAL-2007 / Kit KIT-MAK600-ZAX-7 (Makino-Aero-T600) | PM Interval: 250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-20-1.7: PN-MAK600-TRQ-STAT-2008 / Kit KIT-MAK600-BAX-8 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK600-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-20-1.8: PN-MAK600-BRK-DISC-2009 / Kit KIT-MAK600-CAX-9 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-20-1.9: PN-MAK600-GLASS-RD-2010 / Kit KIT-MAK600-OPT-10 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK600-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-20-1.10: PN-MAK600-IGBT-MOD-2011 / Kit KIT-MAK600-PLC-11 (Makino-Aero-T600) | PM Interval: 1250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-20-1.11: PN-MAK600-INJ-MET-2012 / Kit KIT-MAK600-LUB-12 (Makino-Aero-T600) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK600-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-20-1.12: PN-MAK600-GASK-VIT-2013 / Kit KIT-MAK600-VAC-13 (Makino-Aero-T600) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-20-1.13: PN-MAK600-HEX-CORE-2014 / Kit KIT-MAK600-THM-14 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK600-THM-14 within tolerance ±0.0060 mm TIR.

BOM-20-1.14: PN-MAK600-BRG-CER-2001 / Kit KIT-MAK600-SPN-1 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-20-1.15: PN-MAK600-CAM-GRIP-2002 / Kit KIT-MAK600-ATC-2 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK600-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T600 (GROUP 2)

Certified Aerospace Bill of Materials for Makino-Aero-T600. Verify part numbers before closing Work Orders.

BOM-20-2.1: PN-MAK600-VLV-PROP-2003 / Kit KIT-MAK600-HYD-3 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-20-2.2: PN-MAK600-LIN-COIL-2005 / Kit KIT-MAK600-XAX-5 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-20-2.3: PN-MAK600-CYL-SEAL-2007 / Kit KIT-MAK600-ZAX-7 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-20-2.4: PN-MAK600-BRK-DISC-2009 / Kit KIT-MAK600-CAX-9 (Makino-Aero-T600) | PM Interval: 1250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-20-2.5: PN-MAK600-IGBT-MOD-2011 / Kit KIT-MAK600-PLC-11 (Makino-Aero-T600) | PM Interval: 1500 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-20-2.6: PN-MAK600-GASK-VIT-2013 / Kit KIT-MAK600-VAC-13 (Makino-Aero-T600) | PM Interval: 250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-20-2.7: PN-MAK600-BRG-CER-2001 / Kit KIT-MAK600-SPN-1 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-20-2.8: PN-MAK600-VLV-PROP-2003 / Kit KIT-MAK600-HYD-3 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-20-2.9: PN-MAK600-LIN-COIL-2005 / Kit KIT-MAK600-XAX-5 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-20-2.10: PN-MAK600-CYL-SEAL-2007 / Kit KIT-MAK600-ZAX-7 (Makino-Aero-T600) | PM Interval: 1250 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-20-2.11: PN-MAK600-BRK-DISC-2009 / Kit KIT-MAK600-CAX-9 (Makino-Aero-T600) | PM Interval: 1500 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-20-2.12: PN-MAK600-IGBT-MOD-2011 / Kit KIT-MAK600-PLC-11 (Makino-Aero-T600) | PM Interval: 250 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-20-2.13: PN-MAK600-GASK-VIT-2013 / Kit KIT-MAK600-VAC-13 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-20-2.14: PN-MAK600-BRG-CER-2001 / Kit KIT-MAK600-SPN-1 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-20-2.15: PN-MAK600-VLV-PROP-2003 / Kit KIT-MAK600-HYD-3 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK600-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Makino-Aero-T600 (GROUP 3)

Certified Aerospace Bill of Materials for Makino-Aero-T600. Verify part numbers before closing Work Orders.

BOM-20-3.1: PN-MAK600-PMP-PLG-2004 / Kit KIT-MAK600-CLN-4 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK600-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-20-3.2: PN-MAK600-CYL-SEAL-2007 / Kit KIT-MAK600-ZAX-7 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 33 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAK600-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-20-3.3: PN-MAK600-GLASS-RD-2010 / Kit KIT-MAK600-OPT-10 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 60 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAK600-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-20-3.4: PN-MAK600-GASK-VIT-2013 / Kit KIT-MAK600-VAC-13 (Makino-Aero-T600) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAK600-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-20-3.5: PN-MAK600-CAM-GRIP-2002 / Kit KIT-MAK600-ATC-2 (Makino-Aero-T600) | PM Interval: 1500 Hrs | Torque: 178 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAK600-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-20-3.6: PN-MAK600-LIN-COIL-2005 / Kit KIT-MAK600-XAX-5 (Makino-Aero-T600) | PM Interval: 250 Hrs | Torque: 205 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAK600-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-20-3.7: PN-MAK600-TRQ-STAT-2008 / Kit KIT-MAK600-BAX-8 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 42 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAK600-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-20-3.8: PN-MAK600-IGBT-MOD-2011 / Kit KIT-MAK600-PLC-11 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAK600-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-20-3.9: PN-MAK600-HEX-CORE-2014 / Kit KIT-MAK600-THM-14 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAK600-THM-14 within tolerance ±0.0060 mm TIR.

BOM-20-3.10: PN-MAK600-VLV-PROP-2003 / Kit KIT-MAK600-HYD-3 (Makino-Aero-T600) | PM Interval: 1250 Hrs | Torque: 187 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAK600-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-20-3.11: PN-MAK600-BSW-NUT-2006 / Kit KIT-MAK600-YAX-6 (Makino-Aero-T600) | PM Interval: 1500 Hrs | Torque: 24 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAK600-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-20-3.12: PN-MAK600-BRK-DISC-2009 / Kit KIT-MAK600-CAX-9 (Makino-Aero-T600) | PM Interval: 250 Hrs | Torque: 51 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAK600-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-20-3.13: PN-MAK600-INJ-MET-2012 / Kit KIT-MAK600-LUB-12 (Makino-Aero-T600) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAK600-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-20-3.14: PN-MAK600-BRG-CER-2001 / Kit KIT-MAK600-SPN-1 (Makino-Aero-T600) | PM Interval: 750 Hrs | Torque: 169 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAK600-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-20-3.15: PN-MAK600-PMP-PLG-2004 / Kit KIT-MAK600-CLN-4 (Makino-Aero-T600) | PM Interval: 1000 Hrs | Torque: 196 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAK600-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i150 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i150. Verify part numbers before closing Work Orders.

BOM-21-1.1: PN-MAZ150-CAM-GRIP-2102 / Kit KIT-MAZ150-ATC-2 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ150-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-21-1.2: PN-MAZ150-VLV-PROP-2103 / Kit KIT-MAZ150-HYD-3 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-21-1.3: PN-MAZ150-PMP-PLG-2104 / Kit KIT-MAZ150-CLN-4 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ150-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-21-1.4: PN-MAZ150-LIN-COIL-2105 / Kit KIT-MAZ150-XAX-5 (Mazak-Integrex-i150) | PM Interval: 1250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-21-1.5: PN-MAZ150-BSW-NUT-2106 / Kit KIT-MAZ150-YAX-6 (Mazak-Integrex-i150) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ150-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-21-1.6: PN-MAZ150-CYL-SEAL-2107 / Kit KIT-MAZ150-ZAX-7 (Mazak-Integrex-i150) | PM Interval: 250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-21-1.7: PN-MAZ150-TRQ-STAT-2108 / Kit KIT-MAZ150-BAX-8 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ150-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-21-1.8: PN-MAZ150-BRK-DISC-2109 / Kit KIT-MAZ150-CAX-9 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-21-1.9: PN-MAZ150-GLASS-RD-2110 / Kit KIT-MAZ150-OPT-10 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ150-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-21-1.10: PN-MAZ150-IGBT-MOD-2111 / Kit KIT-MAZ150-PLC-11 (Mazak-Integrex-i150) | PM Interval: 1250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-21-1.11: PN-MAZ150-INJ-MET-2112 / Kit KIT-MAZ150-LUB-12 (Mazak-Integrex-i150) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ150-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-21-1.12: PN-MAZ150-GASK-VIT-2113 / Kit KIT-MAZ150-VAC-13 (Mazak-Integrex-i150) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-21-1.13: PN-MAZ150-HEX-CORE-2114 / Kit KIT-MAZ150-THM-14 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ150-THM-14 within tolerance ±0.0020 mm TIR.

BOM-21-1.14: PN-MAZ150-BRG-CER-2101 / Kit KIT-MAZ150-SPN-1 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-21-1.15: PN-MAZ150-CAM-GRIP-2102 / Kit KIT-MAZ150-ATC-2 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ150-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i150 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i150. Verify part numbers before closing Work Orders.

BOM-21-2.1: PN-MAZ150-VLV-PROP-2103 / Kit KIT-MAZ150-HYD-3 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-21-2.2: PN-MAZ150-LIN-COIL-2105 / Kit KIT-MAZ150-XAX-5 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-21-2.3: PN-MAZ150-CYL-SEAL-2107 / Kit KIT-MAZ150-ZAX-7 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-21-2.4: PN-MAZ150-BRK-DISC-2109 / Kit KIT-MAZ150-CAX-9 (Mazak-Integrex-i150) | PM Interval: 1250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-21-2.5: PN-MAZ150-IGBT-MOD-2111 / Kit KIT-MAZ150-PLC-11 (Mazak-Integrex-i150) | PM Interval: 1500 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-21-2.6: PN-MAZ150-GASK-VIT-2113 / Kit KIT-MAZ150-VAC-13 (Mazak-Integrex-i150) | PM Interval: 250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-21-2.7: PN-MAZ150-BRG-CER-2101 / Kit KIT-MAZ150-SPN-1 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-21-2.8: PN-MAZ150-VLV-PROP-2103 / Kit KIT-MAZ150-HYD-3 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-21-2.9: PN-MAZ150-LIN-COIL-2105 / Kit KIT-MAZ150-XAX-5 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-21-2.10: PN-MAZ150-CYL-SEAL-2107 / Kit KIT-MAZ150-ZAX-7 (Mazak-Integrex-i150) | PM Interval: 1250 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-21-2.11: PN-MAZ150-BRK-DISC-2109 / Kit KIT-MAZ150-CAX-9 (Mazak-Integrex-i150) | PM Interval: 1500 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-21-2.12: PN-MAZ150-IGBT-MOD-2111 / Kit KIT-MAZ150-PLC-11 (Mazak-Integrex-i150) | PM Interval: 250 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-21-2.13: PN-MAZ150-GASK-VIT-2113 / Kit KIT-MAZ150-VAC-13 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-21-2.14: PN-MAZ150-BRG-CER-2101 / Kit KIT-MAZ150-SPN-1 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-21-2.15: PN-MAZ150-VLV-PROP-2103 / Kit KIT-MAZ150-HYD-3 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ150-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i150 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i150. Verify part numbers before closing Work Orders.

BOM-21-3.1: PN-MAZ150-PMP-PLG-2104 / Kit KIT-MAZ150-CLN-4 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ150-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-21-3.2: PN-MAZ150-CYL-SEAL-2107 / Kit KIT-MAZ150-ZAX-7 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 173 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ150-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-21-3.3: PN-MAZ150-GLASS-RD-2110 / Kit KIT-MAZ150-OPT-10 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 200 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ150-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-21-3.4: PN-MAZ150-GASK-VIT-2113 / Kit KIT-MAZ150-VAC-13 (Mazak-Integrex-i150) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ150-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-21-3.5: PN-MAZ150-CAM-GRIP-2102 / Kit KIT-MAZ150-ATC-2 (Mazak-Integrex-i150) | PM Interval: 1500 Hrs | Torque: 128 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ150-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-21-3.6: PN-MAZ150-LIN-COIL-2105 / Kit KIT-MAZ150-XAX-5 (Mazak-Integrex-i150) | PM Interval: 250 Hrs | Torque: 155 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ150-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-21-3.7: PN-MAZ150-TRQ-STAT-2108 / Kit KIT-MAZ150-BAX-8 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 182 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ150-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-21-3.8: PN-MAZ150-IGBT-MOD-2111 / Kit KIT-MAZ150-PLC-11 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ150-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-21-3.9: PN-MAZ150-HEX-CORE-2114 / Kit KIT-MAZ150-THM-14 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ150-THM-14 within tolerance ±0.0020 mm TIR.

BOM-21-3.10: PN-MAZ150-VLV-PROP-2103 / Kit KIT-MAZ150-HYD-3 (Mazak-Integrex-i150) | PM Interval: 1250 Hrs | Torque: 137 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ150-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-21-3.11: PN-MAZ150-BSW-NUT-2106 / Kit KIT-MAZ150-YAX-6 (Mazak-Integrex-i150) | PM Interval: 1500 Hrs | Torque: 164 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ150-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-21-3.12: PN-MAZ150-BRK-DISC-2109 / Kit KIT-MAZ150-CAX-9 (Mazak-Integrex-i150) | PM Interval: 250 Hrs | Torque: 191 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ150-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-21-3.13: PN-MAZ150-INJ-MET-2112 / Kit KIT-MAZ150-LUB-12 (Mazak-Integrex-i150) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ150-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-21-3.14: PN-MAZ150-BRG-CER-2101 / Kit KIT-MAZ150-SPN-1 (Mazak-Integrex-i150) | PM Interval: 750 Hrs | Torque: 119 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ150-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-21-3.15: PN-MAZ150-PMP-PLG-2104 / Kit KIT-MAZ150-CLN-4 (Mazak-Integrex-i150) | PM Interval: 1000 Hrs | Torque: 146 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ150-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i200 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i200. Verify part numbers before closing Work Orders.

BOM-22-1.1: PN-MAZ200-CAM-GRIP-2202 / Kit KIT-MAZ200-ATC-2 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ200-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-22-1.2: PN-MAZ200-VLV-PROP-2203 / Kit KIT-MAZ200-HYD-3 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-22-1.3: PN-MAZ200-PMP-PLG-2204 / Kit KIT-MAZ200-CLN-4 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ200-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-22-1.4: PN-MAZ200-LIN-COIL-2205 / Kit KIT-MAZ200-XAX-5 (Mazak-Integrex-i200) | PM Interval: 1250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-22-1.5: PN-MAZ200-BSW-NUT-2206 / Kit KIT-MAZ200-YAX-6 (Mazak-Integrex-i200) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ200-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-22-1.6: PN-MAZ200-CYL-SEAL-2207 / Kit KIT-MAZ200-ZAX-7 (Mazak-Integrex-i200) | PM Interval: 250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-22-1.7: PN-MAZ200-TRQ-STAT-2208 / Kit KIT-MAZ200-BAX-8 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ200-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-22-1.8: PN-MAZ200-BRK-DISC-2209 / Kit KIT-MAZ200-CAX-9 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-22-1.9: PN-MAZ200-GLASS-RD-2210 / Kit KIT-MAZ200-OPT-10 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ200-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-22-1.10: PN-MAZ200-IGBT-MOD-2211 / Kit KIT-MAZ200-PLC-11 (Mazak-Integrex-i200) | PM Interval: 1250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-22-1.11: PN-MAZ200-INJ-MET-2212 / Kit KIT-MAZ200-LUB-12 (Mazak-Integrex-i200) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ200-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-22-1.12: PN-MAZ200-GASK-VIT-2213 / Kit KIT-MAZ200-VAC-13 (Mazak-Integrex-i200) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-22-1.13: PN-MAZ200-HEX-CORE-2214 / Kit KIT-MAZ200-THM-14 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ200-THM-14 within tolerance ±0.0040 mm TIR.

BOM-22-1.14: PN-MAZ200-BRG-CER-2201 / Kit KIT-MAZ200-SPN-1 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-22-1.15: PN-MAZ200-CAM-GRIP-2202 / Kit KIT-MAZ200-ATC-2 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ200-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i200 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i200. Verify part numbers before closing Work Orders.

BOM-22-2.1: PN-MAZ200-VLV-PROP-2203 / Kit KIT-MAZ200-HYD-3 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-22-2.2: PN-MAZ200-LIN-COIL-2205 / Kit KIT-MAZ200-XAX-5 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-22-2.3: PN-MAZ200-CYL-SEAL-2207 / Kit KIT-MAZ200-ZAX-7 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-22-2.4: PN-MAZ200-BRK-DISC-2209 / Kit KIT-MAZ200-CAX-9 (Mazak-Integrex-i200) | PM Interval: 1250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-22-2.5: PN-MAZ200-IGBT-MOD-2211 / Kit KIT-MAZ200-PLC-11 (Mazak-Integrex-i200) | PM Interval: 1500 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-22-2.6: PN-MAZ200-GASK-VIT-2213 / Kit KIT-MAZ200-VAC-13 (Mazak-Integrex-i200) | PM Interval: 250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-22-2.7: PN-MAZ200-BRG-CER-2201 / Kit KIT-MAZ200-SPN-1 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-22-2.8: PN-MAZ200-VLV-PROP-2203 / Kit KIT-MAZ200-HYD-3 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-22-2.9: PN-MAZ200-LIN-COIL-2205 / Kit KIT-MAZ200-XAX-5 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-22-2.10: PN-MAZ200-CYL-SEAL-2207 / Kit KIT-MAZ200-ZAX-7 (Mazak-Integrex-i200) | PM Interval: 1250 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-22-2.11: PN-MAZ200-BRK-DISC-2209 / Kit KIT-MAZ200-CAX-9 (Mazak-Integrex-i200) | PM Interval: 1500 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-22-2.12: PN-MAZ200-IGBT-MOD-2211 / Kit KIT-MAZ200-PLC-11 (Mazak-Integrex-i200) | PM Interval: 250 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-22-2.13: PN-MAZ200-GASK-VIT-2213 / Kit KIT-MAZ200-VAC-13 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-22-2.14: PN-MAZ200-BRG-CER-2201 / Kit KIT-MAZ200-SPN-1 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-22-2.15: PN-MAZ200-VLV-PROP-2203 / Kit KIT-MAZ200-HYD-3 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ200-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i200 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i200. Verify part numbers before closing Work Orders.

BOM-22-3.1: PN-MAZ200-PMP-PLG-2204 / Kit KIT-MAZ200-CLN-4 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ200-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-22-3.2: PN-MAZ200-CYL-SEAL-2207 / Kit KIT-MAZ200-ZAX-7 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 123 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ200-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-22-3.3: PN-MAZ200-GLASS-RD-2210 / Kit KIT-MAZ200-OPT-10 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 150 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ200-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-22-3.4: PN-MAZ200-GASK-VIT-2213 / Kit KIT-MAZ200-VAC-13 (Mazak-Integrex-i200) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ200-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-22-3.5: PN-MAZ200-CAM-GRIP-2202 / Kit KIT-MAZ200-ATC-2 (Mazak-Integrex-i200) | PM Interval: 1500 Hrs | Torque: 78 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ200-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-22-3.6: PN-MAZ200-LIN-COIL-2205 / Kit KIT-MAZ200-XAX-5 (Mazak-Integrex-i200) | PM Interval: 250 Hrs | Torque: 105 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ200-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-22-3.7: PN-MAZ200-TRQ-STAT-2208 / Kit KIT-MAZ200-BAX-8 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 132 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ200-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-22-3.8: PN-MAZ200-IGBT-MOD-2211 / Kit KIT-MAZ200-PLC-11 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ200-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-22-3.9: PN-MAZ200-HEX-CORE-2214 / Kit KIT-MAZ200-THM-14 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ200-THM-14 within tolerance ±0.0040 mm TIR.

BOM-22-3.10: PN-MAZ200-VLV-PROP-2203 / Kit KIT-MAZ200-HYD-3 (Mazak-Integrex-i200) | PM Interval: 1250 Hrs | Torque: 87 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ200-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-22-3.11: PN-MAZ200-BSW-NUT-2206 / Kit KIT-MAZ200-YAX-6 (Mazak-Integrex-i200) | PM Interval: 1500 Hrs | Torque: 114 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ200-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-22-3.12: PN-MAZ200-BRK-DISC-2209 / Kit KIT-MAZ200-CAX-9 (Mazak-Integrex-i200) | PM Interval: 250 Hrs | Torque: 141 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ200-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-22-3.13: PN-MAZ200-INJ-MET-2212 / Kit KIT-MAZ200-LUB-12 (Mazak-Integrex-i200) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ200-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-22-3.14: PN-MAZ200-BRG-CER-2201 / Kit KIT-MAZ200-SPN-1 (Mazak-Integrex-i200) | PM Interval: 750 Hrs | Torque: 69 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ200-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-22-3.15: PN-MAZ200-PMP-PLG-2204 / Kit KIT-MAZ200-CLN-4 (Mazak-Integrex-i200) | PM Interval: 1000 Hrs | Torque: 96 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ200-CLN-4 within tolerance ±0.0050 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i250 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i250. Verify part numbers before closing Work Orders.

BOM-23-1.1: PN-MAZ250-CAM-GRIP-2302 / Kit KIT-MAZ250-ATC-2 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ250-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-23-1.2: PN-MAZ250-VLV-PROP-2303 / Kit KIT-MAZ250-HYD-3 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-23-1.3: PN-MAZ250-PMP-PLG-2304 / Kit KIT-MAZ250-CLN-4 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ250-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-23-1.4: PN-MAZ250-LIN-COIL-2305 / Kit KIT-MAZ250-XAX-5 (Mazak-Integrex-i250) | PM Interval: 1250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-23-1.5: PN-MAZ250-BSW-NUT-2306 / Kit KIT-MAZ250-YAX-6 (Mazak-Integrex-i250) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ250-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-23-1.6: PN-MAZ250-CYL-SEAL-2307 / Kit KIT-MAZ250-ZAX-7 (Mazak-Integrex-i250) | PM Interval: 250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-23-1.7: PN-MAZ250-TRQ-STAT-2308 / Kit KIT-MAZ250-BAX-8 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ250-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-23-1.8: PN-MAZ250-BRK-DISC-2309 / Kit KIT-MAZ250-CAX-9 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-23-1.9: PN-MAZ250-GLASS-RD-2310 / Kit KIT-MAZ250-OPT-10 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ250-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-23-1.10: PN-MAZ250-IGBT-MOD-2311 / Kit KIT-MAZ250-PLC-11 (Mazak-Integrex-i250) | PM Interval: 1250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-23-1.11: PN-MAZ250-INJ-MET-2312 / Kit KIT-MAZ250-LUB-12 (Mazak-Integrex-i250) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ250-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-23-1.12: PN-MAZ250-GASK-VIT-2313 / Kit KIT-MAZ250-VAC-13 (Mazak-Integrex-i250) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-23-1.13: PN-MAZ250-HEX-CORE-2314 / Kit KIT-MAZ250-THM-14 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ250-THM-14 within tolerance ±0.0060 mm TIR.

BOM-23-1.14: PN-MAZ250-BRG-CER-2301 / Kit KIT-MAZ250-SPN-1 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-23-1.15: PN-MAZ250-CAM-GRIP-2302 / Kit KIT-MAZ250-ATC-2 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ250-ATC-2 within tolerance ±0.0060 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i250 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i250. Verify part numbers before closing Work Orders.

BOM-23-2.1: PN-MAZ250-VLV-PROP-2303 / Kit KIT-MAZ250-HYD-3 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-23-2.2: PN-MAZ250-LIN-COIL-2305 / Kit KIT-MAZ250-XAX-5 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-23-2.3: PN-MAZ250-CYL-SEAL-2307 / Kit KIT-MAZ250-ZAX-7 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-23-2.4: PN-MAZ250-BRK-DISC-2309 / Kit KIT-MAZ250-CAX-9 (Mazak-Integrex-i250) | PM Interval: 1250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-23-2.5: PN-MAZ250-IGBT-MOD-2311 / Kit KIT-MAZ250-PLC-11 (Mazak-Integrex-i250) | PM Interval: 1500 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-23-2.6: PN-MAZ250-GASK-VIT-2313 / Kit KIT-MAZ250-VAC-13 (Mazak-Integrex-i250) | PM Interval: 250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-23-2.7: PN-MAZ250-BRG-CER-2301 / Kit KIT-MAZ250-SPN-1 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-23-2.8: PN-MAZ250-VLV-PROP-2303 / Kit KIT-MAZ250-HYD-3 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-23-2.9: PN-MAZ250-LIN-COIL-2305 / Kit KIT-MAZ250-XAX-5 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-23-2.10: PN-MAZ250-CYL-SEAL-2307 / Kit KIT-MAZ250-ZAX-7 (Mazak-Integrex-i250) | PM Interval: 1250 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-23-2.11: PN-MAZ250-BRK-DISC-2309 / Kit KIT-MAZ250-CAX-9 (Mazak-Integrex-i250) | PM Interval: 1500 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-23-2.12: PN-MAZ250-IGBT-MOD-2311 / Kit KIT-MAZ250-PLC-11 (Mazak-Integrex-i250) | PM Interval: 250 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-23-2.13: PN-MAZ250-GASK-VIT-2313 / Kit KIT-MAZ250-VAC-13 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-23-2.14: PN-MAZ250-BRG-CER-2301 / Kit KIT-MAZ250-SPN-1 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-23-2.15: PN-MAZ250-VLV-PROP-2303 / Kit KIT-MAZ250-HYD-3 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ250-HYD-3 within tolerance ±0.0065 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i250 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i250. Verify part numbers before closing Work Orders.

BOM-23-3.1: PN-MAZ250-PMP-PLG-2304 / Kit KIT-MAZ250-CLN-4 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ250-CLN-4 within tolerance ±0.0010 mm TIR.

BOM-23-3.2: PN-MAZ250-CYL-SEAL-2307 / Kit KIT-MAZ250-ZAX-7 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 73 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ250-ZAX-7 within tolerance ±0.0025 mm TIR.

BOM-23-3.3: PN-MAZ250-GLASS-RD-2310 / Kit KIT-MAZ250-OPT-10 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 100 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ250-OPT-10 within tolerance ±0.0040 mm TIR.

BOM-23-3.4: PN-MAZ250-GASK-VIT-2313 / Kit KIT-MAZ250-VAC-13 (Mazak-Integrex-i250) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ250-VAC-13 within tolerance ±0.0055 mm TIR.

BOM-23-3.5: PN-MAZ250-CAM-GRIP-2302 / Kit KIT-MAZ250-ATC-2 (Mazak-Integrex-i250) | PM Interval: 1500 Hrs | Torque: 28 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ250-ATC-2 within tolerance ±0.0060 mm TIR.

BOM-23-3.6: PN-MAZ250-LIN-COIL-2305 / Kit KIT-MAZ250-XAX-5 (Mazak-Integrex-i250) | PM Interval: 250 Hrs | Torque: 55 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ250-XAX-5 within tolerance ±0.0015 mm TIR.

BOM-23-3.7: PN-MAZ250-TRQ-STAT-2308 / Kit KIT-MAZ250-BAX-8 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 82 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ250-BAX-8 within tolerance ±0.0030 mm TIR.

BOM-23-3.8: PN-MAZ250-IGBT-MOD-2311 / Kit KIT-MAZ250-PLC-11 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ250-PLC-11 within tolerance ±0.0045 mm TIR.

BOM-23-3.9: PN-MAZ250-HEX-CORE-2314 / Kit KIT-MAZ250-THM-14 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ250-THM-14 within tolerance ±0.0060 mm TIR.

BOM-23-3.10: PN-MAZ250-VLV-PROP-2303 / Kit KIT-MAZ250-HYD-3 (Mazak-Integrex-i250) | PM Interval: 1250 Hrs | Torque: 37 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ250-HYD-3 within tolerance ±0.0065 mm TIR.

BOM-23-3.11: PN-MAZ250-BSW-NUT-2306 / Kit KIT-MAZ250-YAX-6 (Mazak-Integrex-i250) | PM Interval: 1500 Hrs | Torque: 64 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ250-YAX-6 within tolerance ±0.0020 mm TIR.

BOM-23-3.12: PN-MAZ250-BRK-DISC-2309 / Kit KIT-MAZ250-CAX-9 (Mazak-Integrex-i250) | PM Interval: 250 Hrs | Torque: 91 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ250-CAX-9 within tolerance ±0.0035 mm TIR.

BOM-23-3.13: PN-MAZ250-INJ-MET-2312 / Kit KIT-MAZ250-LUB-12 (Mazak-Integrex-i250) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ250-LUB-12 within tolerance ±0.0050 mm TIR.

BOM-23-3.14: PN-MAZ250-BRG-CER-2301 / Kit KIT-MAZ250-SPN-1 (Mazak-Integrex-i250) | PM Interval: 750 Hrs | Torque: 209 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ250-SPN-1 within tolerance ±0.0055 mm TIR.

BOM-23-3.15: PN-MAZ250-PMP-PLG-2304 / Kit KIT-MAZ250-CLN-4 (Mazak-Integrex-i250) | PM Interval: 1000 Hrs | Torque: 46 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ250-CLN-4 within tolerance ±0.0010 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i300 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i300. Verify part numbers before closing Work Orders.

BOM-24-1.1: PN-MAZ300-CAM-GRIP-2402 / Kit KIT-MAZ300-ATC-2 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ300-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-24-1.2: PN-MAZ300-VLV-PROP-2403 / Kit KIT-MAZ300-HYD-3 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-24-1.3: PN-MAZ300-PMP-PLG-2404 / Kit KIT-MAZ300-CLN-4 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ300-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-24-1.4: PN-MAZ300-LIN-COIL-2405 / Kit KIT-MAZ300-XAX-5 (Mazak-Integrex-i300) | PM Interval: 1250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-24-1.5: PN-MAZ300-BSW-NUT-2406 / Kit KIT-MAZ300-YAX-6 (Mazak-Integrex-i300) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ300-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-24-1.6: PN-MAZ300-CYL-SEAL-2407 / Kit KIT-MAZ300-ZAX-7 (Mazak-Integrex-i300) | PM Interval: 250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-24-1.7: PN-MAZ300-TRQ-STAT-2408 / Kit KIT-MAZ300-BAX-8 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ300-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-24-1.8: PN-MAZ300-BRK-DISC-2409 / Kit KIT-MAZ300-CAX-9 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-24-1.9: PN-MAZ300-GLASS-RD-2410 / Kit KIT-MAZ300-OPT-10 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ300-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-24-1.10: PN-MAZ300-IGBT-MOD-2411 / Kit KIT-MAZ300-PLC-11 (Mazak-Integrex-i300) | PM Interval: 1250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-24-1.11: PN-MAZ300-INJ-MET-2412 / Kit KIT-MAZ300-LUB-12 (Mazak-Integrex-i300) | PM Interval: 1500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ300-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-24-1.12: PN-MAZ300-GASK-VIT-2413 / Kit KIT-MAZ300-VAC-13 (Mazak-Integrex-i300) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-24-1.13: PN-MAZ300-HEX-CORE-2414 / Kit KIT-MAZ300-THM-14 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ300-THM-14 within tolerance ±0.0020 mm TIR.

BOM-24-1.14: PN-MAZ300-BRG-CER-2401 / Kit KIT-MAZ300-SPN-1 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-24-1.15: PN-MAZ300-CAM-GRIP-2402 / Kit KIT-MAZ300-ATC-2 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ300-ATC-2 within tolerance ±0.0020 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i300 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i300. Verify part numbers before closing Work Orders.

BOM-24-2.1: PN-MAZ300-VLV-PROP-2403 / Kit KIT-MAZ300-HYD-3 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-24-2.2: PN-MAZ300-LIN-COIL-2405 / Kit KIT-MAZ300-XAX-5 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-24-2.3: PN-MAZ300-CYL-SEAL-2407 / Kit KIT-MAZ300-ZAX-7 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-24-2.4: PN-MAZ300-BRK-DISC-2409 / Kit KIT-MAZ300-CAX-9 (Mazak-Integrex-i300) | PM Interval: 1250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-24-2.5: PN-MAZ300-IGBT-MOD-2411 / Kit KIT-MAZ300-PLC-11 (Mazak-Integrex-i300) | PM Interval: 1500 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-24-2.6: PN-MAZ300-GASK-VIT-2413 / Kit KIT-MAZ300-VAC-13 (Mazak-Integrex-i300) | PM Interval: 250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-24-2.7: PN-MAZ300-BRG-CER-2401 / Kit KIT-MAZ300-SPN-1 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-24-2.8: PN-MAZ300-VLV-PROP-2403 / Kit KIT-MAZ300-HYD-3 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-24-2.9: PN-MAZ300-LIN-COIL-2405 / Kit KIT-MAZ300-XAX-5 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-24-2.10: PN-MAZ300-CYL-SEAL-2407 / Kit KIT-MAZ300-ZAX-7 (Mazak-Integrex-i300) | PM Interval: 1250 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-24-2.11: PN-MAZ300-BRK-DISC-2409 / Kit KIT-MAZ300-CAX-9 (Mazak-Integrex-i300) | PM Interval: 1500 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-24-2.12: PN-MAZ300-IGBT-MOD-2411 / Kit KIT-MAZ300-PLC-11 (Mazak-Integrex-i300) | PM Interval: 250 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-24-2.13: PN-MAZ300-GASK-VIT-2413 / Kit KIT-MAZ300-VAC-13 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-24-2.14: PN-MAZ300-BRG-CER-2401 / Kit KIT-MAZ300-SPN-1 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-24-2.15: PN-MAZ300-VLV-PROP-2403 / Kit KIT-MAZ300-HYD-3 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ300-HYD-3 within tolerance ±0.0025 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i300 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i300. Verify part numbers before closing Work Orders.

BOM-24-3.1: PN-MAZ300-PMP-PLG-2404 / Kit KIT-MAZ300-CLN-4 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ300-CLN-4 within tolerance ±0.0030 mm TIR.

BOM-24-3.2: PN-MAZ300-CYL-SEAL-2407 / Kit KIT-MAZ300-ZAX-7 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 23 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ300-ZAX-7 within tolerance ±0.0045 mm TIR.

BOM-24-3.3: PN-MAZ300-GLASS-RD-2410 / Kit KIT-MAZ300-OPT-10 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 50 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ300-OPT-10 within tolerance ±0.0060 mm TIR.

BOM-24-3.4: PN-MAZ300-GASK-VIT-2413 / Kit KIT-MAZ300-VAC-13 (Mazak-Integrex-i300) | PM Interval: 1250 Hrs | Torque: 77 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ300-VAC-13 within tolerance ±0.0015 mm TIR.

BOM-24-3.5: PN-MAZ300-CAM-GRIP-2402 / Kit KIT-MAZ300-ATC-2 (Mazak-Integrex-i300) | PM Interval: 1500 Hrs | Torque: 168 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ300-ATC-2 within tolerance ±0.0020 mm TIR.

BOM-24-3.6: PN-MAZ300-LIN-COIL-2405 / Kit KIT-MAZ300-XAX-5 (Mazak-Integrex-i300) | PM Interval: 250 Hrs | Torque: 195 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ300-XAX-5 within tolerance ±0.0035 mm TIR.

BOM-24-3.7: PN-MAZ300-TRQ-STAT-2408 / Kit KIT-MAZ300-BAX-8 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 32 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ300-BAX-8 within tolerance ±0.0050 mm TIR.

BOM-24-3.8: PN-MAZ300-IGBT-MOD-2411 / Kit KIT-MAZ300-PLC-11 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 59 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ300-PLC-11 within tolerance ±0.0065 mm TIR.

BOM-24-3.9: PN-MAZ300-HEX-CORE-2414 / Kit KIT-MAZ300-THM-14 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 86 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ300-THM-14 within tolerance ±0.0020 mm TIR.

BOM-24-3.10: PN-MAZ300-VLV-PROP-2403 / Kit KIT-MAZ300-HYD-3 (Mazak-Integrex-i300) | PM Interval: 1250 Hrs | Torque: 177 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ300-HYD-3 within tolerance ±0.0025 mm TIR.

BOM-24-3.11: PN-MAZ300-BSW-NUT-2406 / Kit KIT-MAZ300-YAX-6 (Mazak-Integrex-i300) | PM Interval: 1500 Hrs | Torque: 204 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ300-YAX-6 within tolerance ±0.0040 mm TIR.

BOM-24-3.12: PN-MAZ300-BRK-DISC-2409 / Kit KIT-MAZ300-CAX-9 (Mazak-Integrex-i300) | PM Interval: 250 Hrs | Torque: 41 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ300-CAX-9 within tolerance ±0.0055 mm TIR.

BOM-24-3.13: PN-MAZ300-INJ-MET-2412 / Kit KIT-MAZ300-LUB-12 (Mazak-Integrex-i300) | PM Interval: 500 Hrs | Torque: 68 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ300-LUB-12 within tolerance ±0.0010 mm TIR.

BOM-24-3.14: PN-MAZ300-BRG-CER-2401 / Kit KIT-MAZ300-SPN-1 (Mazak-Integrex-i300) | PM Interval: 750 Hrs | Torque: 159 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ300-SPN-1 within tolerance ±0.0015 mm TIR.

BOM-24-3.15: PN-MAZ300-PMP-PLG-2404 / Kit KIT-MAZ300-CLN-4 (Mazak-Integrex-i300) | PM Interval: 1000 Hrs | Torque: 186 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ300-CLN-4 within tolerance ±0.0030 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i350 (GROUP 1)

Certified Aerospace Bill of Materials for Mazak-Integrex-i350. Verify part numbers before closing Work Orders.

BOM-25-1.1: PN-MAZ350-CAM-GRIP-2502 / Kit KIT-MAZ350-ATC-2 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ350-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-25-1.2: PN-MAZ350-VLV-PROP-2503 / Kit KIT-MAZ350-HYD-3 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-25-1.3: PN-MAZ350-PMP-PLG-2504 / Kit KIT-MAZ350-CLN-4 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ350-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-25-1.4: PN-MAZ350-LIN-COIL-2505 / Kit KIT-MAZ350-XAX-5 (Mazak-Integrex-i350) | PM Interval: 1250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-25-1.5: PN-MAZ350-BSW-NUT-2506 / Kit KIT-MAZ350-YAX-6 (Mazak-Integrex-i350) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ350-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-25-1.6: PN-MAZ350-CYL-SEAL-2507 / Kit KIT-MAZ350-ZAX-7 (Mazak-Integrex-i350) | PM Interval: 250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-25-1.7: PN-MAZ350-TRQ-STAT-2508 / Kit KIT-MAZ350-BAX-8 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ350-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-25-1.8: PN-MAZ350-BRK-DISC-2509 / Kit KIT-MAZ350-CAX-9 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-25-1.9: PN-MAZ350-GLASS-RD-2510 / Kit KIT-MAZ350-OPT-10 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ350-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-25-1.10: PN-MAZ350-IGBT-MOD-2511 / Kit KIT-MAZ350-PLC-11 (Mazak-Integrex-i350) | PM Interval: 1250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-25-1.11: PN-MAZ350-INJ-MET-2512 / Kit KIT-MAZ350-LUB-12 (Mazak-Integrex-i350) | PM Interval: 1500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ350-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-25-1.12: PN-MAZ350-GASK-VIT-2513 / Kit KIT-MAZ350-VAC-13 (Mazak-Integrex-i350) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-25-1.13: PN-MAZ350-HEX-CORE-2514 / Kit KIT-MAZ350-THM-14 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ350-THM-14 within tolerance ±0.0040 mm TIR.

BOM-25-1.14: PN-MAZ350-BRG-CER-2501 / Kit KIT-MAZ350-SPN-1 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-25-1.15: PN-MAZ350-CAM-GRIP-2502 / Kit KIT-MAZ350-ATC-2 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ350-ATC-2 within tolerance ±0.0040 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i350 (GROUP 2)

Certified Aerospace Bill of Materials for Mazak-Integrex-i350. Verify part numbers before closing Work Orders.

BOM-25-2.1: PN-MAZ350-VLV-PROP-2503 / Kit KIT-MAZ350-HYD-3 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-25-2.2: PN-MAZ350-LIN-COIL-2505 / Kit KIT-MAZ350-XAX-5 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-25-2.3: PN-MAZ350-CYL-SEAL-2507 / Kit KIT-MAZ350-ZAX-7 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-25-2.4: PN-MAZ350-BRK-DISC-2509 / Kit KIT-MAZ350-CAX-9 (Mazak-Integrex-i350) | PM Interval: 1250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-25-2.5: PN-MAZ350-IGBT-MOD-2511 / Kit KIT-MAZ350-PLC-11 (Mazak-Integrex-i350) | PM Interval: 1500 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-25-2.6: PN-MAZ350-GASK-VIT-2513 / Kit KIT-MAZ350-VAC-13 (Mazak-Integrex-i350) | PM Interval: 250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-25-2.7: PN-MAZ350-BRG-CER-2501 / Kit KIT-MAZ350-SPN-1 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-25-2.8: PN-MAZ350-VLV-PROP-2503 / Kit KIT-MAZ350-HYD-3 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-25-2.9: PN-MAZ350-LIN-COIL-2505 / Kit KIT-MAZ350-XAX-5 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-25-2.10: PN-MAZ350-CYL-SEAL-2507 / Kit KIT-MAZ350-ZAX-7 (Mazak-Integrex-i350) | PM Interval: 1250 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-25-2.11: PN-MAZ350-BRK-DISC-2509 / Kit KIT-MAZ350-CAX-9 (Mazak-Integrex-i350) | PM Interval: 1500 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-25-2.12: PN-MAZ350-IGBT-MOD-2511 / Kit KIT-MAZ350-PLC-11 (Mazak-Integrex-i350) | PM Interval: 250 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-25-2.13: PN-MAZ350-GASK-VIT-2513 / Kit KIT-MAZ350-VAC-13 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-25-2.14: PN-MAZ350-BRG-CER-2501 / Kit KIT-MAZ350-SPN-1 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-25-2.15: PN-MAZ350-VLV-PROP-2503 / Kit KIT-MAZ350-HYD-3 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ350-HYD-3 within tolerance ±0.0045 mm TIR.

SPARE PARTS BOM & PREVENTIVE MAINTENANCE — Mazak-Integrex-i350 (GROUP 3)

Certified Aerospace Bill of Materials for Mazak-Integrex-i350. Verify part numbers before closing Work Orders.

BOM-25-3.1: PN-MAZ350-PMP-PLG-2504 / Kit KIT-MAZ350-CLN-4 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ350-CLN-4 within tolerance ±0.0050 mm TIR.

BOM-25-3.2: PN-MAZ350-CYL-SEAL-2507 / Kit KIT-MAZ350-ZAX-7 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 163 Nm ± 2%
Assembly: Z-Axis Nitrogen Counterbalance & Ram. Required for remediating ERR-MAZ350-ZAX-7 within tolerance ±0.0065 mm TIR.

BOM-25-3.3: PN-MAZ350-GLASS-RD-2510 / Kit KIT-MAZ350-OPT-10 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 190 Nm ± 2%
Assembly: Pressurized Optical Glass Linear Scales. Required for remediating ERR-MAZ350-OPT-10 within tolerance ±0.0020 mm TIR.

BOM-25-3.4: PN-MAZ350-GASK-VIT-2513 / Kit KIT-MAZ350-VAC-13 (Mazak-Integrex-i350) | PM Interval: 1250 Hrs | Torque: 27 Nm ± 2%
Assembly: Workpiece Vacuum Fixture & Seal Manifold. Required for remediating ERR-MAZ350-VAC-13 within tolerance ±0.0035 mm TIR.

BOM-25-3.5: PN-MAZ350-CAM-GRIP-2502 / Kit KIT-MAZ350-ATC-2 (Mazak-Integrex-i350) | PM Interval: 1500 Hrs | Torque: 118 Nm ± 2%
Assembly: Servo Tool Magazine & Gripper Arm. Required for remediating ERR-MAZ350-ATC-2 within tolerance ±0.0040 mm TIR.

BOM-25-3.6: PN-MAZ350-LIN-COIL-2505 / Kit KIT-MAZ350-XAX-5 (Mazak-Integrex-i350) | PM Interval: 250 Hrs | Torque: 145 Nm ± 2%
Assembly: X-Axis Linear Motor & Hydrostatic Guideway. Required for remediating ERR-MAZ350-XAX-5 within tolerance ±0.0055 mm TIR.

BOM-25-3.7: PN-MAZ350-TRQ-STAT-2508 / Kit KIT-MAZ350-BAX-8 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 172 Nm ± 2%
Assembly: B-Axis Direct-Drive Torque Swivel Head. Required for remediating ERR-MAZ350-BAX-8 within tolerance ±0.0010 mm TIR.

BOM-25-3.8: PN-MAZ350-IGBT-MOD-2511 / Kit KIT-MAZ350-PLC-11 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 199 Nm ± 2%
Assembly: Profinet Safety I/O & IGBT Servo Drive. Required for remediating ERR-MAZ350-PLC-11 within tolerance ±0.0025 mm TIR.

BOM-25-3.9: PN-MAZ350-HEX-CORE-2514 / Kit KIT-MAZ350-THM-14 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 36 Nm ± 2%
Assembly: Active Glycol Chiller & Frame Compensation. Required for remediating ERR-MAZ350-THM-14 within tolerance ±0.0040 mm TIR.

BOM-25-3.10: PN-MAZ350-VLV-PROP-2503 / Kit KIT-MAZ350-HYD-3 (Mazak-Integrex-i350) | PM Interval: 1250 Hrs | Torque: 127 Nm ± 2%
Assembly: Proportional Hydraulic Clamping Manifold. Required for remediating ERR-MAZ350-HYD-3 within tolerance ±0.0045 mm TIR.

BOM-25-3.11: PN-MAZ350-BSW-NUT-2506 / Kit KIT-MAZ350-YAX-6 (Mazak-Integrex-i350) | PM Interval: 1500 Hrs | Torque: 154 Nm ± 2%
Assembly: Y-Axis Master-Slave Gantry Ballscrew. Required for remediating ERR-MAZ350-YAX-6 within tolerance ±0.0060 mm TIR.

BOM-25-3.12: PN-MAZ350-BRK-DISC-2509 / Kit KIT-MAZ350-CAX-9 (Mazak-Integrex-i350) | PM Interval: 250 Hrs | Torque: 181 Nm ± 2%
Assembly: C-Axis Hydro-Mechanical Rotary Table. Required for remediating ERR-MAZ350-CAX-9 within tolerance ±0.0015 mm TIR.

BOM-25-3.13: PN-MAZ350-INJ-MET-2512 / Kit KIT-MAZ350-LUB-12 (Mazak-Integrex-i350) | PM Interval: 500 Hrs | Torque: 208 Nm ± 2%
Assembly: Air-Oil Spindle Micro-Dosage Lubrication. Required for remediating ERR-MAZ350-LUB-12 within tolerance ±0.0030 mm TIR.

BOM-25-3.14: PN-MAZ350-BRG-CER-2501 / Kit KIT-MAZ350-SPN-1 (Mazak-Integrex-i350) | PM Interval: 750 Hrs | Torque: 109 Nm ± 2%
Assembly: Electro-Spindle & Drawbar Assembly. Required for remediating ERR-MAZ350-SPN-1 within tolerance ±0.0035 mm TIR.

BOM-25-3.15: PN-MAZ350-PMP-PLG-2504 / Kit KIT-MAZ350-CLN-4 (Mazak-Integrex-i350) | PM Interval: 1000 Hrs | Torque: 136 Nm ± 2%
Assembly: 120-Bar Through-Tool Cryo Coolant Loop. Required for remediating ERR-MAZ350-CLN-4 within tolerance ±0.0050 mm TIR.